

Model-based Battery Management Systems

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Li-ion batteries

PRESS RELEASE

9 October 2019

The Nobel Prize in Chemistry 2019

The Royal Swedish Academy of Sciences has decided to award the Nobel Prize in Chemistry 2019 to

John B. Goodenough

The University of Texas at Austin, USA

M. Stanley Whittingham

Binghamton University, State University of
New York, USA

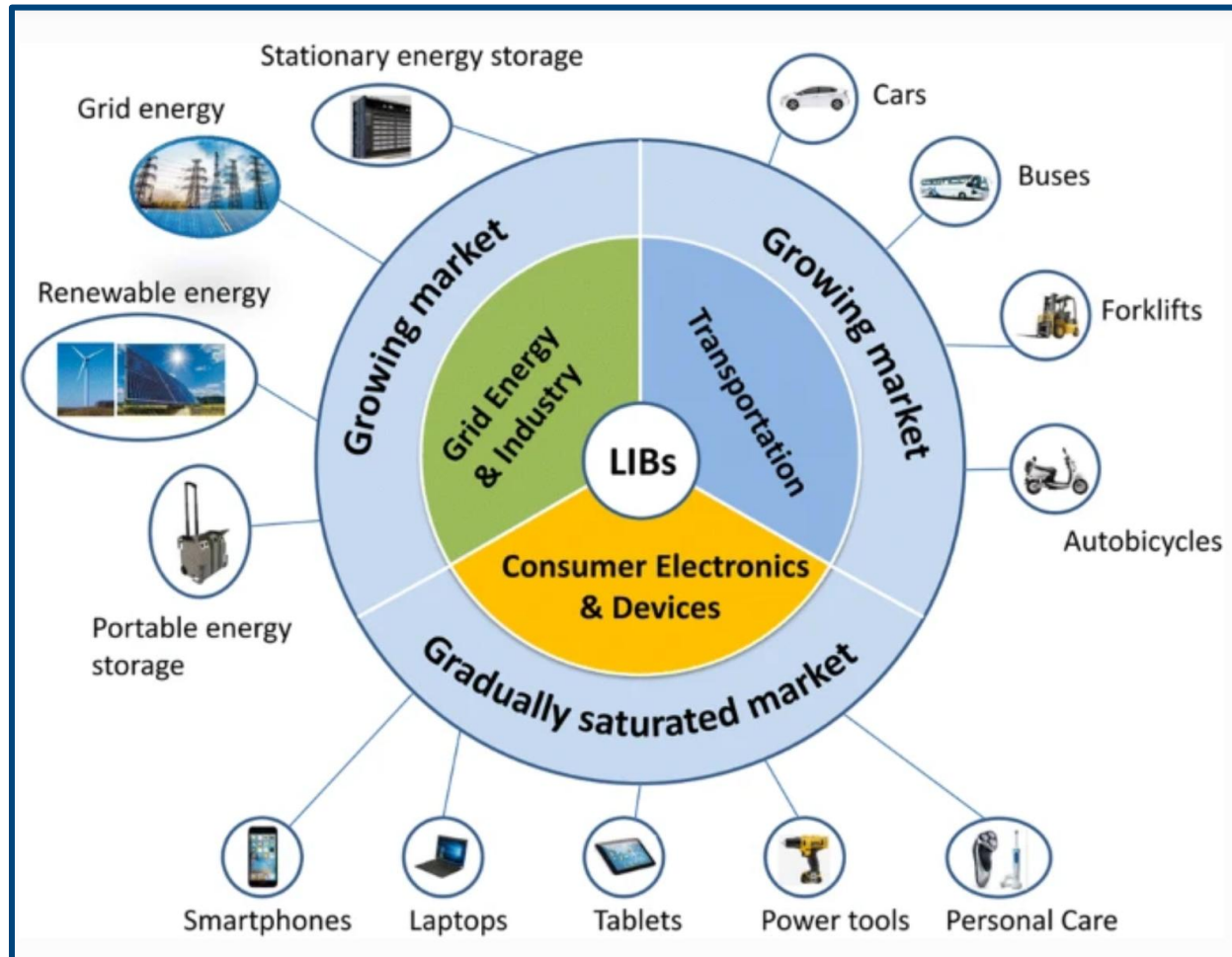
Akira Yoshino

Asahi Kasei Corporation, Tokyo, Japan
Meijo University, Nagoya, Japan

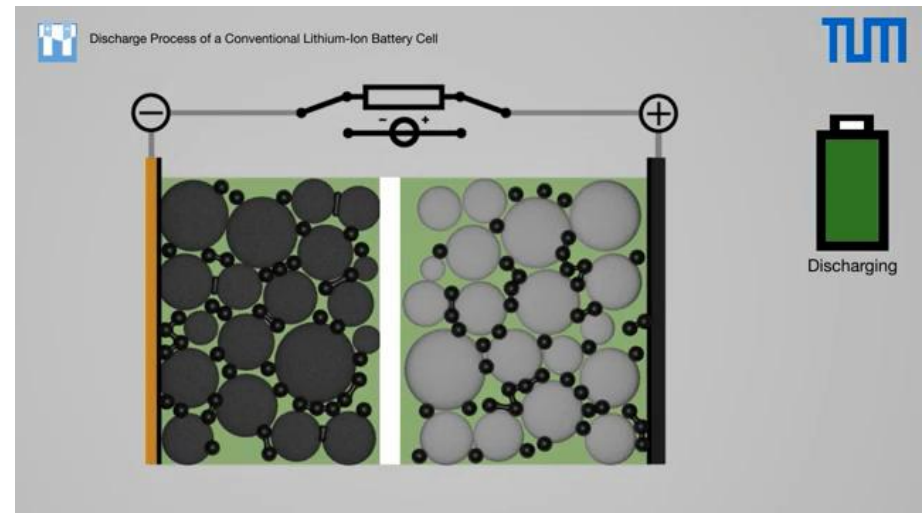
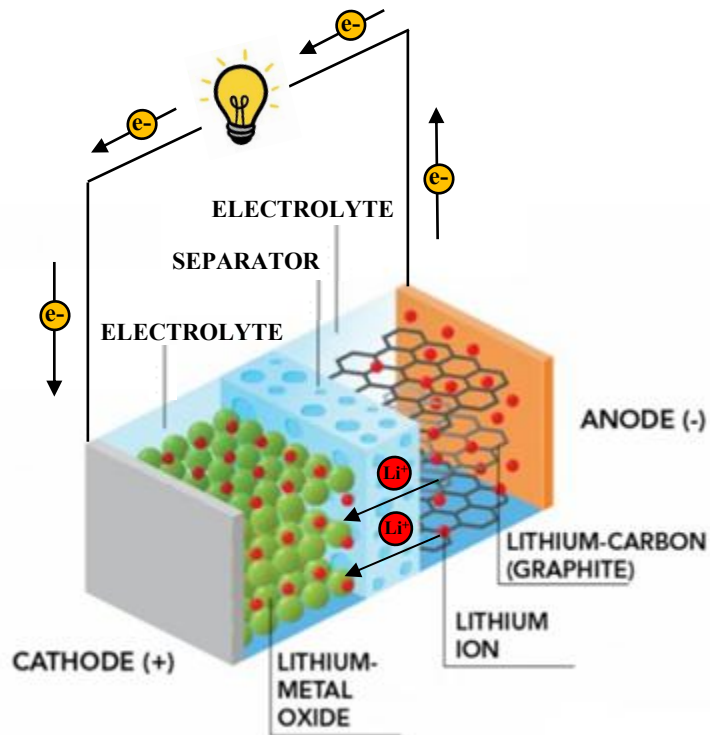
“for the development of lithium-ion batteries”

They created a rechargeable world

Battery applications



Li-ion cell: how does it work?

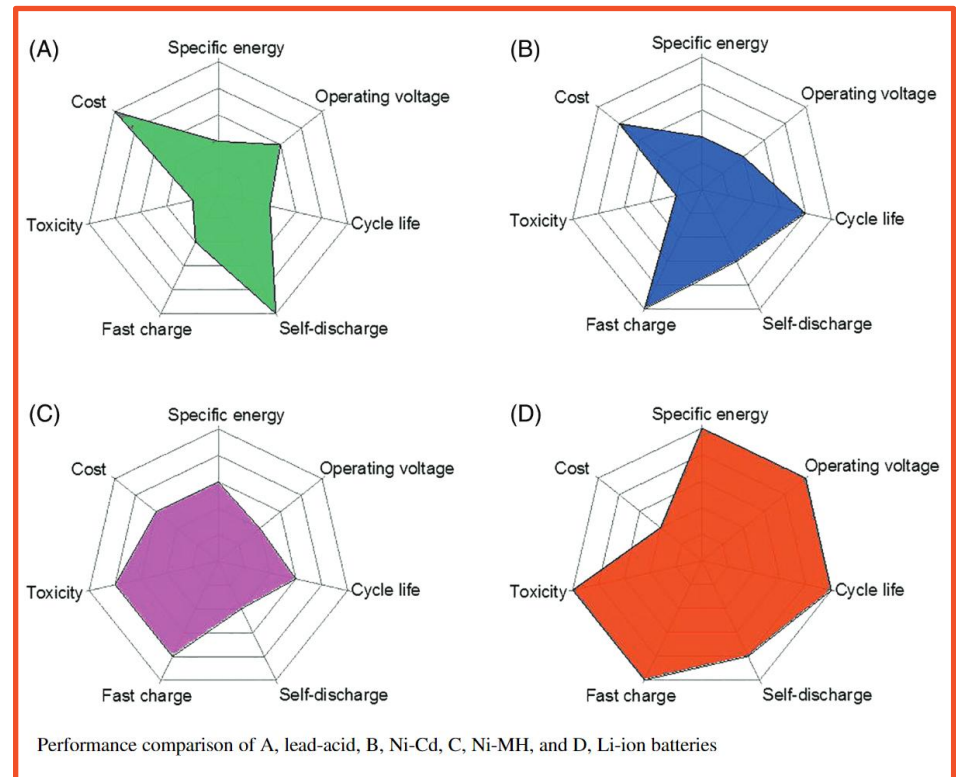


Li-ion cell

With respect to other chemistries, Li-ion batteries provide:

- High energy density
- Low weight
- Slow material degradation
- High per-cell voltage
- Low self discharge

Performance comparison with other chemistries



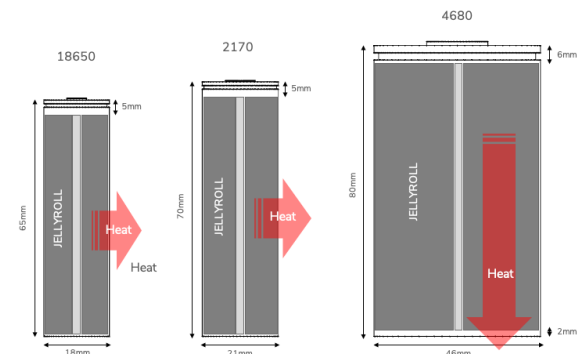
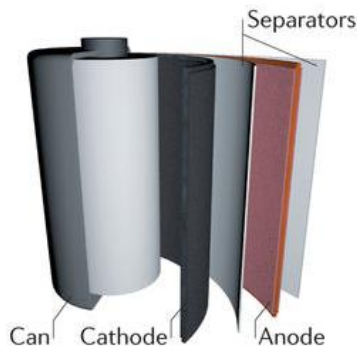
Battery type	Li-based?	Reason
(A) Lead-acid	✗ No	Uses lead (Pb) electrodes and sulfuric acid (H₂SO₄) electrolyte; the electrochemical reactions involve Pb/PbO ₂ .
(B) Ni-Cd	✗ No	Uses nickel oxyhydroxide (NiOOH) and cadmium (Cd) electrodes; charge transfer is based on Ni/Cd redox reactions.
(C) Ni-MH	✗ No	Uses nickel oxyhydroxide and a hydrogen-absorbing metal alloy ; energy storage is based on hydrogen exchange reactions.
(D) Li-ion	✓ Yes	Lithium ions (Li⁺) move between the anode and cathode during charge and discharge through intercalation processes.

Li-ion battery cell formats

Commonly classified according to cell packaging mechanical format.

Cylindrical Cells

- Rigid metal cylindrical casing with high mechanical sealing and robustness
- Highly standardized formats (18650, 21700, 4680). Mature technology (since 1992)
- High energy density, relatively low cost, easier thermal management at pack level
- Widely used in consumer electronics and electric vehicles
- Poor pack space utilization and more complex electrical connections in large packs

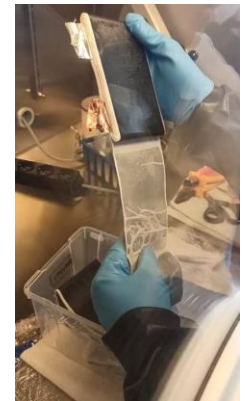
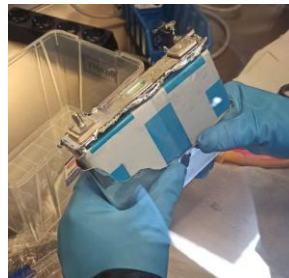
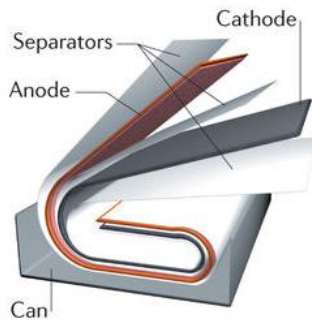


Li-ion battery cell formats

Commonly classified according to cell packaging mechanical format.

Prismatic Cells

- Rectangular rigid enclosure (e.g. aluminum, steel) with high mechanical sealing
- Efficient use of pack volume and greater flexibility in pack design
- High capacity and energy density, widely used in EVs and ESSs
- Mechanical swelling must be carefully managed during operation and aging
- Poorer thermal dissipation and higher manufacturing cost vs cylindrical cells

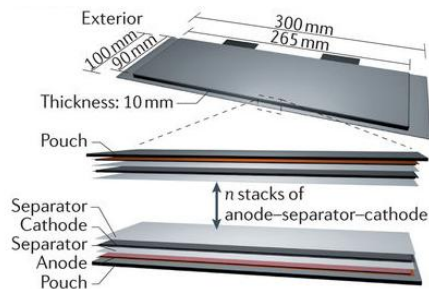


Li-ion battery cell formats

Commonly classified according to cell packaging mechanical format.

Pouch Cells

- Flexible laminated polymer–aluminum pouch, extremely thin and lightweight
- High energy density with medium capacity and flexible geometries
- Relatively easy manufacturing compared to rigid cell formats
- Typically requires external compression/support in the battery pack
- Limitations: needs external protection to ensure safety and structural integrity



Nature Reviews | [Materials](#)



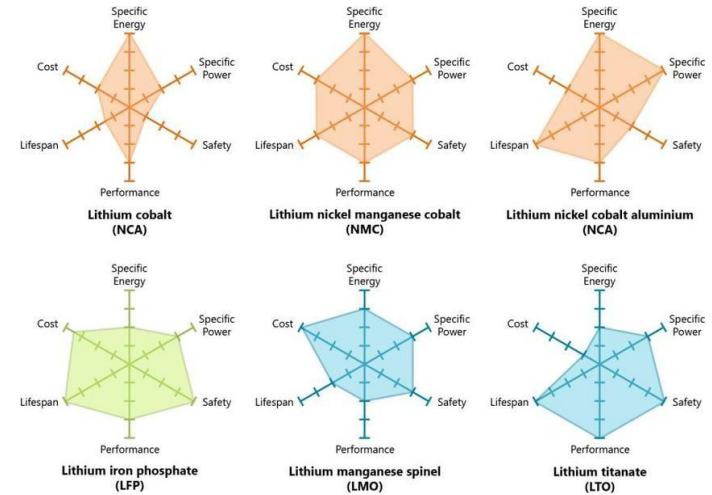
Li-ion chemistries

• Different trade-offs

- NCA and NMC emphasize high energy density
- LFP prioritizes safety and lifespan
- LTO high power capability+cycle life but low spec. energy.

• No single chemistry optimizes all metrics.

Choice depends on **application requirements** (e.g., EVs, energy storage, high-power systems).



LI-ION BATTERY CHEMISTRY	NOMINAL VOLTAGE (V)	CYCLE (LIFE)	CHARGE CURRENT RATE (C)	DISCHARGE CURRENT RATE (C)	THERMAL RUNAWAY (°C)	PACKAGING (TYPICAL)	SPECIFIC ENERGY (Wh/Kg)	APPLICATIONS	REMARKS	COMPATIBLE WITH ION's BMS
Nickel Manganese Cobalt Oxide (NMC)	3.6 (3.0-4.2) range	1000+	0.7-1C	1-2C	210°C (410°F)	18650, 21700	150-220	E-Bikes, Medical Devices, EVs, Industrial	High-specific energy, Low self-heating rate	Yes
Lithium Iron Phosphate (LFP)	3.2 (2.5-3.65) range	2000+	1C	1C	270 °C (518°F)	18650, 32650, prismatic	90-120	Stationary Applications with high capacity, EV	Flat discharge voltage, high power, low capacity, safe	Yes
Lithium Nickel Cobalt Aluminium Oxide (NCA)	3.6 (3.0-4.2) range	500-1000	0.7C	1C	150 °C (302°F)	18650	200-260	Medical, Industrial, Electric Powertrain	Long life, fast charge, wide temperature range, safe & expensive	Yes
Lithium Titanate Oxide (LTO)	2.4 (1.8-2.85) range	3000-7000	1C	10C	Highest	Prismatic	50-80	Electric Vehicle and Energy Storage Systems	Highest capacity with moderate power	Yes
Lithium Cobalt Oxide (LCO)	3.6 (3.0-4.2) range	500-1000	0.7- 1C	1C	150 °C (302°F)	18650 Prismatic & pouch cell	150-200	Laptops, Mobile Phones, Tablets, Cameras	High energy, limited power	Yes
Lithium Manganese Oxide (LMO)	3.7 (3.0-4.2) range	300-700	0.7-1C	1C	250 °C (482 °F)	Prismatic	100-150	Medical Devices, Electric Powertrains, Power Tools	High power, less capacity; safer than LCO	Yes

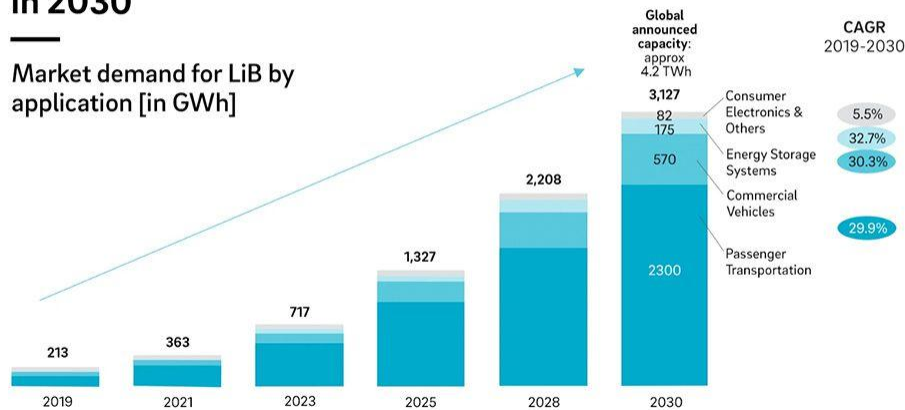
Raw materials trends in Li-ion batteries

Growing Demand for Battery Materials

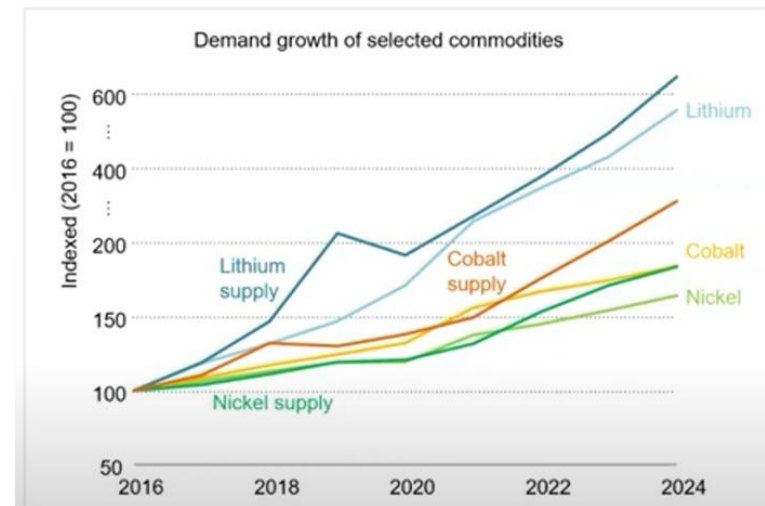
- Rapid expansion of EVs is driving exponential growth in raw material demand by 2040 (~20× nickel, 19× graphite, 14× lithium)
- Supply risk arises from **rapid demand growth & geographically concentrated production**, which may lead to structural deficits (especially for lithium).
- Increasing efforts in **recycling** and **material diversification**.

Global demand for lithium-ion batteries will be over 3,100 GWh in 2030

Market demand for LiB by application [in GWh]



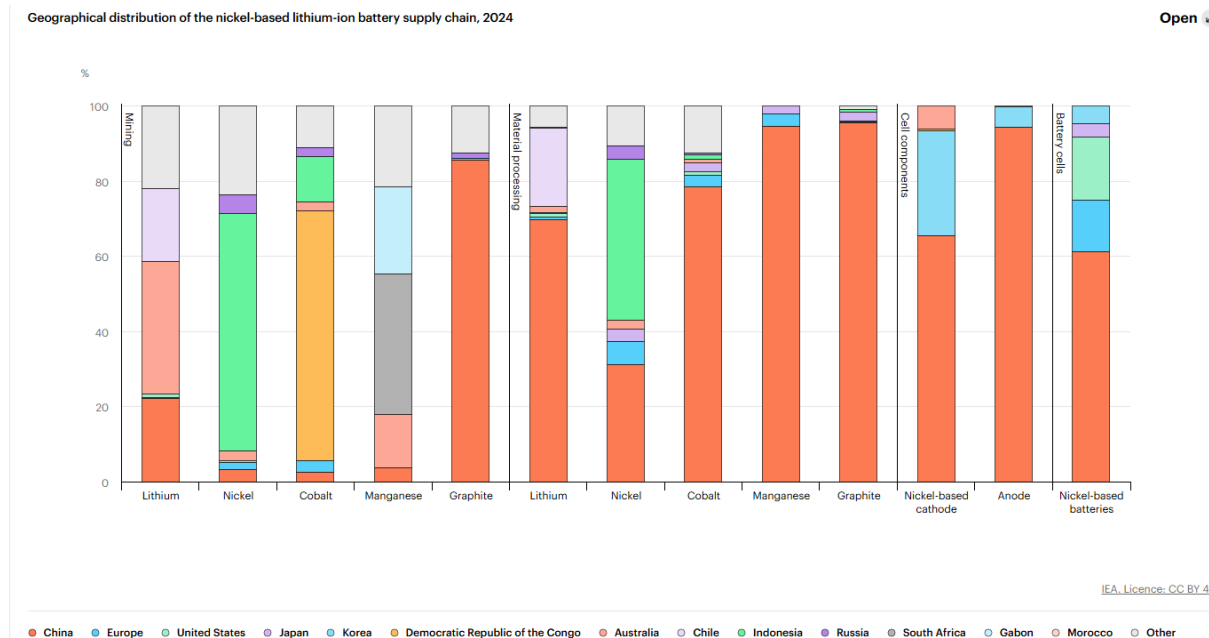
Sources: Avicenne, Fraunhofer, IHS Markit, Interviews with market participants, Roland Berger



Raw materials trends in Li-ion batteries

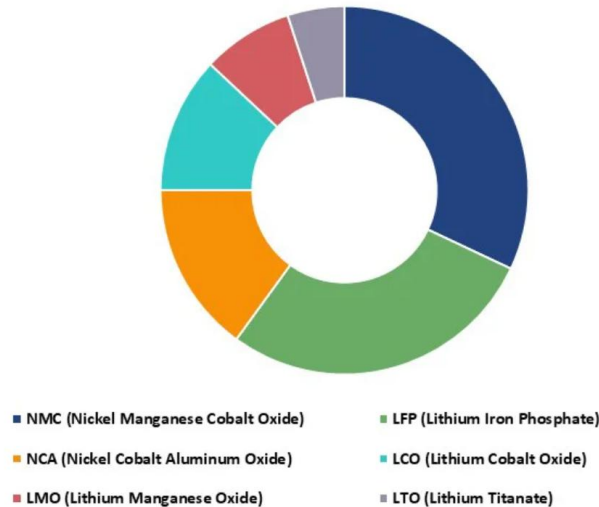
Key Materials and Market Trends

- **Lithium (Li):** most critical element; demand frequently exceeds supply.
- **Nickel (Ni) & Cobalt (Co):** used in high-energy cathodes (NMC); industry aims to **reduce cobalt due to cost and supply risk**.
- **Graphite (C):** dominant anode material with supply highly concentrated in China.
- **Silicon (Si):** emerging additive to increase anode energy density.



Li-ion chemistries

Global Lithium-ion Battery Market Outlook, by Cell Chemistry, 2025 – 2032



Source: www.persistencemarketresearch.com

Global Market Size, 2025
US\$64.1 Bn

NMC
Leading Cell Chemistry
Segment, 2025

LFP
Fast-Growing Cell
Chemistry Segment, 2025

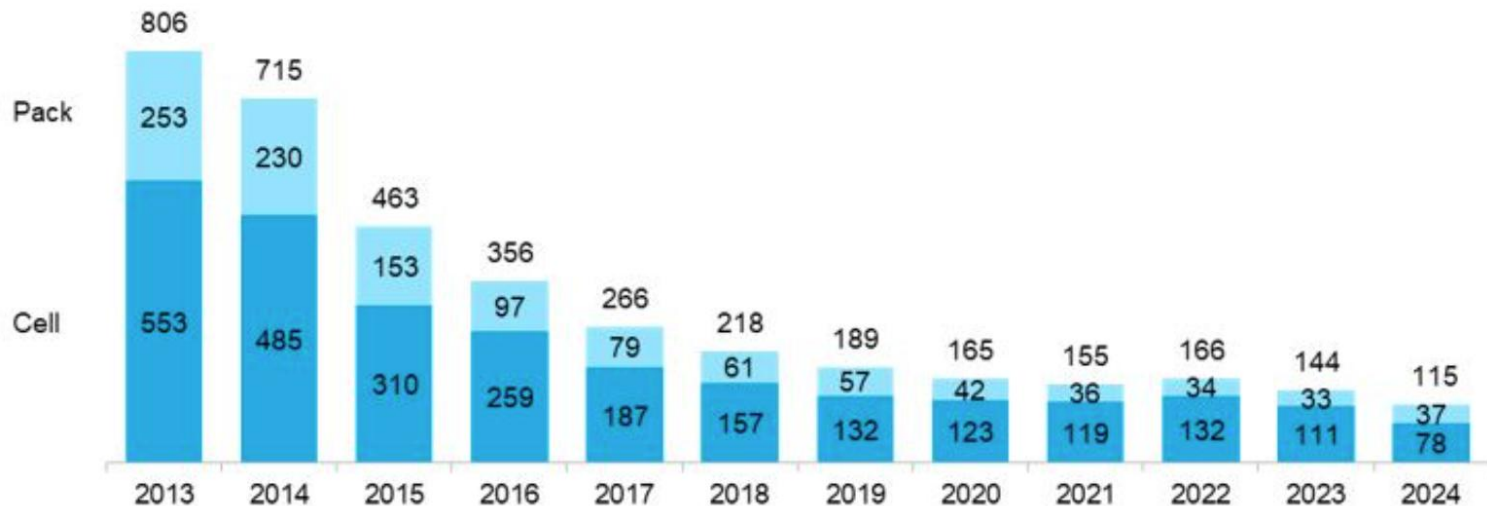
Impact on Battery Chemistries

- **LFP**: cobalt- and nickel-free → **lower cost and lower sensitivity to raw material volatility**, rapidly increasing market share (~40% of EVs in 2023).
- **NMC**: higher energy density for long-range vehicles, with a trend toward **high-nickel / low-cobalt compositions**.

Battery prices

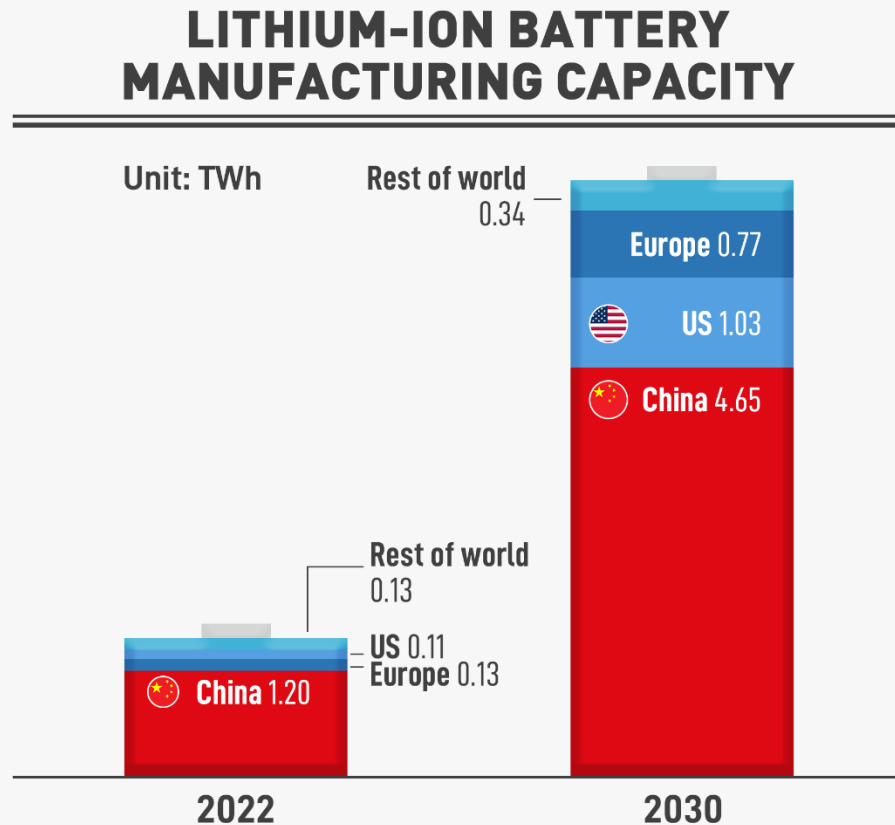
Figure 1: Volume-weighted average lithium-ion battery pack and cell price split, 2013-2024

Real 2024 \$/kWh



Source: BloombergNEF. Note: Historical prices have been updated to reflect real 2024 dollars. Weighted average survey value includes 343 data points from passenger cars, buses, commercial vehicles and stationary storage.

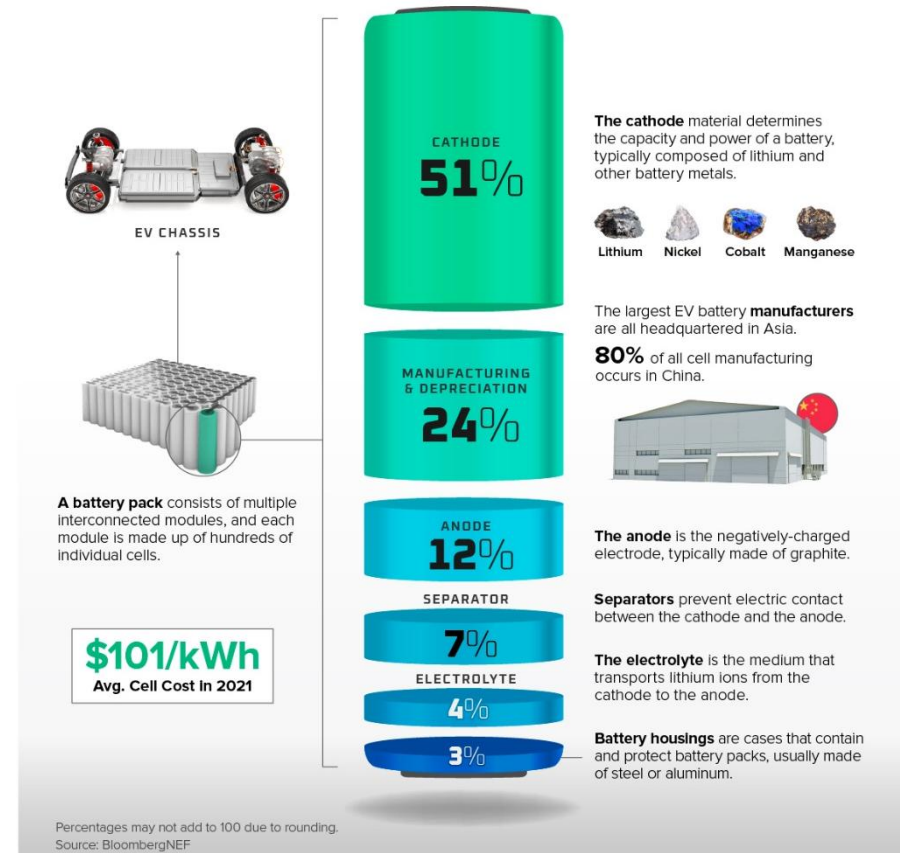
Battery manufacturers



EV Battery cell cost breakdown

The average cost of lithium-ion batteries has declined by 89% since 2010.

What makes up the cost of lithium-ion cells?



Vehicle cost development ICE vs BEV

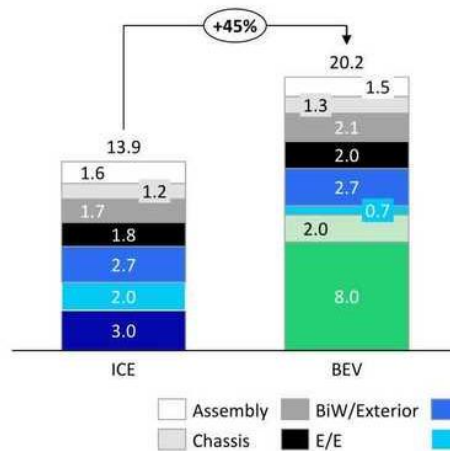
VEHICLE COST DEVELOPMENT

Battery electric vehicles are expected to almost close the gap to internal combustion engine powered vehicles by 2030

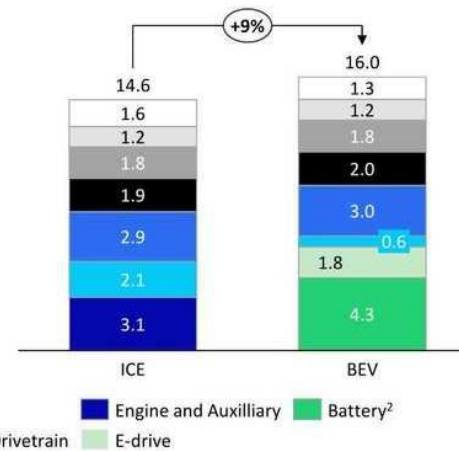
Comparison of direct costs ICE vs. BEV – European compact-class vehicle¹

In thousand EUR

2020



2030



¹ Does not include indirect costs (e.g. ramp-up, CAPEX, relative SG&A etc.)

² Equals a 50kWh battery with 160 €/kWh in 2020 and Equals with 85 €/kWh in 2030

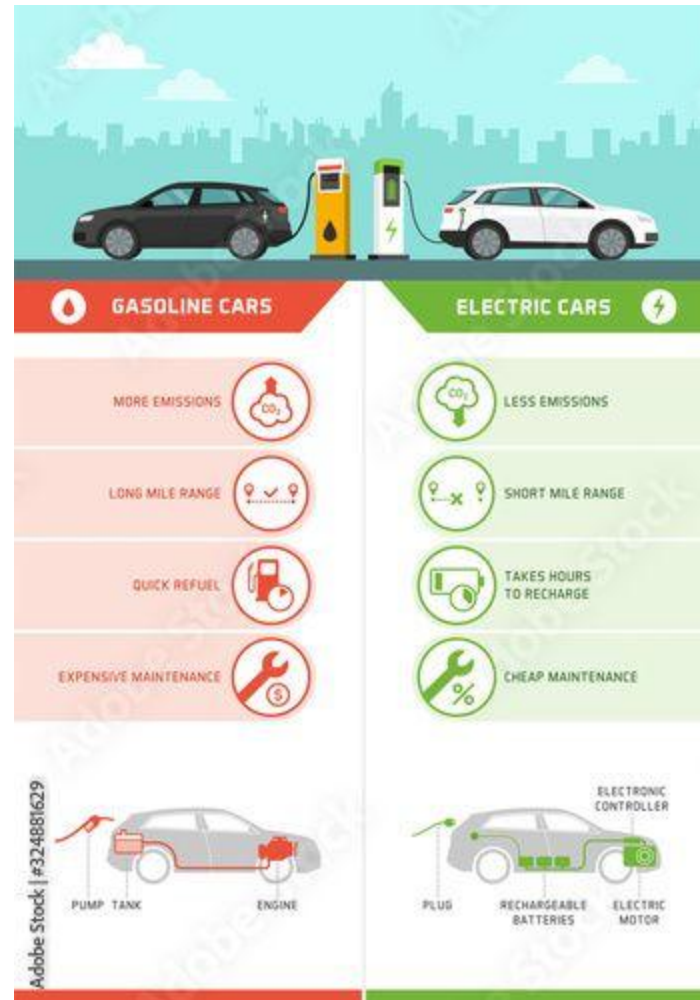
Sources: Oliver Wyman FAST 2030 proprietary model, Bank of America, IEA Global EV Outlook, expert interviews, Oliver Wyman Research

© Oliver Wyman

1

- **Batteries main cost driver** in BEVs (expensive materials and complex manufacturing).
- **Simpler mechanics** and **fewer moving parts** reduce **maintenance** and **operating costs** over vehicle lifetime.
- **Upfront cost** gap with ICE vehicles is **decreasing** thanks to battery improvements and economies of scale, while BEVs offer **higher drivetrain efficiency**, **lower energy** (fuel) **cost** per km, and **lower local emissions**.

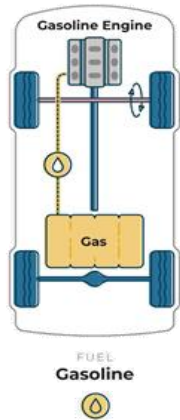
ICE vs xEV



Types of xEV

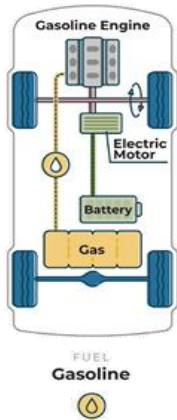
CONVENTIONAL VEHICLES

Use internal combustion engines. Fuel is injected into the engine, mixing with air before being ignited to start the engine.



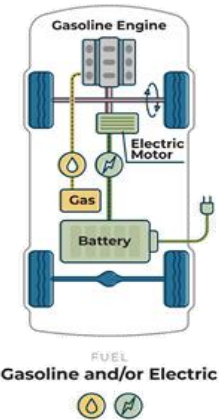
HYBRID ELECTRIC VEHICLES

Powered by both engine and electric motor. The battery is charged internally through the engine.



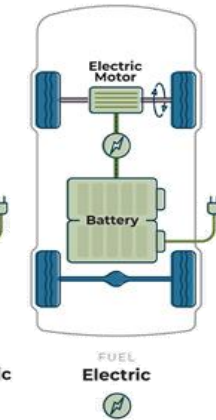
PLUG-IN HYBRID ELECTRIC VEHICLES

Battery can be charged both internally and externally through outlets. Run on electric power before using the engine.



ALL-ELECTRIC VEHICLES

Powered only by electric motor with no engine. Have large traction battery and must be plugged externally to charge.



Consumption: Fuels

Driven by: Engines

Advantages: Easy to refuel, long driving range and high speed

Disadvantages: More emissions, high cost of fuel

Fuels

Engines primarily, motors secondarily

Easy to refuel, less fuel consumption, less emissions

Less power, heavier weight of the car

Fuels and electricity

Motors primarily, engines secondarily

Easy to refuel, less fuel consumption, less emissions

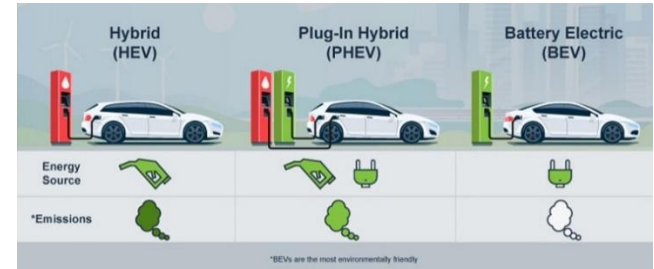
High price, limited models to choose from, heavier weight

Electricity

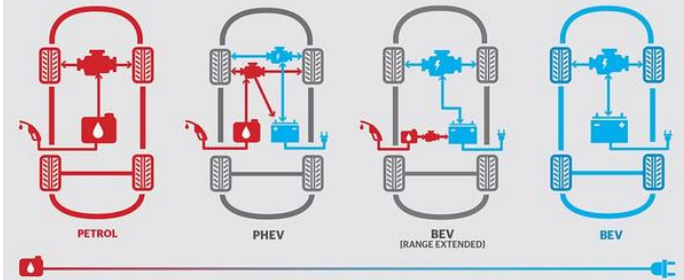
Motors

Environmentally friendly, low maintenance, government support

Lack of charging stations, short driving range and low speed, heavier weight



From petrol to battery electric

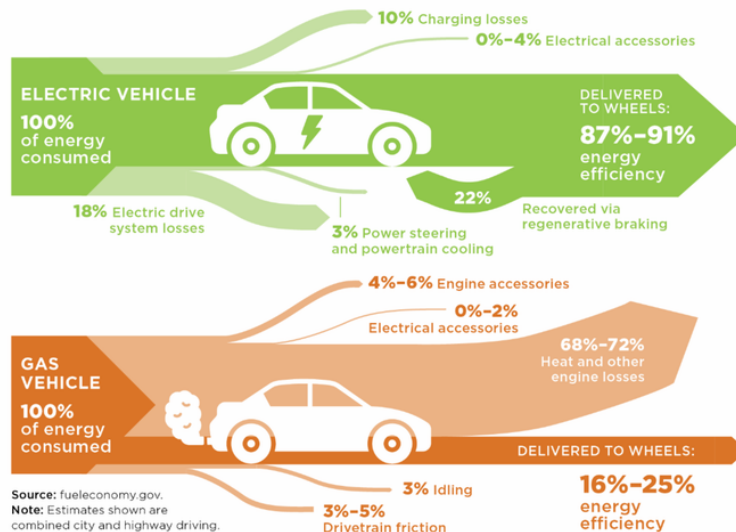


Efficiency and costs

Typical Energy Prices (Input Energy) before drivetrain efficiency Early 2026

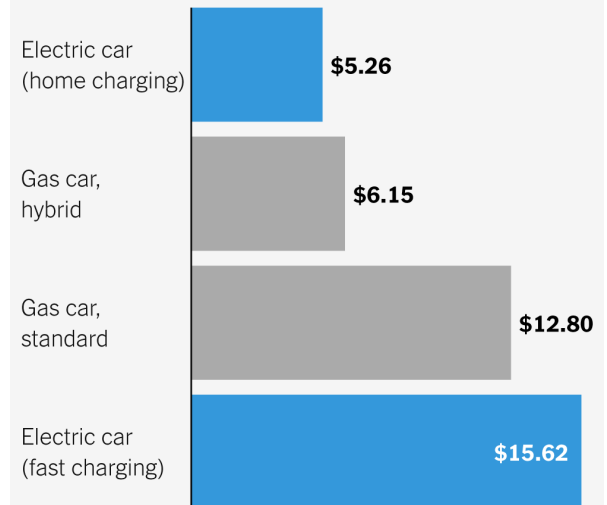
Gasoline (energy equivalent)	~\$0.19–0.23 / kWh
Electricity (home charging)	~\$0.23–0.38 / kWh

Where the energy goes: Electric vehicles vs gas vehicles



ENERGY ACTION NETWORK

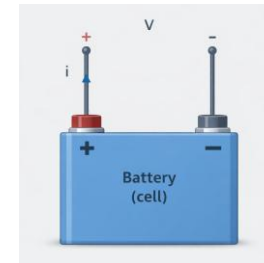
Cost to drive 100 miles



Battery cell: key characteristics

- **Chemistry** (e.g., Lead-acid, NiCd, NiMH, Li-ion)
- **Primary vs. secondary cells** (non-rechargeable vs. rechargeable)
- **Voltage levels:** nominal, maximum, and cutoff voltage (V)
- **Capacity and energy:** capacity Q (Ah) and stored energy E (Wh)
- **Power capability:** maximum charge/discharge current or power (A / W)
- **Energy and power density:** Wh/kg and W/kg
- **Operating temperature range** and thermal limitations
- **Lifetime:** cycle life (number of charge–discharge cycles)
- **Self-discharge rate**

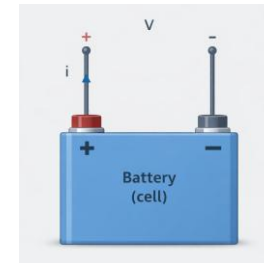
Definitions and key terms



- **Nominal voltage** V_N [V]: average operating voltage of the cell during discharge (between maximum and cut-off voltage).
- **Terminal voltage** v [V]: instantaneous voltage measured at the cell terminals.
- **Open-circuit voltage (OCV)** [V]: voltage at the cell terminals when current $i = 0$ after sufficient rest. Typically $OCV = f(\text{remaining capacity})$.
- **Current** i [A]: rate of charge flow ($1A=1C/s$). $i > 0$ discharging, $i < 0$ charging.
- **C-rate** [h^{-1}]: charge or discharge current normalized by the nominal capacity of the cell.

Note: In practice, the measurable quantities are voltage, current, and temperature.

Definitions and key terms



- **Fully charged:** condition where the cell voltage reaches the maximum charge voltage (typically 4.2 V at 25°C for many Li-ion chemistries).
- **Fully discharged:** condition where the cell voltage reaches the cut-off voltage (typically ≈ 3.0 V at 25°C, depending on the chemistry).
- **Q_N Nominal capacity** [Ah or As]: nominal amount of electric charge that can be stored in the cell.
- **Q_R Remaining capacity (remaining charge)** [Ah or As]: amount of charge that can still be delivered by the cell.

General case

$$Q(t) = Q_0 - \int_{t_0}^t i(\tau) d\tau \quad [As]$$

where

- t_0 = initial time
- t = current time

In ampere-hours

$$Q(t) = Q_0 - \frac{1}{3600} \int_{t_0}^t i(\tau) d\tau \quad [Ah]$$

Constant current $i(t) = I$

$$Q(t) = Q_0 - \frac{I}{3600} (t - t_0)$$

Definitions and key terms

State of Charge (SOC) and Depth of Discharge (DOD)

State of Charge (SOC)

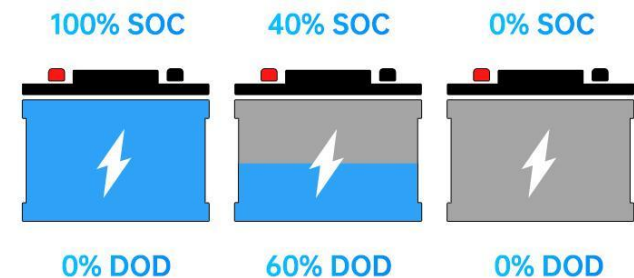
$$SOC = \frac{Q_R}{Q_N}$$

- Q_R : remaining charge
- Q_N : nominal capacity

$$Q_R = SOC \cdot Q_N$$

Depth of Discharge (DOD)

$$DOD = 1 - SOC$$



Relationship

- SOC = available charge in the battery
- DOD = fraction of capacity already used

Practical implication: Limiting the DOD increases battery lifetime.
(BMS often operate batteries within a restricted SOC window.)

Definitions and key terms

Operational SOC (BMS)

Battery Management System restricts usable SOC range limits to protect battery.

Typical usable SOC window: $SOC_{true} \approx 10\% - 90\%$

Buffers avoid:

- deep discharge
- high-SOC degradation
- safety risks

SOC displayed to the user: remapped to 0–100%.

<i>Displayed SOC</i>	<i>Approx. true SOC</i>
0%	~5– 10%
100%	~90– 95%

Practical recommendation

To reduce degradation, many manufacturers recommend operating mainly within:

$20\% - 80\%$ (*displayed SOC*)

This range limits both **DOD stress** and **high-SOC degradation**.

Definitions and key terms

Battery Cycling

Refers to the repeated **discharge and recharge** of a battery during operation.

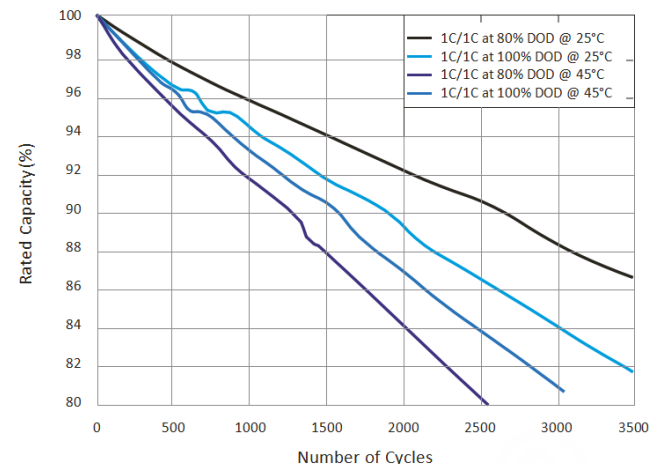
Full cycle: transferring a charge equal to the nominal capacity Q_N .

Example: 100% → 0% → 100% SOC = 1 full cycle

Partial cycles: 100% → 50% → 100% SOC (0.5 cycle)

Factors affecting cycling **aging**

- Depth of discharge (DOD)
- C-rate
- Temperature
- SOC operating window



Definitions and key terms

State of Health (SOH)

SOH quantifies battery aging and long-term degradation.

$$SOH = \frac{Q_N}{Q_{N,0}}$$

- $Q_{N,0}$: nominal capacity of a new battery
- Q_N : current available capacity

Relation with SOC

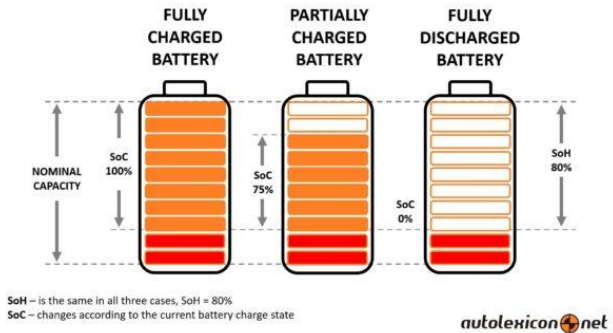
$$Q_R = SOC \cdot Q_N$$

Since

$$Q_N = SOH \cdot Q_{N,0}$$

then

$$Q_R = SOC \cdot SOH \cdot Q_{N,0}$$



End of Life (EOL)

A battery is typically considered **End-of-Life** when

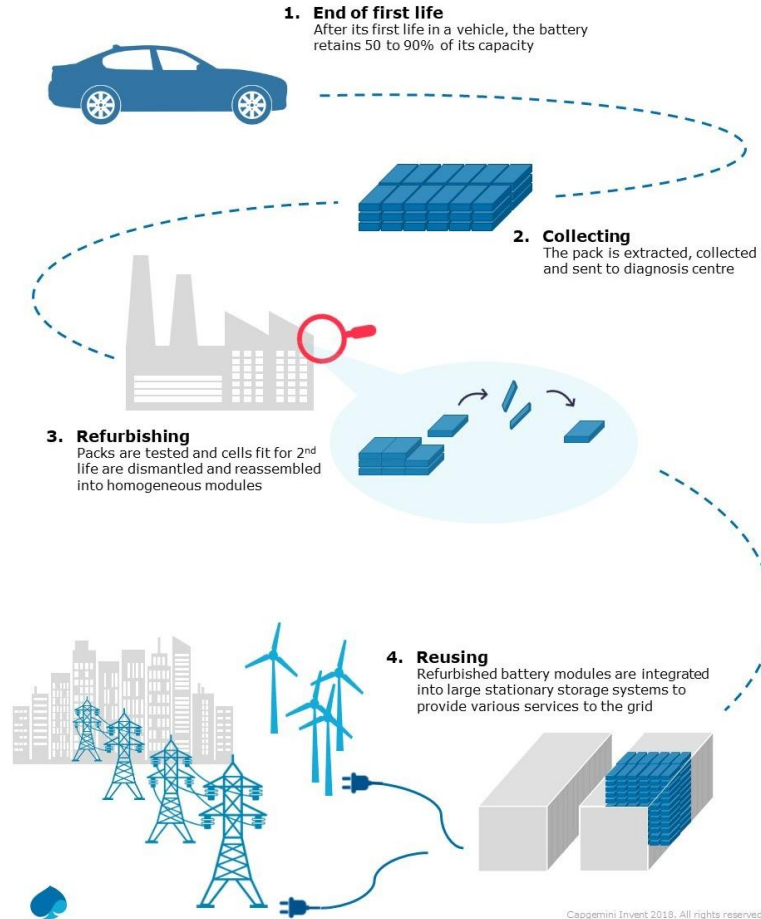
$$SOH < 80\%$$

(capacity reduced to 80% of the initial value)

What affects SOH

- Capacity fade
- Increase of internal resistance

2nd life batteries and vehicles to grid



Remaining Useful Life (RUL)

Estimates **remaining time, cycles, or kilometers** before a battery reaches its **End-of-Life (EOL)** threshold (typically SOH $\approx 80\%$).

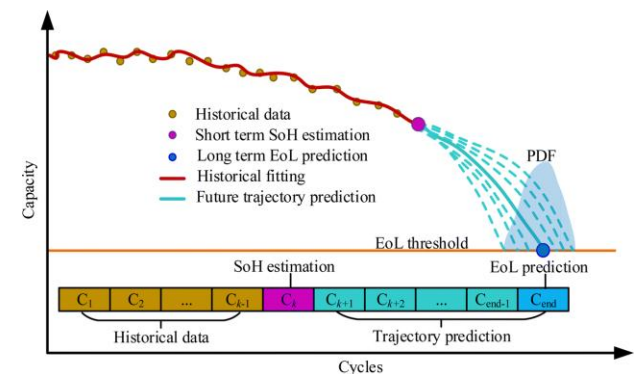
$$RUL = t_{EOL} - t_{now}$$

or equivalently

$$RUL = \text{remaining cycles/mileage until SOH} \approx 80\%$$

RUL prediction combines:

- battery aging models
- BMS measurements (current, voltage, temperature)
- usage history (SOC window, charging power, temperature)
- future usage projection based on driving patterns



The degradation model is **projected forward** until SOH reaches the EOL threshold.

Why RUL matters

- **Insurance:** driving behavior affects battery degradation → potential usage-based premiums.
- **Used EV market:** battery health strongly influences vehicle residual value.

LG Chem ICR18650 C2 2800mAh

Item	Condition / Note	Specification
2.1 Capacity	Std. charge / discharge	Nominal 2800mAh (C_{nom}) Minimum 2700mAh (C_{min})
2.2 Nominal Voltage	Average	3.72V
2.3 Standard Charge (Refer to 4.1.1)	Constant current Constant voltage End current(Cut off)	0.5C (1350mA) 4.30V 50mA
2.4 Max. Charge Voltage		4.30V
2.5 Max. Charge Current		1.0C (2700mA)
2.6 Standard Discharge (Refer to 4.1.2)	Constant current End voltage(Cut off)	0.2C (540mA) 3.0V
2.7 Max. Discharge Current	-20 ~ 5℃	0.5C (1350mA)
	5 ~ 45℃	2.0C (5400mA)
	45 ~ 60℃	1.5C (4050mA)
2.8 Weight	Approx.	Max. 50.0 +/- 3.0 g
2.9 Operating Temperature	Charge	0 ~ 45℃
	Discharge	-20 ~ 60℃
2.10 Storage Temperature (for shipping state)	1 month	-20 ~ 60℃
	3 month	-20 ~ 45℃
	1 year	-20 ~ 20℃
2.11 Cell Voltage (for shipping state)	Voltage range	3.7 ~ 3.9V

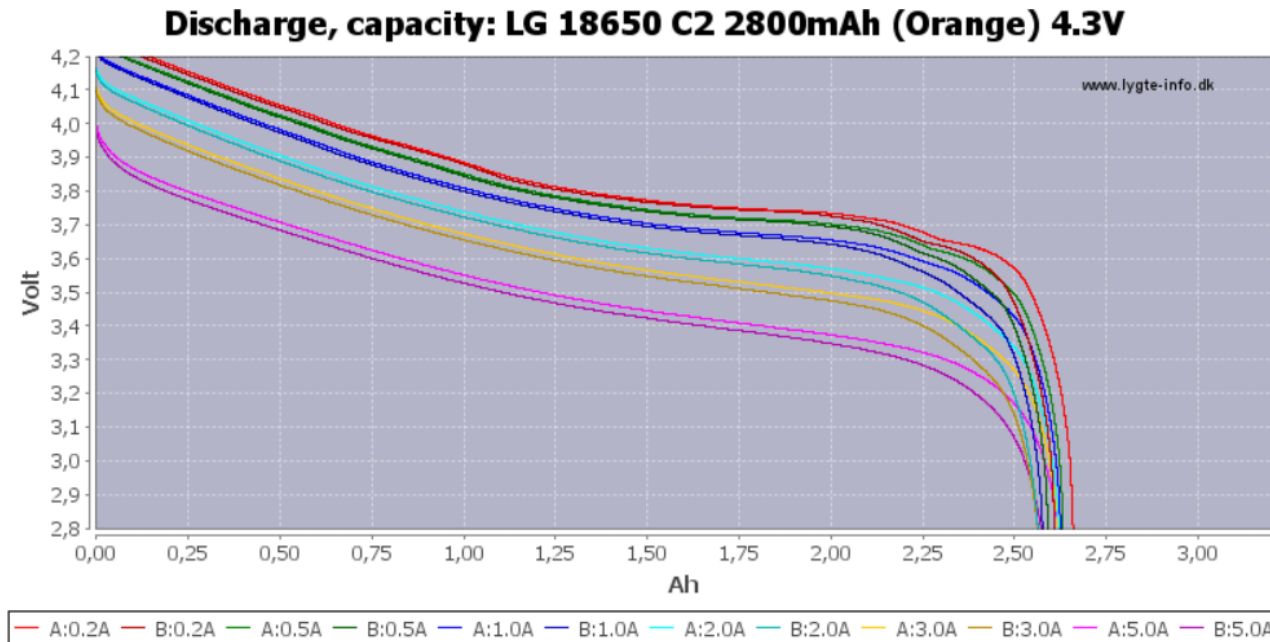


$$\begin{aligned}
 V_N &= 3.72 \text{ V} \\
 V_{max} &= 4.30 \text{ V} \\
 V_{cutoff} &= 3.0 \text{ V} \\
 Q_N &= 2800 \text{ mAh} \\
 C_{max} &= 1 \text{ C}
 \end{aligned}$$

$$\begin{aligned}
 E_N &= V_N \cdot Q_N \approx 10 \text{ Wh} \\
 I_{max} &= Q_N \cdot C_{max} = 2.8 \text{ A} \\
 P_{max} &= E_N \cdot C_{max} \approx 10 \text{ W}
 \end{aligned}$$

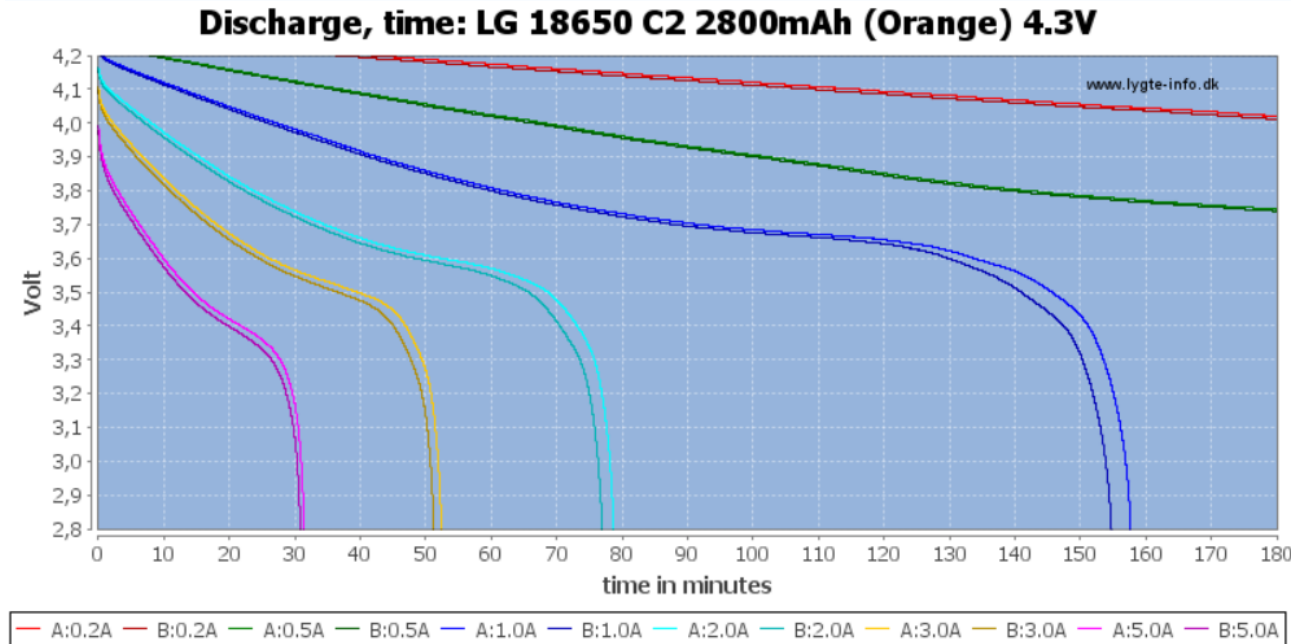
$$\frac{10 \text{ Wh}}{0.05 \text{ kg}} = 200 \text{ Wh/kg}$$

LG Chem ICR18650 C2 2800mAh



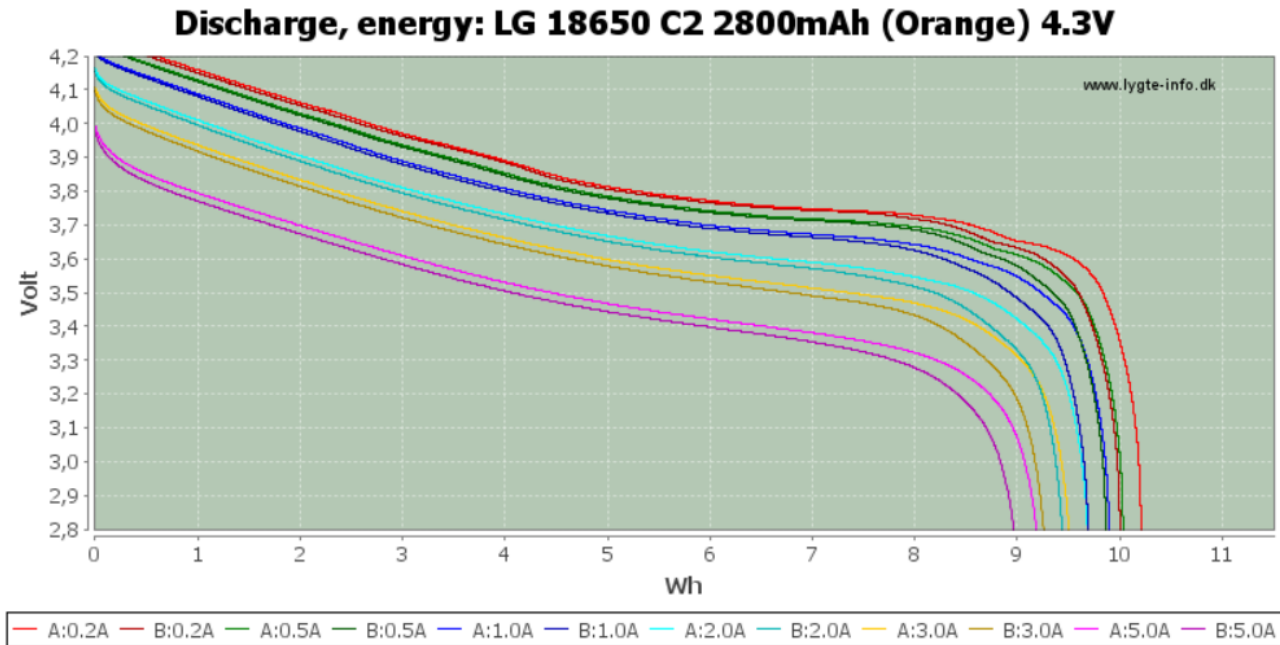
- **Constant-current discharge (0.2–5 A)** from 4.3 V to ~2.8 V.
- **A / B: repeated tests at same current** → measurement repeatability (measurement uncertainty, temperature variations, slight SOC differences...).
- Nominal capacity = 2800 mAh; at 1C discharge, **extracted capacity < nominal**.
- **Higher current** → **larger initial voltage drop** due to internal resistance (ohmic drop) and electrochemical polarization (reaction kinetics and ion diffusion limits).
- Test goes below 3.0V (cut-off ≈2.8V). Likely chosen to extract near-full capacity for comparison.

LG Chem ICR18650 C2 2800mAh



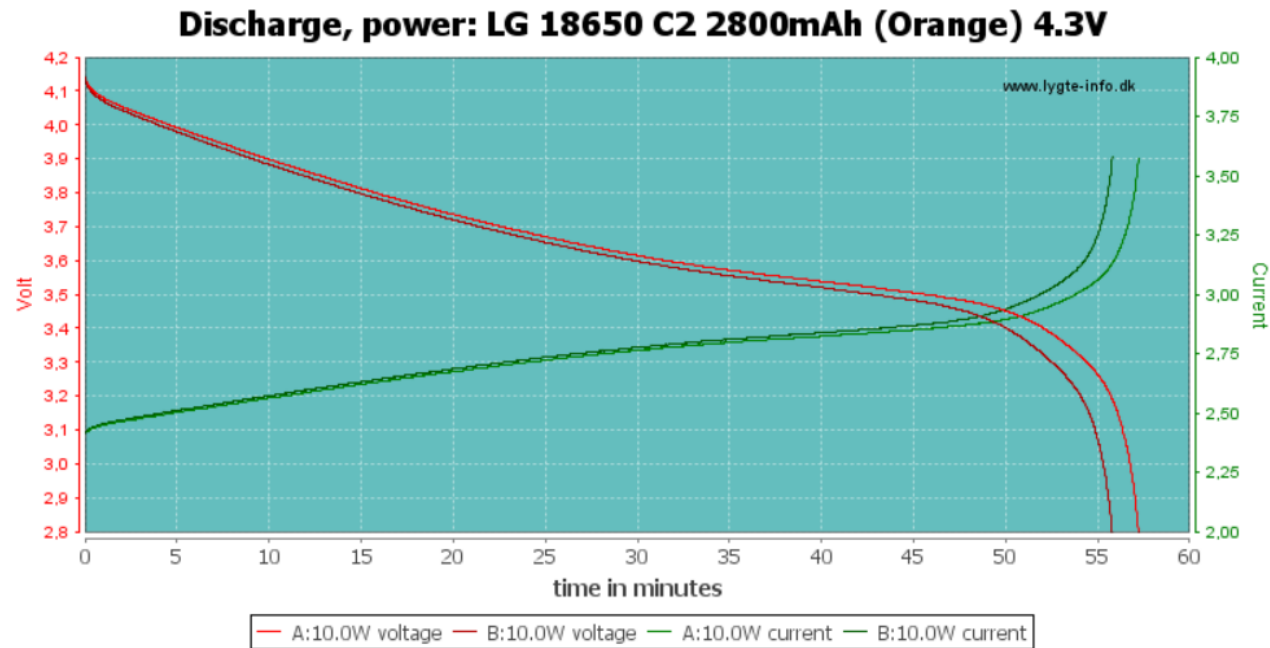
- Constant-current discharge (0.2–5 A). **x-axis** shows **discharge duration**.
- Discharge time **inversely proportional to C-rate** (ideal case $t \approx 1/C_{\text{rate}}$).
- For 2.8 Ah cell, expected discharge times approximately:
 - 1C (2.8 A) → ~1 h
 - 0.5C (1.4 A) → ~2 h
 - 0.2C (~0.56 A) → ~5 h
- Voltage plateau followed by rapid drop near end of discharge, typical Li-ion behavior.

LG Chem ICR18650 C2 2800mAh



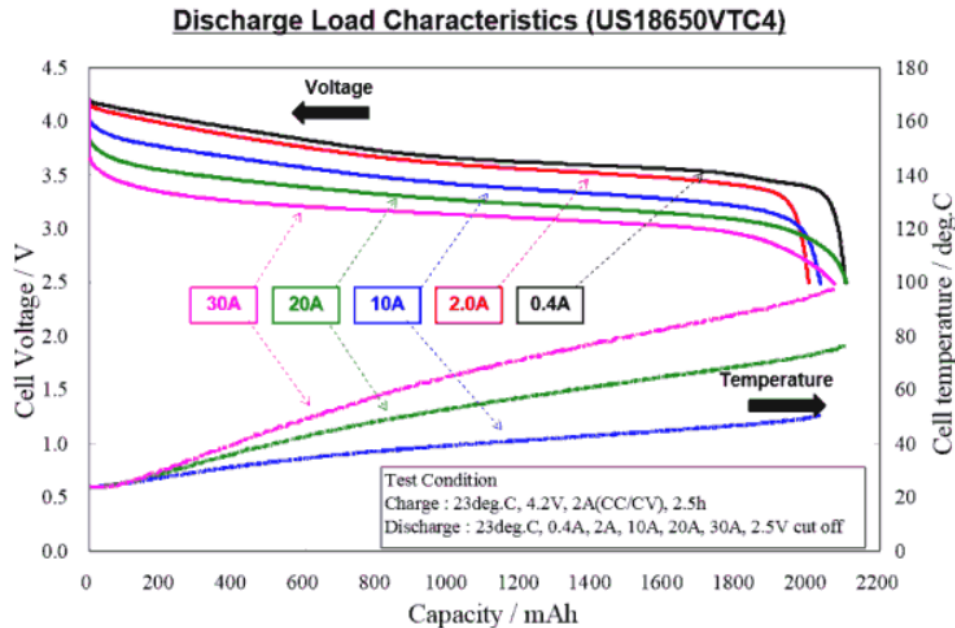
- **x-axis** shows **energy delivered** (Wh) during constant-current discharge.
- Nominal energy ≈ 10 Wh, derived from $V_N \approx 3.7$ V and $Q_N = 2800$ mAh.
- At low discharge currents, extracted energy approaches the nominal value; higher currents reduce usable energy.
- Reduction due to lower average discharge voltage under higher current (internal losses).
- Curves show typical Li-ion behavior: **long voltage plateau** (~ 3.6 – 3.7 V) followed by steep drop near end of discharge.

LG Chem ICR18650 C2 2800mAh



- **Constant power discharge (10 W):** load adjusts current to maintain $P = v(t)i(t)$.
- Compared to constant-current discharge, as **voltage decreases**, **current increases** to keep power constant.
- Near end of discharge, current rises sharply as voltage approaches cut-off (~2.8–3.0 V).

US18650VTC4 (Sony/Murata)



- Cell: US18650VTC4 (Sony/Murata), $Q_N = 2100$ mAh, high-power 18650 Li-ion cell.
- Constant-current discharge at 0.4 A, 2 A, 10 A, 20 A, 30 A (23 °C ambient, cut-off = 2.5 V).
- **Cell temperature increases during discharge:** at high currents (20–30 A) temperature rises significantly due to ohmic losses $I^2 R$.

Discharge current influences voltage behavior, extracted capacity, and cell temperature.

Battery system



Battery cell

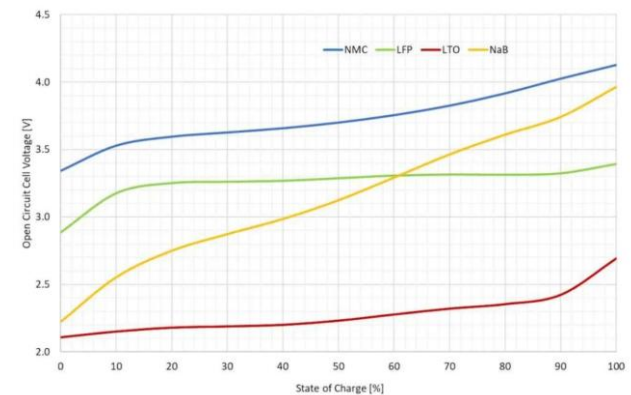
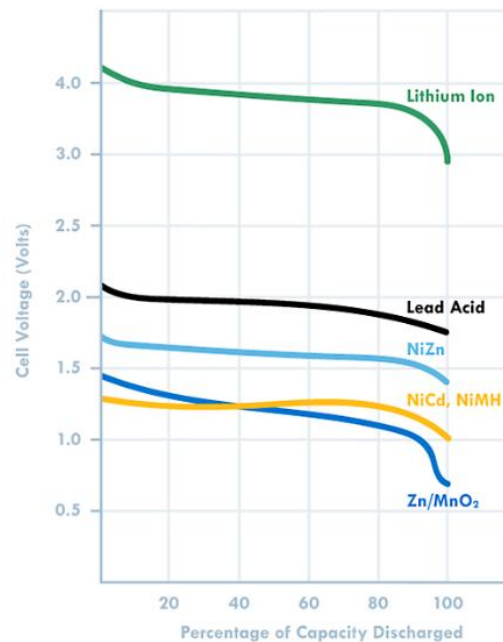
Battery system



Battery cell

- **Single-cell voltage is limited by electrochemical stability**
Typically $\approx 2\text{--}4$ V depending on chemistry.
- Different chemistries show **different nominal voltages and OCV curves**.
Example: Li-ion $\approx 3.6\text{--}3.7$ V, LFP ≈ 3.2 V, lead-acid ≈ 2 V, Ni-based ≈ 1.2 V.
- **Higher voltages** require connecting multiple **cells in series**.

	Half Reaction	Standard Potential (V)
 stronger oxidizing agent	$\text{F}_2 + 2\text{e}^- \rightleftharpoons 2\text{F}^-$	+2.87
	$\text{Pb}^{4+} + 2\text{e}^- \rightleftharpoons \text{Pb}^{2+}$	+1.67
	$\text{Cl}_2 + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-$	+1.36
	$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}$	+1.23
	$\text{Ag}^+ + 1\text{e}^- \rightleftharpoons \text{Ag}$	+0.80
	$\text{Fe}^{3+} + 1\text{e}^- \rightleftharpoons \text{Fe}^{2+}$	+0.77
	$\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu}$	+0.34
	$2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2$	0.00
	$\text{Pb}^{2+} + 2\text{e}^- \rightleftharpoons \text{Pb}$	-0.13
	$\text{Fe}^{2+} + 2\text{e}^- \rightleftharpoons \text{Fe}$	-0.44
 stronger reducing agent	$\text{Zn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Zn}$	-0.76
	$\text{Al}^{3+} + 3\text{e}^- \rightleftharpoons \text{Al}$	-1.66
	$\text{Mg}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mg}$	-2.36
	$\text{Li}^+ + 1\text{e}^- \rightleftharpoons \text{Li}$	-3.05

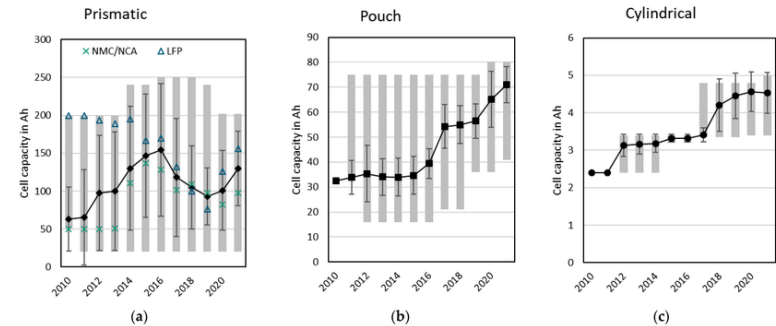


Battery system



Battery cell

- Typical Li-ion **cell capacities**
 - 18650: ~2–3.5 Ah
 - 21700: ~4–6 Ah
 - Large prismatic / pouch:
 - tens of Ah (larger format cells)



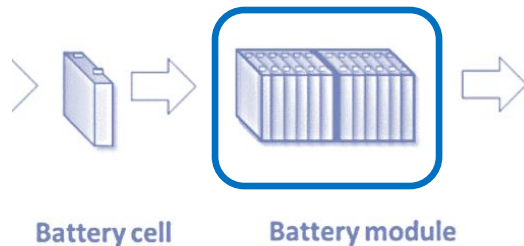
Prismatic and pouch cells can reach larger capacities because their stacked electrode geometry improves cooling, shortens current paths, and allows larger cell formats.

Why capacity cannot grow arbitrarily:

- **Material limits:** active materials have finite specific capacity (e.g. graphite ≈ 372 mAh/g, typical cathodes ≈ 160 – 200 mAh/g)
- **Electrode thickness:** increasing capacity $\rightarrow$ more active material, typically thicker electrodes.
- **Transport limitations:** thicker electrodes increase Li^+ diffusion distances in electrolyte and particles $\rightarrow$ concentration gradients and higher overpotential.
- **Thermal effects:** higher current density $\rightarrow I^2R$ losses and heating.
- **Safety constraints:** large overpotential at the anode $\rightarrow$ Li-plating and accelerated degradation.

Increasing capacity requires more active material, but this worsens transport, thermal and safety constraints $\rightarrow$ so battery packs use many **cells in parallel**.

Battery system



Module: a group of cells assembled with electrical interconnections, mechanical housing, sensors, and optionally local monitoring electronics

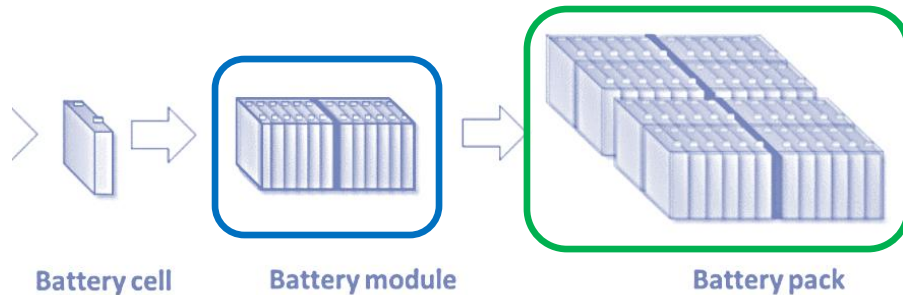
- several cells (**series and/or parallel**)
- busbars and electrical connections
- voltage and temperature sensors
- mechanical housing and supports that keep cells in place
- thermal interface to the cooling system

Typical characteristics:

- 10–50 cells
- ~20–60 V

Modules are designed as **repeatable building blocks**.

Battery system



Pack: assembly of battery modules connected in series and/or parallel, typically including electrical interconnections, protective housing, and basic protection devices.

- integrates modules and busbars
- may include voltage sensing connections
- provides electrical energy to the higher-level system

The pack itself is primarily the **electrochemical energy storage assembly**.

Cell Arrangement

1. Parallel connection (capacity and current)

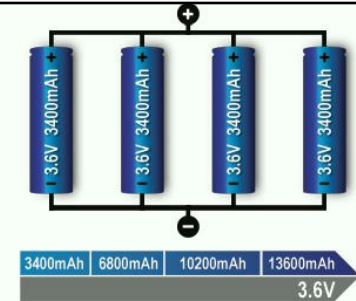
Cells are first connected in parallel: $Q_{group} = N_P \cdot Q_{cell}$

Advantages:

- Higher capacity
- Current sharing

Example: $3 \times 3000 \text{ mAh} = 9000 \text{ mAh}$

If pack current = **30 A**, **each cell** in a 3P group provides: **10 A**
→ **lower heating and longer lifetime.**



2. Series connection (voltage)

Parallel groups are then connected in series.

$$V_{pack} = N_S \cdot V_{cell}$$

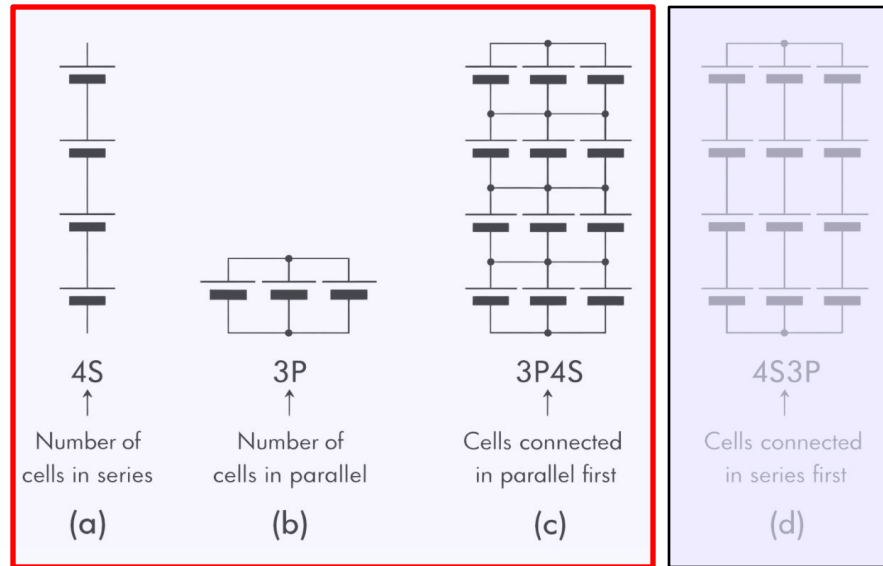
Example:

$$10 \times 3.6 \text{ V} \approx 36 \text{ V}$$

Higher voltage improves power delivery and efficiency.



Cell Arrangement



Standard architecture (industry practice)

Battery packs usually built as **series of parallel cell groups**. Industry notation: **series number first** (NS NP) since pack voltage is primary design parameter.

Example: 10S3P

3 cells in parallel → one group, 10 groups in series → pack voltage

Configuration widely used in **power tools, e-bikes, and electric vehicles**.

Advantages of standard architecture

Feature

Explanation

Scalability

Standard cells (e.g. 18650 or 21700) can build packs for many applications.

Redundancy

If one cell in a parallel group has lower capacity/higher resistance:
other cells in same group provide more current, reducing impact of the cell.

BMS simplification

BMS measures **voltage of each parallel group** (one sensor/group).
Cells in parallel behave electrically like **one larger cell**.

Application	Cell type	Cell capacity	Example configuration	Modules in pack	Nominal pack voltage	Pack energy
Smartphone	pouch	~3–5 Ah	1S1P	—	~3.7 V	~10–20 Wh
Laptop	pouch	~4–8 Ah	3S1P	—	~11 V	~40–80 Wh
Power tools	18650 / 21700	~2–5 Ah	5S2P	usually none	~18 V	~80–150 Wh
E-bike	18650 / 21700	~3 Ah	13S4P	sometimes modular	~48 V	~500–700 Wh
EV pack	cylindrical / prismatic	varies	~96S–400S with P groups	6–16 modules (typ.)	~350–800 V	~40–120 kWh

Nominal pack voltage

$$V_{pack,nom} = N_S \cdot V_{cell,nom}$$

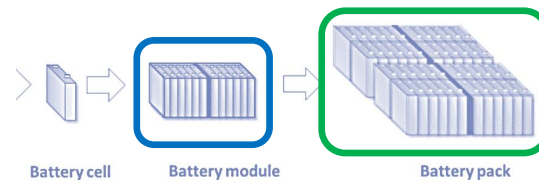
Pack capacity

$$Q_{pack} = N_P \cdot Q_{cell}$$

Approximate pack energy

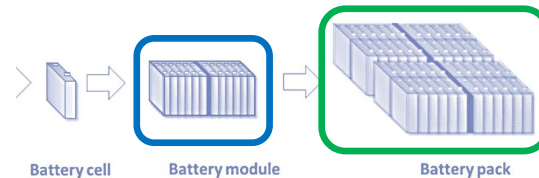
$$E_{pack} \approx V_{pack,nom} \cdot Q_{pack}$$

Why modules?



- **Manufacturing & scalability:** standard building blocks
 - easier assembly and automation
 - same module reused across different pack configurations
- **Serviceability & fault containment:** modules can be replaced instead of the whole pack
 - reduced maintenance cost and downtime
 - faults can be localized and contained
- **Thermal design & management:** smaller, repeatable units
 - easier cooling design and validation
 - better control of temperature distribution at module level

Why modules? Electrical segmentation



High-voltage pack (400–800 V)

- assembled from low-voltage modules (~20–60V, **Safety** Extra-Low Voltage SELV range)
- safe interaction with subsystems (reduced shock hazard)
- less insulation required and no need for large separation distances
- more compact system design

Modules are connected in series to form the pack voltage

- **full voltage** appears **only** across the **pack terminals**

Pack terminals are **enclosed and protected**

- no direct access to both ends of the series string

Contactors isolate the pack when inactive

- the HV bus is electrically disconnected

Why a high-voltage pack?

Most **applications** require a given **power** ($P = VI$), **not a specific voltage**.

For the same power:

- **higher voltage** → lower current
- lower current → **lower losses** $P_{loss} = I^2 R$

Implications:

- smaller cables and connectors
- lower thermal losses
- higher system efficiency

Typical voltage levels depend on the **application**:

- low power systems → low voltage (simplicity, safety)
- high power systems → high voltage (efficiency, current reduction)

e.g.:

- e-bikes ~48 V
- electric vehicles 400–800 V
- grid/storage systems up to kV range

The application sets the power. The designer chooses the voltage

Why not arbitrarily high voltage?

Power electronics limitations

Power electronic devices are designed to operate up to a maximum voltage.

- higher system voltage requires different (higher-voltage) devices
- increased cost and often higher losses

Insulation requirements

Higher voltage increases the risk of electrical discharge.

- more insulation and larger distances between components
- the system becomes physically larger and more difficult to design

Switching and EMI challenges

Higher voltage leads to faster voltage changes during switching.

- increased electromagnetic interference
- additional components and design effort needed to keep the system stable and reliable

Safety and fault severity

Higher voltage → more severe fault conditions

- higher risk and more demanding protection systems

Increasing voltage improves efficiency, but also increases **cost and complexity**

Battery packs for xEVs



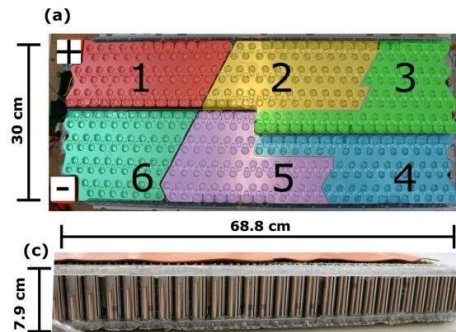
Model S's 84kWh battery

7104 Li-ion cells (18650 size)



16 modules connected in series

Each module contains 6 groups (in series)



Each group: 74 cells connected in parallel

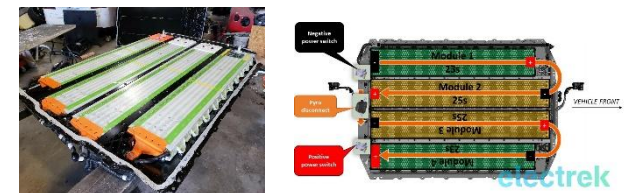


Model 3's 75kWh battery

4416 Li-ion cells (2170 size)



4 modules connected in series



Large modules: 25 blocks in series

Small modules: 23 blocks in series

Each block: 46 cells connected in parallel

Important features: **power** and **range**

Battery packs for xEVs

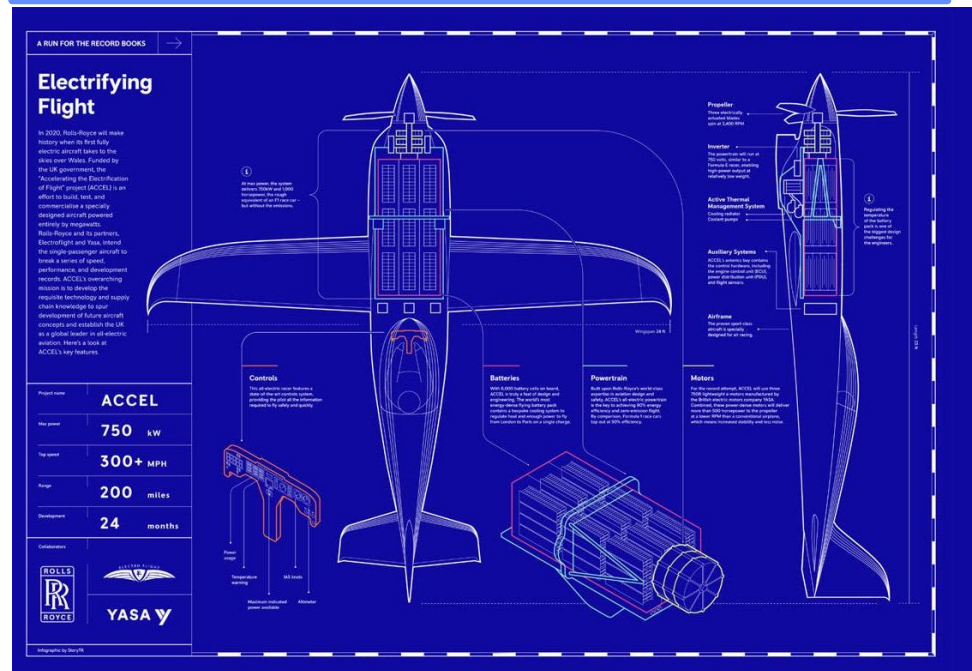
HOME » INTERNATIONAL » ROLLS-ROYCE IS DESIGNING THE 'WORLD'S

Rolls-Royce is designing the 'world's fastest all-electric airplane' named ACCEL that can reach up to 300 mph

Brittany Chang

🕒 27 Dez 2019

That's thanks to “an energy-dense flying battery pack” according to Rolls-Royce. The aircraft battery consists of 3 packs in parallel (180S12P each), totaling ~6480 Li-ion cells, ~648 V nominal voltage, and ~72.9 kWh energy per pack. Total weight: 450 Kg



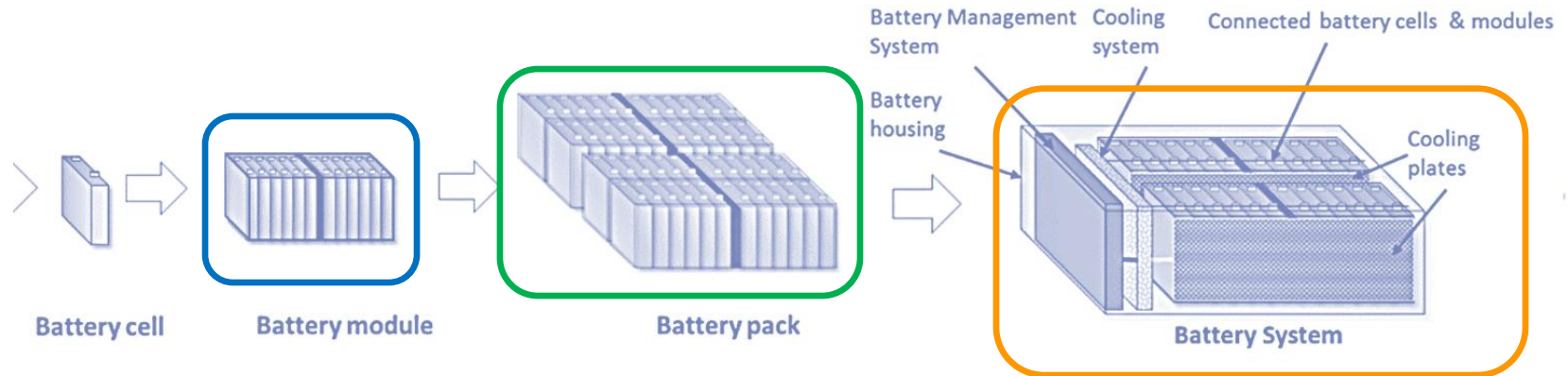
Battery packs for xEVs

Electric Vertical Take-Off and Landing (**eVTOL**) battery packs typically use thousands of individual cells, often 21700 or 18650 cylindrical cells, to achieve required energy storage and power output.

Typical eVTOL battery: 5,000 - 9,000 cylindrical cells organized into roughly 20–50 modules, with multiple packs used in parallel to **provide redundancy**.



Battery system



Images source <https://www.isi.fraunhofer.de/>

Battery system: complete functional energy storage system consisting of the battery pack(s) together with monitoring, protection, control and thermal management subsystems.

- battery pack(s)
- Battery Management System (BMS)
- thermal management (cooling/heating)
- contactors and protection circuits
- current sensors
- system interfaces to the load or grid

Battery cooling system

Maintains cells within a safe temperature range to ensure performance, lifetime, and safety.

Typical operating temperature range: 20°C – 40°C

Why is it so important?



Battery cooling system

Maintains cells within a safe temperature range to ensure performance, lifetime, and safety.

Typical operating temperature range: $20^{\circ}\text{C} - 40^{\circ}\text{C}$

Air cooling

Heat removed by natural or forced airflow.

Characteristics:

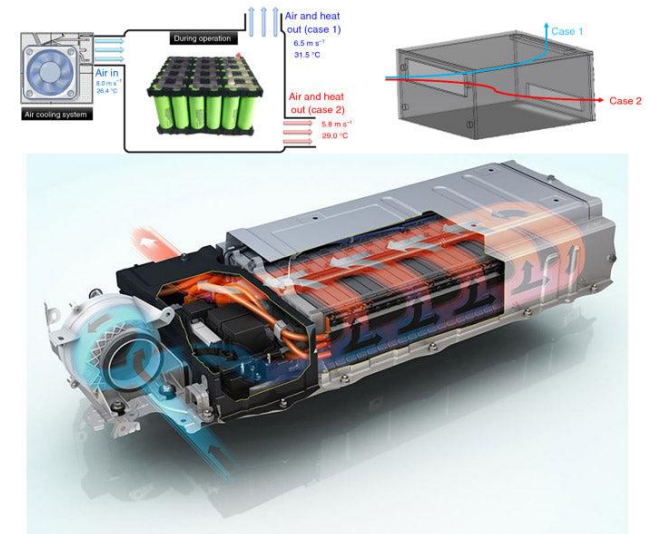
- lowest cost
- very lightweight
- simple architecture

Limitations:

- low heat transfer capability
- poor temperature uniformity
- not suitable for high-power packs.

Typical applications:

- consumer electronics
- small battery packs
- early EV designs.



Battery cooling system

Maintains cells within a safe temperature range to ensure performance, lifetime, and safety.

Typical operating temperature range: 20°C – 40°C

Liquid cooling

Coolant (usually water–glycol mixture) circulates through cooling plates or channels near the cells.

Characteristics:

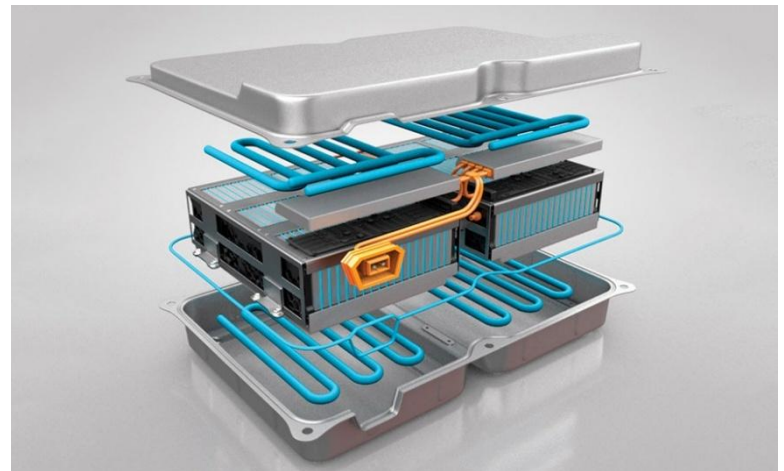
- higher cooling performance
- good temperature uniformity
- medium system complexity

Trade-offs:

- heavier system (coolant, pumps, piping)
- higher cost than air cooling.

Applications:

- electric vehicles
- large battery packs.



Battery cooling system

Maintains cells within a safe temperature range to ensure performance, lifetime, and safety.

Typical operating temperature range: 20°C – 40°C

Immersion cooling

Cells are directly immersed in dielectric fluid.

Characteristics:

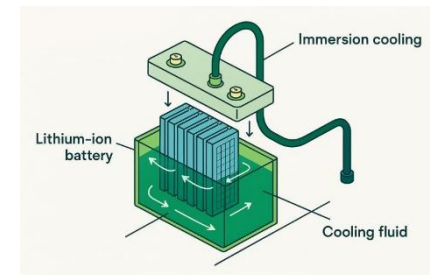
- very high heat transfer
- excellent thermal uniformity

Trade-offs:

- expensive dielectric fluids
- heavier and more complex system.

Applications:

- Aviation
- High-power EV systems
- Fast-charging batteries



Battery cooling system

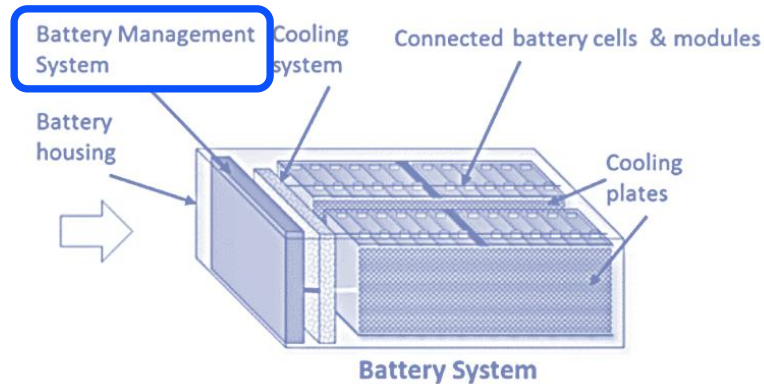
Maintains cells within a safe temperature range to ensure performance, lifetime, and safety.

Typical operating temperature range: 20°C – 40°C

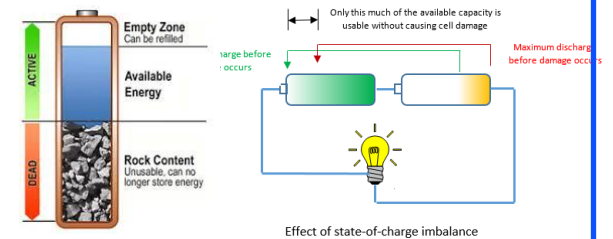
Cooling type	Principle	Cost & weight	Typical use
Air cooling	airflow removes heat from cells	low cost, very lightweight	electronics, small packs
Liquid cooling	coolant flows through plates or channels	medium cost, heavier due to pumps and coolant	modern EV battery packs
Immersion cooling	cells immersed in dielectric liquid	high cost, heavy system	aviation, high-power EV
Phase-change cooling (PCM)	material absorbs heat during melting	medium cost, moderate weight	hybrid cooling systems
Heat pipes / hybrid systems	heat transported through high-conductivity pipes	medium–high cost, moderate weight	high-performance packs

Battery management system (BMS)

Li-ion battery safe operation requires a BMS to **maximize performance, prolong life, avoid damage,** and, in extreme cases, **prevent serious health and safety threats.**



Safety



Degradation

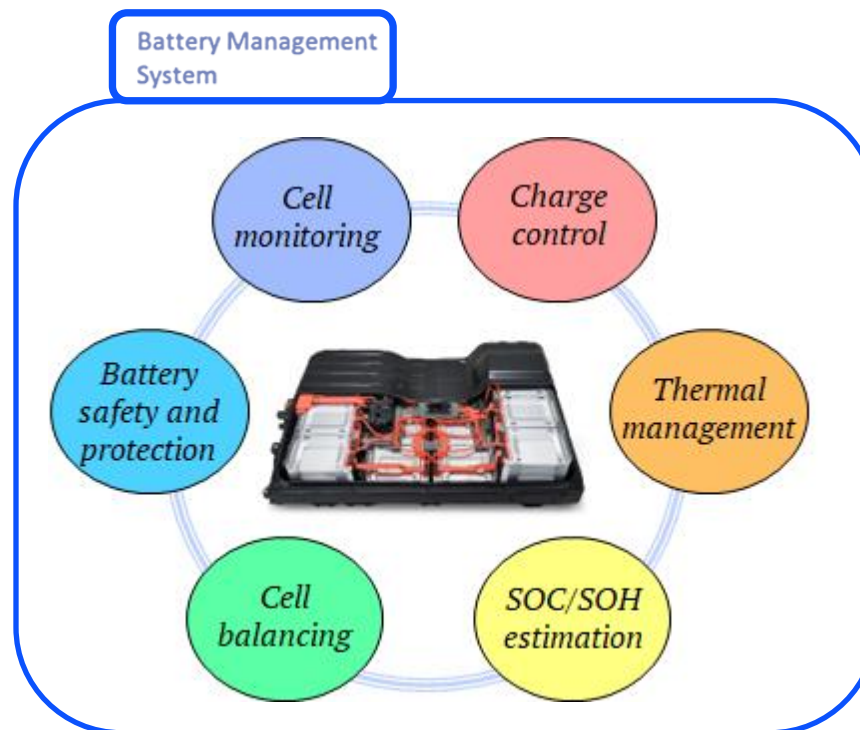


Anxiety

BMS Functional Overview

The Battery Management System (BMS) interfaces with all key components of the battery pack and the host system (e.g., vehicle, grid storage, etc.).

Its responsibilities can be grouped into six main functions:



BMS Functional Overview

The Battery Management System (BMS) interfaces with all key components of the battery pack and the host system (e.g., vehicle, grid storage, etc.).

Its responsibilities can be grouped into six main functions:

1. Cell Monitoring

- Continuously measures voltage, current, and temperature of cells
- Detects abnormal conditions such as ground faults (unintended paths to chassis or earth)
- Manages high-voltage switches to safely connect/disconnect the battery
- Handles pre-charge circuits to limit inrush current during system startup

BMS Functional Overview

The Battery Management System (BMS) interfaces with all key components of the battery pack and the host system (e.g., vehicle, grid storage, etc.).

Its responsibilities can be grouped into six main functions:

2 Charge Control

- Regulates charging current and voltage based on cell conditions
- Communicates with external chargers to enforce safe charging limits
- Inhibits charging under unsafe conditions (e.g., low temperatures)
- Applies charging strategies that reduce cell degradation and extend battery life

BMS Functional Overview

The Battery Management System (BMS) interfaces with all key components of the battery pack and the host system (e.g., vehicle, grid storage, etc.).

Its responsibilities can be grouped into six main functions:

3 Thermal Management

- Monitors battery temperatures across modules or zones
- Coordinates with cooling or heating systems to maintain safe operating range
- Prevents thermal runaway by responding to overheating or rapid temperature rise

BMS Functional Overview

The Battery Management System (BMS) interfaces with all key components of the battery pack and the host system (e.g., vehicle, grid storage, etc.).

Its responsibilities can be grouped into six main functions:

4. SOC / SOH Estimation

- Estimates State of Charge (SOC) to indicate usable energy
- Tracks battery aging through State of Health (SOH) metrics such as capacity and resistance
- Supports energy and power predictions for system control

BMS Functional Overview

The Battery Management System (BMS) interfaces with all key components of the battery pack and the host system (e.g., vehicle, grid storage, etc.).

Its responsibilities can be grouped into six main functions:

5. Cell Balancing

- Detects imbalance among cell voltages
- Applies passive or active balancing techniques to equalize cells
- Enhances overall energy usage and contributes to longer pack lifespan

BMS Functional Overview

The Battery Management System (BMS) interfaces with all key components of the battery pack and the host system (e.g., vehicle, grid storage, etc.).

Its responsibilities can be grouped into six main functions:

6. Battery Safety and Protection

- Detects and mitigates over-voltage, under-voltage, over-current, short circuits, and isolation faults
- Acts quickly to disconnect the battery in hazardous conditions
- Ensures safe operation across a range of applications and environments

What does a BMS look like?

A **BMS** is an **embedded system** typically consisting of one or more electronic boards (printed circuit board - PCB) inside the battery pack, including:

- **Processing unit** (microcontroller, microprocessor, or FPGA)
- **Measurement circuits** for voltage, current, and temperature
- **Protection and safety circuits**
- **Communication interfaces** with the vehicle

The BMS interfaces with:

- the battery pack
- the vehicle control unit
- other vehicle subsystems

The processor executes algorithms for:

- battery monitoring
- battery protection
- battery control

Embedded system:
purpose-built electronics
and processing designed
for a specific application.

Microcontroller (MCU)	CPU + memory + peripherals on one chip	Most common in BMS
Microprocessor (MPU)	CPU requiring external memory/peripherals	More complex embedded systems
FPGA	Programmable digital hardware	Parallel and customized processing

Tesla Model 3 battery pack



Tesla Model 3 battery BMS

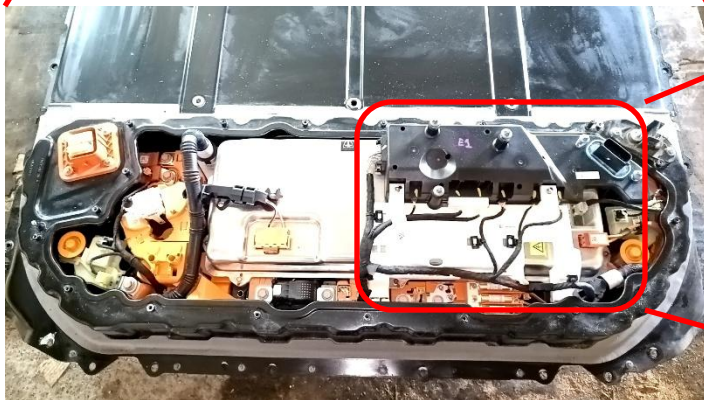


Figure. Model 3 BMS main board cover and connecting bracket@zhixiauto

Tesla Model 3 battery BMS

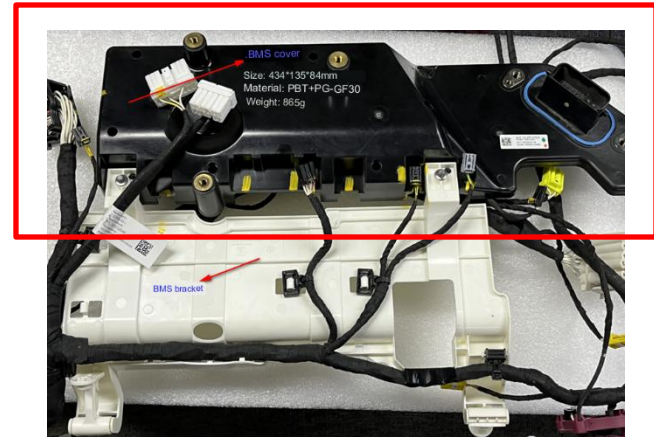
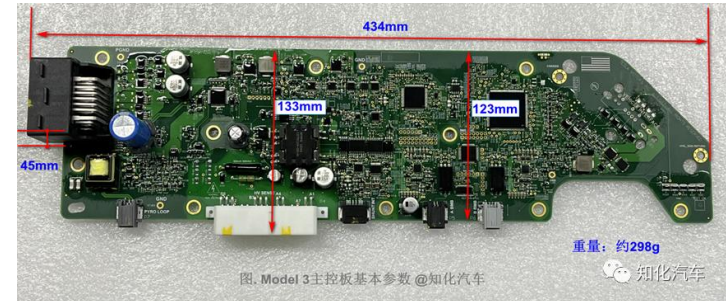


Figure. Model 3 BMS main board cover and connecting bracket@zhihuauto Zhihua Automobile

Tesla Model 3 battery BMS

This board implements the main functions of a BMS.

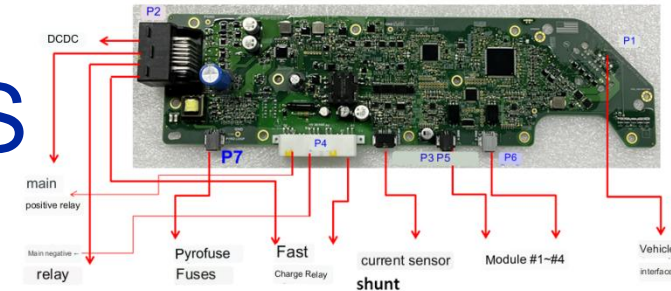


Figure. Model 3 BMS main control board external interface relationship diagram @zhuhuaautomobilezhuhuaautomobile

1. Computation (BMS "brain"):

Microcontrollers

- run the BMS algorithms (estimate SOC / SOH, perform diagnostics and protection..)

2. Communication with the vehicle:

CAN transceiver (CAN is the standard communication network used in vehicles)

- interface to the CAN bus
- exchanges data with other controllers (vehicle control unit, charger, etc.)

3. Control of the high-voltage battery

Relay driver circuits

- drive the high-voltage contactors
- electrically controlled switches that connect or isolate the battery pack

4. Safety functions

HVIL (High Voltage Interlock Loop)

- detects if HV connectors are unplugged or the battery pack is opened. In these cases the BMS disconnects the battery

5. Measurement and protection

- current sensor (shunt) for pack current measurement
- fuses / pyrofuse for fault protection
- TVS (Transient Voltage Suppressor) and protection diodes against voltage spikes

Tesla Model 3 battery BMS

Multiple microcontrollers:

- **BATMAN – Main BMS controller**
 - executes the main battery management algorithms
 - performs SOC / SOH estimation
 - supervises protection and system operation
- **ROBIN – Safety / monitoring controller**
 - supervises critical safety functions (contactors, current limits, HVIL) and cross-checks the BATMAN for functional safety.
- **MCU-LP – Low-power controller**
 - operates in low-power / standby modes
 - manages sleep and wake-up of the battery system

Fast-charge relay: dedicated relay used during DC fast charging. provides a separate current path from the charger to the battery. Allows the BMS to safely control high charging currents.

Current sensor (shunt): measures the total battery pack current. Used for SOC estimation, power limits, fault detection Only the current sensor is located on the main BMS board.

Other measurements (cell voltage, temperature sensors) and cell balancing circuits are performed inside the **battery modules** which are connected to the main BMS via module interfaces (Module #1–#4).

The BMS uses a **distributed architecture**:

- main control board → computation and pack-level control
- module electronics → cell-level sensing and balancing

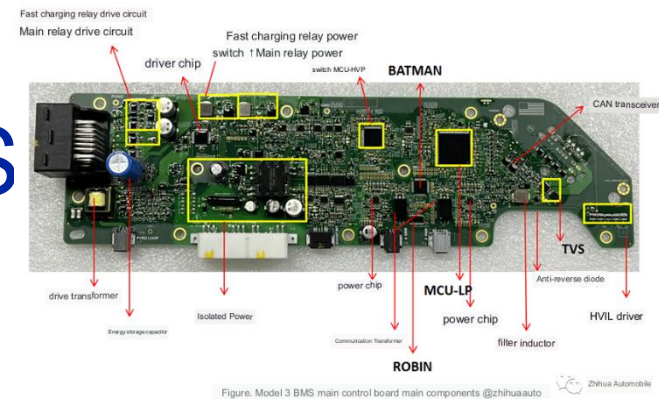


Figure. Model 3 BMS main control board main components @zhihuauto

Cell monitoring – sensing

Voltage

All cell voltages must be measured:

- Key indicator of cell balance
- Critical input for SOC and SOH estimation
- Essential for safety: overcharging can cause thermal runaway

Measurement Electronics

Specialized chipsets assist with high-voltage monitoring:

- Low-cost “dumb” chips near cells for basic sensing
- High-end processing units located centrally

Advanced A/D chips offer:

- High accuracy
- Robust performance in Electromagnetic Interference (EMI), heat, and vibration

Often designed with parallel redundancy: enables fault-tolerant monitoring

Cell monitoring – sensing

Temperature

Battery temperature strongly affects performance and degradation.

- Low temperatures: avoid charging to prevent damage.
- High or sudden temperature changes: indicate faults or safety risks.
- *Ideally*, measure each cell's internal temperature
- *Alternatively*, place external sensors on representative cells and estimate internal temperatures using a calibrated thermal model

Sensors measure temperature indirectly **through voltage**:

- **Thermocouple** directly produces a (very small) voltage, which can be amplified and measured. Best suited for lab tests.
- **Thermistor** (NTC/PTC) easier to use in products.
 - Resistance changes significantly with temperature.
 - Lookup tables convert measured voltage back to temperature in software

Cell monitoring – sensing

Current

Essential for safety monitoring and abuse detection & SOC and SOH estimation

Two main sensing methods:

1. Shunt Sensor

- A low-value precision **resistor** (e.g., 0.1 mΩ) placed in series with the battery, typically at the low-voltage end
- Current is calculated from the voltage drop: $I = V_{\text{shunt}} / R_{\text{shunt}}$

Advantages:

- High accuracy and zero offset at no current

Challenges:

- Sensitive to temperature: resistance must be calibrated
- Produces a very small signal, requiring EMI protection and amplification
- Not electrically isolated from the battery:
 - The shunt shares electrical potential with the battery terminals.
 - If the BMS electronics require isolation (e.g., for safety, noise immunity, or grounding strategy), additional isolation circuitry (e.g., isolated amplifiers or ADCs) is required

Cell monitoring – sensing

Current

Essential for safety monitoring and abuse detection & SOC and SOH estimation

Two main sensing methods:

2. Hall-Effect Sensor

Measures the **magnetic field** generated by current flowing through a conductor

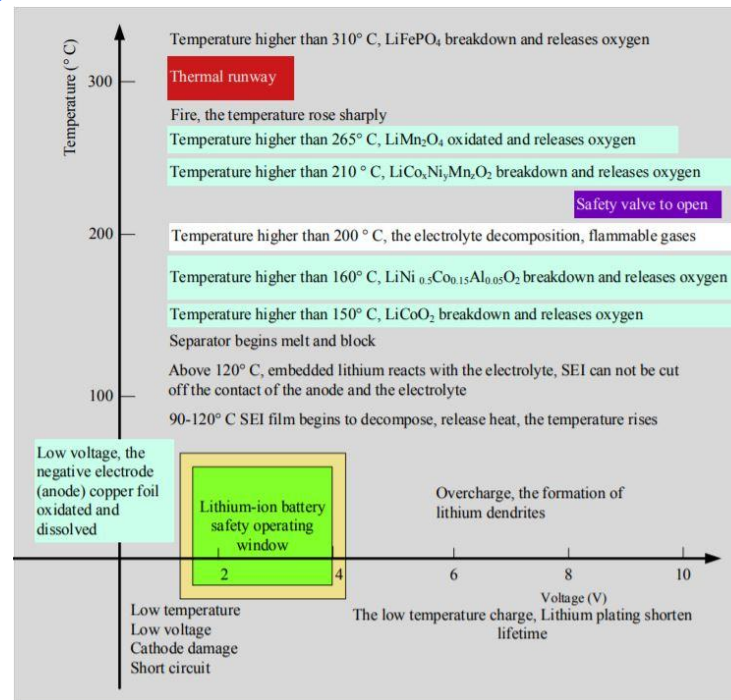
Advantages:

- Provides galvanic isolation from battery current path (no direct electrical contact)
- Easier to integrate into systems requiring isolated signal paths

Challenges:

- Prone to offset errors that vary with temperature
- Requires calibration and feedback circuitry to reduce magnetic hysteresis and improve accuracy

Thermal/voltage operating window



Manufacturers recommend a safe and reliable operating area, which is limited by **temperature and voltage windows**.

“...exceeding the restrictions of these windows will lead to rapid attenuation of battery performance and even result in safety problems.”

State-of-Charge Estimation: Why?

xEVs must know:

- How much energy is available (important for EV → driving range)
- How much power is available (important for HEV → acceleration/braking)

To compute these:

- Energy → needs each cell's SOC (z_k) and capacity (Q_k)
- Power → needs each cell's SOC (z_k) and resistance (R_k)

These quantities **cannot be measured** directly and must be estimated
Available inputs: cell voltages, pack current, cell/module temperature

Impact of Poor Estimation

- Simple methods are easier to code but less accurate
Consequences include:
 - Abrupt current/voltage limit violations → poor drivability
 - Overcharge/over-discharge → cell damage
 - Oversized packs to compensate → added cost, weight, volume

Key takeaway: Investing in advanced estimation algorithms enables smaller, cheaper, and safer battery packs.

Understanding State of Charge (SOC)

SOC reflects lithium concentration in the negative electrode.

Defined using stoichiometry: $\theta = c_{s,avg} / c_{s,max}$

SOC computed as: $z = (\theta - \theta_{0\%}) / (\theta_{100\%} - \theta_{0\%})$

Voltage $\neq$ SOC:

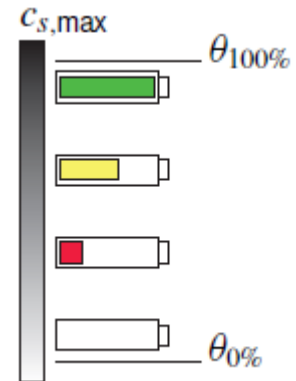
- Voltage depends on surface concentration & temperature
- SOC depends on average concentration in particles
- Voltage changes (rest, temp) do not mean SOC changes
- Hysteresis also causes mismatch between V and SOC

SOC only changes with current flow — charge/discharge or self-discharge

$$z(t) = z(0) - \frac{1}{Q} \int_0^t \eta i(\tau) d\tau$$

Pack SOC is ill-defined:

- Example: one cell full, one cell empty — what's the SOC?
- Highlights the need for cell balancing
- Average SOC may help for control, but energy is what really matters



Battery Diagnostics: SOH and SOL

The BMS must estimate State-of-Health (SOH) of the battery pack.

SOH is not precisely defined — generally tracks aging effects.

Key measurable indicators:

- Capacity fade: 20–30% reduction over life
- Power fade: 50–100% increase in internal resistance

Estimating cell resistance (R_k) and capacity (Q_k) during operation helps track SOH.

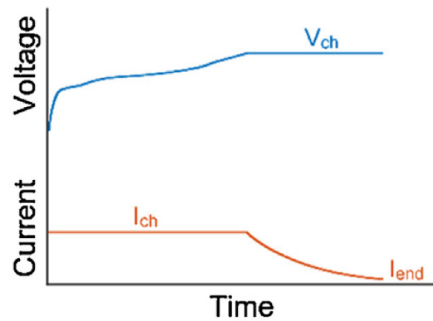
Some also define State-of-Life (SOL): estimates remaining usable life (in % or time).

But aging rates are not constant — future cell stress may differ from past.

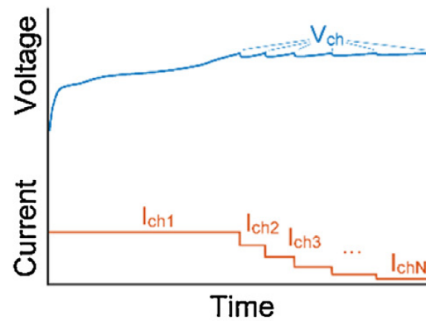
Better insight comes from tracking internal degradation mechanisms, not just R_k and Q_k .

Charge control: common strategies

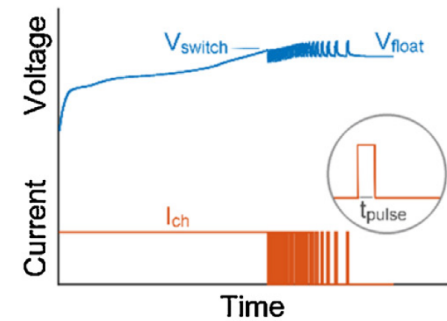
(a) Constant Current Constant Voltage



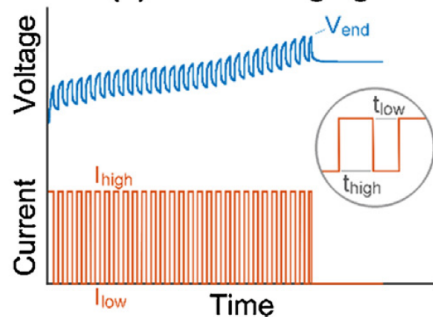
(b) Multistage Constant Current



(c) Constant Current + Pulsed Charging



(d) Pulse Charging



(e) Boost Charging

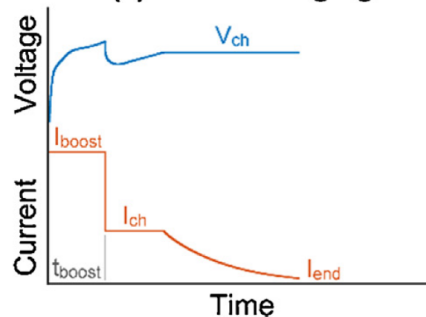
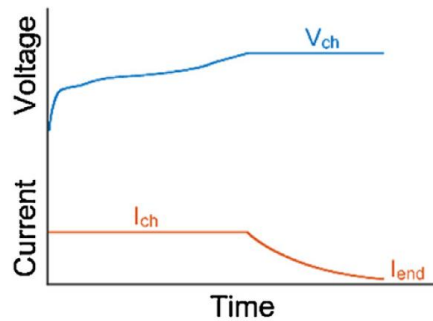


Fig. 1. Schematic illustration of different charging protocols: (a) CCCV, (b) MSCC, (c) CCPC, (d) PC, (e) BC

Charge control: CC-CV

(a) Constant Current Constant Voltage

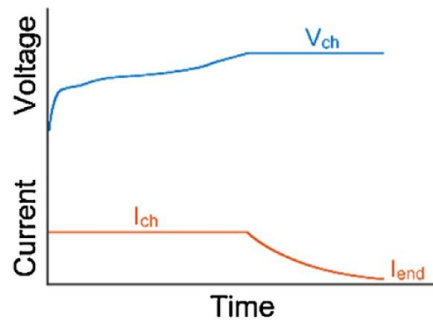


The CC-CV charging circuit is easy to design, implement and operate.

However, it poses various **issues**.

Charge control: CC-CV

(a) Constant Current Constant Voltage



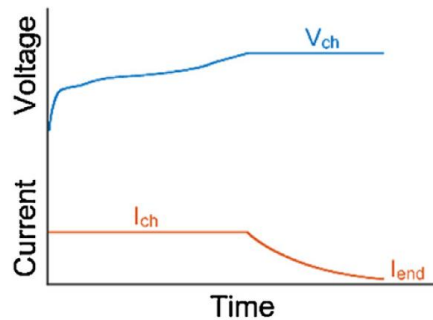
The CC-CV charging circuit is easy to design, implement and operate.

However, it poses various **issues**.

1. the **CV phase is very time-consuming**, which is often considered unsuitable for a quick charging process. A relatively **high current** in the CC phase can be used to **reduce charging time**, but could **exacerbate side reactions** inside the charged battery and **affects its cycle life**.

Charge control: CC-CV

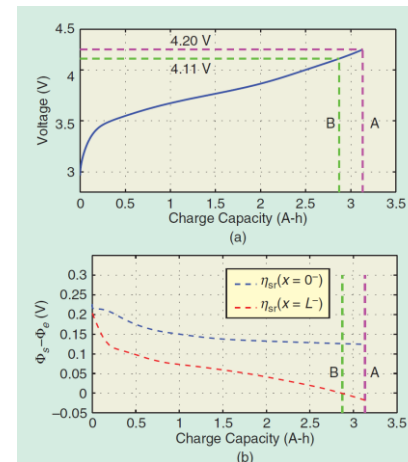
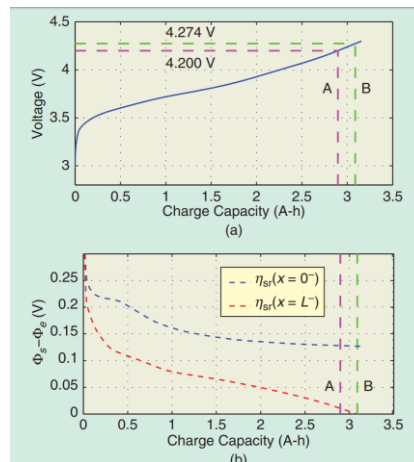
(a) Constant Current Constant Voltage



The CC-CV charging circuit is easy to design, implement and operate.

However, it poses various **issues**.

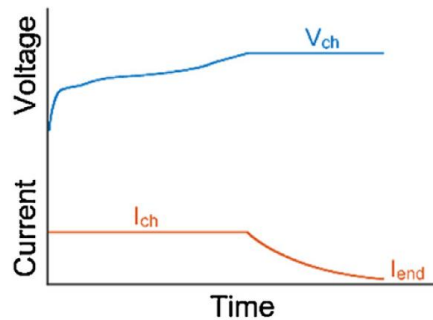
- the **Voltage bound** restriction can **unnecessarily limit the performance**. Additionally, this **bound may not guarantee safety** as the battery **ages** and its characteristics change.



One could constrain the **overpotential for a side reaction** instead.

Charge control: CC-CV

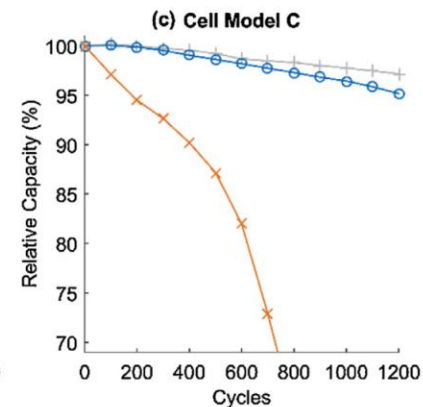
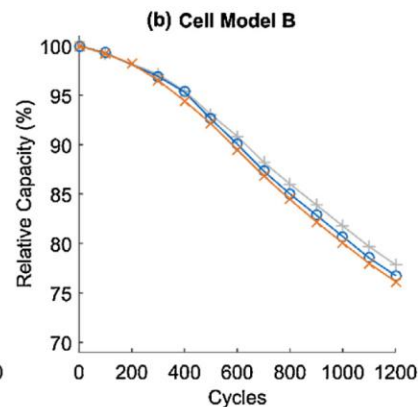
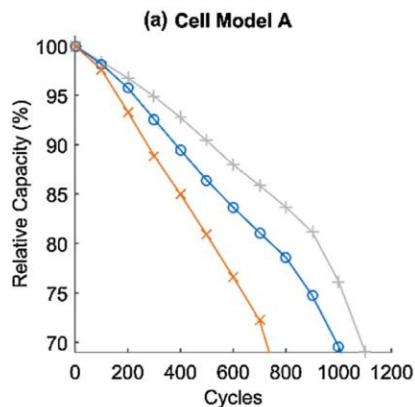
(a) Constant Current Constant Voltage



The CC-CV charging circuit is easy to design, implement and operate.

However, it poses various **issues**.

- the standard method **is independent of battery models** so that it **cannot distinguish between individual cells**.



Charge control: CC-CV

How can we provide an **optimal CC-CV tuning** for each **individual cell** which **adapts** with its **ageing**?

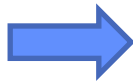
Using input/output data and a mathematical model

If we use a **data/model**, can we come up with a better strategy than the CC-CV?

Optimal model-based charging strategies

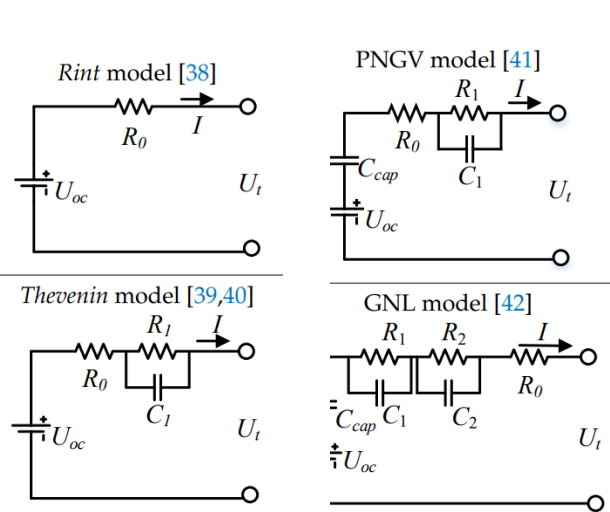
Advanced BMS

Advanced Battery Management Systems (ABMSs) rely on mathematical models and data to achieve high performance



Battery models

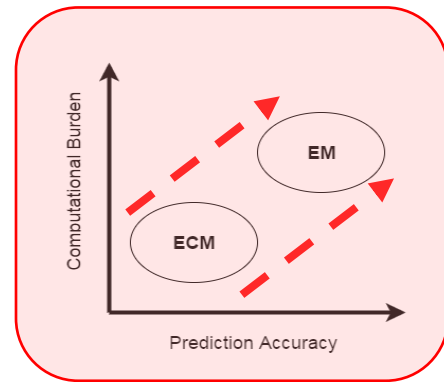
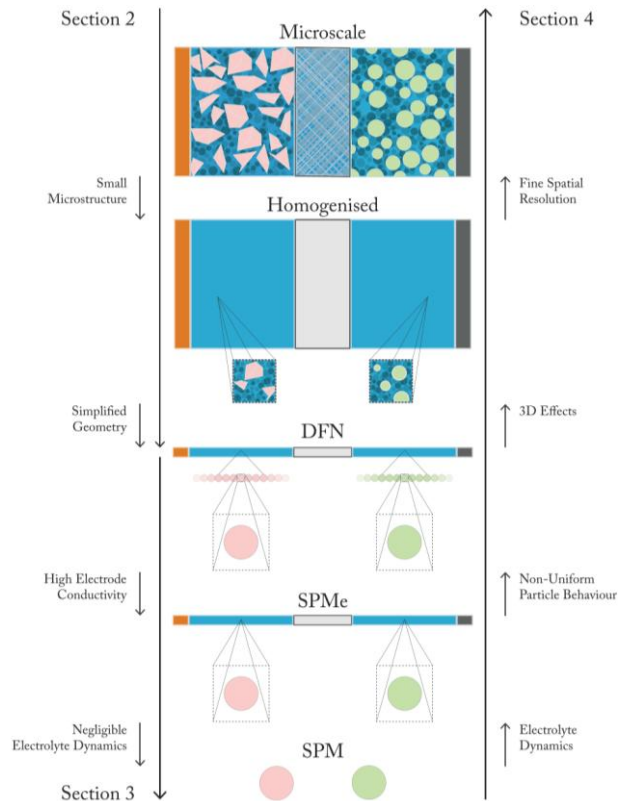
Equivalent circuit models (ECMs)



S. Nejad, D.T. Gladwin, D.A. Stone, Journal of Power Sources 316, 183-196, 2016

Could also have parameters depending on SOC and T, augmented with thermal and aging mechanisms...

Electrochemical models (EMs)



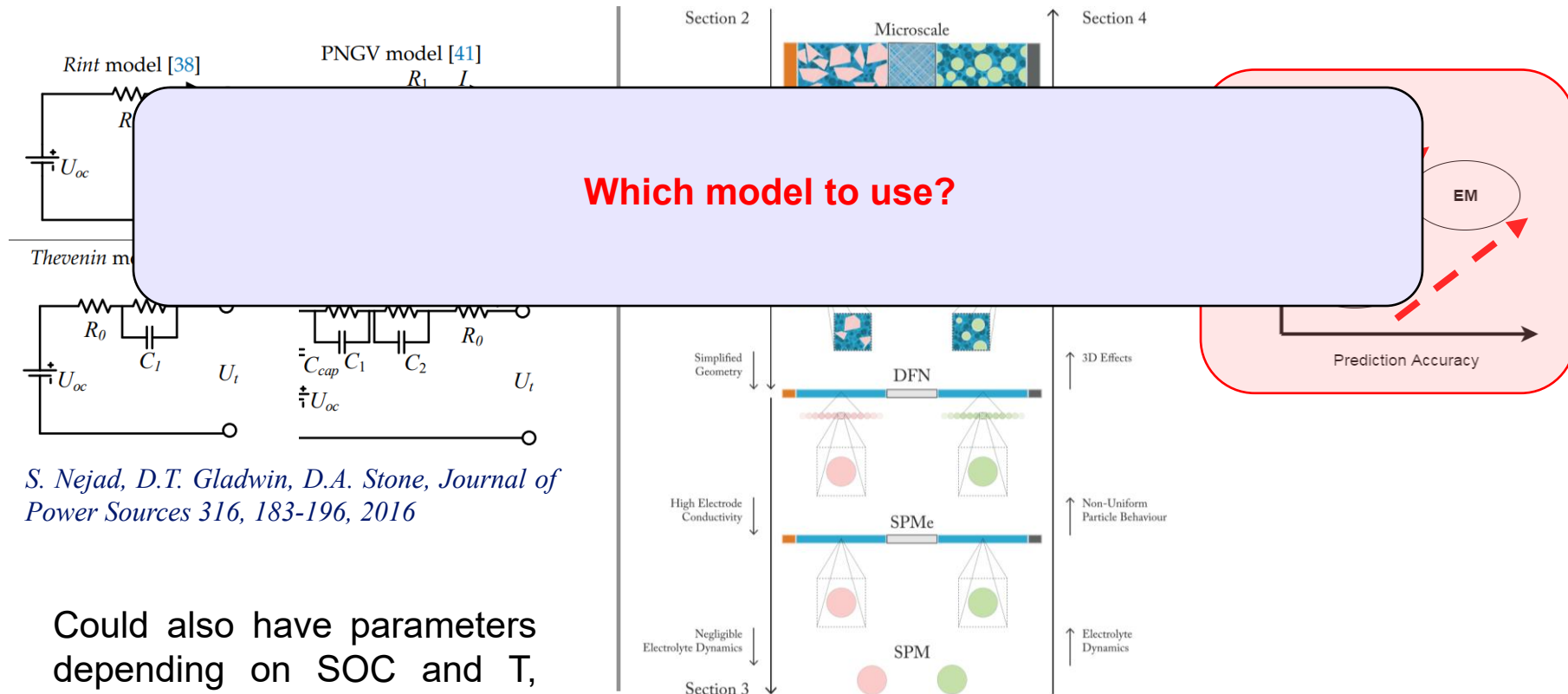
F. Brosa Planella et al, Prog. Energy 4, 042003, 2022

Also coupled with thermal and aging dynamics..

Battery models

Equivalent circuit models (ECMs)

Electrochemical models (EMs)



S. Nejad, D.T. Gladwin, D.A. Stone, Journal of Power Sources 316, 183-196, 2016

Could also have parameters depending on SOC and T, augmented with thermal and aging mechanisms...

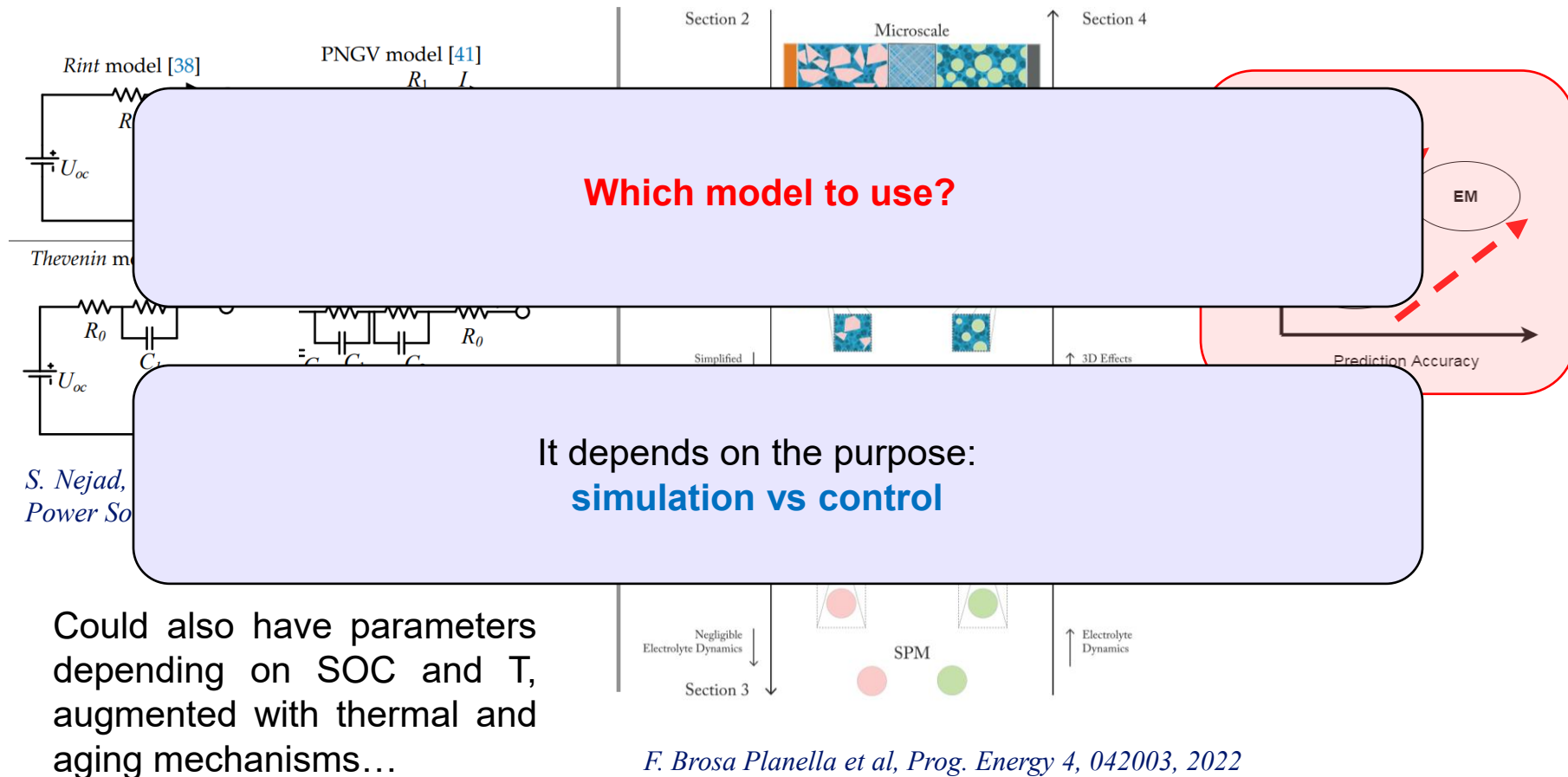
F. Brosa Planella et al, Prog. Energy 4, 042003, 2022

Also coupled with thermal and aging dynamics..

Battery models

Equivalent circuit models (ECMs)

Electrochemical models (EMs)

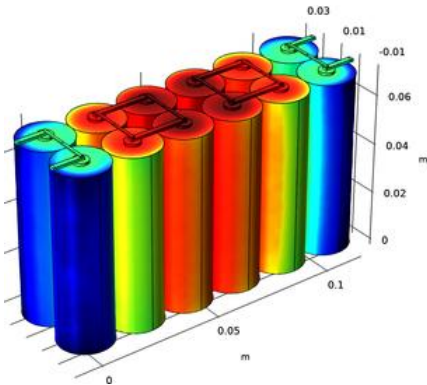


Also coupled with thermal and aging dynamics..

Digital twin



For accelerated product development, many industries have established a **modeled approach** to accelerating design iterations and **increasing understanding** gained from formalizing empirical know-how.



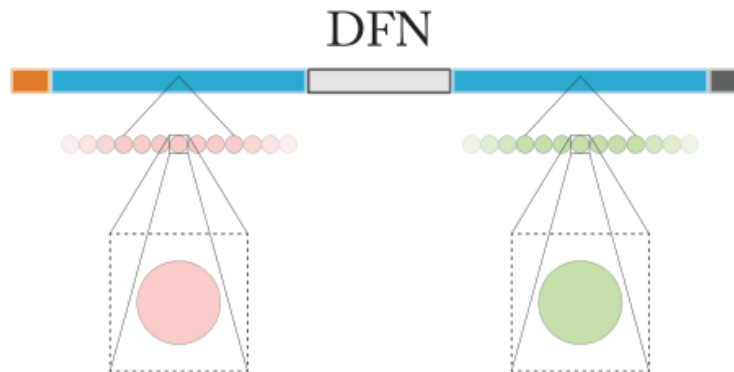
In the context of **battery modeling**, **physics-based models** can provide an improved **understanding** of battery behavior and a modelled approach can assist in **accelerating the pace of battery development**.

The Pseudo two-dimensional model

The Pseudo Two Dimensional (P2D) model is the most widely used **electrochemical model** for Li-ion cells.

It is described by a set of **tightly coupled and highly nonlinear partial differential-algebraic equations** (PDAEs) that define the conservation of **mass** and **charge** in the cell.

M Doyle, TF Fuller, J Newman, Modeling of galvanostatic charge and discharge of the lithium/polymer/insertion cell, Journal of The Electrochemical Society, 140 (6), 1526, 1993



A P2D-based simulator: LIONSIMBA



Li-ION SIMulation BAttery Toolbox

A Matlab framework based on a **finite volume** model suitable for Li-ion battery **design**, **simulation**, and **control** for **single elements** or **series-connected packs**

- ✓ Software available on Github

<https://github.com/lionsimbatoolbox/LIONSIMBA>

- Alternative link

<http://sisdin.unipv.it/labsisdin/lionsimba.php>¹

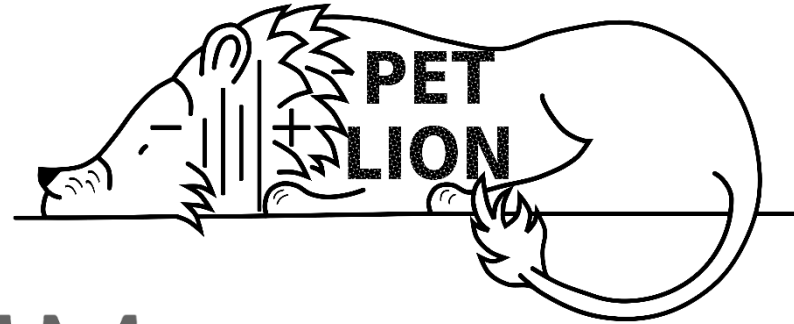
M. Torchio, L. Magni, R.B. Gopaluni, R.D. Braatz, and D.M. Raimondo, LIONSIMBA: A Matlab Framework Based on a Finite Volume Model Suitable for Li-Ion Battery Design, Simulation, and Control, Journal of The Electrochemical Society, 163(7), A1192-A1205. 2016

>290 citations and >17000 visualizations worldwide

Battery simulators



TOOFAB



PyBaMM

fastDFN

SPMeT

MULTIBAT



Dandelion

● + ● - BEST

Parameterization

Parameter identification

Accurate parameters identification is necessary in order models to be useful in practice.

For battery models two approaches are available:

- i. Parameterization from **experiments**.
- ii. Parameterization from **input-output data**.

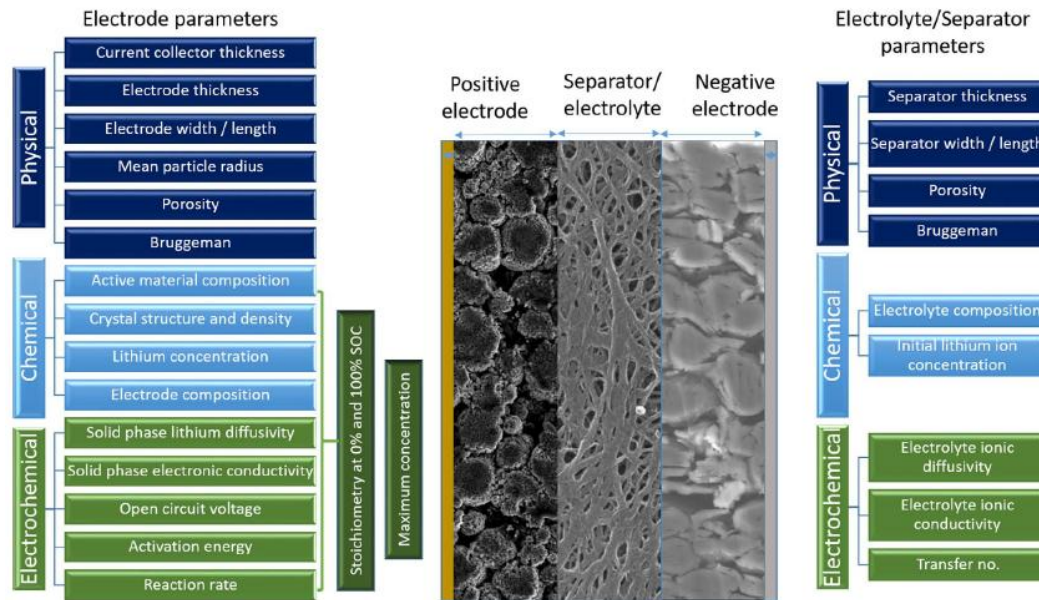
Parameterization from experiments

- It is a **time consuming** procedure.



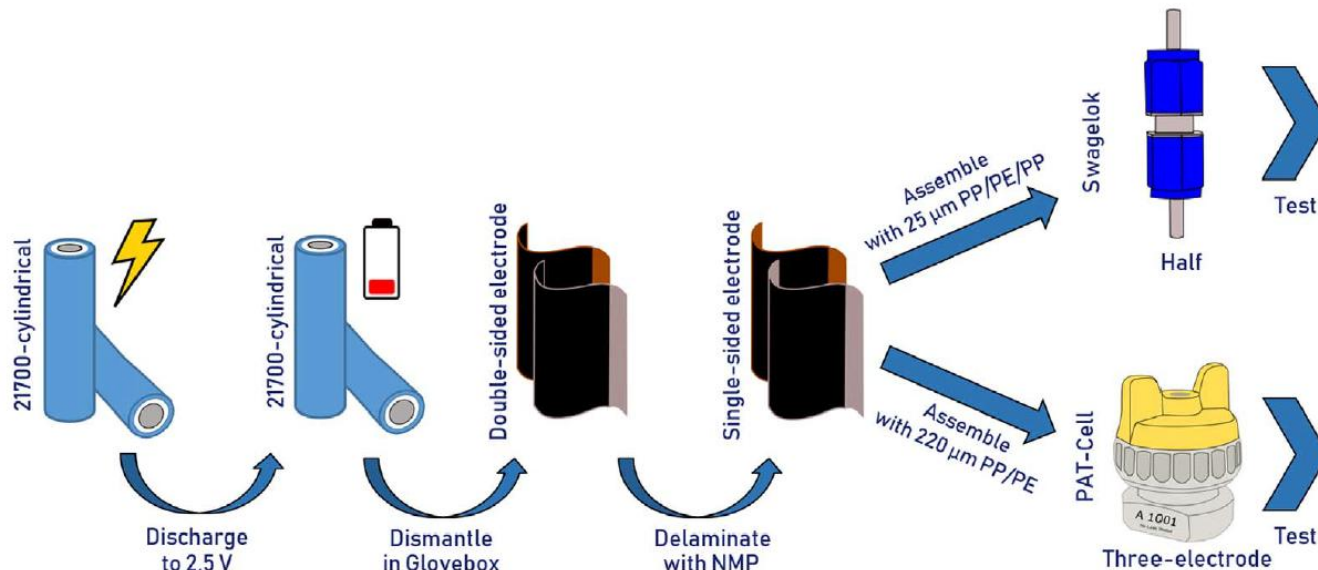
Parameterization from experiments

- It requires **expensive equipment** for performing **experiments** to capture the cell **physical**, **chemical** and **electrochemical** properties (X-ray tomography, scanning electron microscopy, X-ray diffraction, X-ray spectroscopy, inductively coupled plasma optical emission spectroscopy, electrochemical impedance spectroscopy,...).



Parameterization from experiments

- It involves **cell teardown** and **assembly of half cells** for **electrochemical characterization**.



Are the parameters obtained from the half cells well representing the original cell?

Are the obtained parameters valid also for other cells of the same type?

Parameterization from experiments

- The obtained parameters may require a **posterior adjustments** to match the output voltage profile → it is unclear how this affects the internal states behavior.

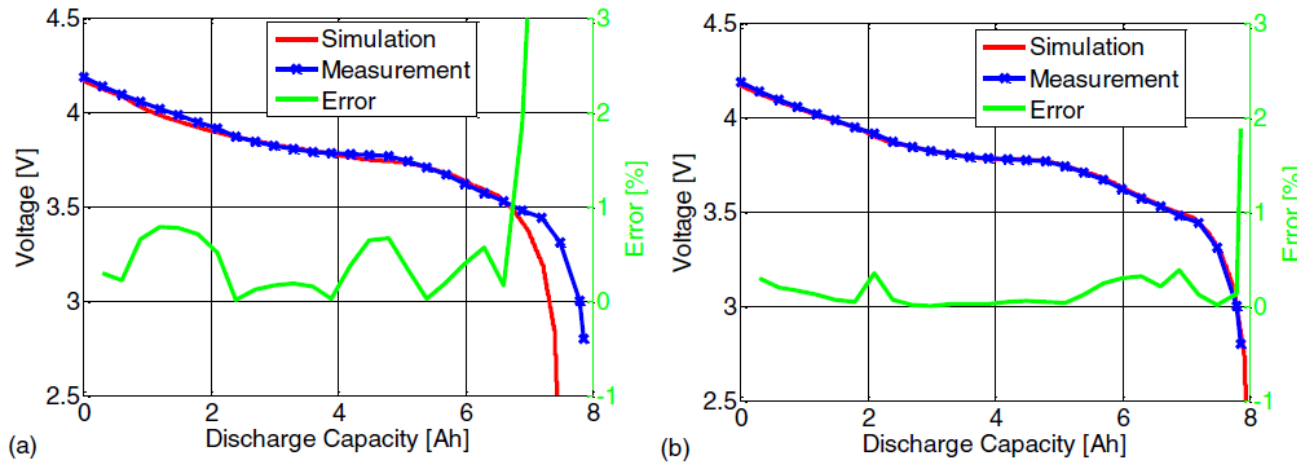


Figure 2. Comparison of a simulated and measured OCV curve of a 7.5 Ah Kokam cell at 25°C. The simulation error is displayed on the right axis. (a): Simulation performed with the measured set of model parameters, (b): Simulation performed with an adapted set of model parameters determining the balancing (see Table II). The inactive parts of anode and cathode are increased from 39.17% to 42% and from 40.37% to 44.5%, respectively, the oversize of the anode is neglected and the capacity loss due to SEI is decreased from 14% to 6.8%.

Parameterization from experiments

- The obtained set of parameters has a **physical meaning**.
- The parameter **accuracy is higher** than in the input-output case.

Suitable for digital twins!



Parameterization from input–output data

- Only the input **current**, the output **voltage** and, sometimes, the external cell **temperature** are usually available

Parameterization from input–output data

Which model?

- Simple models are easier to parametrize but exhibit higher **structural errors**
 - non-negligible **bias** between the optimally **parametrized model** and the **true system**.
 - the optimized parameters tend to **lose** their **physical meaning**.
- Complex models have **identifiability issues** and tend to have **high parameter variance**

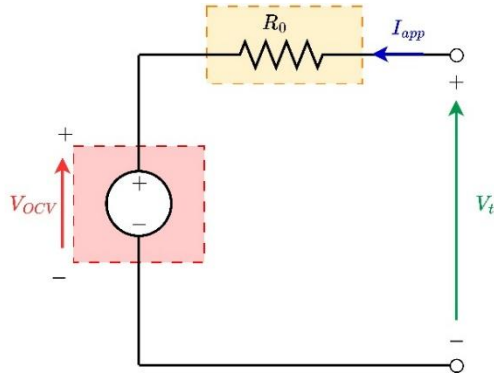
Electrochemical models are preferable but **simplifications** are usually necessary in order to obtain a **good balance of bias and variance**

Equivalent Circuit Models

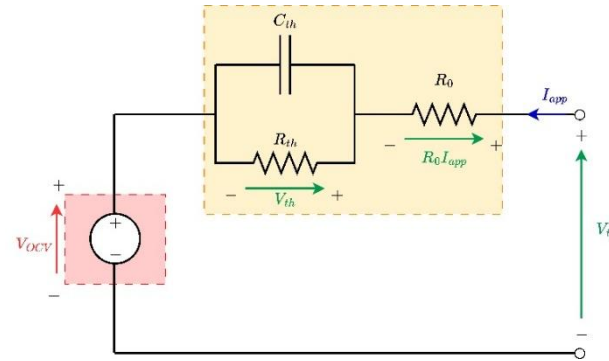
Equivalent Circuit Models (examples)

Aim to reproduce battery terminal voltage under different inputs using standard electrical components.

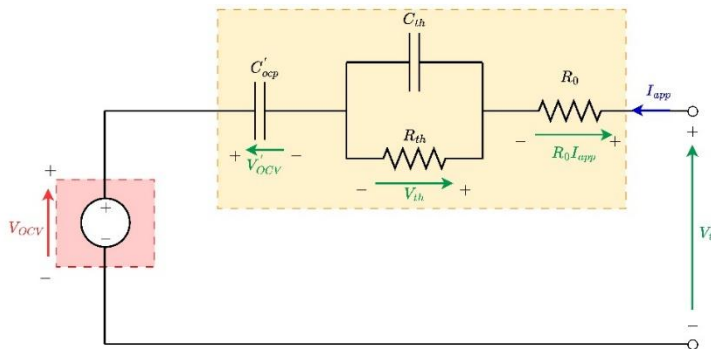
● Internal Resistance Model



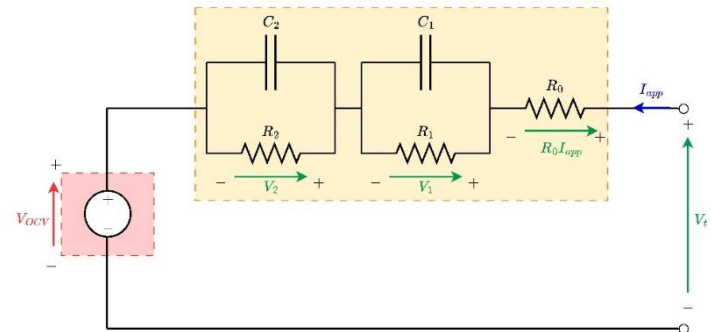
● Thevenin Model



● PNGV Model



● DP Model



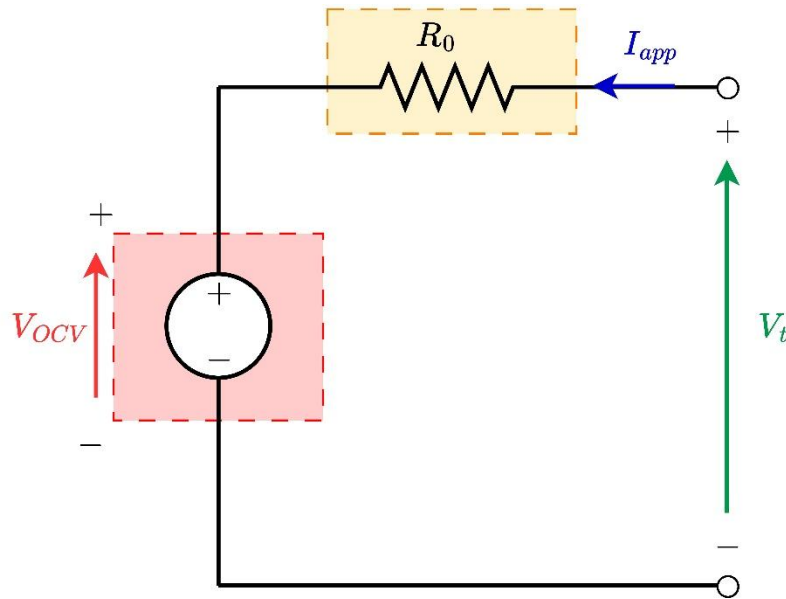
Note: the model elements do not correspond to the cell's internal physical structure.

Increasing model complexity generally improves ability to capture battery terminal-voltage behavior.

Equivalent Circuit Models

● Internal Resistance Model

The internal resistance model uses an ideal voltage source to model the terminal voltage V_t as a function of the applied current I_{app} . It is the simplest model of all but also the least precise.



Mathematical Model:

$$V_t = V_{OCV} + R_0 I_{app}$$

V_{OCV} : the voltage we measure when there is no load connected and no current is flowing through the circuit.

I_{app} : the current flowing through the cell.

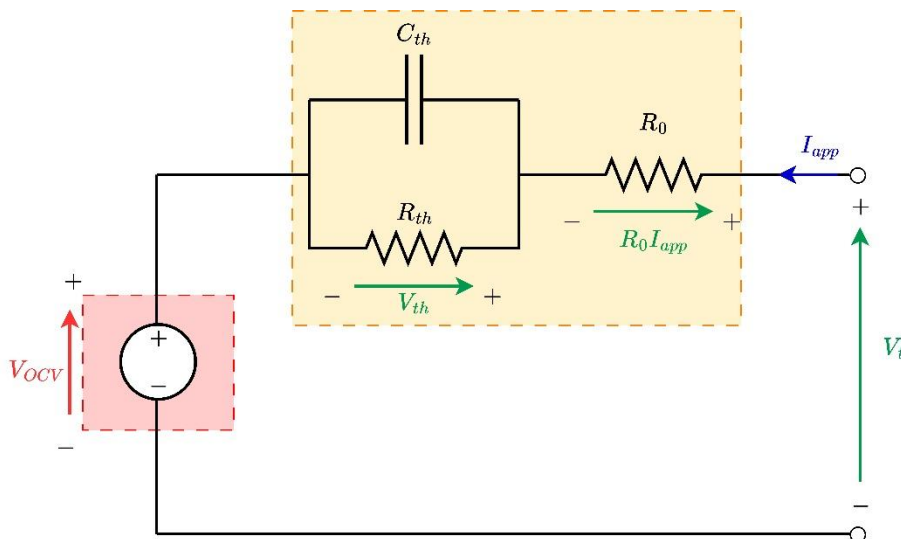
$R_0 I_{app}$: the component modeling the internal voltage drop.

In this model, the voltage source V_{OCV} is assumed to be constant, whereas it is inherently dependent on the SoC.

Equivalent Circuit Models

• Thevenin Model

The Thevenin model connects to the *internal resistance model* a resistor capacitor (RC) parallel describing the dynamics of the cell, in particular the *Recovery effect*. Accuracy is better than with internal resistance model.



Mathematical Model:

$$\begin{cases} \dot{V}_{th} = \frac{-1}{R_{th}C_{th}}V_{th} + \frac{1}{C_{th}}I_{app} \\ V_t = V_{OCV} + V_{th} + R_0 I_{app} \end{cases}$$

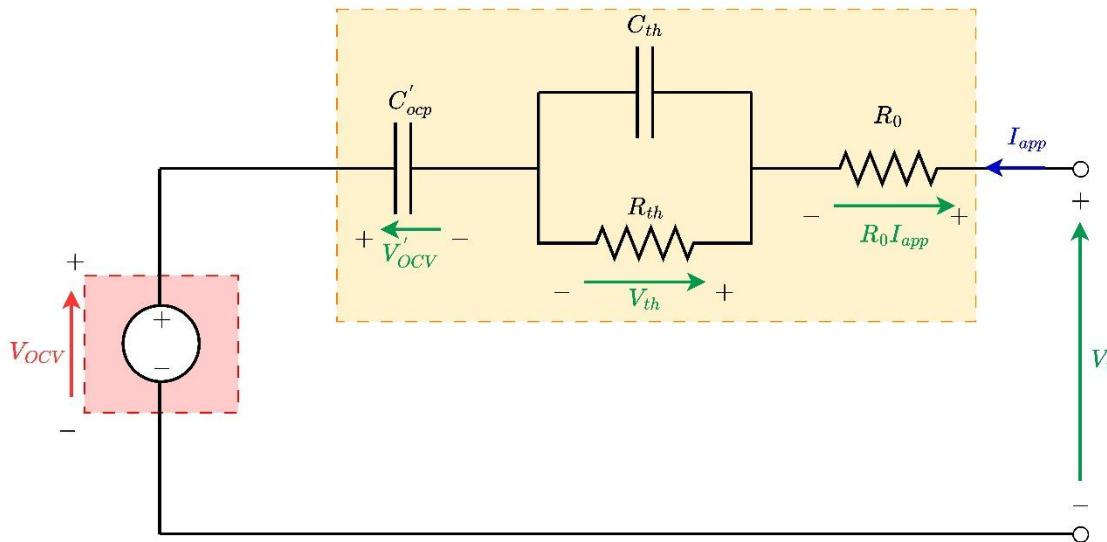
R_{th}, C_{th} : the polarization resistance and capacity used to describe the transient response during charge and discharge.

In this model, the voltage source V_{OCV} is assumed to be constant, whereas it is inherently dependent on the SoC.

Equivalent Circuit Models

● PNGV Model

The partnership for a new generation of vehicle (PNGV) model is obtained by adding a capacitor C'_{ocp} in series to the *Thevenin model* to describe the changing of open circuit potential due to the accumulation of load current.



Mathematical Model:

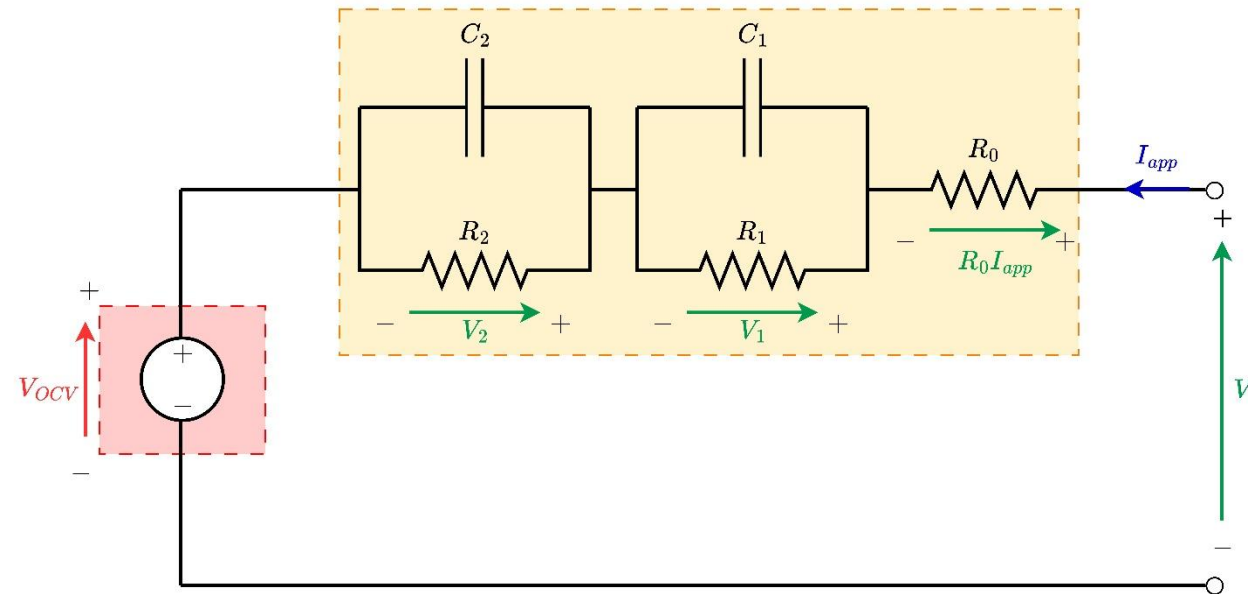
$$\begin{cases} \dot{V}'_{ocp} = \frac{1}{C'_{ocp}} I_{app} \\ \dot{V}_{th} = \frac{-1}{R_{th} C_{th}} V_{th} + \frac{1}{C_{th}} I_{app} \\ V_t = V_{OCV} + V_{th} - V'_{ocp} + R_0 I_{app} \end{cases}$$

In this model, the voltage source V_{OCV} is assumed to be constant, whereas it is inherently dependent on the SoC.

Equivalent Circuit Models

• DP Model

The *Dual Polarization (DP) model* is realized starting from the *Thevenin model* by adding a second *RC* parallel to refine the description of polarization characteristics and simulate the concentration polarization and the electrochemical polarization separately. The transients associated to the polarization capacitors are different, a long one and a short one, with different time constants. In this way the transient response is more precise.



Mathematical Model:

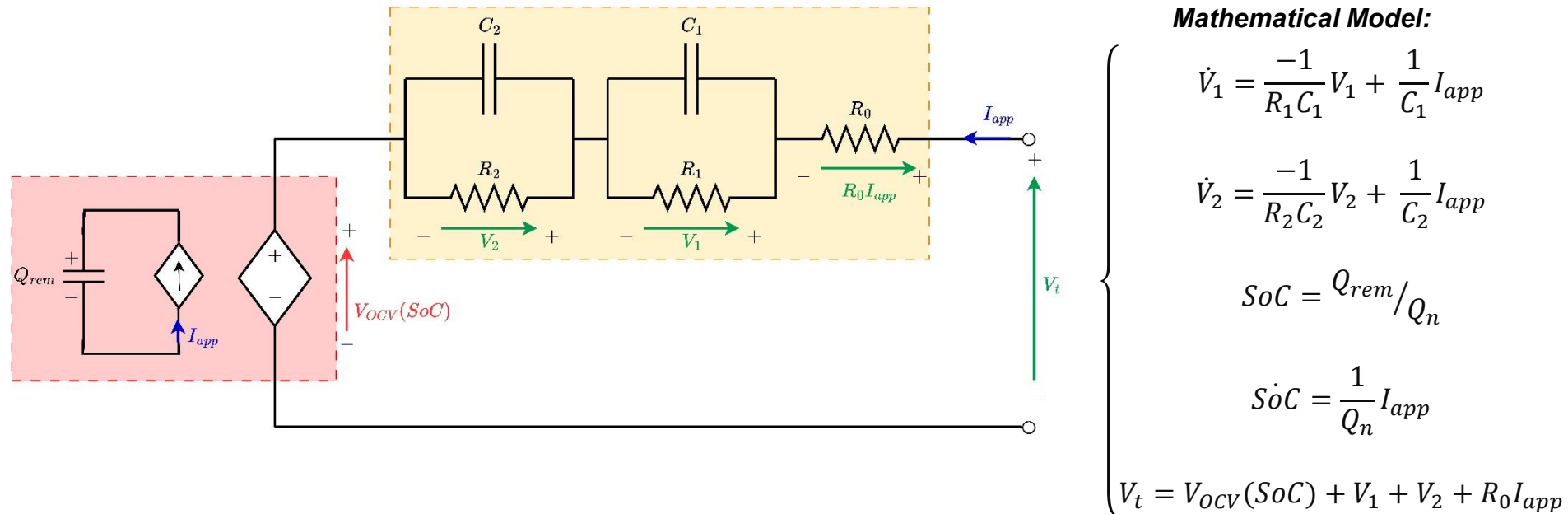
$$\begin{cases} \dot{V}_1 = \frac{-1}{R_1 C_1} V_1 + \frac{1}{C_1} I_{app} \\ \dot{V}_2 = \frac{-1}{R_2 C_2} V_2 + \frac{1}{C_2} I_{app} \\ V_t = V_{OCV} + V_1 + V_2 + R_0 I_{app} \end{cases}$$

In this model, the voltage source V_{OCV} is assumed to be constant, whereas it is inherently dependent on the SoC.

Equivalent Circuit Models

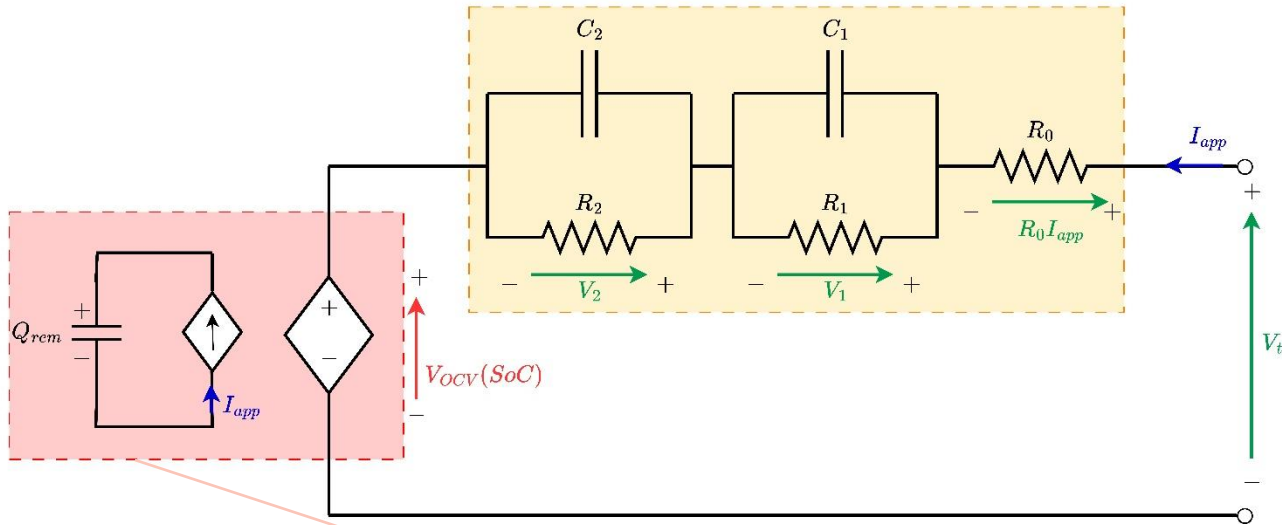
In reality, V_{OCV} is not constant for all levels of SoC. To better say, it is SoC-dependent, i.e., $V_{OCV} = V_{OCV}(SoC)$.

We can consider the SoC-dependent voltage source V_{OCV} in all introduced models: Internal Resistance, Thevenin, PNGV, and DP Model. Here, we just illustrate the DP model with SoC-dependent V_{OCV} .

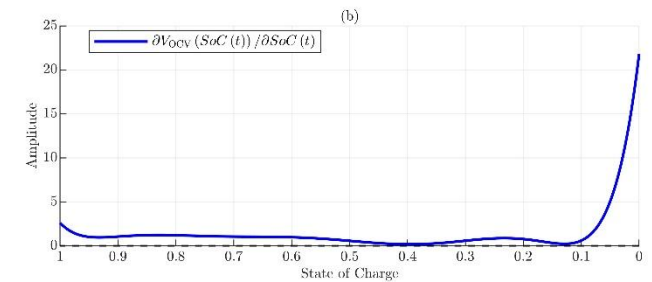
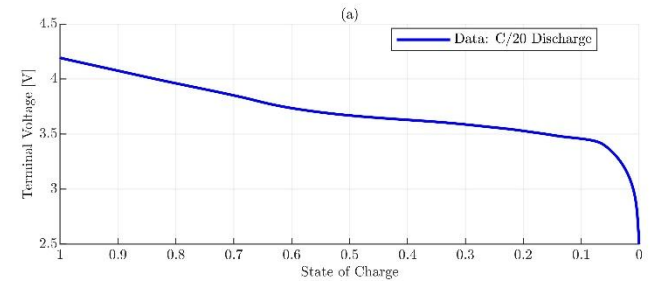


Note: in order to further improve the accuracy of the ECMs, resistances and capacitors in all models can be made SOC, Temperature and SOH-dependent.

Equivalent Circuit Models

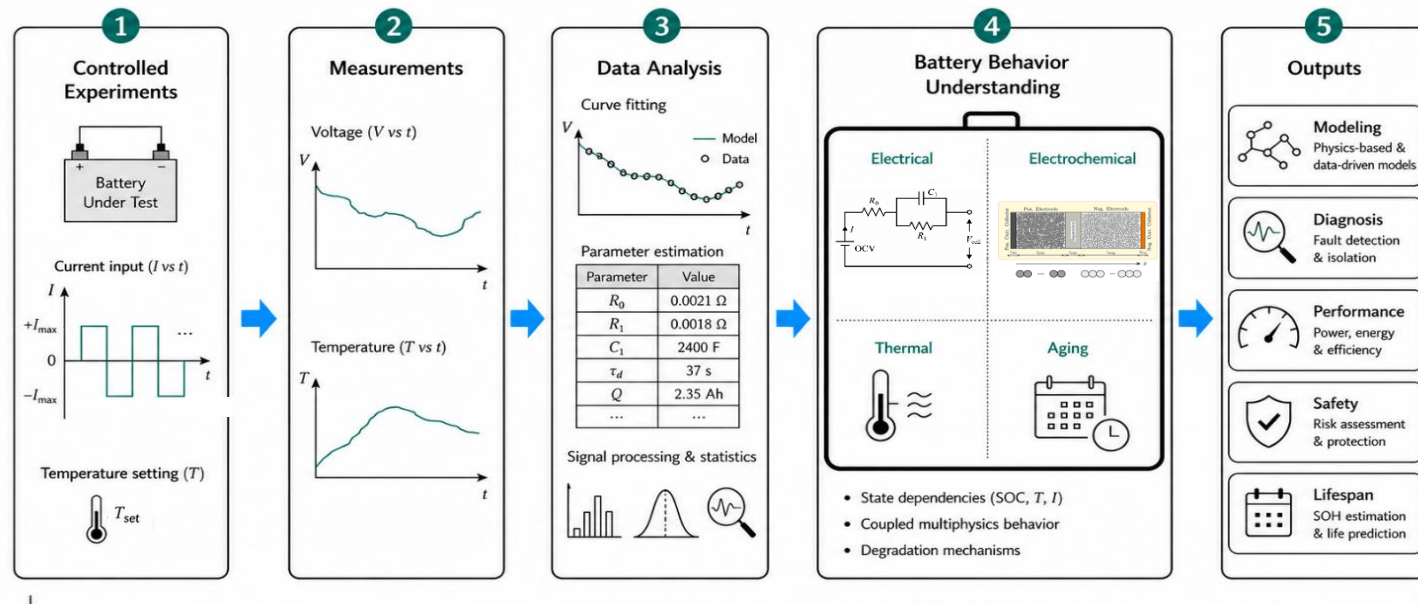


Open Circuit Voltage of the Battery Cell



Battery Characterization Tests for Modeling

Battery characterization refers to the process of performing controlled experiments, measurements, and data analysis to understand the behavior of a battery and to obtain the information required for modeling, diagnosis, and performance evaluation. It provides a more comprehensive understanding of battery operation, supporting improved functionality, safety, and lifespan.



Battery characterization: experiments, measurements, and analysis to understand battery behavior and obtain information for modeling, diagnosis, and performance evaluation—ultimately supporting improved functionality, safety, and lifespan.

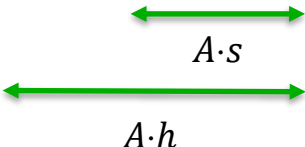
Battery Characterization Tests for Modeling

Classification criterion	Possible categories	Meaning / examples
Physical impact	Non-destructive, destructive / post-mortem	Some tests can be repeated on the same cell without damaging it. Other tests require opening or damaging the cell after use.
Test environment	Offline / laboratory, online / on-board	Some tests are performed in the laboratory under controlled conditions. Other procedures are performed during battery operation, for example inside a battery management system.
Analysis domain	Time-domain, frequency-domain	Time-domain tests study the evolution of measured quantities over time in response to controlled input profiles, such as current, voltage, power, or temperature profiles. Frequency-domain tests study the response to small periodic excitations at different frequencies.
Physical domain	Electrical, electrochemical, thermal, mechanical, aging / degradation	The test may focus on voltage-current behavior, internal electrochemical processes, heat generation and temperature dynamics, deformation or swelling, or long-term degradation.
Excitation type	Constant-current, constant-voltage, pulse-current, realistic load profiles, sinusoidal / frequency sweep, thermal excitation	The battery can be excited using slow charge/discharge currents, voltage-controlled phases, current pulses, application-like current or power profiles, small sinusoidal signals, or controlled temperature changes.
Model representation	Physics-based, equivalent-circuit, data-driven, hybrid	The measured data can be used to build models based on physical principles, circuit analogies, data-driven input-output maps, or combinations of these approaches.
Scale	Cell, module, pack	Tests can be performed on a single cell, on a group of connected cells, or on a complete battery pack including sensing, cooling, electronics, and system-level constraints.

Battery Characterization Tests for Modeling

C/20 discharge test

Capacity test: consists of slowly discharging the battery, for example at a $C/20$ rate, using a constant current amplitude (the C -rate is defined with respect to the nominal battery capacity). In this test, the 100% and 0% SoC values *are operationally defined* through voltage limits: 100% SoC corresponds to the fully charged condition at the upper cutoff voltage, for example $V_{max} = 4.2V$, while 0% SoC corresponds to the fully discharged condition at the lower cutoff voltage, for example $V_{min} = 2.5V$. The discharge starts at $t = 0$, with the battery at 100% SoC and ends at $t = t_f$, when the battery reaches the lower cutoff voltage and is therefore assigned 0% SoC. The total charge extracted during the test is referred to as the discharged capacity, denoted by C_{bat}^{disch} , and is computed as

$$C_{bat}^{disch} = \frac{1}{3600} \int_0^{t_f} I(\tau) d\tau = \frac{1}{3600} I \cdot t_f$$


If I is expressed in $[A]$ and t_f in $[s]$, the capacity is expressed in ampere-hours $[Ah]$.

Battery Characterization Tests for Modeling

C/20 discharge test

Note I. Under the same operational voltage limits, the extracted charge depends on the applied C-rate. Higher C-rates generally lead to a lower measured discharge capacity because ohmic drops and polarization effects cause the terminal voltage to reach the lower cutoff voltage earlier.

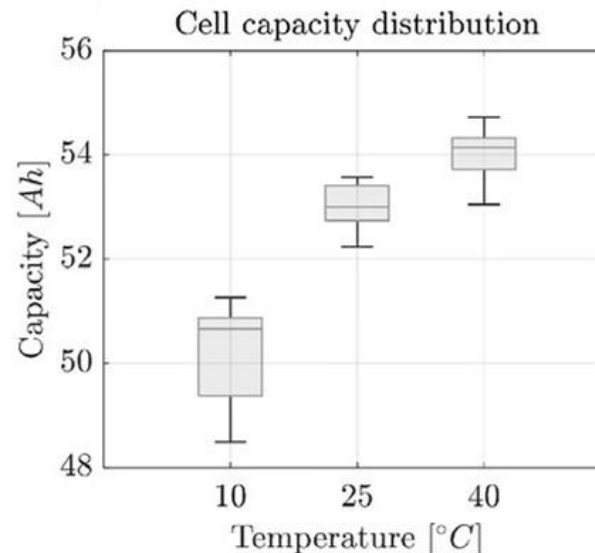
Which capacity should be used? For SoC definition and model calibration, a low-rate capacity test is commonly used as reference. For performance evaluation and range/energy prediction, the usable capacity under realistic operating conditions may be more relevant.

C-rate discharge	Extracted capacity (mAh)
1C	1551 mAh
C/2	1646 mAh
C/5	1954 mAh
C/10	1853 mAh
C/20	2382 mAh

Battery Characterization Tests for Modeling

C/20 discharge test

Note II. Temperature dependence: the extracted capacity is temperature-dependent. Therefore, to characterize $C_{bat}^{disch}(T)$, capacity tests are performed at the same discharge rate and with the same cutoff voltages, while varying only the controlled ambient or cell temperature, for example 10°C, 25°C, and 40°C. This makes it possible to separate the effect of temperature from other protocol-dependent effects.

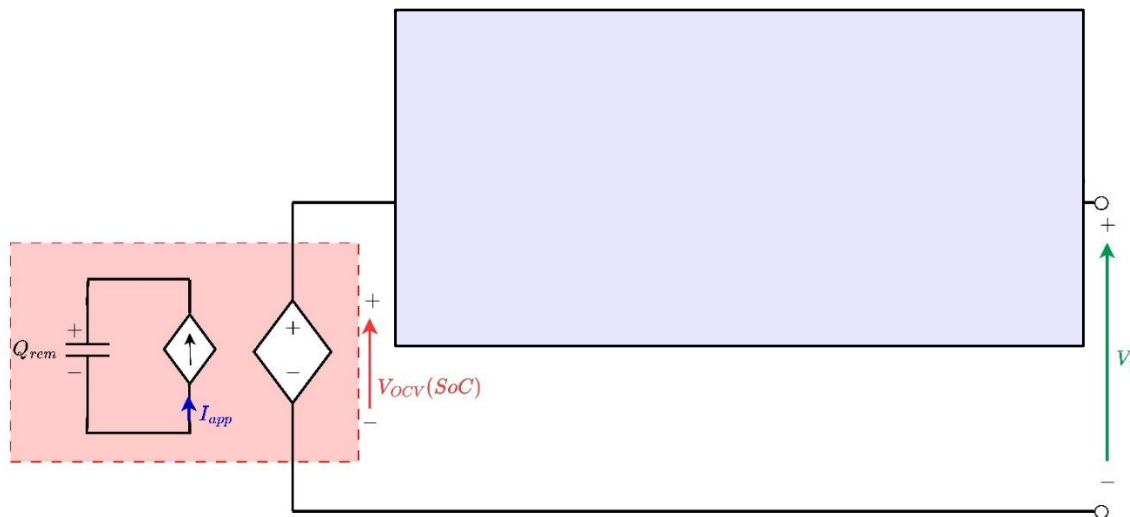


Battery Characterization Tests for Modeling

Open-Circuit Voltage Tests

Open-circuit voltage (OCV) is the battery voltage measured under no-load conditions, i.e., when the applied current is zero, $I = 0$, after a sufficiently long rest period. During this rest period, voltage transients due to ohmic drops and polarization effects decay, so that the terminal voltage approaches the equilibrium voltage of the cell: $V_t \approx V_{OCV}$.

Since V_{OCV} depends primarily on the state of charge (SoC), OCV tests are used to characterize the static relationship $V_{OCV}(\text{SoC})$. This relation is fundamental for battery modeling, SoC estimation, and BMS calibration.



Battery Characterization Tests for Modeling

Open-Circuit Voltage Tests

1. Galvanostatic Intermittent Titration Technique — GITT

Principle: short low-current charge/discharge pulses are followed by long rest periods. The terminal voltage measured at the end of each rest period is used as an estimate of the equilibrium OCV.

Main features

- ✓ More accurate OCV–SoC curve
- ✓ Reduced influence of ohmic drop and polarization
- ✗ Long test duration
- ✗ Requires careful choice of pulse amplitude and rest time

2. Low-rate discharge test, e.g. C/20

Principle: the battery is discharged at a very low constant current. The measured terminal voltage is used as a *pseudo-OCV curve*.

Main features

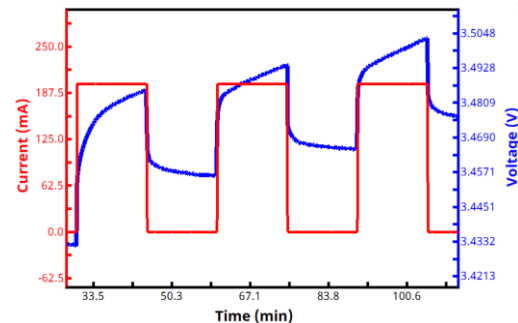
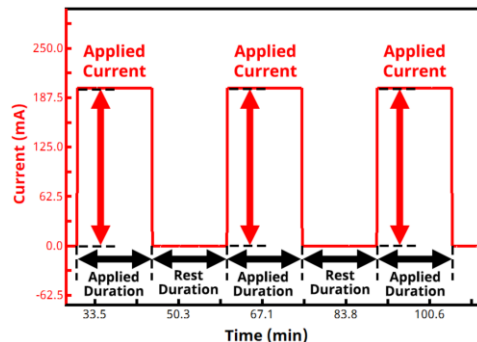
- ✓ Simpler and faster than GITT
- ✓ Useful for practical OCV–SoC characterization
- ✗ Less accurate than GITT
- ✗ Voltage still includes residual overpotentials

GITT provides a more accurate OCV–SoC curve by measuring the relaxed terminal voltage after current interruptions. In contrast, a low-rate discharge test, such as C/20, provides a faster pseudo-OCV curve, but the measured voltage still includes residual polarization effects.

Battery Characterization Tests for Modeling

Open-Circuit Voltage Tests: Galvanostatic Intermittent Titration (GITT)

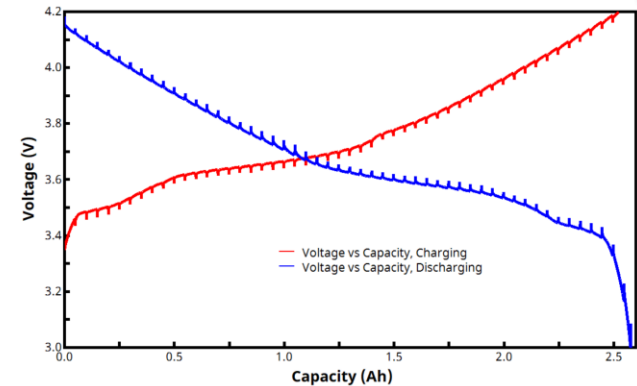
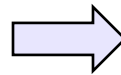
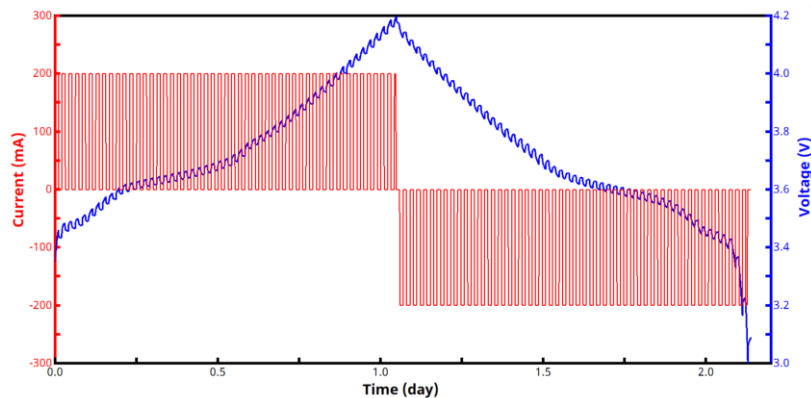
In a GITT test, a small constant current, e.g. C/20, C/25, or even lower, is applied to the battery cell for a short period of time, typically lasting a few minutes. This active step produces only a small variation in SoC and terminal voltage. During each current pulse, a known amount of charge is inserted into or extracted from the cell. If the applied current and pulse duration are fixed, the charge variation is $\Delta Q = I_{app}\Delta t$. The corresponding SoC variation can then be obtained by **Coulomb counting**. When the current pulse is terminated, I_{app} becomes zero. During this rest period, no additional charge is exchanged: the SoC remains approximately constant, while the terminal voltage relaxes as polarization effects decay and approaches a new equilibrium value. The cell is left **under open-circuit conditions**. In practice, equilibrium is assumed when the variation of the terminal voltage becomes smaller than a threshold, or equivalently when $|dV/dt|$ is sufficiently small.



Battery Characterization Tests for Modeling

Open-Circuit Voltage Tests: Galvanostatic Intermittent Titration (GITT)

The relaxed terminal voltage is then taken as an estimate of the open-circuit voltage, V_{OCV} , at the corresponding SoC. The SoC value associated with this point is obtained by Coulomb counting during the applied current pulse. By repeating this sequence of current pulses and rest periods over the desired SoC range, the OCV–SoC relationship can be characterized.



During a GITT current pulse, charge changes according to $\Delta Q = I_{app} \Delta t$. During the subsequent rest period, $I_{app} = 0$, so the charge/SoC remains constant while the terminal voltage relaxes. Therefore, in a voltage–capacity plot, **rest periods** appear as **vertical voltage relaxations**.

Battery Characterization Tests for Modeling

Open-Circuit Voltage Tests: Galvanostatic Intermittent Titration (GITT)

Note I. GITT starts from fully charged condition, defined as SoC=100% at the upper cutoff voltage, and terminates at the lower cutoff voltage, conventionally assigned to SoC=0%.

Note II. Each pulse step should produce only a small voltage and SoC change → pulse duration is kept relatively short, usually between 5 and 30 minutes. A sufficiently long rest period is then imposed after each pulse to allow voltage relaxation. This makes it possible to obtain a more reliable estimate of the V_{OCV} -SoC relationship.

Note III. GITT steps duration and number is not exactly known a priori. Even if the applied current is chosen from the nominal capacity these depend on the cell response.

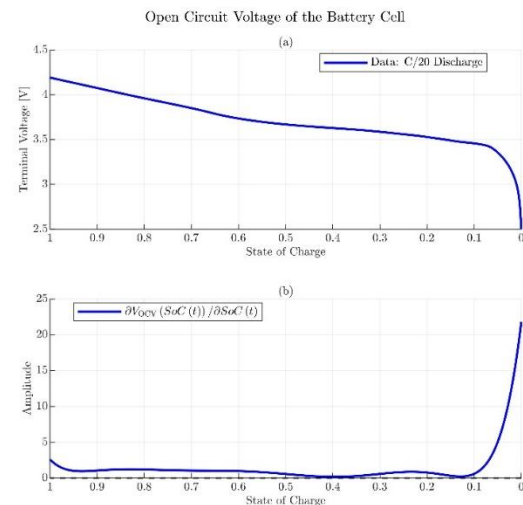
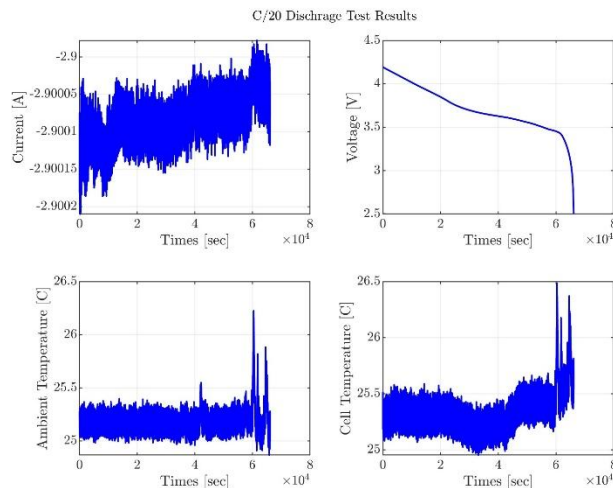
Note IV. GITT tests are **time-consuming**: each current pulse is followed by a rest period. If both discharge & charge branches are characterized over the full SoC range, the complete measurement can take several days and, in some cases, even longer than a month.

Battery Characterization Tests for Modeling

Open-Circuit Voltage Tests: C/20 discharge test

The test starts from the fully charged condition, operationally defined as SoC=100% at the upper cutoff voltage. A **small constant discharge current**, e.g. C/20 or C/25, is **continuously applied** to the battery cell, without intermediate rest periods, until the terminal voltage reaches the lower cutoff voltage, conventionally assigned to SoC=0%.

Since the current is not zero during the test, the measured terminal voltage does not correspond exactly to the equilibrium OCV. Instead, it provides a **pseudo-OCV curve**, whose accuracy improves as the applied current is reduced.



Battery Characterization Tests for Modeling

Hybrid Pulse Power Characterization (HPPC)

Used to characterize **short-term voltage response** and power capability of a cell **as a function of SoC**. At each prescribed SoC level, the cell is subjected to short constant-current discharge (used to evaluate the voltage drop and discharge power capability) and charge/regenerative pulses (used to evaluate voltage rise and charge acceptance capability) usually separated by rest periods. The cell is then moved to the next SoC level, and the procedure is repeated over the desired SoC range.

Compared with low-rate capacity tests, HPPC uses **short current pulses**, typically at **higher C-rates** → it emphasizes ohmic drops and short-term polarization effects → **very useful for estimating pulse resistance, identifying ECM parameters**, and assessing the ability of the cell/pack to deliver/absorb power under realistic operating conditions.

Compared with GITT, HPPC is generally **less suited for accurate OCV estimation**. HPPC usually applies shorter and higher-current pulses, with rest periods that are often not long enough to guarantee full voltage relaxation. Therefore, the voltage measured at the end of an HPPC rest period can be interpreted **only as an approximate OCV**.

GITT data **can** be used to estimate ECM parameters **but HPPC is more suited** especially because it is specifically designed to excite the short-term electrical dynamics of the cell.

Battery Characterization Tests for Modeling

Hybrid Pulse Power Characterization (HPPC)

Test Procedure (Detailed)

Initially, the battery is fully charged using a $C/4$ Constant Current-Constant Voltage (CCCV) cycle.

Step 1: A sequence of 10-second discharge and charge constant current (CC) pulses is applied, with a 10-minute rest period after each pulse. Note that the amplitude of the charging pulse is selected to be 70% of the $1C$ -rate current value.

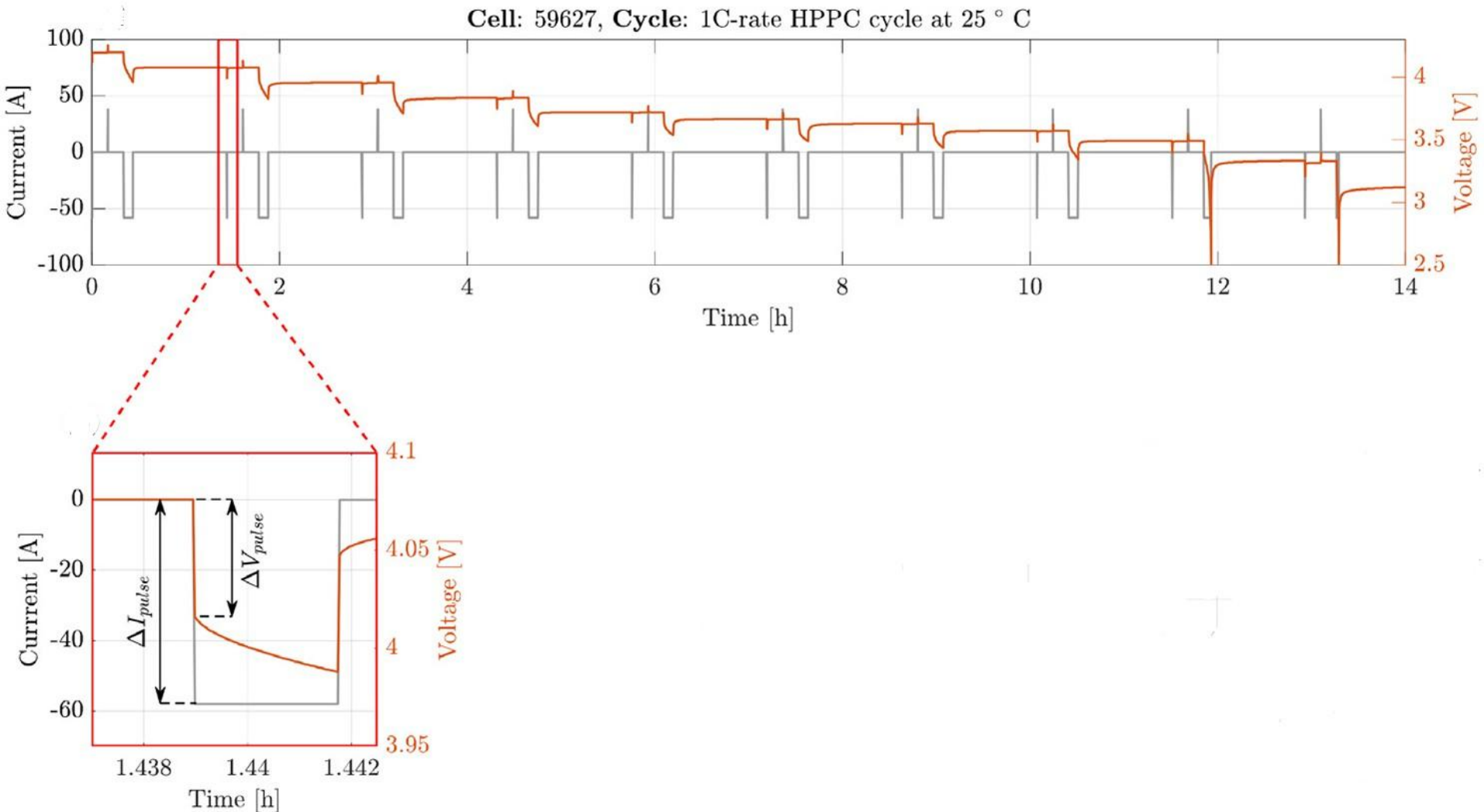
Step 2: Subsequently, the battery is discharged at a $1C$ rate for 6 minutes to establish the next desired SoC.

Step 3: It is followed by a resting period of 60 minutes to allow the battery voltage to relax.

Step 4: Finally, Steps 1-3 are repeated 10 times in a loop. Note that this procedure is repeated using two different C-rates for the discharging pulses.

Battery Characterization Tests for Modeling

Hybrid Pulse Power Characterization (HPPC)



Battery Characterization Tests for Modeling

Hybrid Pulse Power Characterization (HPPC)

Internal Resistance/Impedance Tests

Initially, the battery is fully charged using a $C/4$ Constant Current-Constant Voltage (CCCV) cycle.

Step 1: A sequence of 10-second discharge and charge constant current (CC) pulses is applied, with a 10-minute rest period after each pulse. Note that the amplitude of the charging pulse is selected to be 70% of the $1C$ -rate current value.

Step 2: Subsequently, the battery is discharged at a $1C$ rate for 6 minutes to establish the next desired SoC.

Step 3: It is followed by a resting period of 60 minutes to allow the battery voltage to relax.

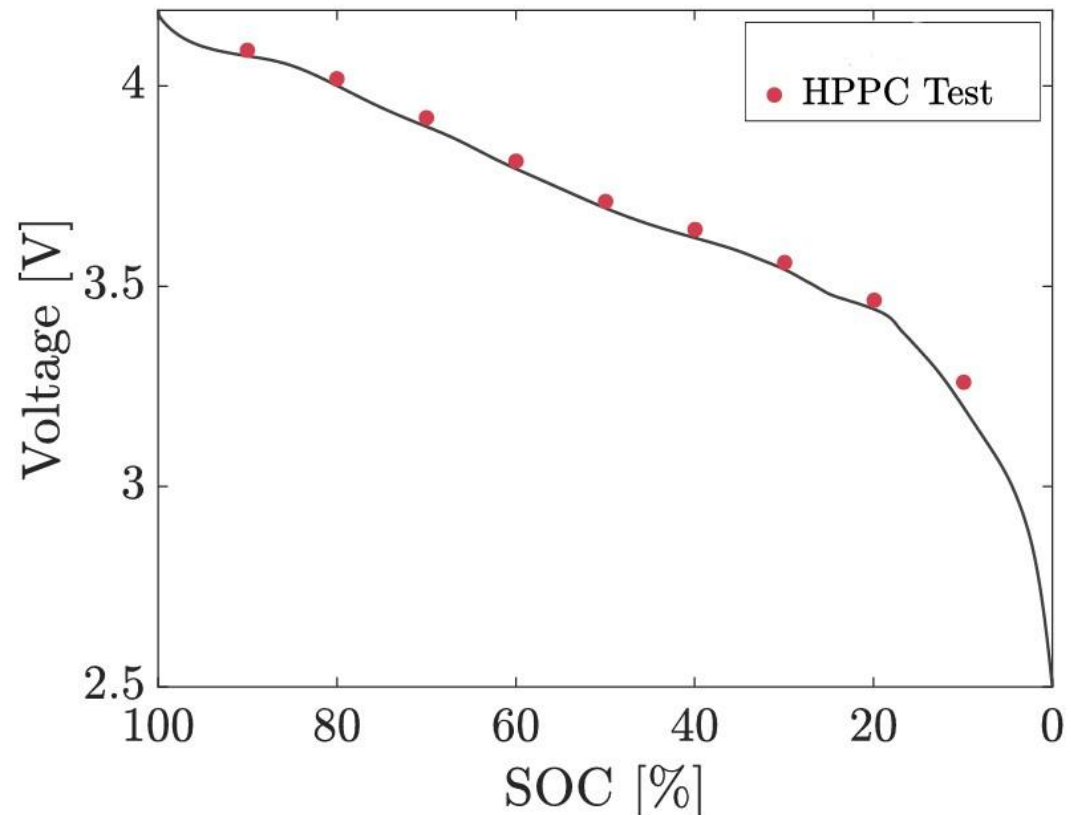
Step 4: Finally, Steps 1-3 are repeated 10 times in a loop. Note that this procedure is repeated using two different C-rates for the discharging pulses.

Battery Characterization Tests for Modeling

Hybrid Pulse Power Characterization (HPPC)

Open Circuit Voltage

Although the $C/20$ discharge test, commonly referred to as the capacity test, is often used to obtain a pseudo-OCV–SOC curve, the HPPC test can also be employed for OCV curve identification. Specifically, each OCV–SOC point is extracted at the end of a sufficiently long rest period, where the terminal voltage has relaxed. Consequently, the number of identified points corresponds to the number of charge/discharge cycles in the HPPC test.

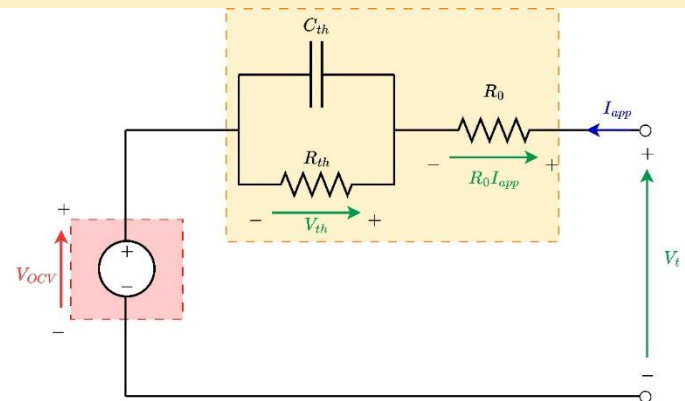


Battery Characterization Tests for Modeling

Hybrid Pulse Power Characterization (HPPC)

Internal impedance/resistance

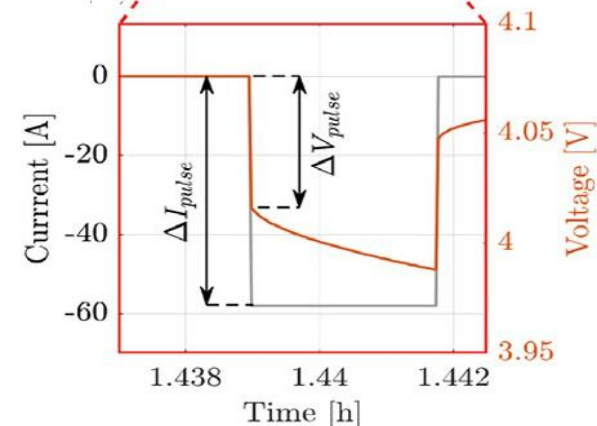
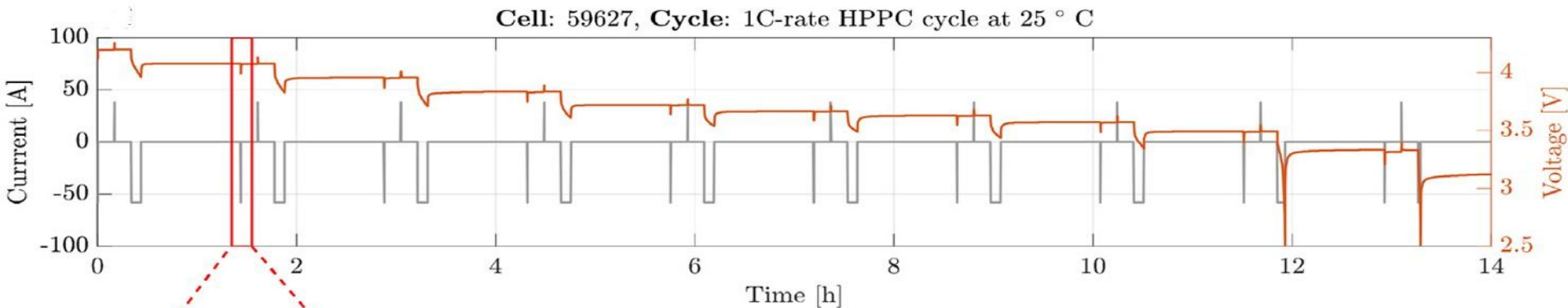
The power capability of a cell is intrinsically related to its impedance characteristics. In simple terms, the impedance is the voltage drop of a cell under a given current, which depends on the frequency and duration of the applied current. The impedance will depend on the condition of a cell because a cell is an electrochemical system: SoC, temperature, current magnitude and ageing history. The real part of complex impedance (i.e. the resistive part) is directly responsible for the dissipative heat generation within a cell and thus the temperature rise of a cell while it is being used. The resistive values of impedance have different contributions arising from pure ohmic resistance, charge transfer resistance and entropy change



Battery Characterization Tests for Modeling

Hybrid Pulse Power Characterization (HPPC)

Internal impedance/resistance



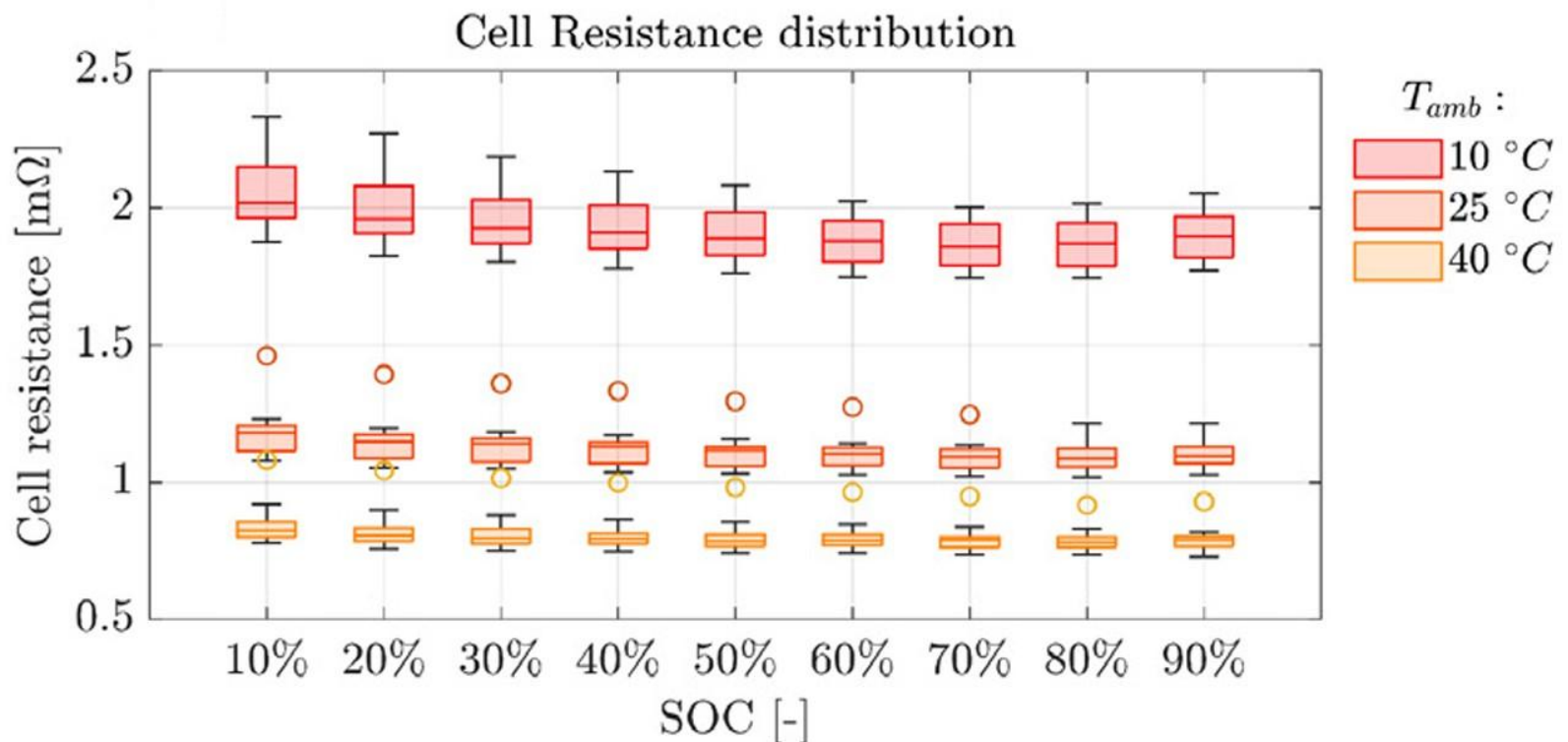
$$R_0 = \frac{\Delta V_{pulse}}{\Delta I_{pulse}}$$

Battery Characterization Tests for Modeling

Hybrid Pulse Power Characterization (HPPC)

Internal impedance/resistance

R_0 is a function of SoC at the different ambient temperature.



Battery Characterization Tests for Modeling

Driving Cycles

Why Driving Cycles?

Each cycle represents a standardized speed profile over time and is typically used to compare fuel consumption and performance in conventional combustion vehicles. In the context of electrical vehicles, time-varying current profiles are generated from an electric vehicle simulator to simulate the battery pack input current in real driving conditions. These profiles are then scaled for a single cell based on battery pack topology, providing a realistic representation of actual current patterns.

In the laboratory, batteries are often tested with constant current charge and discharge cycles, but in reality, electric vehicles batteries are subjected to highly dynamic profiles: acceleration, braking, energy recovery, and stops. To reproduce these conditions, driving cycles are used, i.e., time sequences of power or current derived from real driving data or regulatory standards. The goal is to verify how the cell or battery pack behaves in realistic: (1) energy efficiency, (2) actual capacity available in use, (3) heating during rapid transients, (4) the effects of accelerated degradation due to pulsating currents.

Furthermore, it is a good way to deteriorate batteries if you want to age them

Battery Characterization Tests for Modeling

Driving Cycles

Why Driving Cycles? What are variants?

Classification

UDDS–Urban Dynamometer Driving Schedule

Duration: 1370 [s]
Average Speed: 31 [km/h]
Origin: U.S. Environmental Protection Agency

NEDC–New European Driving Cycle

Duration: 1180 [s]
Average Speed: 34 [km/h]
Origin: European Union

WLTP–Worldwide Harmonised Light Vehicle Test Procedure

Duration: 1800 [s]
Average Speed: 46 [km/h] (max: 135 [km/s])
Origin: UN/UNECE

FUDS /LA92 (Federal Urban Driving Schedule

Duration: 1435 [s]
Average Speed: 32 [km/h] (max: 135 [km/s])
Origin: successor to the UDDS

US06–Supplemental Federal Test Procedure (SFTP)

Duration: 600 [s]
Average Speed: 130 [km/h]
Origin: U.S. Environmental Protection Agency

DST–Dynamic Stress Test

Duration: 1180 [s]
Average Speed: ---
Origin: U.S. Department of Energy (USABC procedure)
(This is not real driving, more for stressing the battery).

Battery Characterization Tests for Modeling

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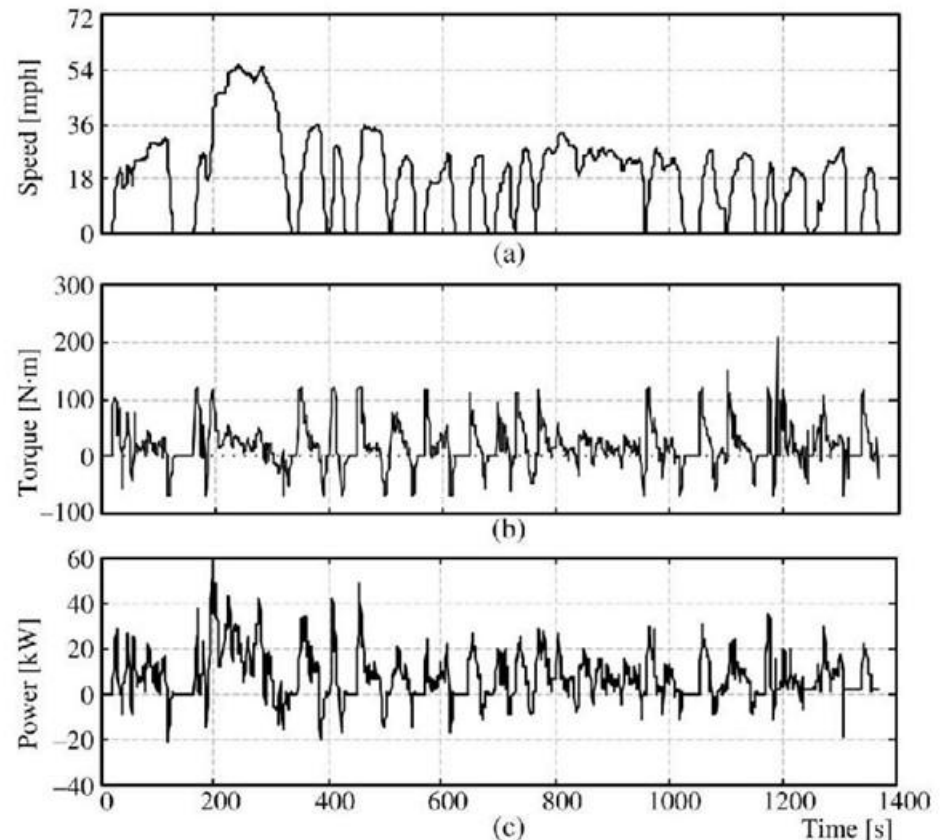
Battery Characterization Tests for Modeling

Driving Cycles

UDDS—Urban Dynamometer Driving Schedule

Profile: numerous stop-and-go phases, moderate acceleration and deceleration.

Uses: Reproduces the thermal and power conditions typical of an urban environment. Good for evaluating energy efficiency and available capacity at low speeds. Used for power capability and energy efficiency tests in “soft duty” conditions



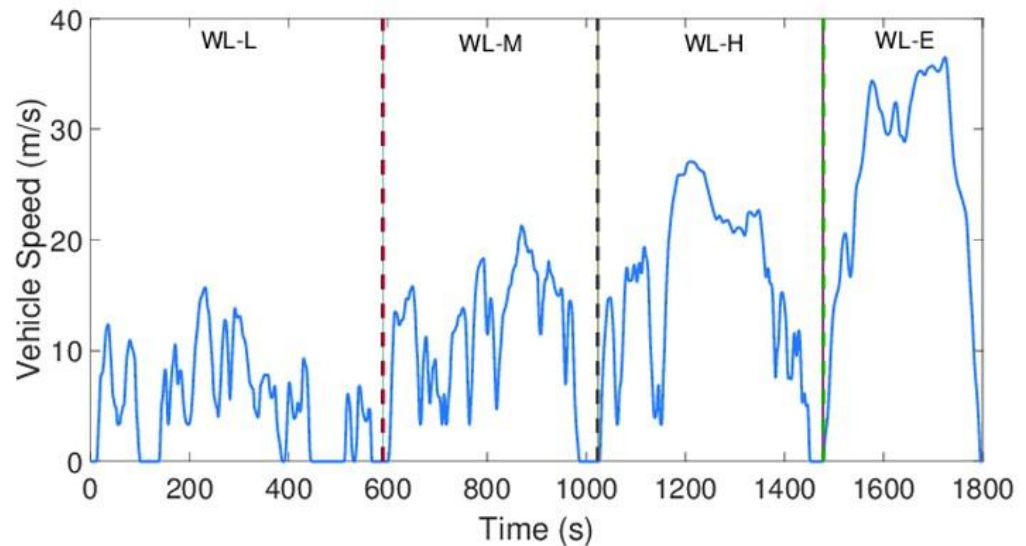
Battery Characterization Tests for Modeling

Driving Cycles

WLTP—Worldwide Harmonised Light Vehicle Test Procedure

Profile: four phases (low, medium, high, extra-high), with realistic acceleration and deceleration transitions

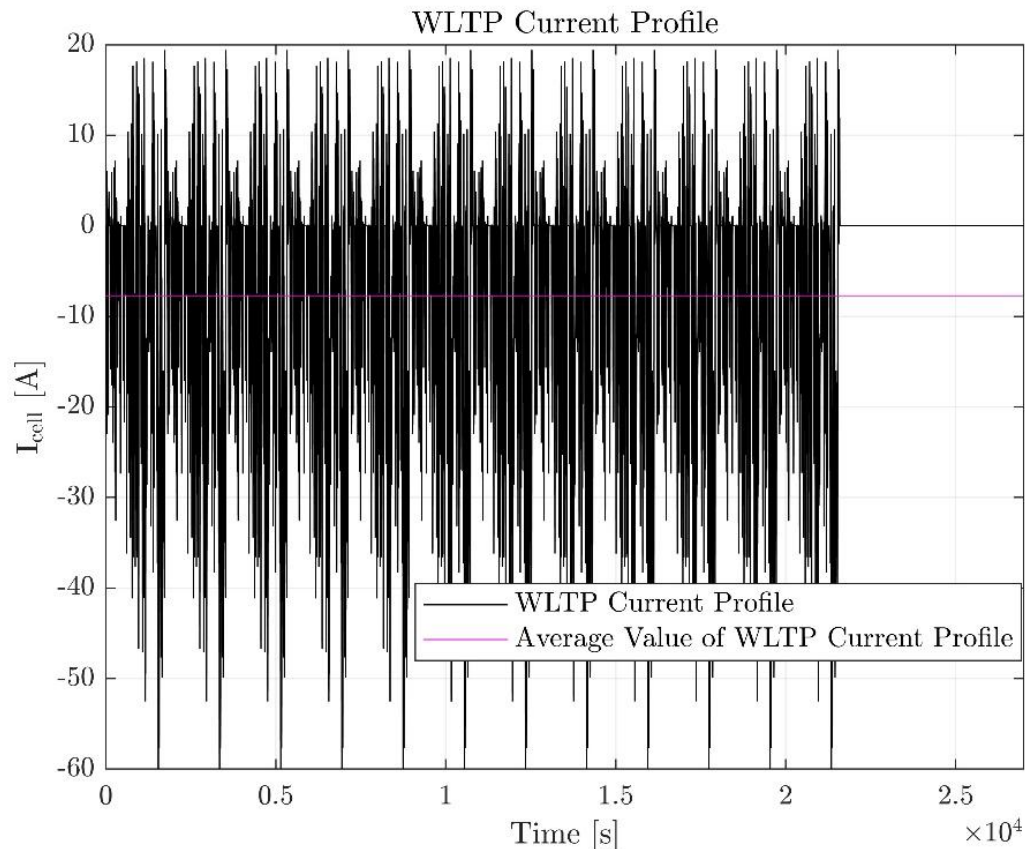
Uses: Today it is the main reference for assessing real energy performance. Excellent for studying efficiency, energy recovery during braking and thermal variations in mixed driving conditions. It allows for more representative degradation analyses than constant tests.



Battery Characterization Tests for Modeling

Driving Cycles

WLTP—Worldwide Harmonised Light Vehicle Test Procedure



Battery Lab

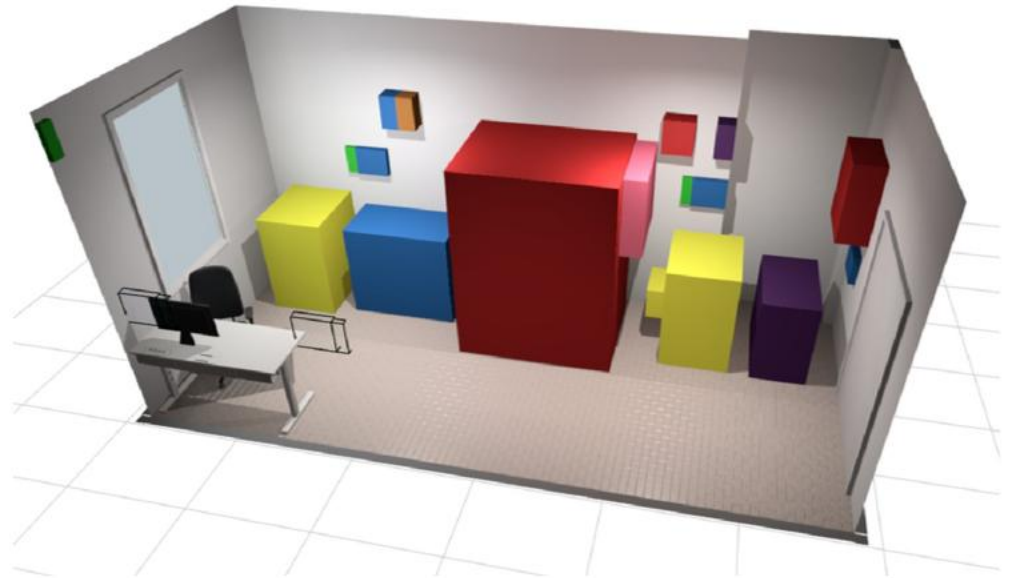


ELISA

Energy Laboratory for Interdisciplinary Storage Applications

Laboratory equipment:

- Cell cycler
- EIS instrumentation
- Module/pack cycler
- Embedded PC solutions and HIL
- 1x climatic chambers
- 2x thermal chambers
- glove box





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Standalone

- Potentiostat/Galvanostat/Zero Resistance Ammeter
- Current range: 0 to 3A
- Booster: up to 30A
- EIS: 10uHz – 1MHz



Interfaceable with cell cycler (EIS can be performed during cycling)

- 8 channels
- channel voltage: 0 – 5V
- channel current: 0 – 60A
- current ranges: 60A / 5A / 500mA / 20mA
- parallelisable channels for charge/discharge up to **480A**
- auxiliary system for temperature measurement (16 channels)
- interfacing with climatic chambers

EIS device



Cell level cycler 5V/60A 8 channels

+

Thermocouples
input board
16 ch.





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2 parallelizable channels, each:

- voltage range: 0 - 80 V
- current range: 0 - 450 A
- power range: 0 - 15 kW
- sampling time: 10us
- custom profile cycling function (up to 10M profile points)
- Battery Simulation Software (**battery module emulation**)
 - CAN, LAN, digital I/O, Analog and RS323 interface

1 channels

- voltage range: 0 - 800 V
- Current range: 0 - 75 A
- power range: 0 - 18 kW
- sampling time: 10us
- custom profile cycling function (up to 10M profile points)
- Battery Simulation Software (**battery pack emulation**)
 - CAN, LAN, digital I/O, Analog and RS323 interface

ITECH
YOUR POWER TESTING SOLUTION



ITECH
YOUR POWER TESTING SOLUTION





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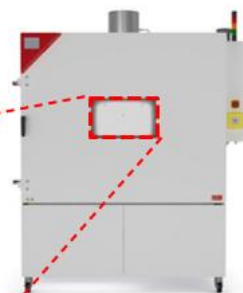
- Cell cycler
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- Hazard level EUCAR 6
- temperature range:
- humidity range:
- internal volume:
- internal dimensions:
- thermal power:
- internal datalogger
- remote control
- RS232 interface
- CO, H₂ and O₂ monitor
- CO₂ inertization

40° / + 180° T → 4.8 K/min
10 – 98 % rH
734 lt
1200x1020x600 mm
5.8 kW

Climate chamber



2 x thermal chambers

- temperature range:
- internal volume:
- internal dimensions:
- internal datalogger
- remote control
- RS232 interface

-20° / + 85° T
105 lt
500x600x390 mm



Battery data

Data in Brief 58 (2025) 111301



Contents lists available at [ScienceDirect](https://www.sciencedirect.com)

Data in Brief

journal homepage: www.elsevier.com/locate/dib



Data Article

A dataset for large prismatic lithium-ion battery cells (CALB L148N58A): Comprehensive characterization and real-world driving cycles



Simone Fasolato^a, Nicola Blasuttigh^{b,c}, Giacomo Galuppini^a,
Davide M. Raimondo^{b,c,*}

^a Department of Electrical, Computer and Biomedical Engineering, University of Pavia, Pavia, Italy

^b Department of Engineering and Architecture, University of Trieste, Trieste, Italy

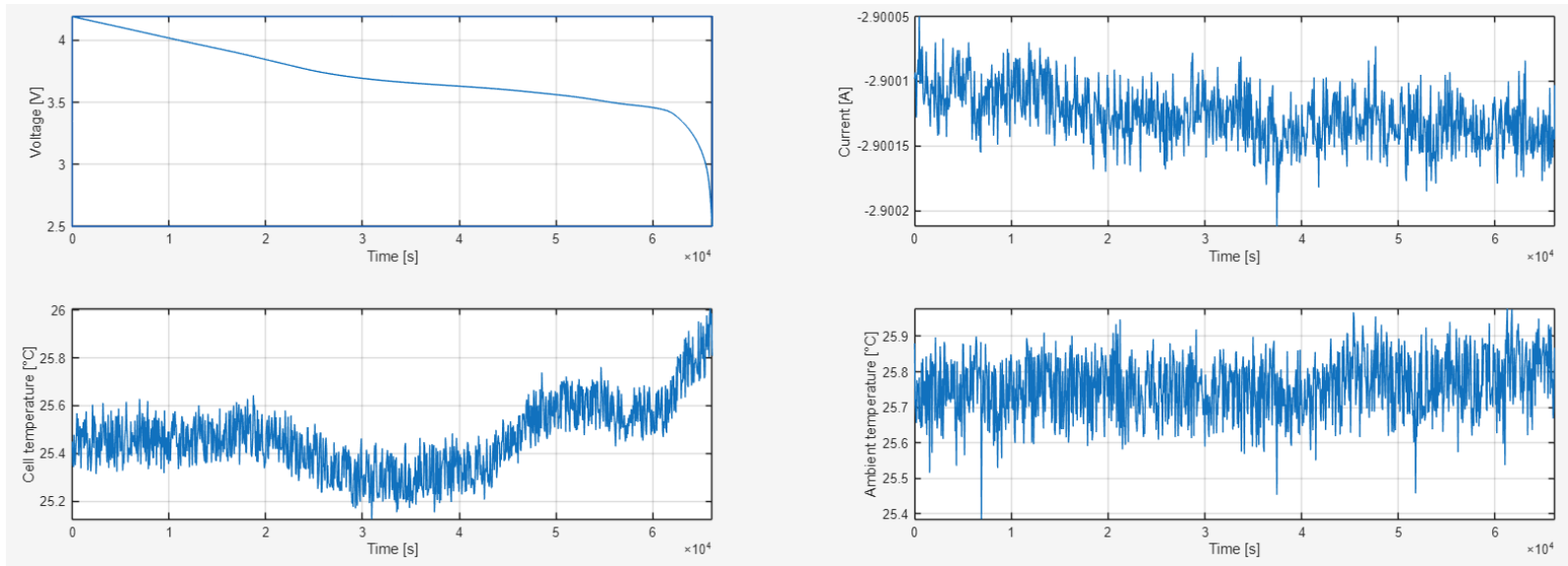
^c Center for Energy, Environment and Transport Giacomo Ciamician, University of Trieste, Trieste, Italy

Lab 1

1. Download data from <https://data.mendeley.com/datasets/ycx459r5c3/1>
2. Install curve fitting toolbox for Matlab
3. Load data from directory Processed Data/Temperature_25C/C20_discharge
4. Choose one of the battery cell e.g. 59294 and load the corresponding file 59294.mat
5. Plot data (Voltage vs Time, Current vs. Time, Temperature vs Time, Amb. Temp. vs Time)
6. Extract capacity (be careful about the sampling time!)
7. Extract SOC vs Time
8. Plot OCV Curve (SOC vs Voltage)
9. Use Curve Fitting to obtain an approximation of the OCV curve

Lab 1: results (contained in Lab1234_in_class.m)

Battery cell 59294 data with a C/20 profile at 25°C



Assuming 100% SOC corresponds to 4.2V and 0% SOC to 2.5V under a C/20 discharge, the capacity can be estimated by integrating the current between these voltage limits. Data are sampled at nonuniform time instants. Capacity is approximated numerically as:

$$C_{\text{bat}} \approx \sum_{i=i_{4.2\text{V}}}^{i_{2.5\text{V}}-1} I(i) \Delta t_i, \quad \Delta t_i = t(i+1) - t(i)$$

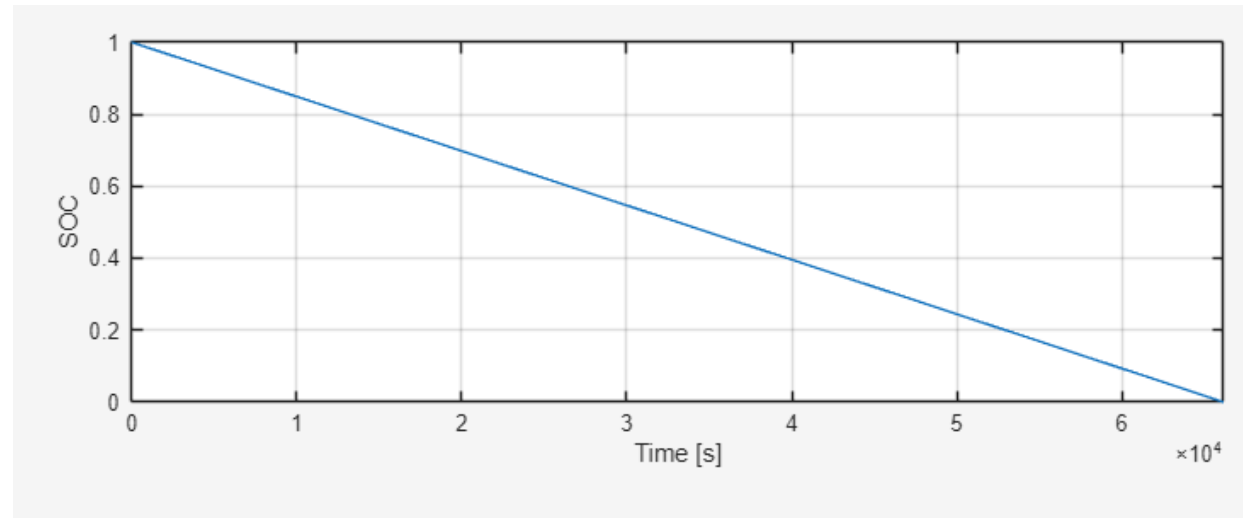
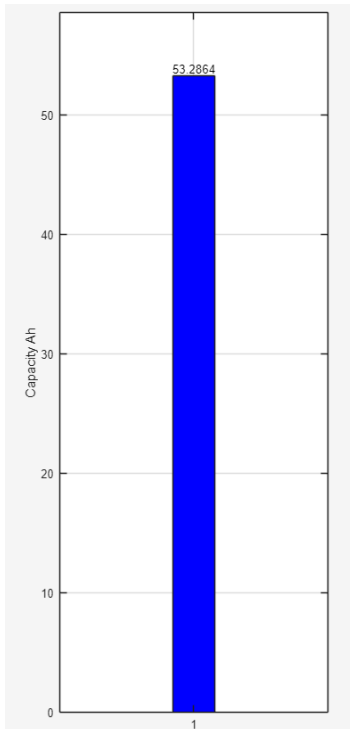
where $i_{4.2\text{V}}$ and $i_{2.5\text{V}}$ denote the samples at which the terminal voltage is 4.2 V and 2.5 V.

Lab 1: results (contained in Lab1234_in_class.m)

Once the battery capacity has been estimated, the SOC profile can be reconstructed over time via **Coulomb counting**. Assuming $SOC=100\%$ at the sample where the voltage is 4.2 V, the SOC is updated as:

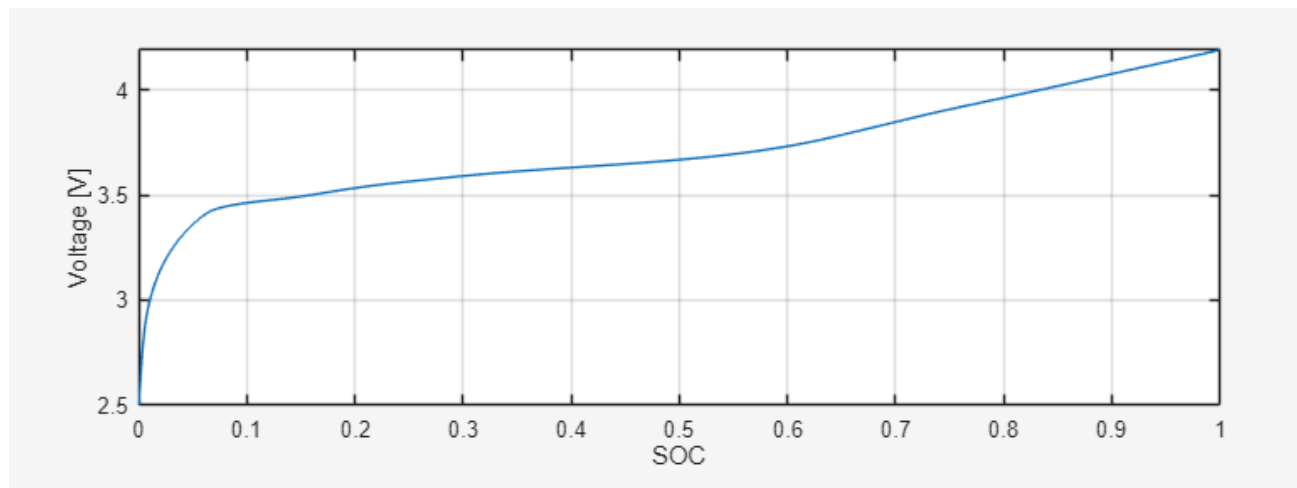
$$SOC(i+1) = SOC(i) + \frac{I(i) \Delta t_i}{C_{bat}}, \quad \Delta t_i = t(i+1) - t(i),$$

where $I(i)$ is negative during discharge and C_{bat} is expressed in ampere-seconds (As). The corresponding capacity in ampere-hours (Ah) is obtained as $C_{bat}/3600$.



Lab 1: results (contained in Lab1234_in_class.m)

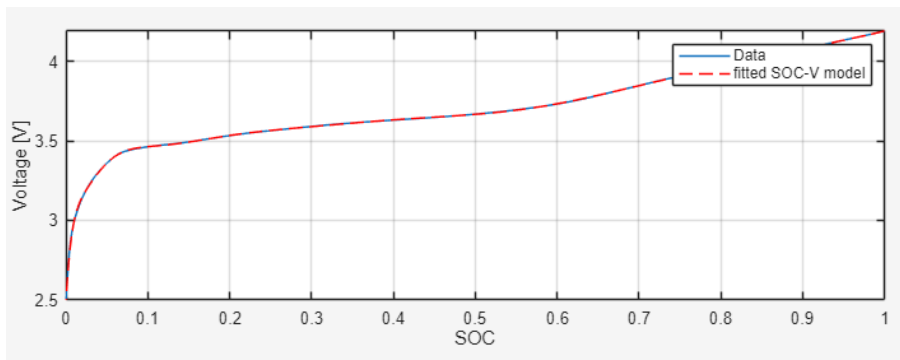
Once the SOC profile has been reconstructed, the C/20 discharge data can be used to identify the **OCV–SOC relationship**. Specifically, **each voltage measurement** along the **discharge curve** is **paired with** the corresponding **SOC value** computed via Coulomb counting, thereby yielding an empirical mapping between OCV and SOC. The resulting characteristic can then be represented by plotting OCV as a function of SOC.



What we obtain from a C/20 discharge is **not a true equilibrium** OCV–SOC curve, but rather a **pseudo-OCV** curve measured under a small but nonzero current.

Lab 1: results (contained in Lab1234_in_class.m)

An **analytical expression** for the **OCV-SOC** relationship was identified from data. This is **compact and smooth** and can be employed in **simulation, control, and state-estimation algorithms**, without resorting to lookup tables defined only at discrete SOC values. The model parameters were identified through a **linear least-squares procedure**, while the functional structure was fixed a priori as



$$\begin{aligned} \text{OCV}(z) = & 2.5 + bz + cz^2 + dz^3 + ez^4 + fz^5 \\ & + gz^6 + hz^7 + iz^8 + lz^9 \\ & + m \frac{z}{0.1+z} + n \frac{z}{(0.1+z)^2} + o \frac{z}{(0.1+z)^3} + p \frac{z}{(0.1+z)^4} \\ & + q \log(1+z) + r \log^2(1+z), \end{aligned}$$

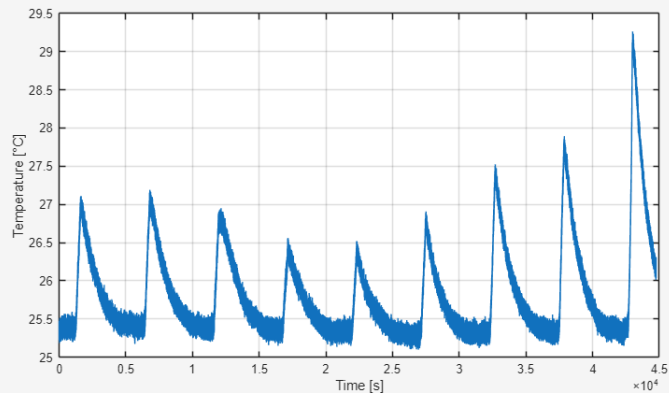
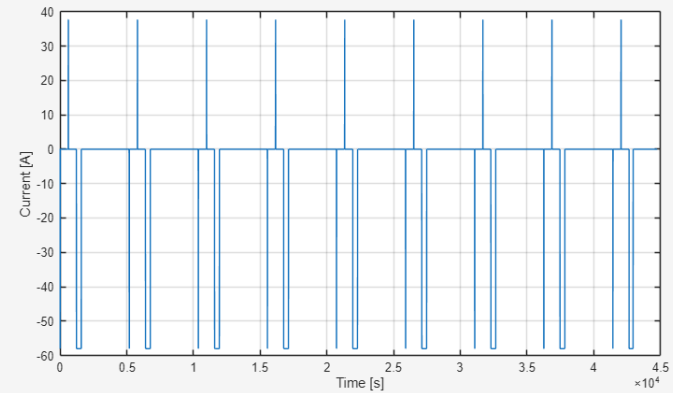
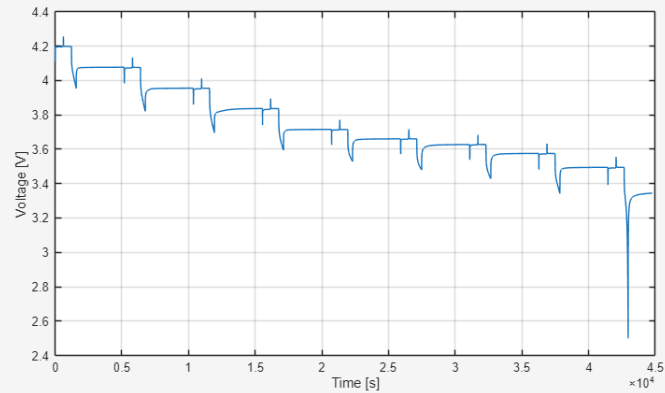
where z denotes the SOC. The constant term was fixed at 2.5 to ensure consistency with the adopted SOC reference, i.e., $\text{OCV}(0)=2.5\text{V}$. This structure was selected because a purely polynomial approximation did not provide a sufficiently accurate fit of the experimental data and was less effective in preserving the **monotonicity** of the OCV-SOC characteristic. The quality of the fit was assessed through the RMSE (0.0018).

Lab 2

1. Load data from directory Processed Data/Temperature_25C/HPPC_1C/
2. Choose one of the battery cell e.g. 59294 and load the corresponding file 59294.mat
3. Plot data (Voltage vs Time, Current vs. Time, Temperature vs Time)
4. Find times where current switches from 0 to a negative value
5. Find the duration of this discharging times
6. Select only the short spikes
7. Compute the internal resistance by looking at the voltage drop from the beginning of the spike to the next time instant divided by the applied current. Average the value
8. Simulate the obtained mathematical model with OCV+Rint and compare with the HPPC profiles

Lab 2: results (contained in Lab1234_in_class.m)

Battery cell 59294 data with a HPPC profile at 25°C

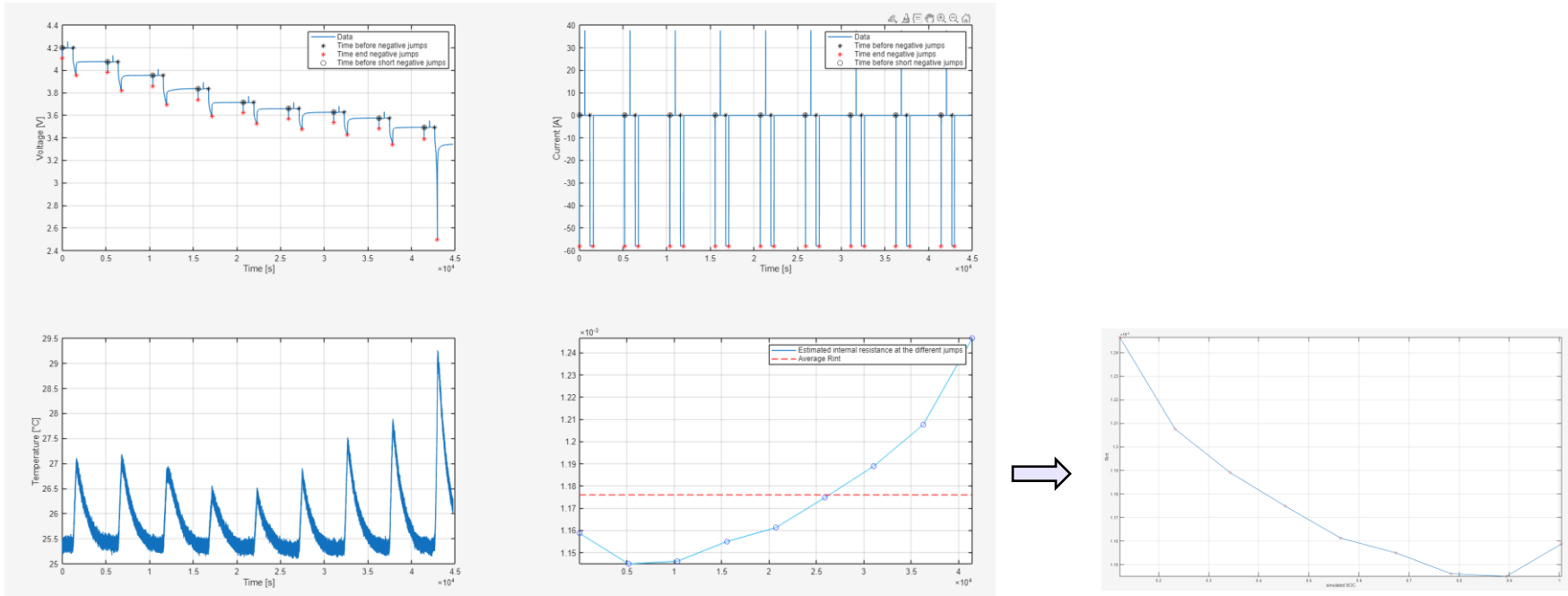


Lab 2: results (contained in Lab1234_in_class.m)

From HPPC data, the time instants corresponding to **negative current steps** were identified. Only **short-duration pulses** were retained as they are preceded by **longer rest** intervals and are more consistent with the estimation of the instantaneous **ohmic contribution**. The internal resistance was estimated from the instantaneous voltage drop at the beginning of each pulse, divided by the applied current:

$$R_{\text{int}} = \frac{V(t^+) - V(t^-)}{I(t^+)},$$

where t^- and t^+ denote the instants immediately before and after the current step.

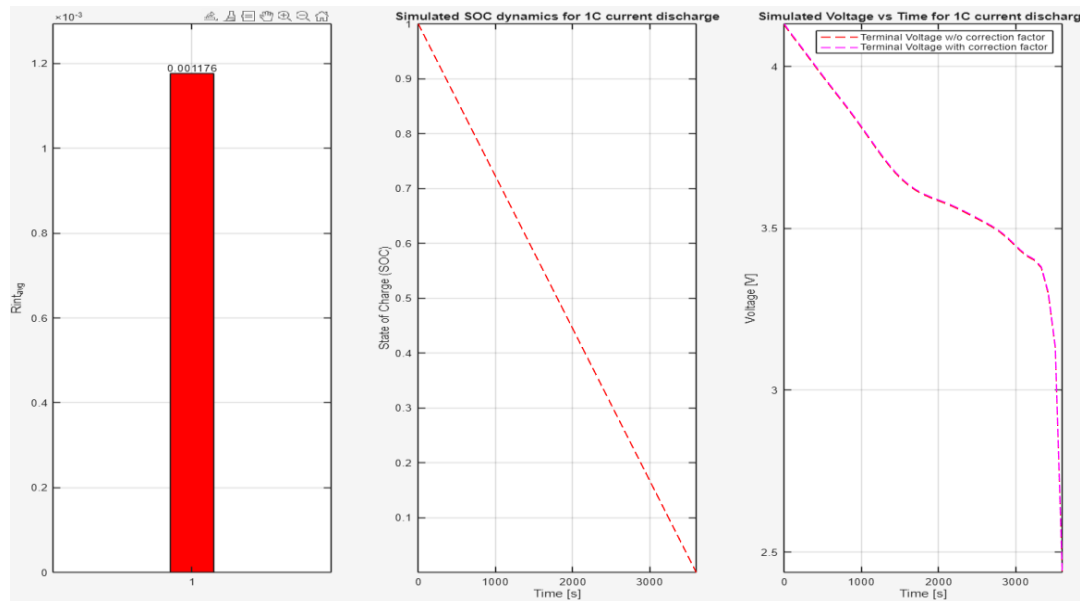


Lab 2: results (contained in Lab1234_in_class.m)

Using the estimated battery capacity C_{bat} , the identified OCV-SOC relationship, and the average internal resistance R_{int_avg} , the cell behavior can now be simulated (using ode45 and SOC_dynamics) under a 1C discharge, assuming $SOC(0) = 1$. The terminal voltage is evaluated both **with and without correction of the pseudo-OCV curve** identified from the C/20 test. In particular, the corrected OCV is computed as

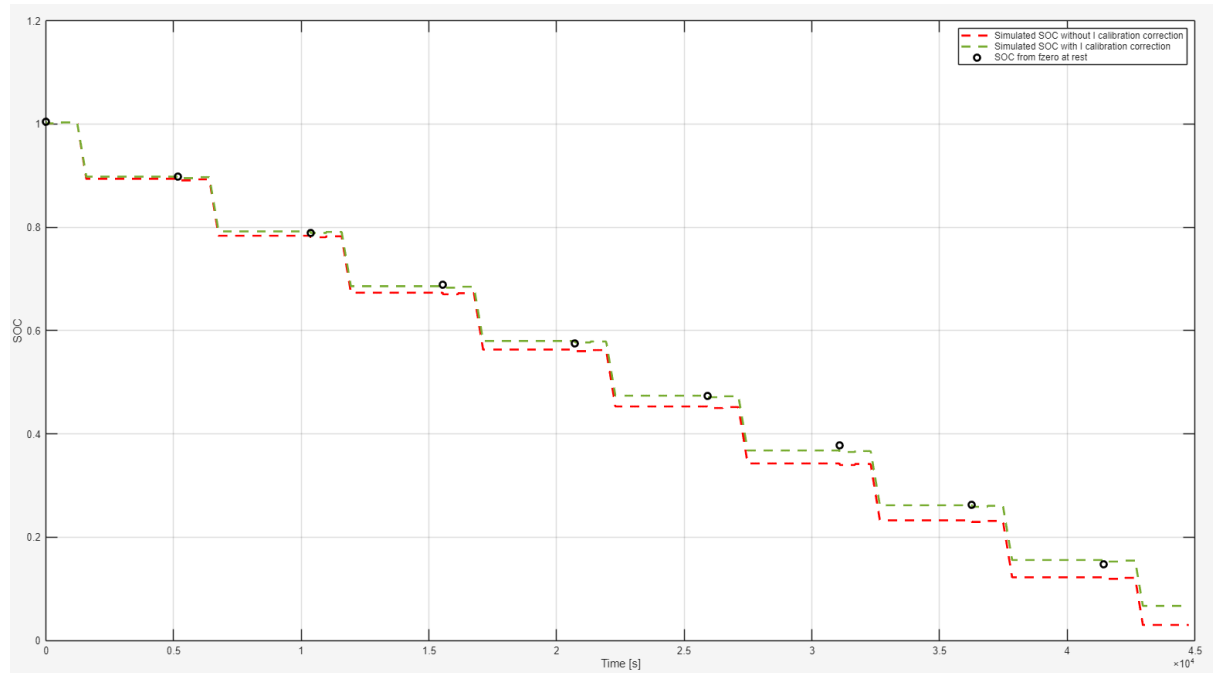
$$y_{corr} = \text{fittedmodel}(x) + R_{int,avg} \left(I + \frac{C_{bat,Ah}}{20} \right),$$

where the additional term compensates for the fact that the identified OCV-SOC curve is obtained from a C/20 discharge and therefore represents a pseudo-OCV rather than an equilibrium OCV.



Lab 2: results (contained in Lab1234_in_class.m)

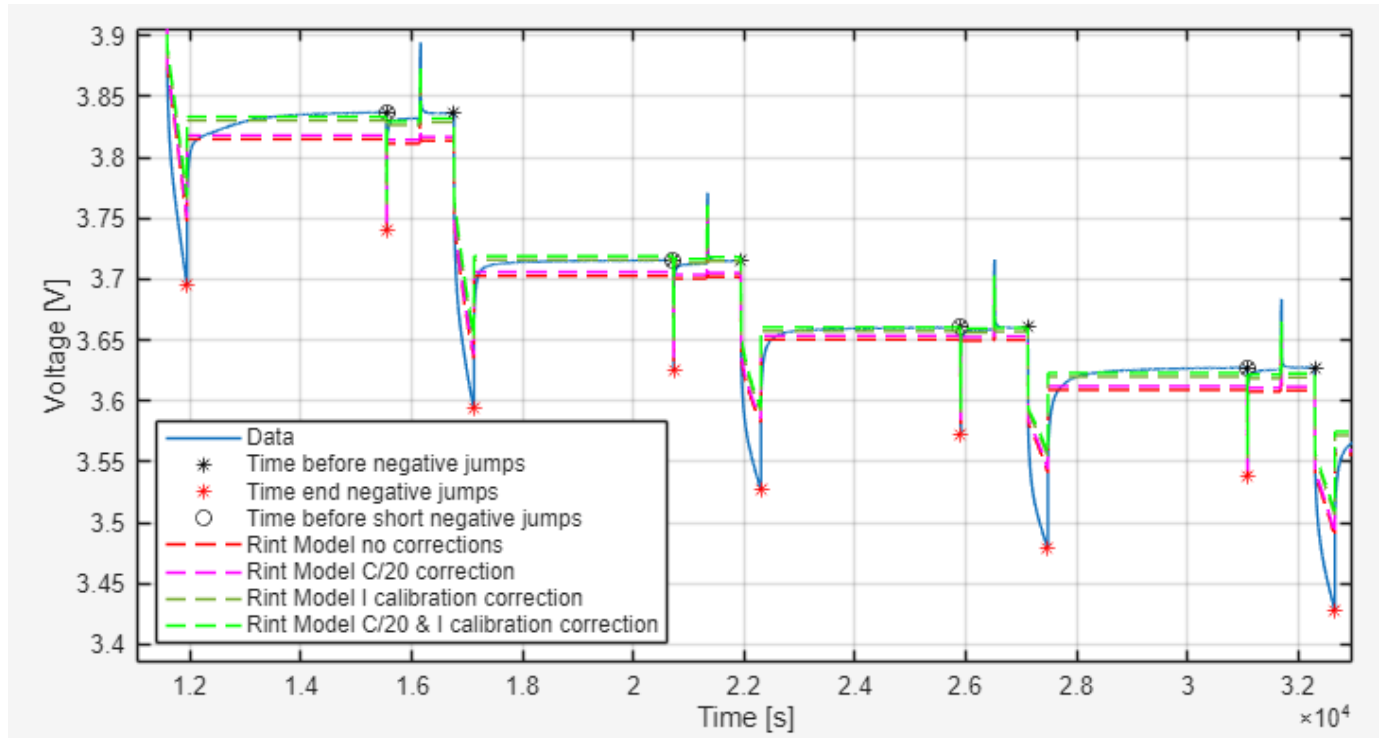
HPPC data: **current sensor calibration error** evident in the SOC behavior.



- Circle markers: SOC obtained at rest by inverting the OCV-SOC relationship with f_{zero} .
- Dashed curves obtained by simulating the model from $SOC(0)=1$:
 - using the injected current,
 - using the injected current corrected by a **calibration factor** $I_{real} = 0.962I$.
- The corrected-current simulation shows **better agreement** with the SOC values inferred at rest, although a **residual mismatch is still present**.

Lab 2: results (contained in Lab1234_in_class.m)

Voltage comparison on HPPC data



- The current **calibration correction** reduces the **steady-state error**.
- Part of the **remaining mismatch** in terminal voltage **may also be due** (for the terminal voltage not for the SOC) to the use of an **average value of Rint**, rather than an SOC-dependent resistance.
- The **present model** is not able to **accurately reproduce the transient behavior**, where the discrepancy between model and experimental data remains significant.

Lab 3

Simulate a CC-CV charging protocol with the model obtained during Lab 2

1. Initialize SOC at 0%.
2. Constant Current (CC) stage
 - Apply constant current (0.5C)
 - Integrate using ode45
 - **odeset('Events', ...)** to define event function: stop when terminal voltage ≥ 4.2 V
3. Constant Voltage (CV) stage
 - Voltage fixed at 4.2V. **Current not imposed, but computed** from: $V = OCV(SOC) + R \cdot I$
 - Formulate as a Differential Algebraic Equations (**DAEs**):
 - differential state: SOC
 - algebraic variable: current
 - Solve using **ode15i** Use **decic** to compute consistent initial conditions
 - Use **odeset('Events', ...)**: stop when current < 50 mA

Lab 3: results (contained in Lab1234_in_class.m)

CC–CV charge simulation

- Model from Lab 2, initial SOC = 0%
- CC phase: $I(t) = 0.5C$ solved with ode45
- Stop CC when $V_t(t) = OCV(SOC(t)) + R \left(I(t) + \frac{C_{Ah}}{20} \right) = 4.2 \text{ V}$
- CV phase: $V_t(t) = 4.2 \text{ V}$, with current computed from $V_t(t) = OCV(SOC(t)) + R \left(I(t) + \frac{C_{Ah}}{20} \right)$

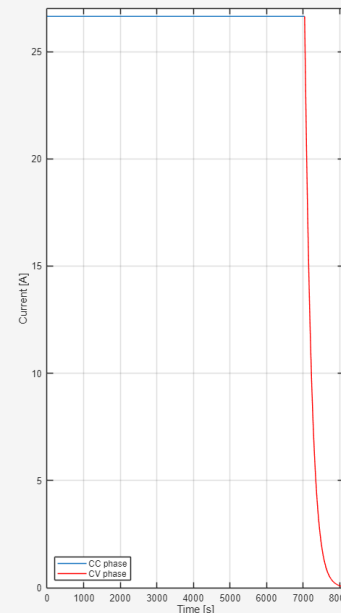
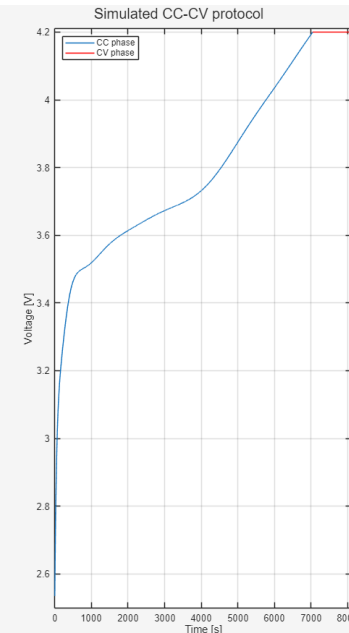
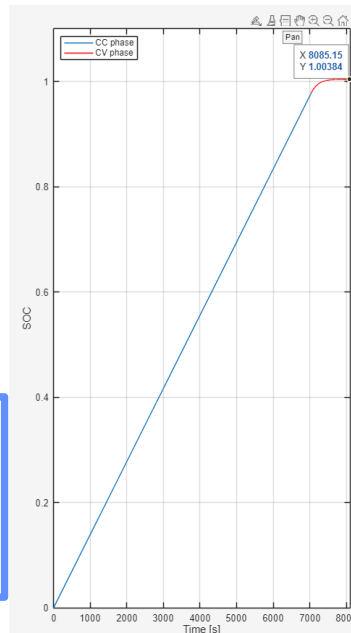
DAE (Differential Algebraic Equations) formulation (solved with ode15i + decic):

- differential state: $SOC(t)$
- algebraic variable: $I(t)$

Stop CV when $I(t) < 50 \text{ mA}$

$SOC(t)$ increases during CC,
 $V_t(t)$ reaches 4.2 V, and $I(t)$
decreases during CV.

SOC goes above 100%. 100% SOC was defined at 4.2 V for $C/20$. In CV the current falls below $C/20 \rightarrow$ the model can still hold 4.2V at slightly higher SOC.



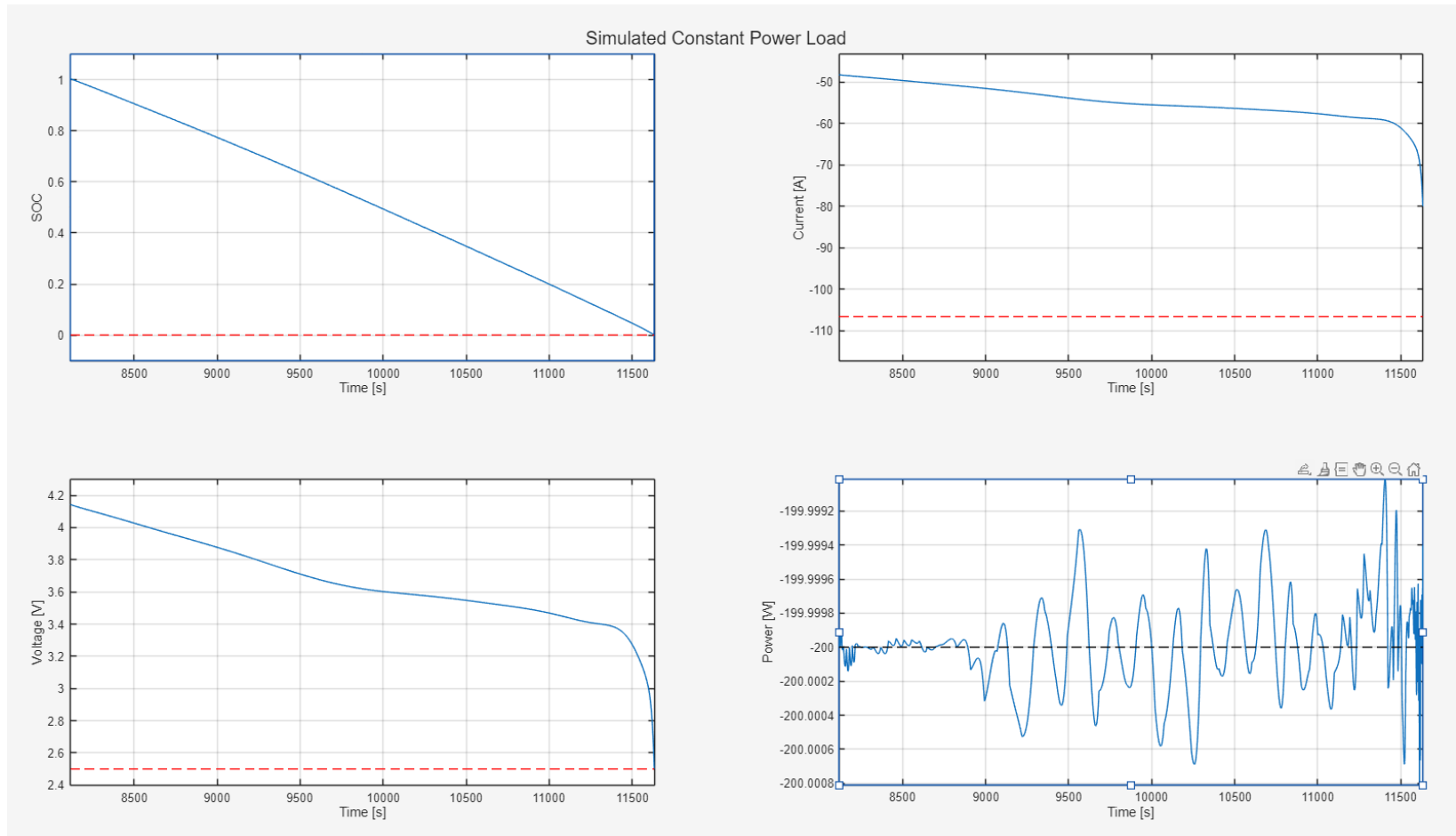
Lab 4

Constant-Power Discharge Simulation

1. Simulate battery discharge under a different constant-power loads (negative since I is negative during discharge)
2. Formulate the problem as a DAE and solve it using `ode15i`
3. Impose the algebraic constraint: $P=VI$ =power load
4. Use `odeset('Events', ...)` to stop the simulation when any of the following occurs:
 - current exceeds $2C$
 - terminal voltage reaches the cutoff limit 2.5 V
 - SOC reaches $\text{SOC}= 0\%$
5. Plot SOC, voltage, and current versus time

Lab 4: results (contained in Lab1234_in_class.m)

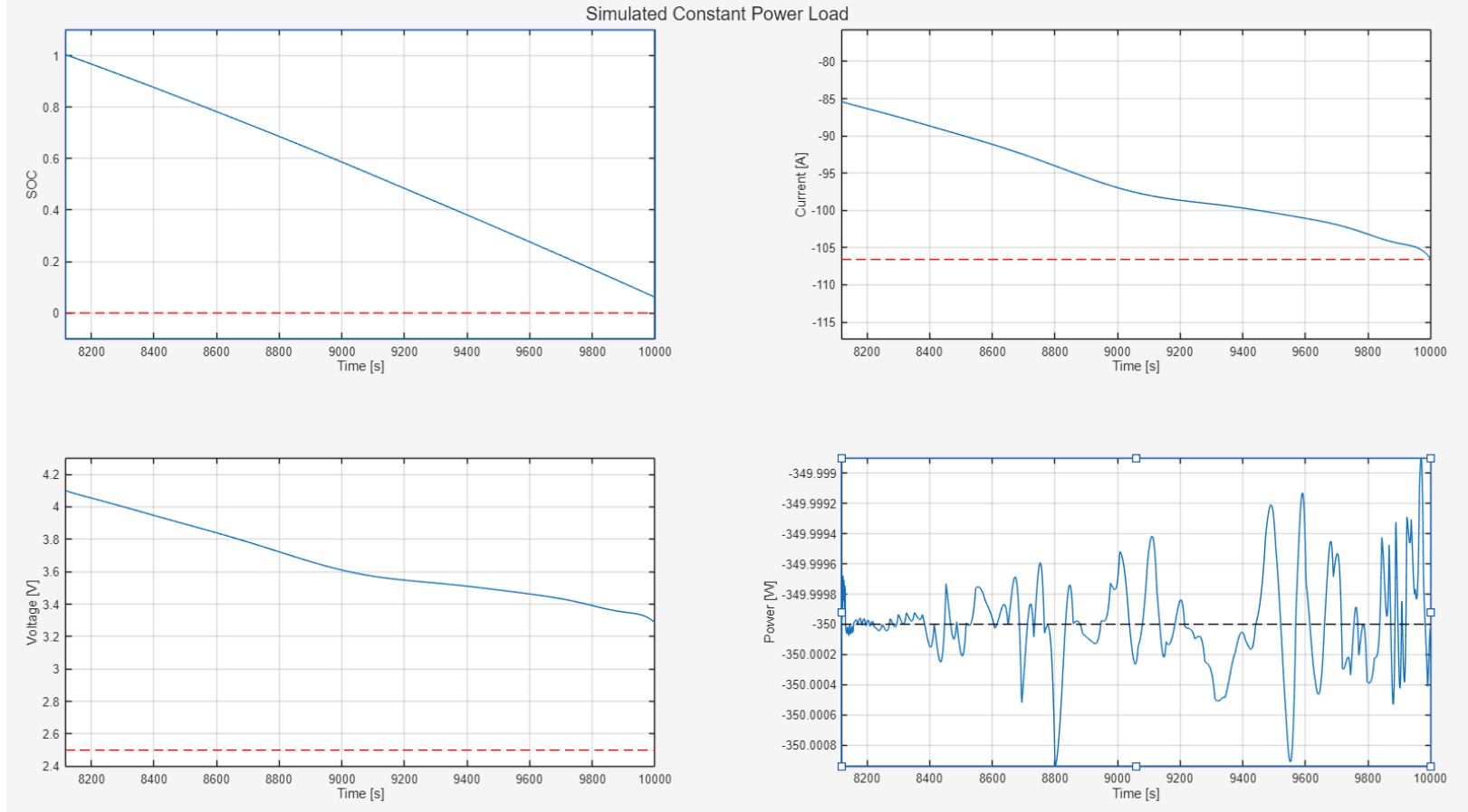
CP simulation (200 W)



In this case, the cell hits first the cutoff voltage 2.5V

Lab 4: results (contained in Lab1234_in_class.m)

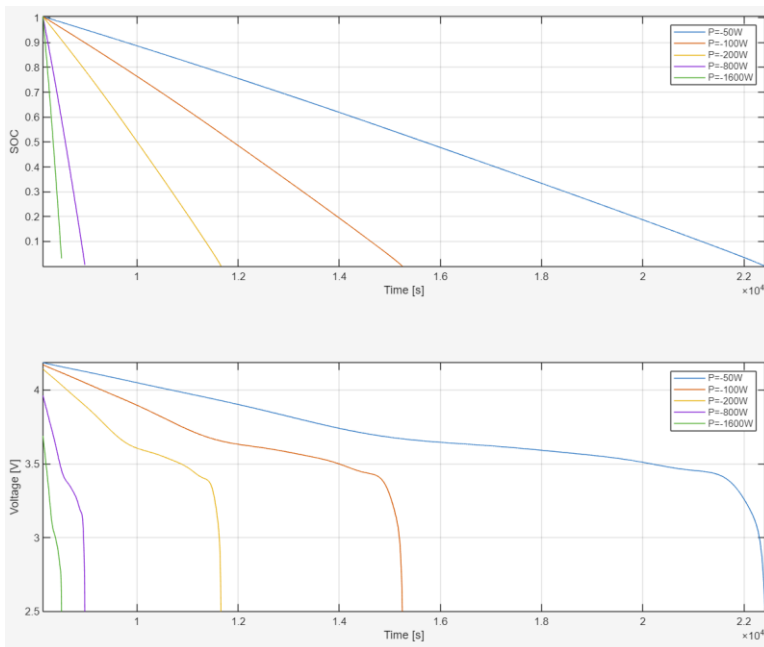
CP simulation (350 W)



Here, with a current limit of 2C, the current limit constraint is the first to be hit.

Lab 4: results (contained in Lab1234_in_class.m)

CP simulation – comparison of different power demands (with current limit at 20C)



When the power demand increases significantly (1600W), assuming very high current limit, we see that the cutoff voltage limit is hit earlier than the SOC. This was not visible even for 800 W.

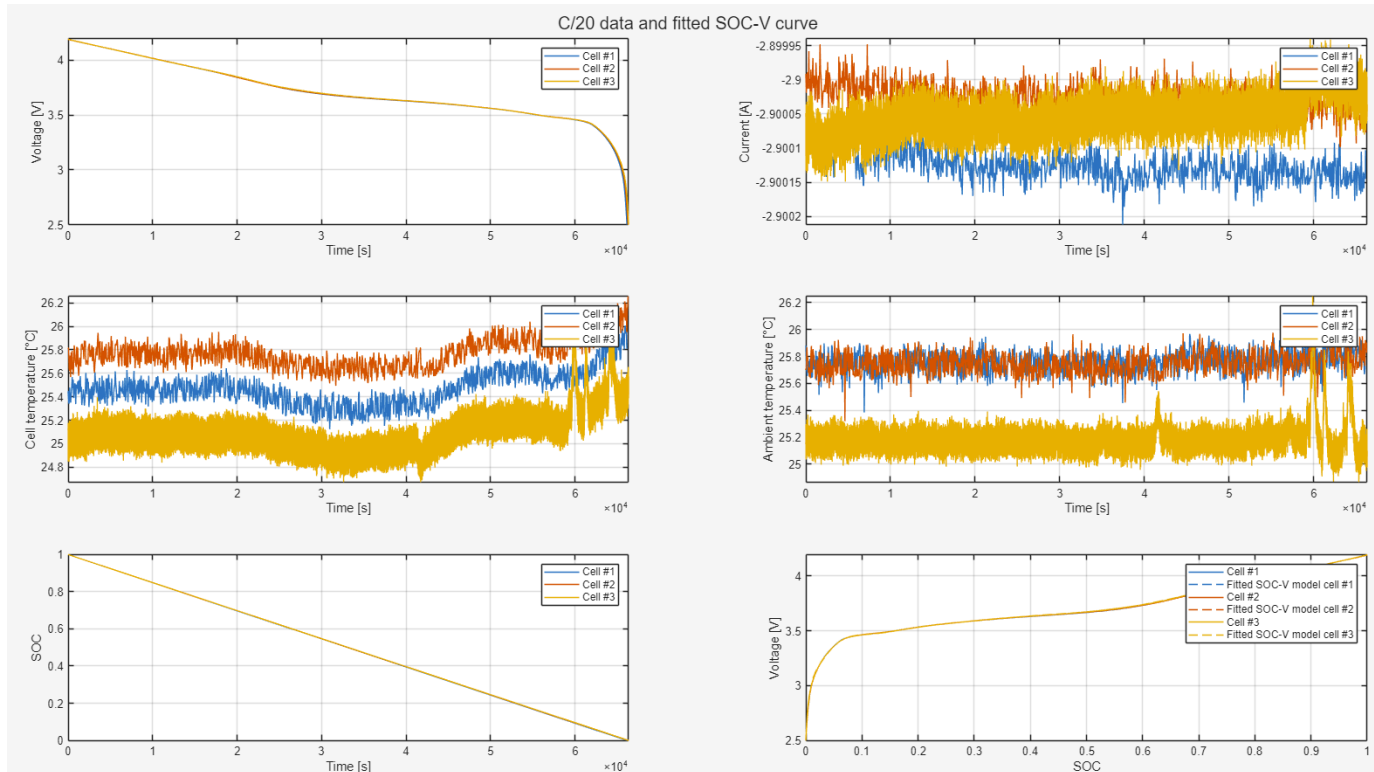
Why?

The main reason is that the **voltage–SOC curve is steep near low SOC**: therefore, even a significant *RI* drop corresponds to only a small residual-SOC difference, until the power becomes very large.

Lab 5 (multiple cells)

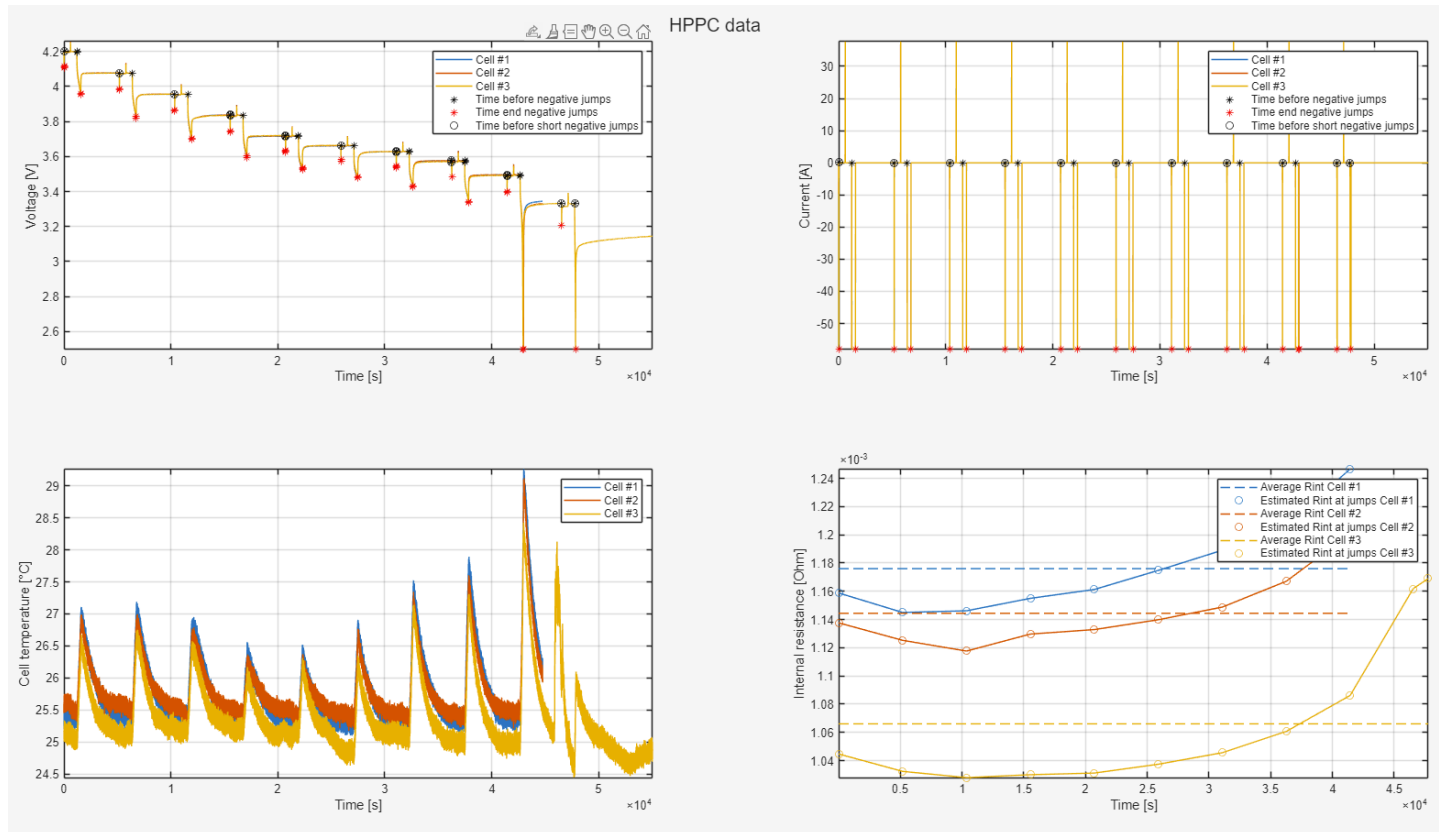
1. Load data from directory Processed Data/Temperature_25C/C20_discharge
2. **Select all cells** by extracting the list of .mat files in the directory. **For all of them:**
3. Plot data (Voltage vs Time, Current vs. Time, Temperature vs Time, Amb. Temp. vs Time)
4. Extract capacity (be careful about the sampling time!)
5. Extract SOC vs Time
6. Plot OCV Curve (SOC vs Voltage)
7. Use Curve Fitting to obtain an approximation of the OCV curve
8. Load data from directory Processed Data/Temperature_25C/HPPC_1C/
9. Plot data (Voltage vs Time, Current vs. Time, Temperature vs Time)
10. Find times where current switches from 0 to a negative value
11. Find the duration of this discharging times
12. Select only the short spikes
13. Compute the internal resistance by looking at the voltage drop from the beginning of the spike to the next time instant divided by the applied current. Average the value

Lab 5: results (contained in Lab5_in_class.m)



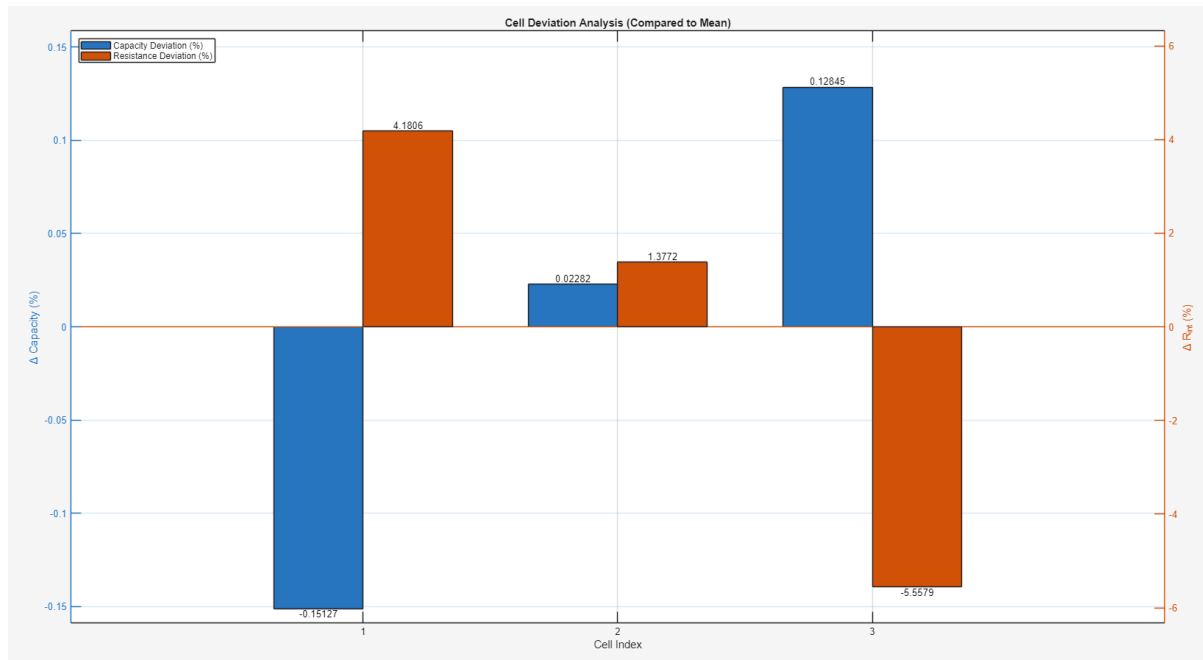
- Experiments have **different sampling densities**; in particular, **Cell 3 contains more data points**.
- Measured **cell temperatures are not identical** across tests. The thermal chamber regulation is based on a **sensor mounted on the chamber wall**, so the **internal temperature is not perfectly uniform** and cells placed at different positions may experience slightly different thermal conditions.
- **Ambient temperature** suggests tests probably **not all performed simultaneously**, but in at least **two different batches**.
- **Small differences in voltage and SOC trajectories** are observed, likely **due both to slightly different thermal conditions and to intrinsic cell-to-cell variability**.
- Despite these differences, the **OCV–SOC curves are very similar** for the three cells.

Lab 5: results (contained in Lab5_in_class.m)



- Similar HPPC behaviour for all three cells. The **tests** do **not** have all the **same duration**.
- Voltage jumps at pulse transitions are used to estimate **Rint**.
- Rint is fairly stable, with a slight increase at lower SOC. Small differences observed across cells: **Cell 3 lowest**, **Cell 1 highest** average resistance.
- Temperature follows the pulse pattern and may contribute to part of the variability.

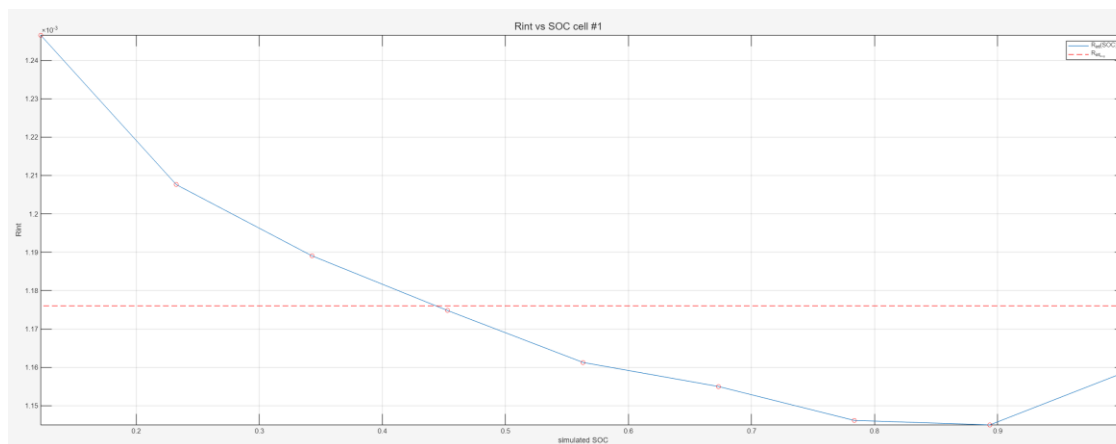
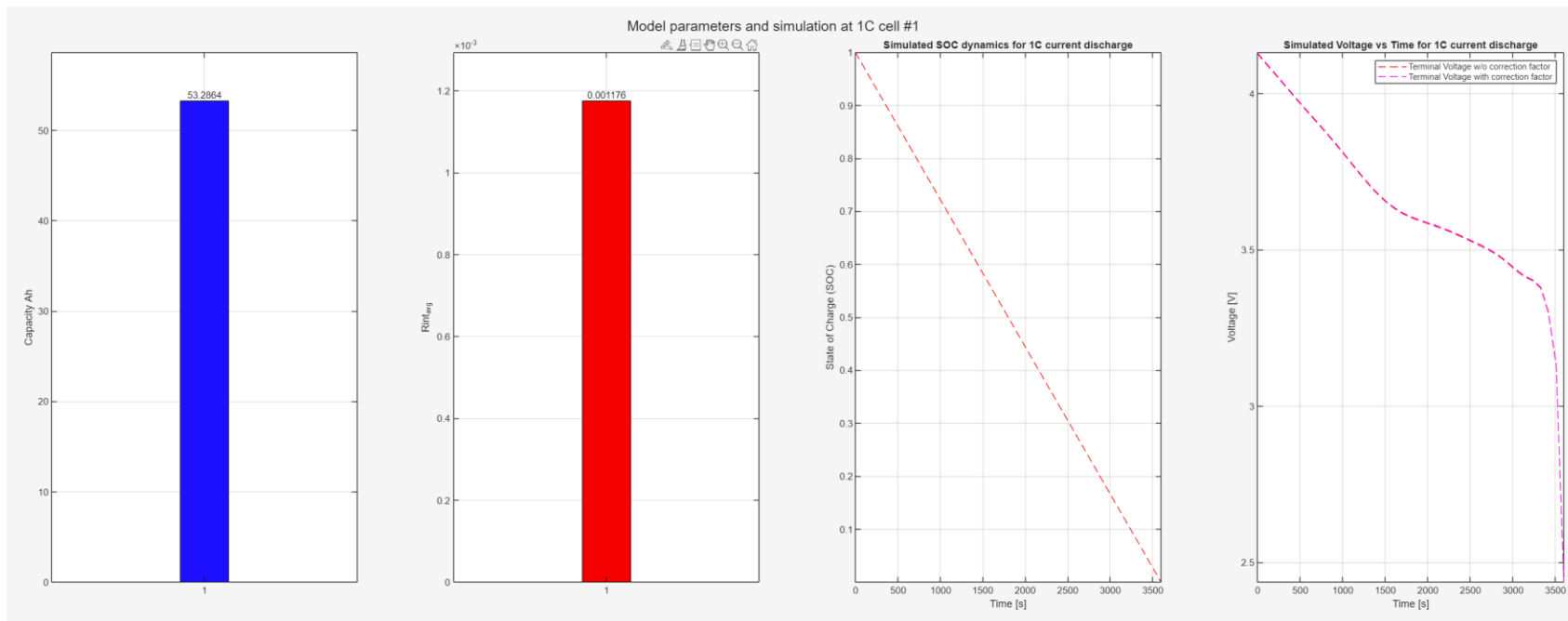
Lab 5: results (contained in Lab5_in_class.m)



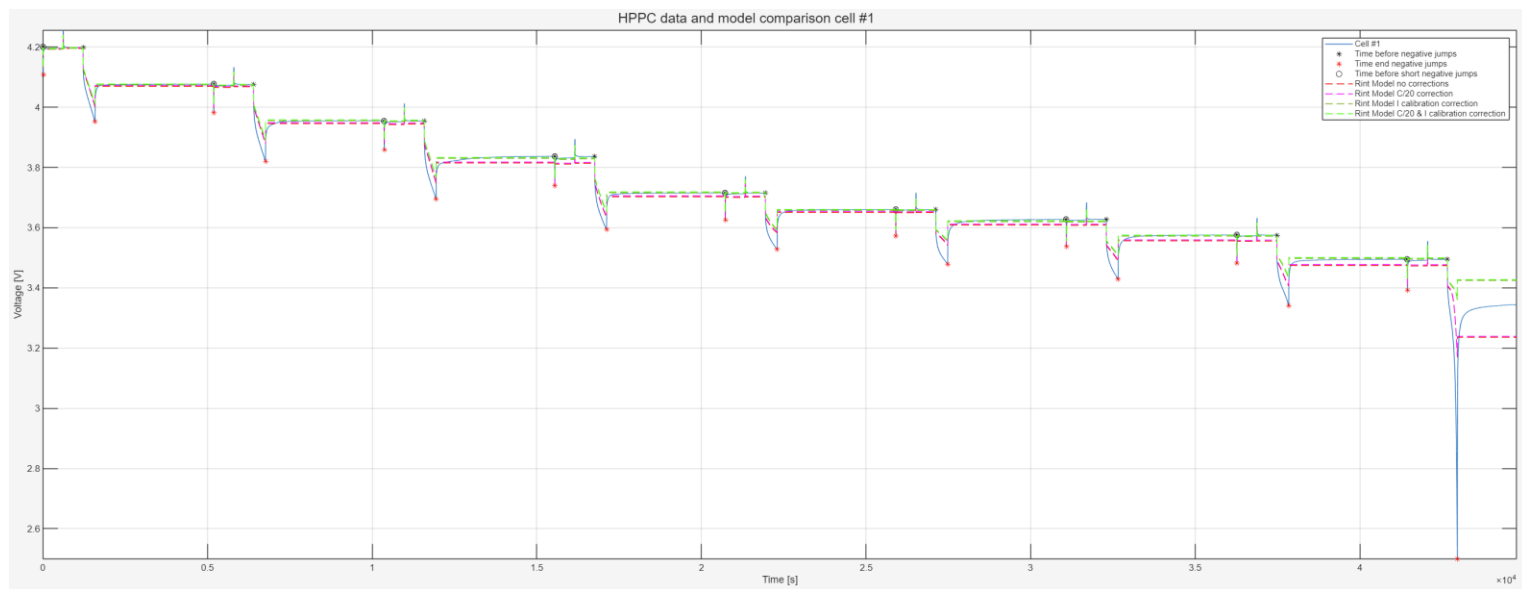
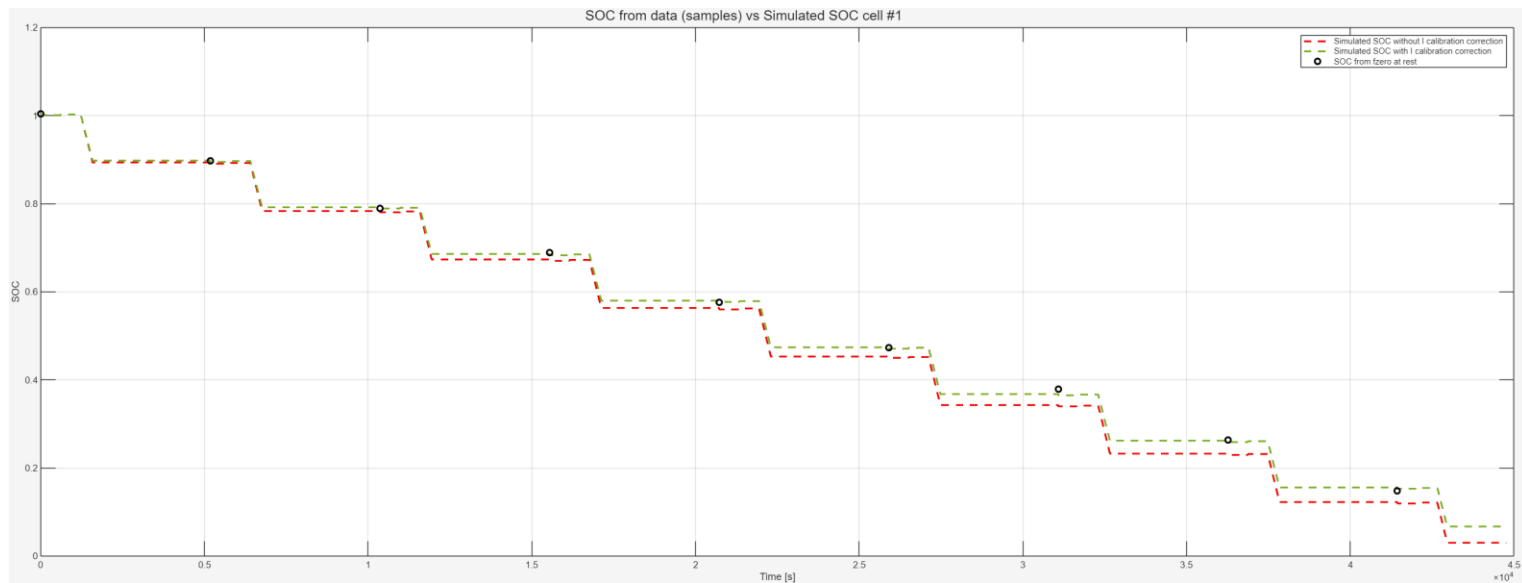
- **Capacity mismatch is negligible:** all cells are within $\pm 0.15\%$ of the mean.
- **Resistance mismatch is larger:** about $\pm 5\text{--}6\%$ around the mean.
- **Cell 1** is the most resistive, **Cell 3** the least resistive; **Cell 2** is closest to the average.

Lab 5: results (contained in Lab5_in_class.m)

The simulations have been performed with the averaged internal resistance R_{int_avg}

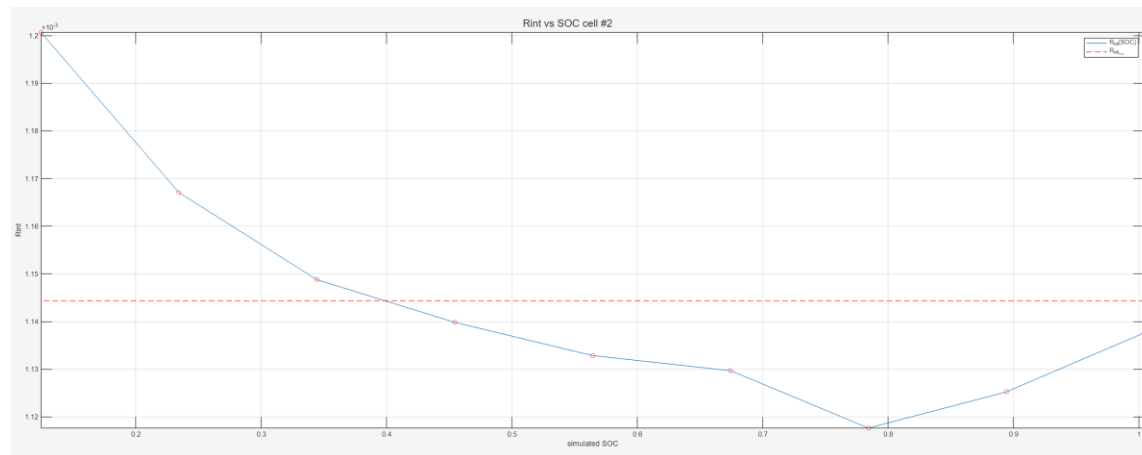
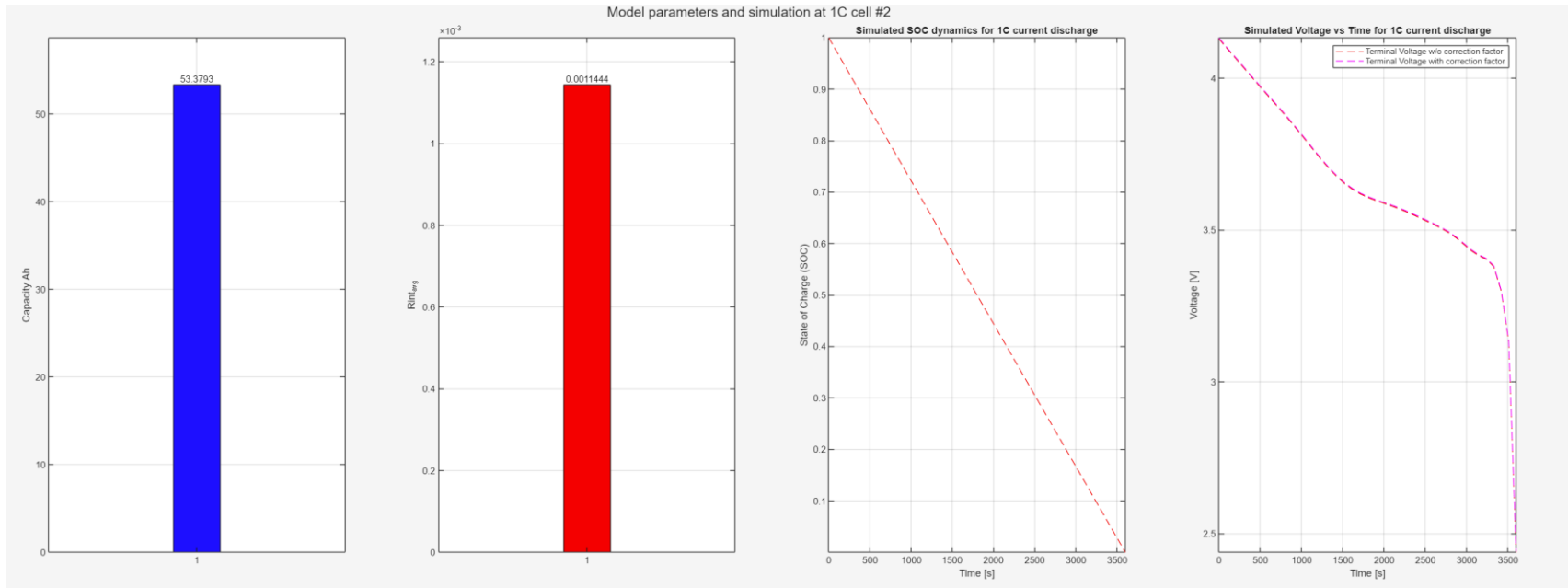


Lab 5: results (contained in Lab5_in_class.m)

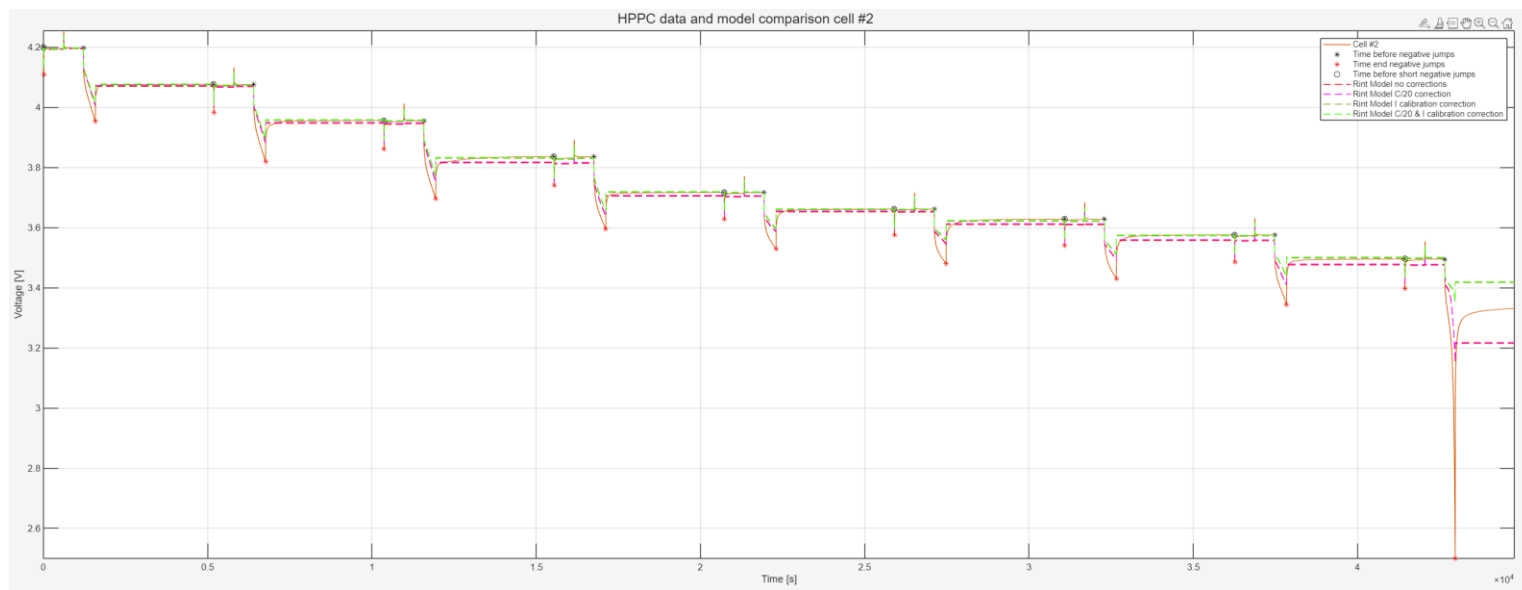
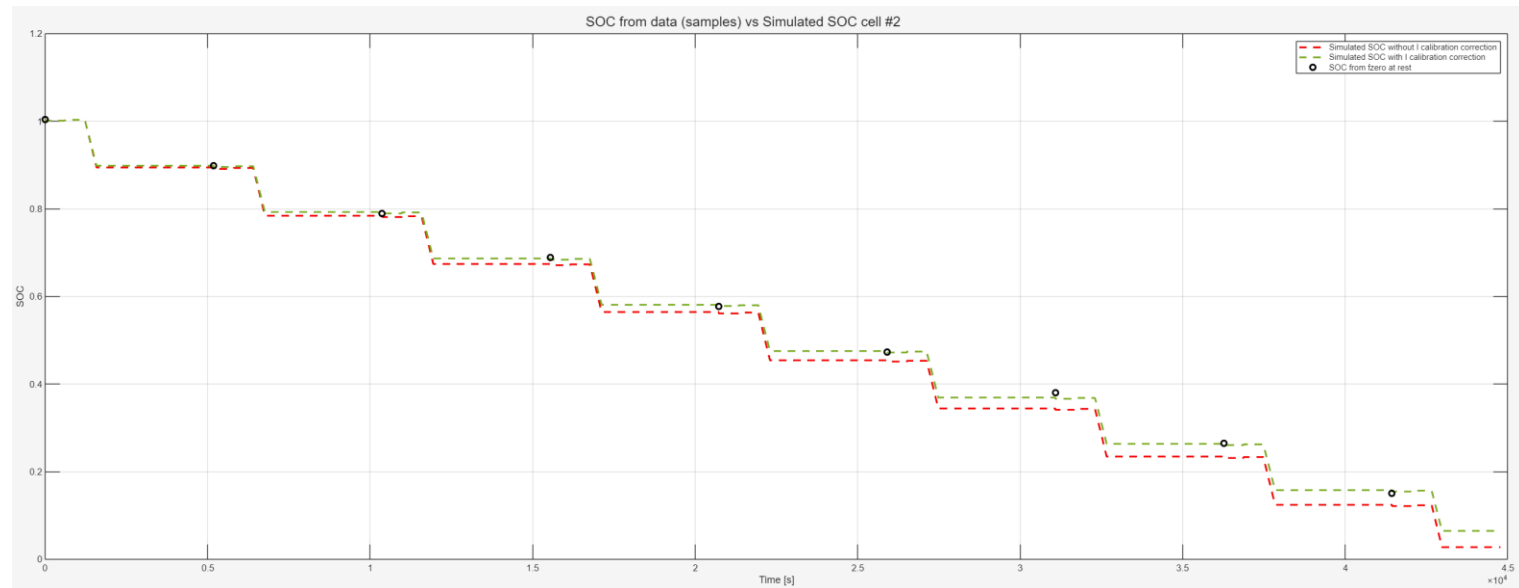


Lab 5: results (contained in Lab5_in_class.m)

The simulations have been performed with the averaged internal resistance R_{int_avg}

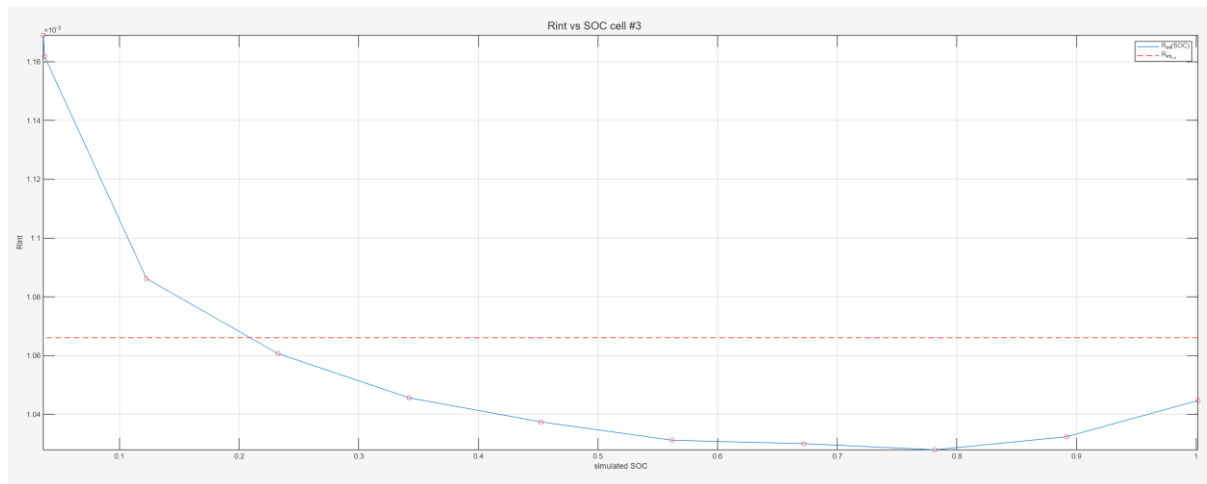
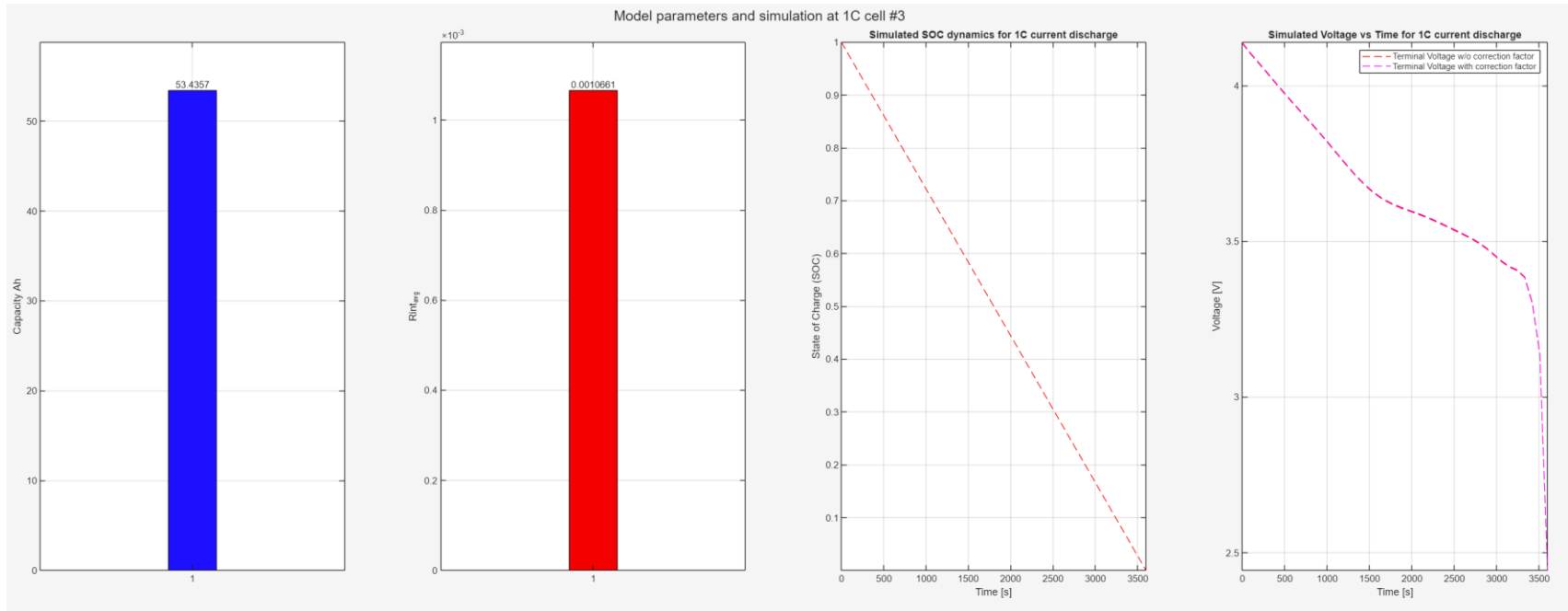


Lab 5: results (contained in Lab5_in_class.m)

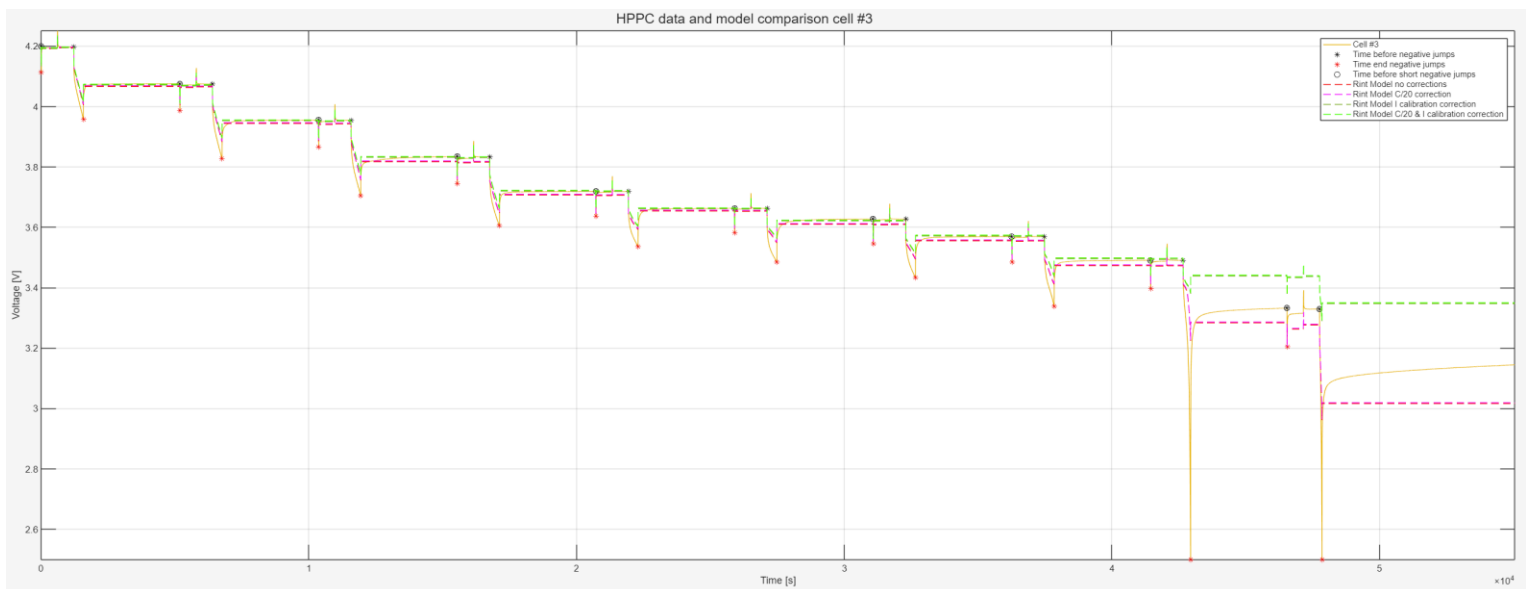
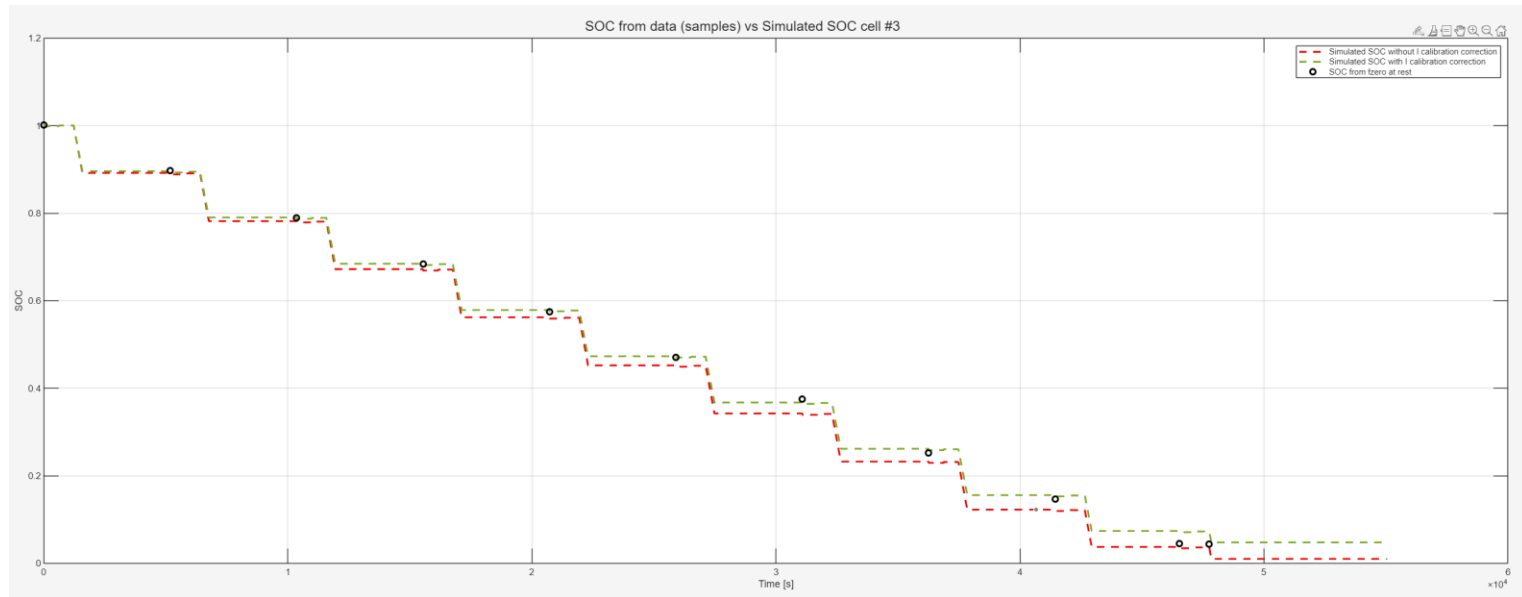


Lab 5: results (contained in Lab5_in_class.m)

The simulations have been performed with the averaged internal resistance R_{int_avg}



Lab 5: results (contained in Lab5_in_class.m)



Lab 6 (series and parallel connections)

Consider a set of cells, e.g. $N_{cell} = 3$

Use the parameters identified in Lab 5: SOC–OCV curve, R_{int} , C_{bat} .

For more visible mismatch effects, generate heterogeneous cells by randomly perturbing:

- C_{bat} ,e.g. 30% variability
- R_{int} ,e.g. 50% variability

Series-connected cells

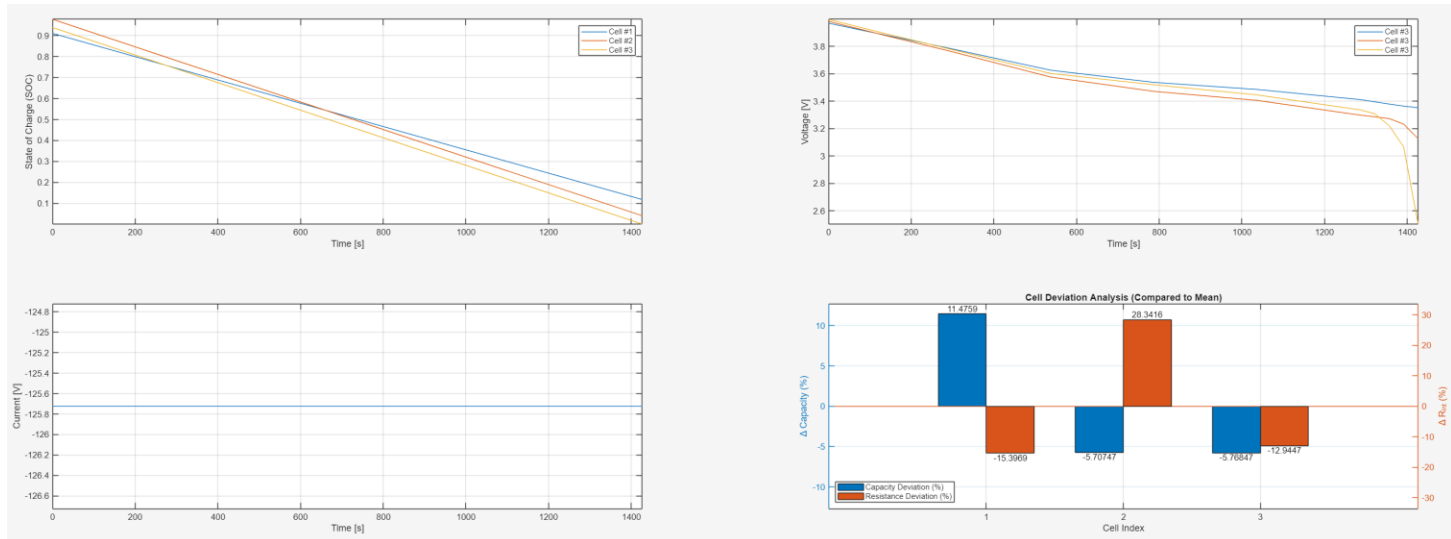
- Apply a 2C discharge (referred to the capacity of cell 1)
- Stop when one cell reaches 2.5 V
- Then apply a CCCV charging protocol:
 - CC phase: charge at 0.2C (referred to capacity of cell 1) until one cell reaches 4.2 V
 - **CV approach 1**: keep the pack voltage fixed at the **total voltage** reached at the end of the CC phase, until the current falls below $C_{min} = 50\text{mA}$
 - **CV approach 2**: keep the voltage of **the first cell** reaching 4.2 V fixed at 4.2 V. If another cell reaches 4.2 V, switch the active CV constraint to that cell. Continue until the current falls below $C_{min} = 50\text{mA}$

Parallel-connected cells

- Connect the N_{cell} cells in parallel. Algebraic constraints are required (Kirchhoff laws):
 - same voltage across all parallel branches
 - total current equal to the sum of branch currents
- Apply a constant discharge current proportional to the number of cells
- Analyze the current partitioning among the cells

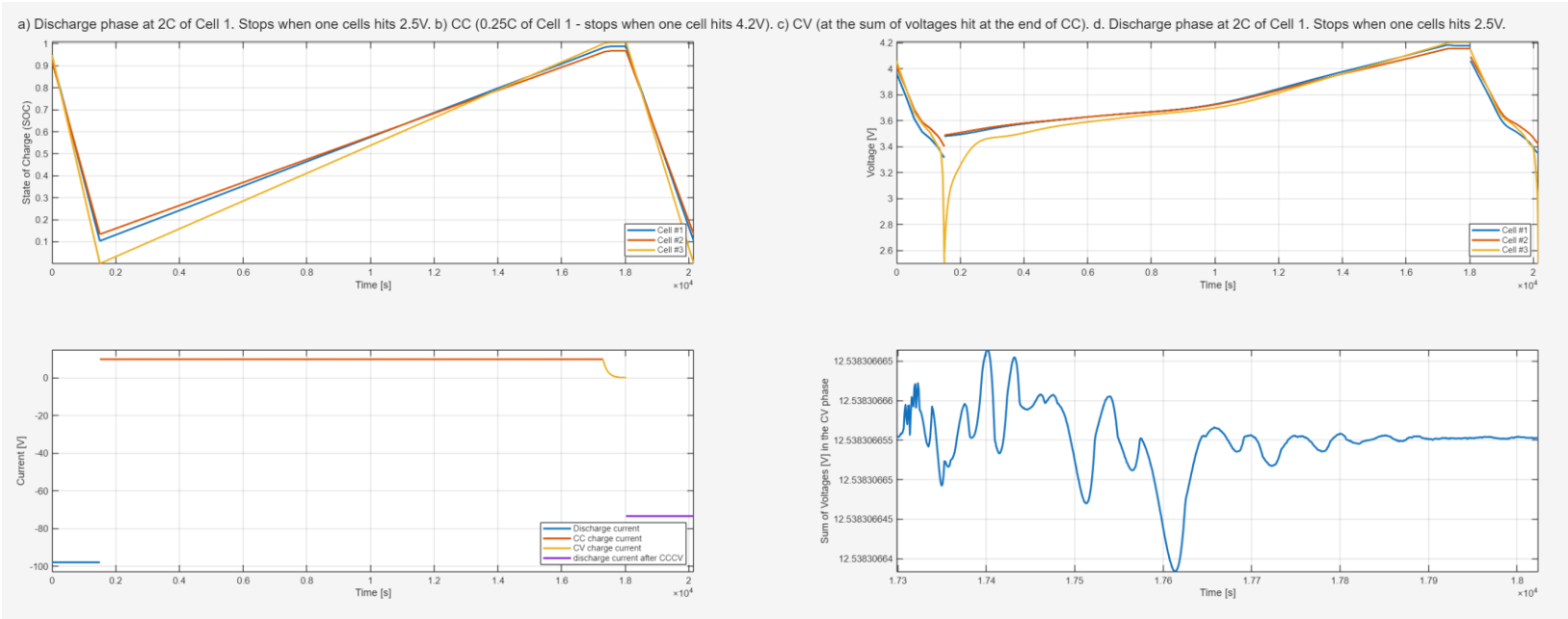
Lab 6: results (series)

- Experimental data for multiple cells show **small capacity dispersion**
- To better highlight imbalance effects, we generate synthetic cells with increased variability:
 - **30% variability** around the average capacity estimated from data
 - **50% variability** around the average $R_{int,avg}$ estimated from data



- The same current flows through all series-connected cells (rate 2C of the first cell)
- The cells start from **different initial SOC values**
- Capacity mismatch affects the **SOC depletion rate**
- Internal-resistance and capacity mismatch affects the **terminal-voltage behavior**
- Both **SOC** and **voltage** trajectories exhibit **crossings** during discharge
- Crossings: combination of **initial SOC differences**, **capacity and resistance mismatch**
- The cell that reaches the cutoff voltage first limits the usable energy of the whole string

Lab 6: results (series)



- New random cell parameters are considered.
- After the initial discharge, the pack is charged with a **CCCV protocol**.
- **CC phase**: charging 0.2C (of first cell) stops when the **first cell reaches 4.2 V**.
- **CV phase**: **total pack voltage** fixed at the sum of the cell voltages at the end of CC phase. CV ends when current drops below a minimum value (e.g. $C_{min} = 50\text{mA}$).
- During charging, **SOC spread is partially reduced**, mainly because the same current flows through all series-connected cells while the cells have different capacities.
- The **CV phase** also helps compensate for the effect of different **internal resistances**, since the cell voltages redistribute while the total pack voltage is fixed.
- Final discharge: **C-rate is different** from initial one → resistive voltage drops are also different, and the **2.5 V limit is reached at different SOC values** than in the first discharge.

Lab 6: results (series)

CV phase: total pack voltage fixed at the value reached at end of CC.

Is this the right approach? To be safe: cell reaching 4.2V first at end of CC should not exceed this limit during CV. Hence, during CV, pack current must satisfy simultaneously:

- **constant pack voltage**
- $V_i \leq 4.2\text{V}$ for all cells
- **CV ends** when current drops below a minimum value (e.g. $C_{min} = 50\text{mA}$).

Even with simplified model $V_i = OCV_i(SOC_i) + R_{int,i}I$ this is **not always trivial** because:

- capacity mismatch affects SOC evolution
- R_{int} mismatch affects voltage drop under same current

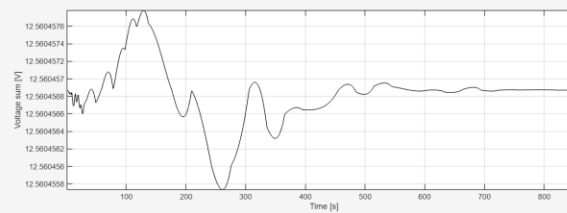
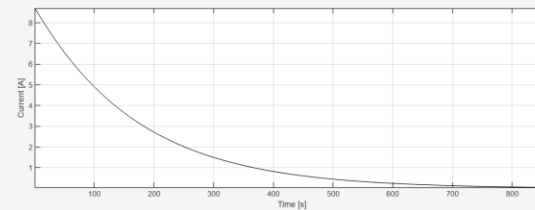
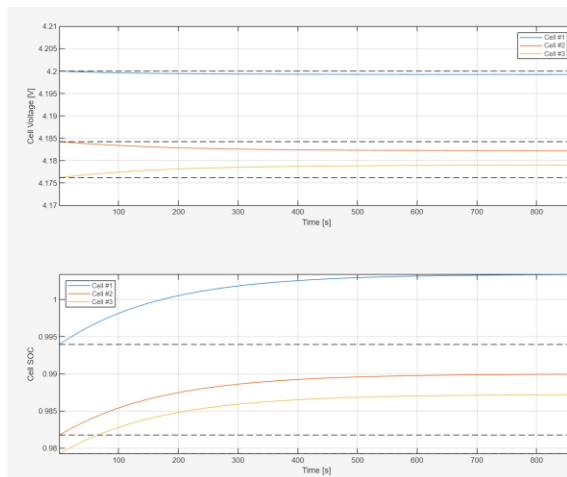
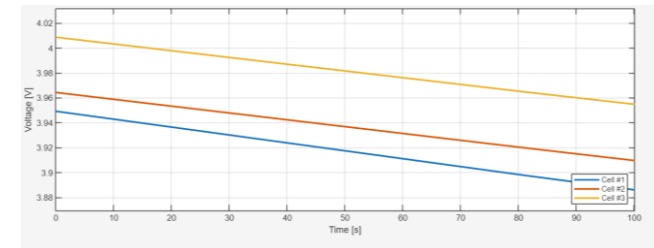
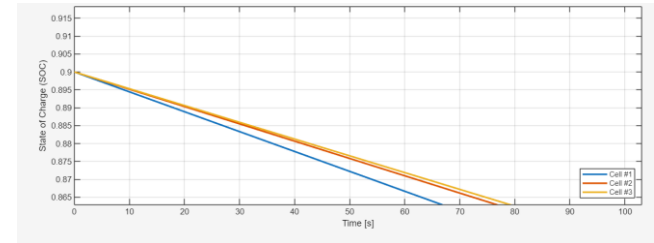
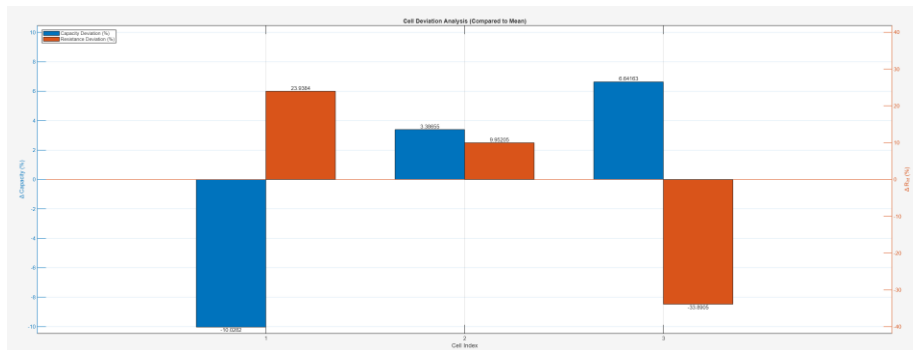
Favorable case: cell reaching 4.2V first also has **relatively larger R_{int}** . As current decreases in CV, its resistive contribution decreases, so its voltage may drop or remain close to 4.2V, while voltages of other cells can increase. Then pack voltage can remain constant without violating cell-voltage limits. This is only one possible case. In general, feasibility depends on combined effect of:

- SOC evolution
- capacity mismatch
- R_{int} mismatch

Pack-level CV can stop due to infeasibility, not only because current reaches C_{min} .

Lab 6: results (series)

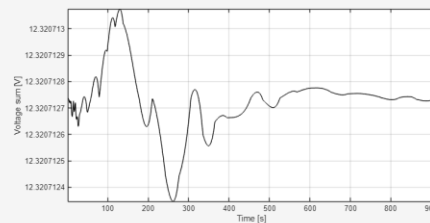
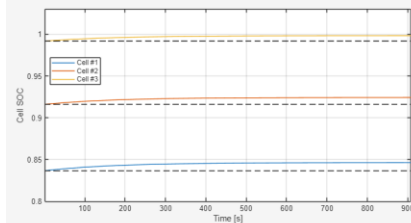
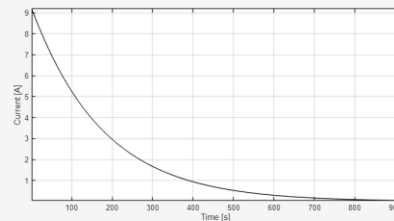
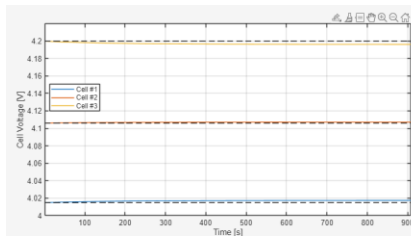
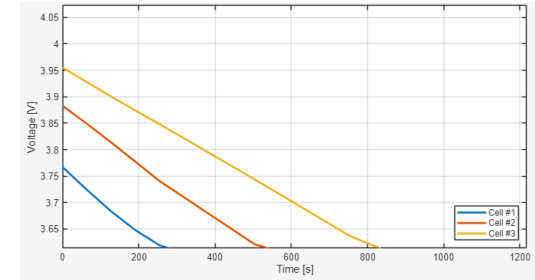
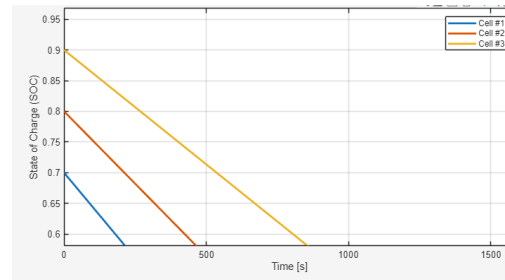
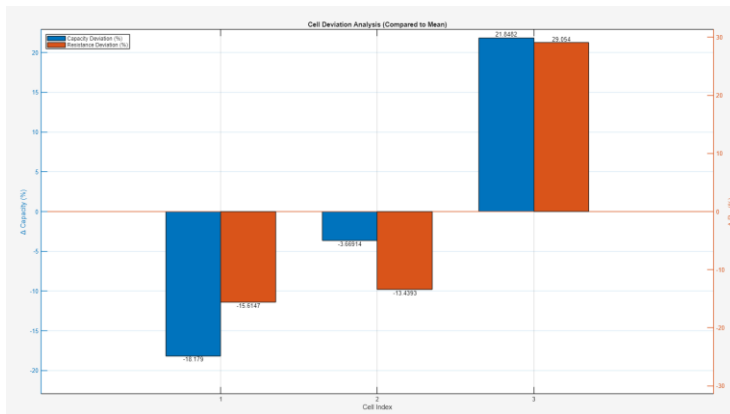
Example 1 (favorable): cell #1 has the highest internal resistance and the lowest capacity. All cells start from the same initial SOC.



CV phase: voltages of cells #1 and #2 decrease, while voltage of Cell #3 increases. SOC of all cells increases. Cell #1 increases the most because of **lower capacity**.

Lab 6: results (series)

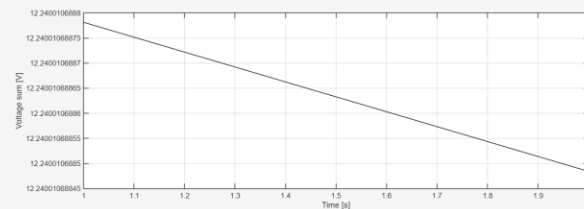
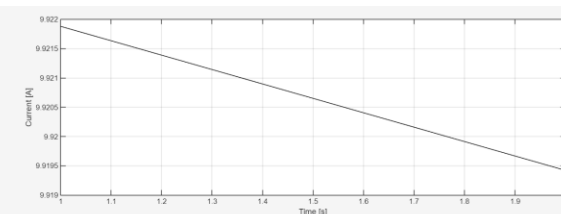
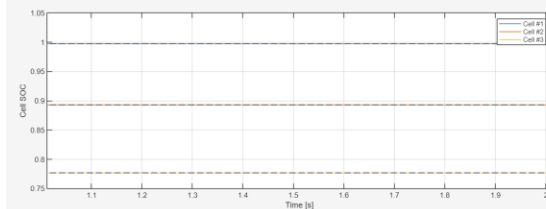
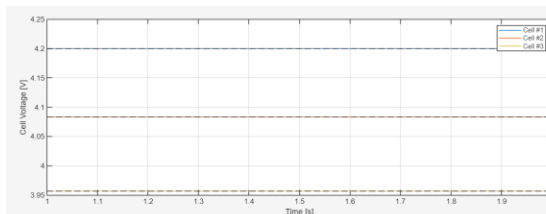
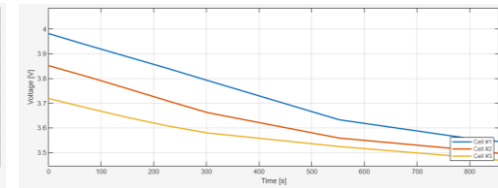
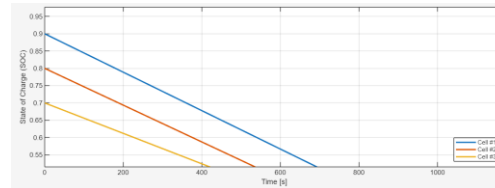
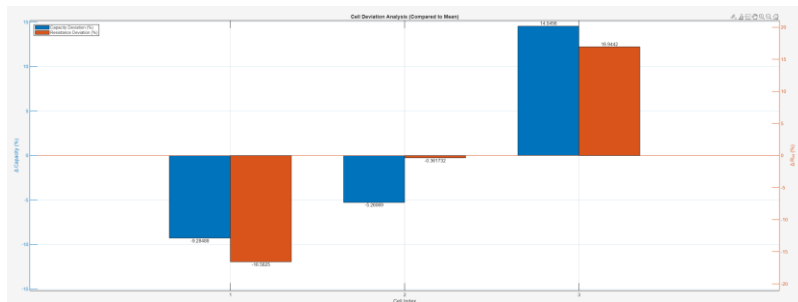
Example 2 (favorable): cell #3 has the highest R_{int} and the highest capacity. Although its larger capacity would, in principle, limit its SOC increase, its higher initial SOC makes it the first cell to reach 4.2 V.



In the **CV phase**, the voltage of cell #3 decreases slightly, whereas the voltages of cells #1 and #2 increase. This is due to their lower resistive drops and to their larger SOC increase, which raises their OCV more than the one of cell #3.

Lab 6: results (series)

Example 3 (unfavorable): cell #1 has the lowest R_{int} , the lowest capacity, and the highest initial SOC. Because of its **higher initial SOC**, together with its smaller capacity, it is the first cell to reach 4.2 V. In the CV phase, since the other cells have higher R_{int} and larger capacity (lower OCV increase over time), **their voltages will drop more as the current decreases than for cell #1**. As a result, a **CV phase based on constant pack voltage is not feasible** in this case, and the CV phase stops prematurely.



Lab 6: results (series)

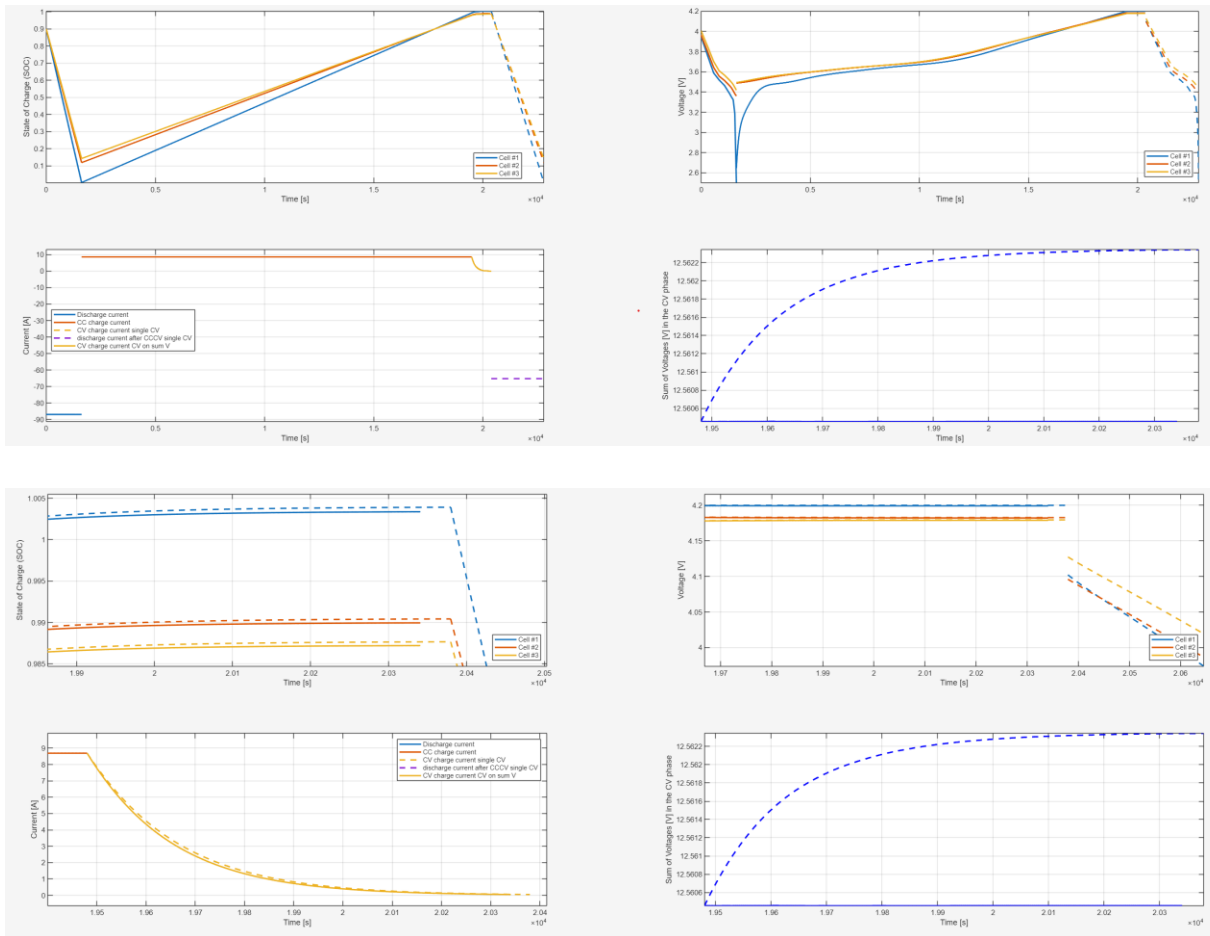
Alternative CV phase: the pack current is adjusted so that the cell reaching 4.2V first is kept at this voltage. If another cell becomes the limiting one, the CV constraint is shifted to that cell. This continues until the current falls below C_{min} or the problem becomes infeasible, i.e. no current can satisfy simultaneously the active 4.2V constraint and the condition $V_i \leq 4.2V$ for all other cells.

This **strategy** is **preferable** to a pack-level CV based on the end-of-CC total voltage, because it directly enforces the relevant safety constraint at cell level ($V_i \leq 4.2 \text{ V}$). With mismatched cells, fixing only the pack voltage may be infeasible or may stop the CV phase prematurely.

Lab 6: results (series)

Comparison (dashed alternative CV, continuous CV on voltage sum)

Example 1: cell #1 highest resistance and lowest capacity. Same initial SOC for all cells.

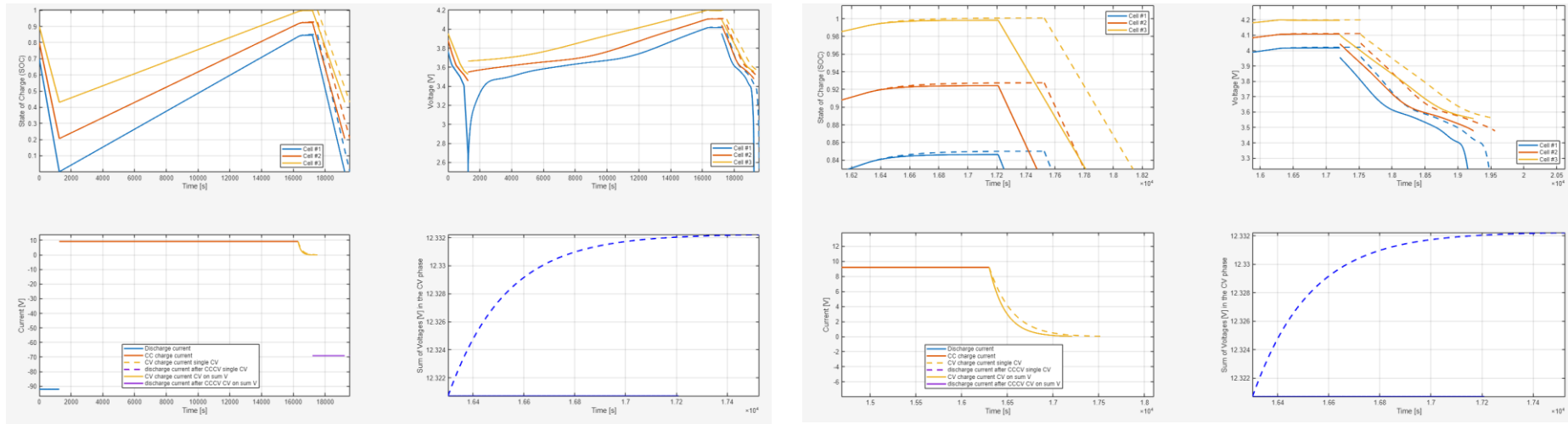


New strategy leads to
slightly higher SOC
(better cell utilization)

Lab 6: results (series)

Comparison (dashed alternative CV, continuous CV on voltage sum)

Example 2: cell #3 highest R_{int} , capacity and initial SOC. R_{int} and this latter make cell #3 reach 4.2 V first at the end of CC.

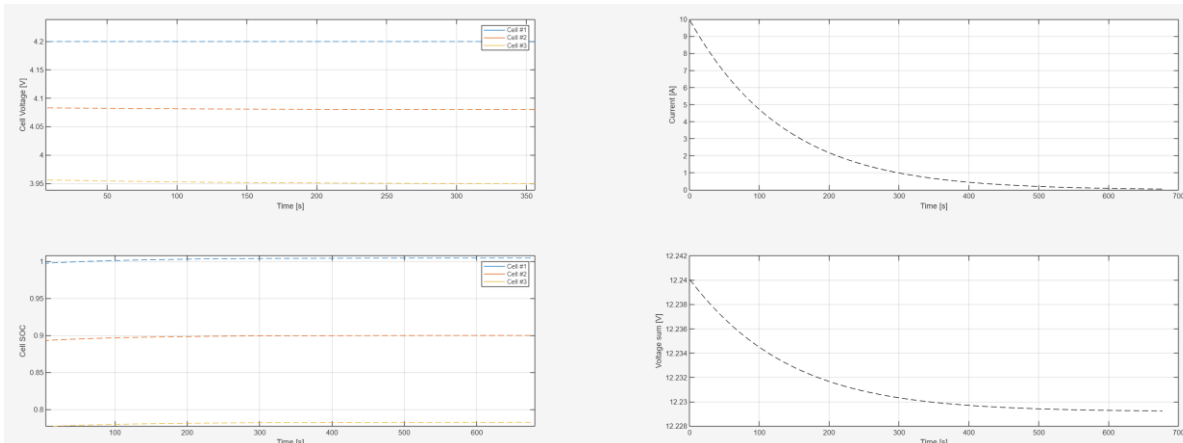
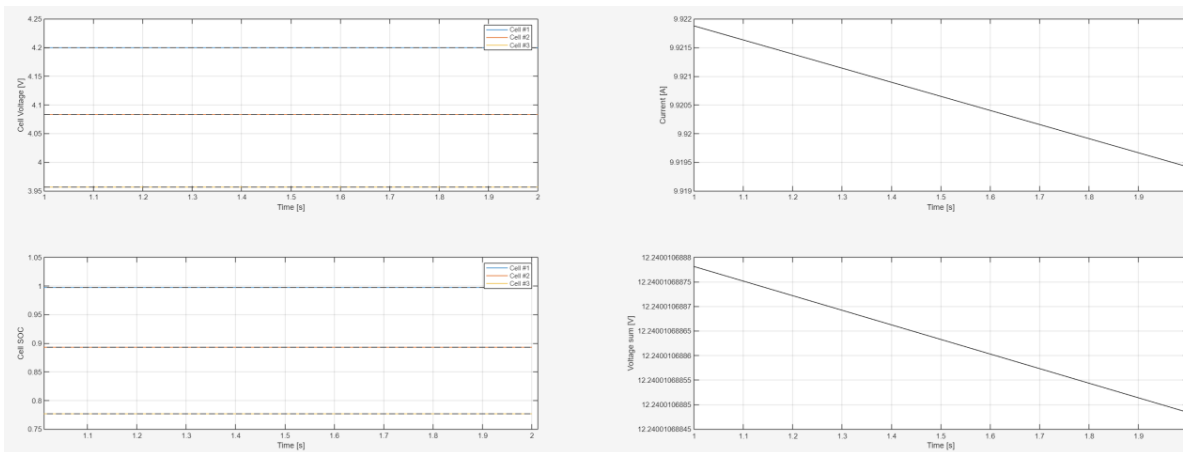


New strategy leads to higher SOC (better cell utilization).

Lab 6: results (series)

Comparison (dashed alternative CV, continuous CV on voltage sum)

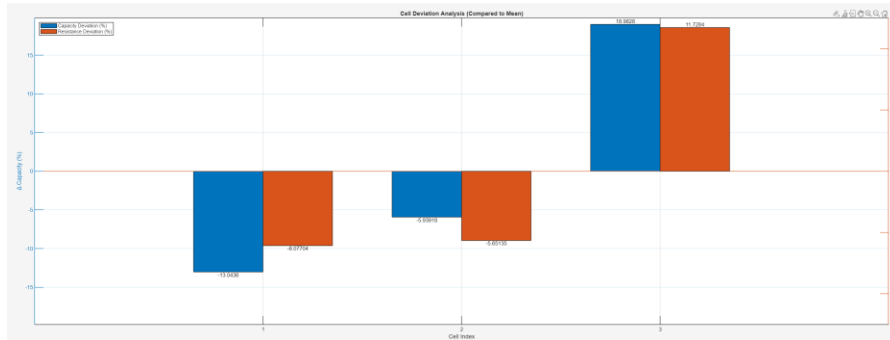
Example 3: cell #3 lowest R_{int} and capacity, highest initial SOC $\rightarrow$ first cell to reach 4.2 V.



Previous CV strategies terminated early because of infeasibility. The new one allowed to further charge the cells thus leading to a higher utilization.

Note: the sum of voltages dropped but the SOC increased!

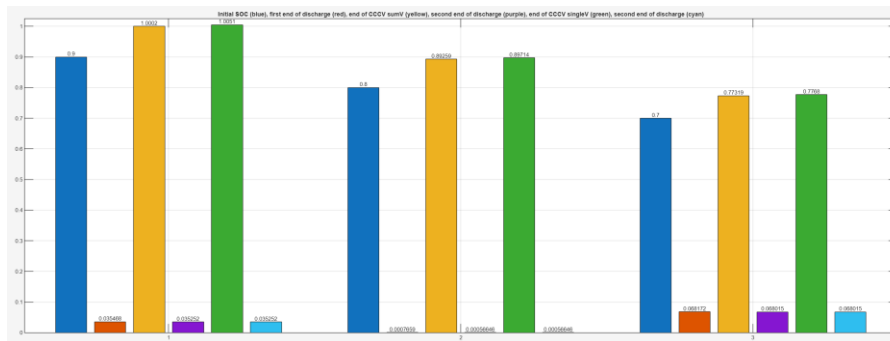
Lab 6: results (series)



← Rint and Cbatt distribution of example 3

Last two figures

SOC at: initial condition, end of 1st discharge, end of CCCV (sum vs one cell), end of 2nd discharge

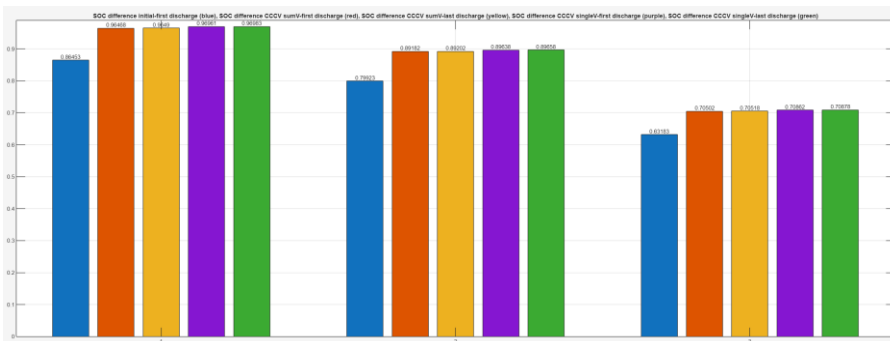


- Initial SOC mismatch randomly generated
- CCCV brings cells close to full charge; SOC can slightly exceed 100% because in CV the current falls below C/20
- After the 2nd discharge, cells reach again similar low SOC values

Bottom figure

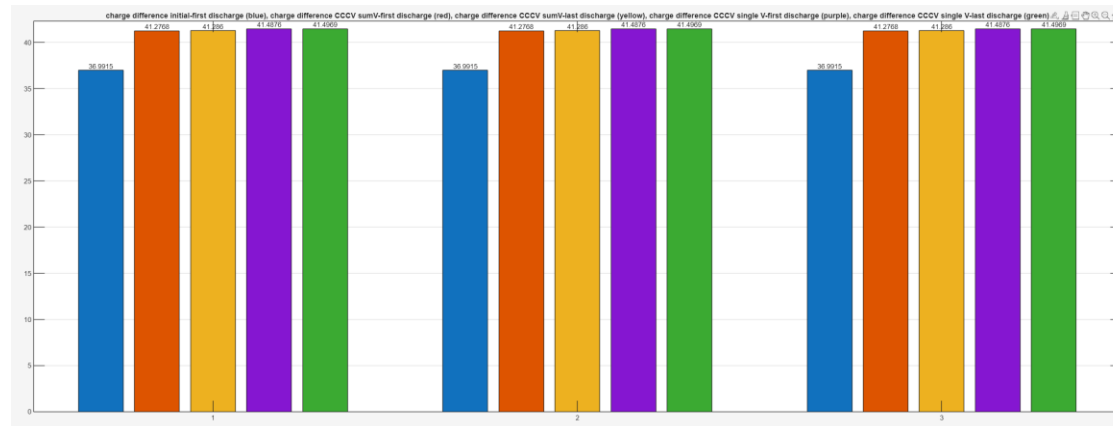
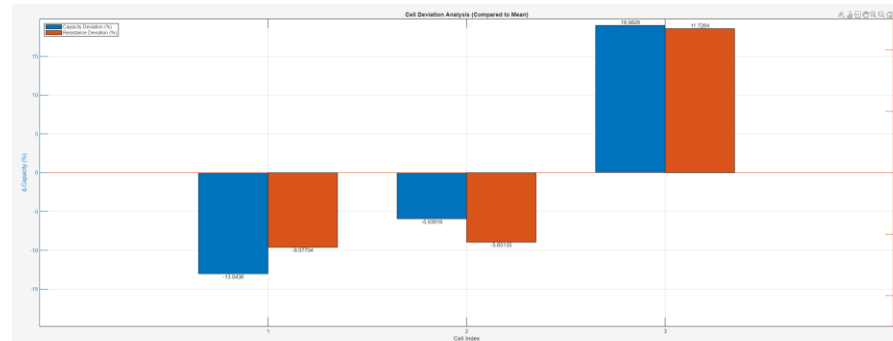
Exploited SOC in the 1st discharge, CCCV, and 2nd discharge

- The lowest-capacity cell is used the most
- Lower-capacity cells undergo larger SOC excursions



New CV strategy allows to make better use of cell capacities.

Lab 6: results (series)



Note

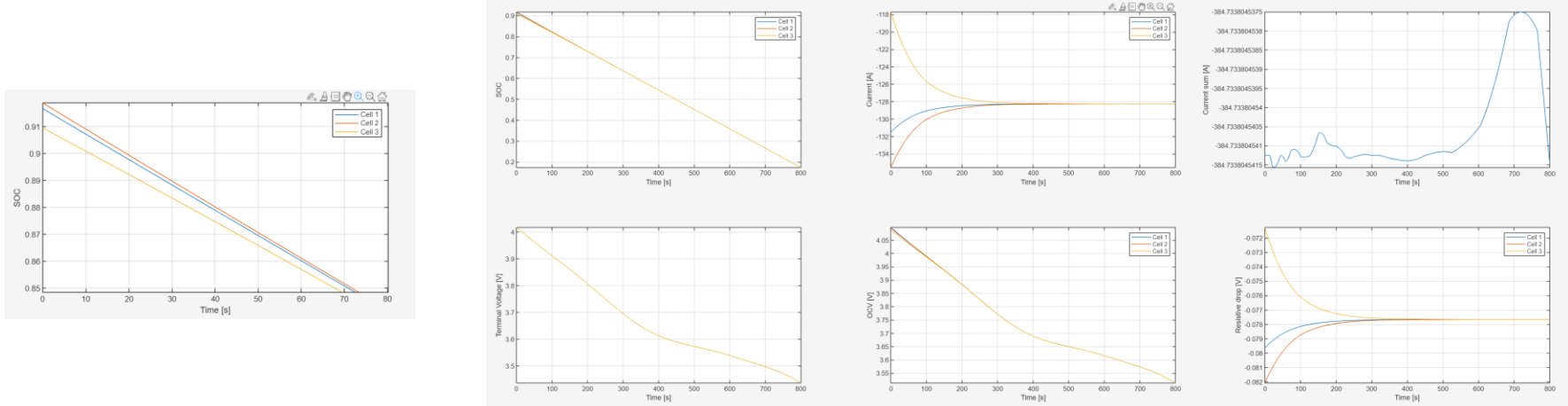
The SOC differences of the previous slide are due to capacity differences. As shown here, the amount of charge extracted is the same for all cells in the different stages (same current flowing through the cells in series). The CV strategy on the single cell allows for a better utilization

Lab 6: results (parallel)

First scenario: same R_{int} , Capacity and SOC-OCV for all cells. Different initial SOC.

All scenarios use an event to stop the simulation when:

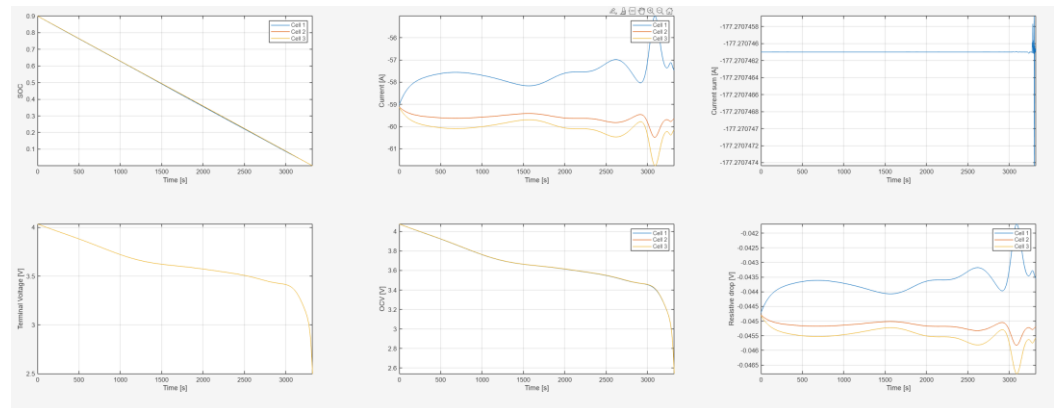
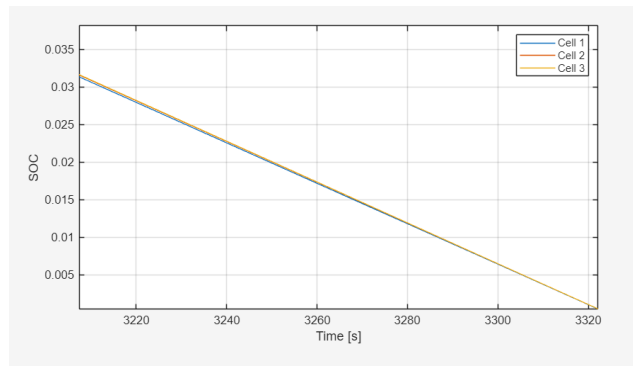
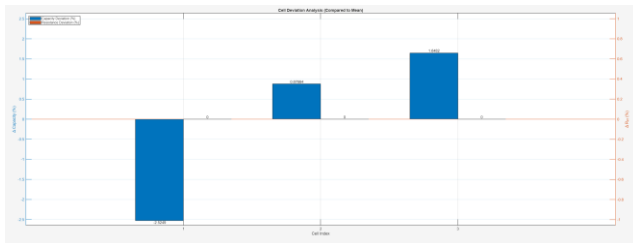
- a C-rate (function of cell capacities) is attained in one of the parallel branches
- Cutoff voltage 2.5V is attained
- 0% SOC is attained by one of the cells



Cells balance over time. More current flows through the cell with higher SOC to match the voltage the other cells (at lower SOC) exhibit.

Lab 6: results (parallel)

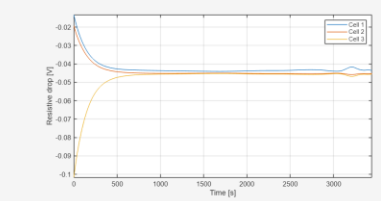
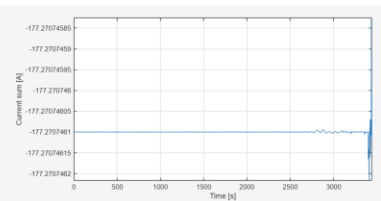
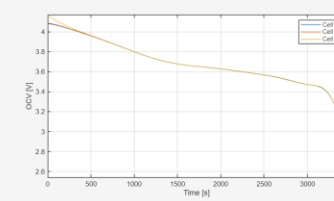
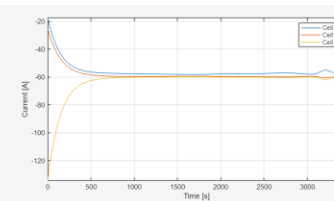
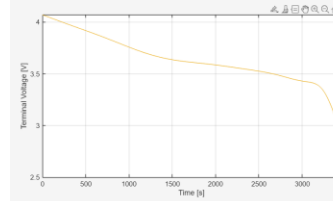
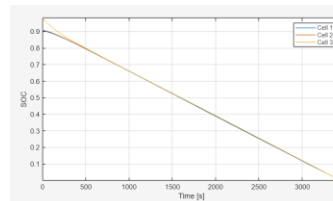
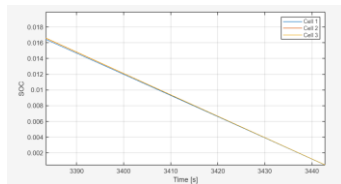
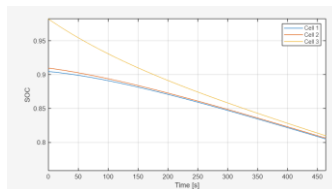
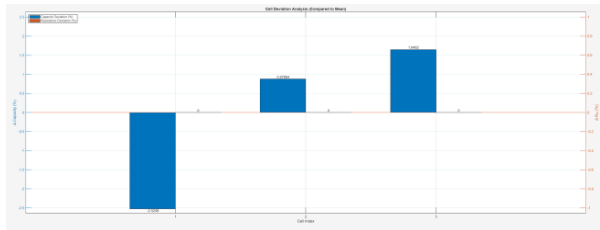
Second scenario: same R_{int} , SOC-OCV and initial SOC for all cells. Different capacities.



Capacity mismatch alone causes **different SOC depletion rates**. Resulting SOC differences $\rightarrow$ OCV differences $\rightarrow$ current redistribution in parallel branches (voltage matching through different resistive drop). Near the end of discharge: **steep OCV-SOC knee** & common-voltage constraint **force SOC values to converge again**.

Lab 6: results (parallel)

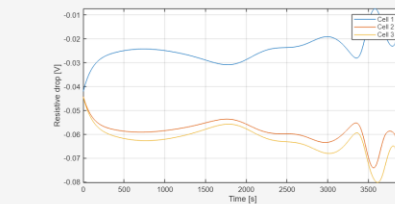
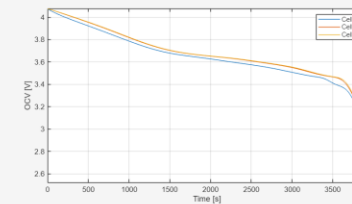
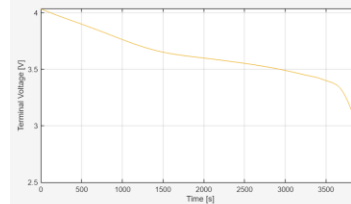
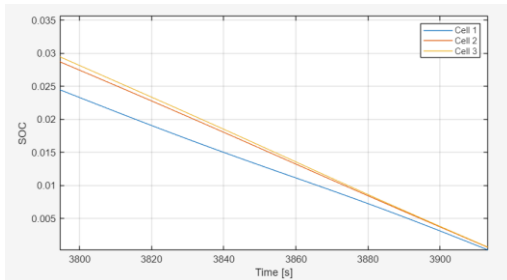
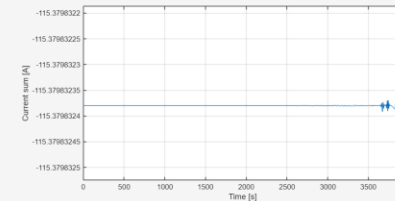
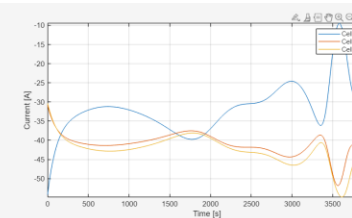
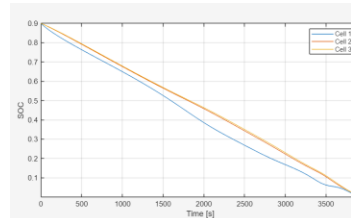
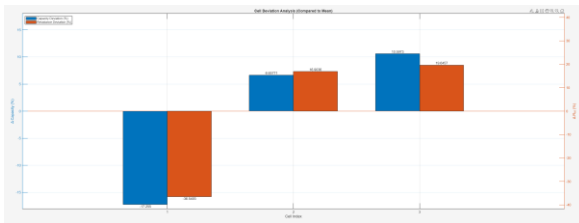
Third scenario: same R_{int} and SOC-OCV for all cells. Different capacities and initial SOC.



As in scenario #1, when the initial SOC difference is significant, current redistribution is required to satisfy common terminal-voltage constraint of parallel connection, i.e. cell with higher initial SOC carries a larger current. This progressively reduces the SOC imbalance. Near the end of discharge, the effect becomes even stronger because of the steep SOC-OCV knee, which further drives the SOC values toward each other.

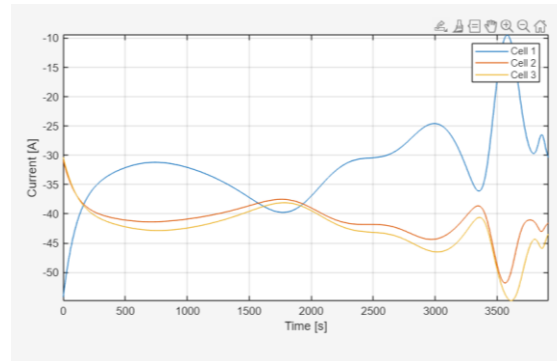
Lab 6: results (parallel)

Fourth scenario: same SOC-OCV and initial SOC for all cells. Different capacities and R_{int} .



The scenario becomes more complex because of R_{int} mismatch. The SOC values still tend to converge near the end of discharge, to compensate for the steep OCV drop at the knee. However, unlike the previous case, a residual SOC mismatch remains, since part of the voltage equalization can also be achieved through different resistive drops.

Lab 6: results (parallel)

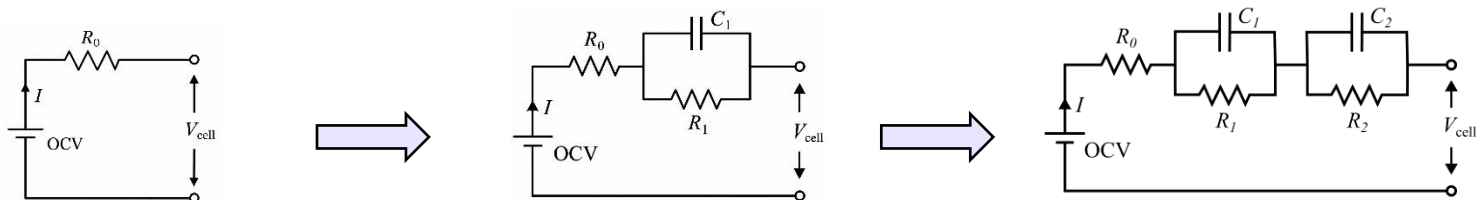


Important: **current sharing** among parallel cells can be **highly uneven**. This may become **critical** for some cells, which may experience very high currents, especially in the presence of mismatch or faults. Since **branch currents are usually not measured** individually, a state/current estimator would be very useful.

In general, the **healthier cell**, typically characterized by higher capacity and lower internal resistance, **carries a larger share of the current**. On one hand, this may accelerate the degradation of the healthier cell. On the other hand, from a pack-level perspective, where individual cells are not expected to be replaced, this **mechanism can be beneficial**: the load is naturally shifted toward the cell that can sustain it better, improving the overall utilization of the pack and **potentially reducing cell heterogeneity** over time.

Lab 7 (contained in Lab7_in_class.m)

1. Load data from directory Processed Data/Temperature_25C/C20_discharge
2. **Select all cells** by extracting the list of .mat files in the directory. **For all of them:**
3. Improve the modelling. Rather than using an Rint model only, use also one RC parallel.
4. Identify first the RC parameters manually (use Rint, Cbat and SOC-OCV as identified in Lab 5) looking at the relaxation times. In particular, $u=0$ allows to identify $\tau=RC$. Once τ is obtained, compute R and C when $u=\text{constant}$. Rint, R and C will have different values at the different jumps/relaxation times. Using average values may not lead to the best performance.
5. In order to obtain the best average values, use the system identification toolbox.
6. Use again the system identification toolbox with 2RC
7. Compare the performance of all the different models



Lab 7: RC time constant τ from voltage relaxation

Given C_{bat} , R_{int} and SOC-OCV model, estimate $\tau = R_1 C_1$ using the voltage relaxation observed after a current pulse, when the applied current becomes zero. The RC polarization voltage $v_1(t)$ satisfies:

$$\dot{v}_1(t) = -\frac{1}{R_1 C_1} v_1(t) + \frac{1}{C_1} u(t)$$

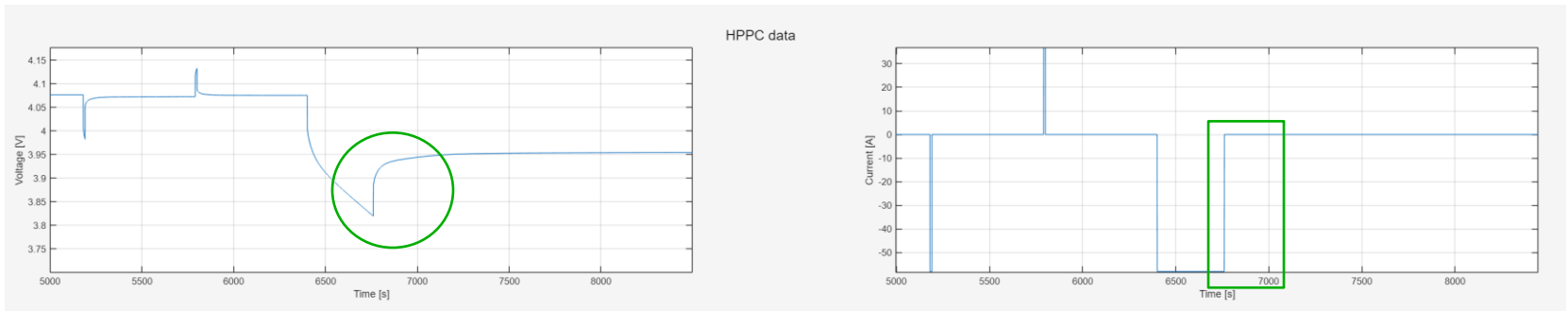
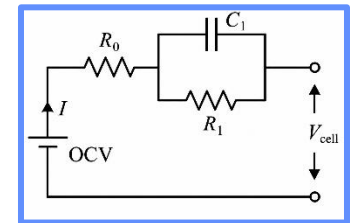
where $u(t)$ is the current. During the relaxation after current pulse $u(t) = 0$. Therefore:

$$\dot{v}_1(t) = -\frac{1}{R_1 C_1} v_1(t) = -\frac{1}{\tau} v_1(t)$$

whose solution is:

$$v_1(t) = e^{-\frac{t-t_0}{\tau}} v_1(t_0)$$

with t_0 being the initial time.



Lab 7: RC time constant τ from voltage relaxation

If $v_1(t)$ were directly measurable, then:

$$\frac{v_1(t_f)}{v_1(t_0)} = e^{-\frac{t_f-t_0}{\tau}} \Rightarrow \ln\left(\frac{v_1(t_f)}{v_1(t_0)}\right) = -\frac{t_f-t_0}{\tau} \Rightarrow \tau = -\frac{t_f-t_0}{\ln\left(\frac{v_1(t_f)}{v_1(t_0)}\right)}$$

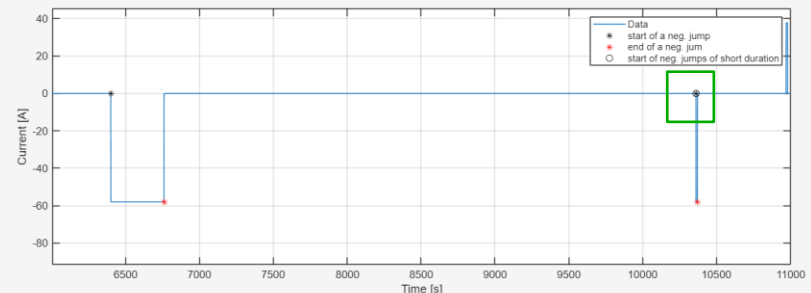
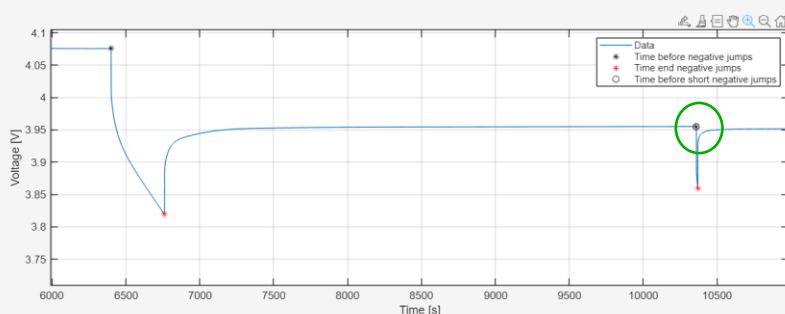
Issue: $v_1(t)$ is not directly measured. The measured quantity is the terminal voltage:

$$v(t) = OCV(SOC(t)) + R_{int} \left(I(t) + \frac{C_{bat,Ah}}{20} \right) + v_1(t)$$

During relaxation interval $I(t) = 0$ and the **SOC is constant: $SOC(t) = SOC(t_0)$** . Therefore:

$$v(t) = OCV(SOC(t_0)) + R_{int} \frac{C_{bat,Ah}}{20} + v_1(t) = K + v_1(t)$$

where $K = OCV(SOC(t_0)) + R_{int,avg} \frac{C_{bat,Ah}}{20}$. After a sufficient rest, the RC transient almost vanishes: $v_1(t) \rightarrow 0$. The terminal voltage approaches: $v_{regime} \approx K$.



Lab 7: RC time constant τ from voltage relaxation

Because the relaxation is observed after a negative-current pulse, the voltage increases toward v_{regime} . Therefore, it is convenient to define:

$$\begin{aligned}v_1(t_0) &= v_{\text{regime}} - v(t_0) \\v_1(t_f) &= v_{\text{regime}} - v(t_f)\end{aligned}$$

where:

- t_0 is the time immediately after the current becomes zero **already after the resistive drop** due to R_{int} .
- t_f is an intermediate time during the relaxation

Finally, as explained before

$$\tau = - \frac{t_f - t_0}{\ln\left(\frac{v_1(t_f)}{v_1(t_0)}\right)}$$

Note: a sufficiently long rest is needed to estimate $v_{\text{regime}} \approx K$ because after approximately 4.6τ the exponential transient has reached about 99% of its final value.

Moreover, t_f should be chosen such that $v_1(t_f) > 0$ and sufficiently larger than the voltage measurement noise. If t_f is too close to the final steady state, then: $v_1(t_f) \approx 0$ and the logarithm becomes numerically unreliable.

Lab 7: Estimating R_1 and C_1 once τ is known

From the relaxation phase we already identified: $\tau = R_1 C_1$. However, **when $u(t) = 0$ only the product $R_1 C_1$ is observable (R_1 and C_1 cannot be identified independently)**. To obtain R_1 and C_1 separately, we need a time window with nonzero current: $u(t) \neq 0$.

The RC branch voltage satisfies: $\dot{v}_1(t) = -\frac{1}{R_1 C_1} v_1(t) + \frac{1}{C_1} u(t) = -\frac{1}{\tau} v_1(t) + \frac{1}{C_1} u(t)$

For a constant input $u(t)$ over the selected window: $u(s) = u, s = t_0, \dots, t$ one has:

$$v_1(t) = e^{-\frac{t-t_0}{\tau}} v_1(t_0) + \int_{t_0}^t e^{-\frac{t-s}{\tau}} \frac{u(s)}{C_1} ds = e^{-\frac{t-t_0}{\tau}} v_1(t_0) + \frac{u}{C_1} e^{-\frac{t}{\tau}} \int_{t_0}^t e^{\frac{s}{\tau}} ds$$

The integral is: $\int_{t_0}^t e^{\frac{s}{\tau}} ds = \tau(e^{\frac{t}{\tau}} - e^{\frac{t_0}{\tau}})$. Therefore: $\frac{u}{C_1} e^{-\frac{t}{\tau}} \int_{t_0}^t e^{\frac{s}{\tau}} ds = \frac{u\tau}{C_1} (1 - e^{-\frac{t-t_0}{\tau}})$ and

$$v_1(t) = e^{-\frac{t-t_0}{\tau}} v_1(t_0) + \frac{u\tau}{C_1} (1 - e^{-\frac{t-t_0}{\tau}})$$

Since: $\tau = R_1 C_1$ we have: $\frac{\tau}{C_1} = R_1$ and therefore:

$$v_1(t) = e^{-\frac{t-t_0}{\tau}} v_1(t_0) + R_1 u (1 - e^{-\frac{t-t_0}{\tau}})$$

Lab 7: Estimating R_1 and C_1 once τ is known

Rearranging the previous expression:

$$v_1(t) - e^{-\frac{t-t_0}{\tau}} v_1(t_0) = R_1 u \left(1 - e^{-\frac{t-t_0}{\tau}}\right)$$

Hence:

$$R_1 = \frac{v_1(t) - e^{-\frac{t-t_0}{\tau}} v_1(t_0)}{u \left(1 - e^{-\frac{t-t_0}{\tau}}\right)}$$

Practical issue: $v_1(t)$ is not directly measured. The measured terminal voltage is:

$$v(t) = OCV(SOC(t)) + R_{\text{int}} \left(I(t) + \frac{C_{\text{bat,Ah}}}{20} \right) + v_1(t)$$

During the constant-current window $I(t) = u \neq 0$ so the SOC is no longer constant. Therefore: $SOC(t) \neq SOC(t_0)$ and the OCV changes during the pulse:

$$OCV(SOC(t)) \neq OCV(SOC(t_0))$$

Thus, before applying the formula for R_1 , we must reconstruct $v_1(t)$ by subtracting the OCV and ohmic contributions from the measured voltage.

Lab 7: Estimating R_1 and C_1 once τ is known

The selected current pulse should start after a long rest. Then, at the beginning of the pulse, the SOC can be initialized accurately by inverting the OCV-SOC relationship with **zero** but during nonzero-current pulse, SOC is **propagated using the SOC dynamics**:

$$\dot{SOC}(t) = -\frac{I(t)}{C_{bat}}$$

It is required because the **OCV contribution varies** during the current pulse as the **SOC changes**. The obtained SOC is **only a model-based estimate** and may be affected by measurement noise, parameter uncertainty, and integration errors $\rightarrow$ the identification window $[t_0, t_f]$ should not be too long to limit the accumulation of SOC estimation errors. Once $SOC(t)$ is reconstructed then one can obtain:

$$R_1 = \frac{v_1(t_f) - e^{-\frac{t_f-t_0}{\tau}} v_1(t_0)}{u \left(1 - e^{-\frac{t_f-t_0}{\tau}}\right)}$$

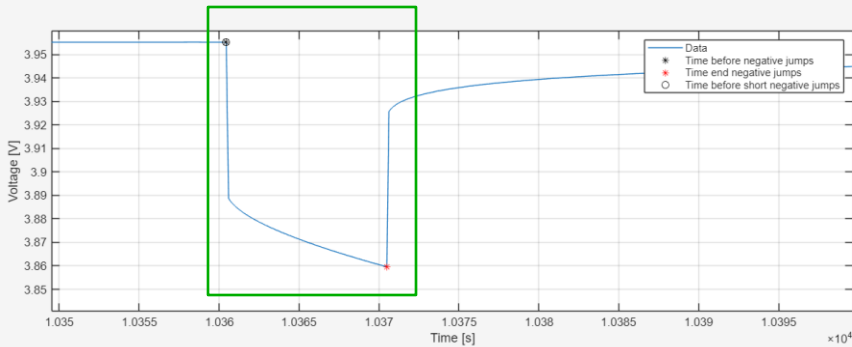
where:

$$v_1(t_0) = v_{meas}(t_0) - OCV(SOC(t_0)) - R_{int} \left(u + \frac{C_{bat,Ah}}{20} \right)$$

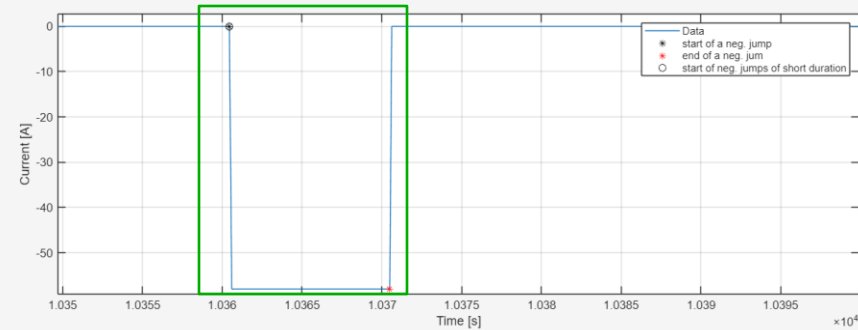
and:

$$v_1(t_f) = v_{meas}(t_f) - OCV(SOC(t_f)) - R_{int} \left(u + \frac{C_{bat,Ah}}{20} \right)$$

Lab 7: Estimating R_1 and C_1 once τ is known



HPPC data

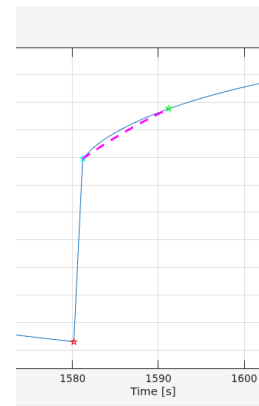
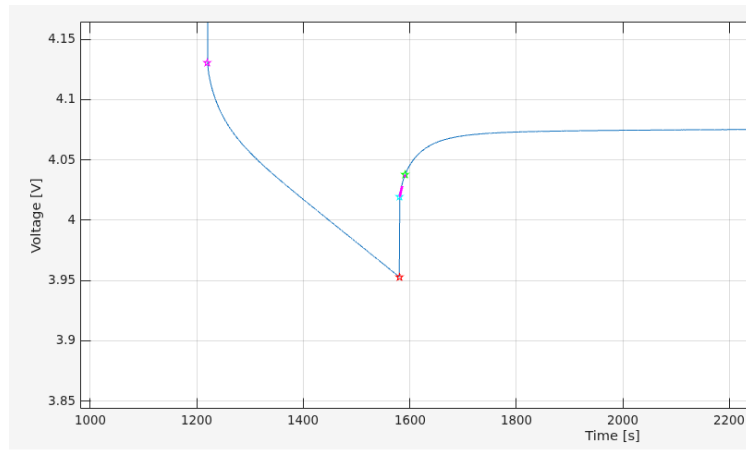


Once R_1 has been obtained, the capacitance follows directly from: $\tau = R_1 C_1$ therefore:

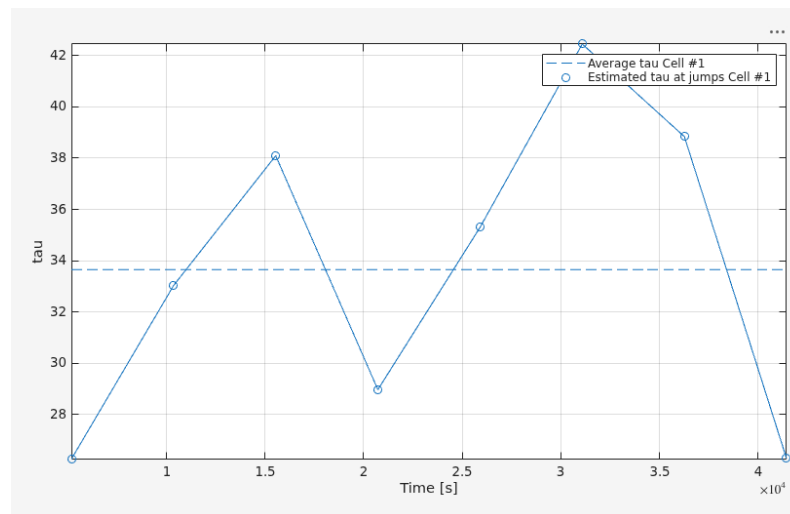
$$C_1 = \frac{\tau}{R_1}$$

Lab 7: results (contained in Lab7_in_class.m)

Computing τ from voltage relaxation (example of cell #1). A different τ is computed at the different jumps

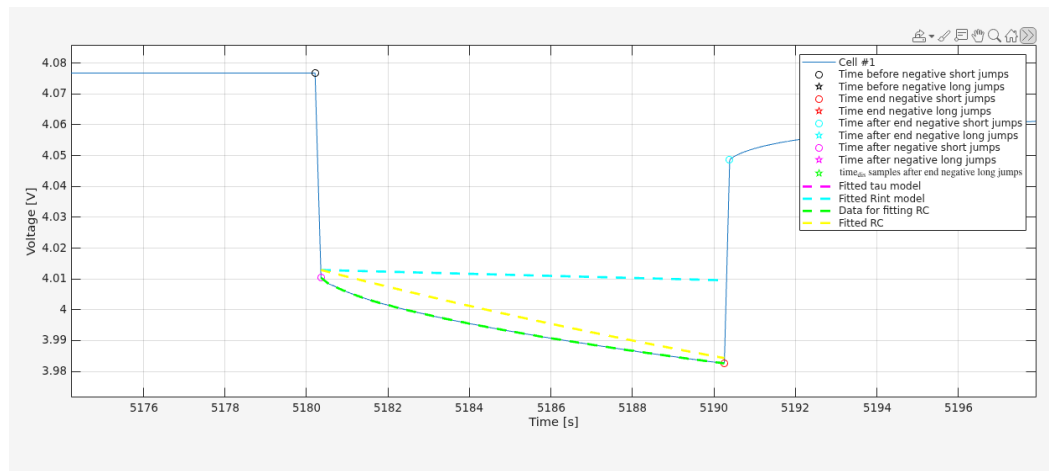


The model matches the extreme data points (cyan and green) but the data behavior is not perfectly reproduced! One RC parallel is maybe not enough?

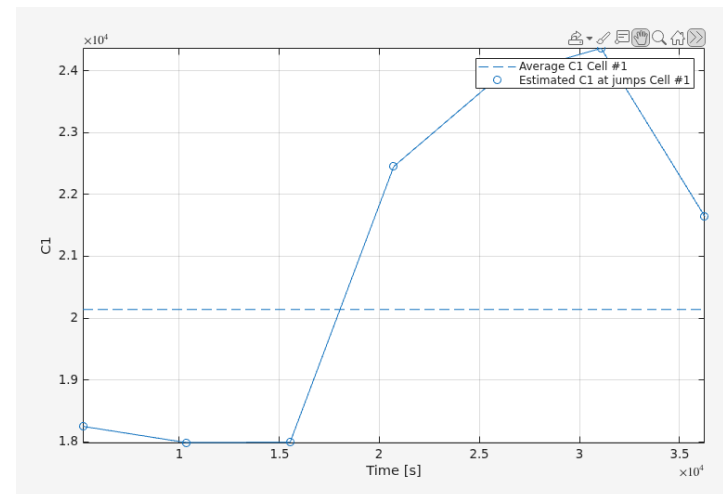
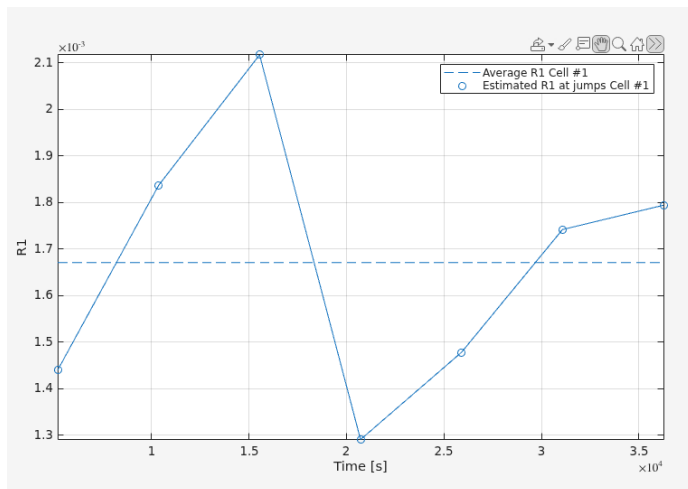


Lab 7: results (contained in Lab7_in_class.m)

Estimating R_1 and C_1 once τ is known from the non-zero current phase



The advantage of using an RC parallel vs Rint only is clear. However, also here, one can see that the RC model does not match the behaviour exactly.



Lab 7: results (contained in Lab7_in_class.m)

Different R_{int} , R_1 and C_1 are identified at different current steps. These values cannot be directly used in practice unless their dependence on SOC and/or temperature is modeled. A simpler option is to use **average values**, at the cost of **some loss in accuracy**.

Alternatively, the Matlab **System Identification Toolbox** can be used to determine the best average values of R_{int} , R_1 and C_1 by solving a least-squares problem that minimizes the mean square error between the HPPC data and the model output.

Let:

- $\theta = [R_{int} \quad R_1 \quad C_1]^\top$
- V_k^{meas} be the measured battery voltage from the HPPC test
- $V_k^{\text{mod}}(\theta)$ be the simulated voltage of the ECM at sample k , obtained with input current I_k

Then the toolbox is effectively trying to find the parameter vector that minimizes the discrepancy between measured and simulated voltage:

$$\theta^* = \arg \min_{\theta} \sum_{k=1}^N (V_k^{\text{meas}} - V_k^{\text{mod}}(\theta))^2, \quad \theta = [R_{int} \quad R_1 \quad C_1]^\top$$

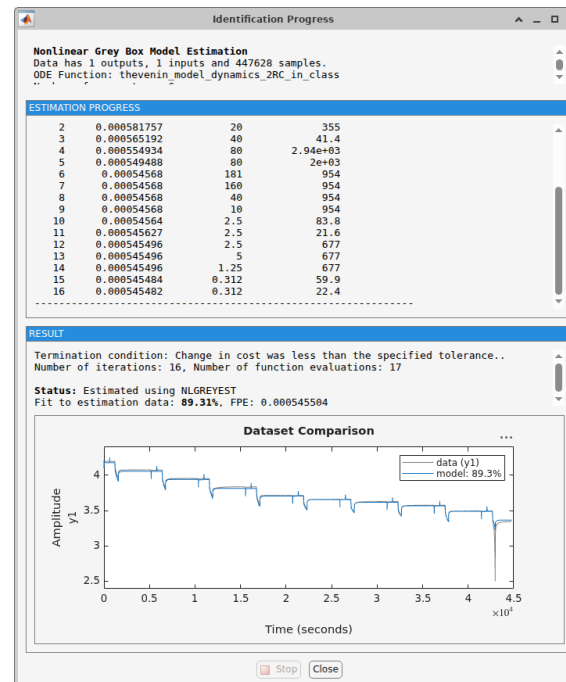
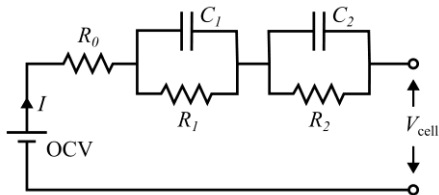
subject to the model dynamics above, and typically with positivity constraints such as $R_{int} > 0, R_1 > 0, C_1 > 0$.

Lab 7: results (contained in Lab7_in_class.m)

The Matlab System Identification Toolbox requires uniformly sampled data. Since HPPC data are not uniformly sampled, they were resampled onto a uniform time grid, while carefully preserving the discontinuities caused by the current steps.

This method allows easily to extend the identification to the case of 2RC parallel.

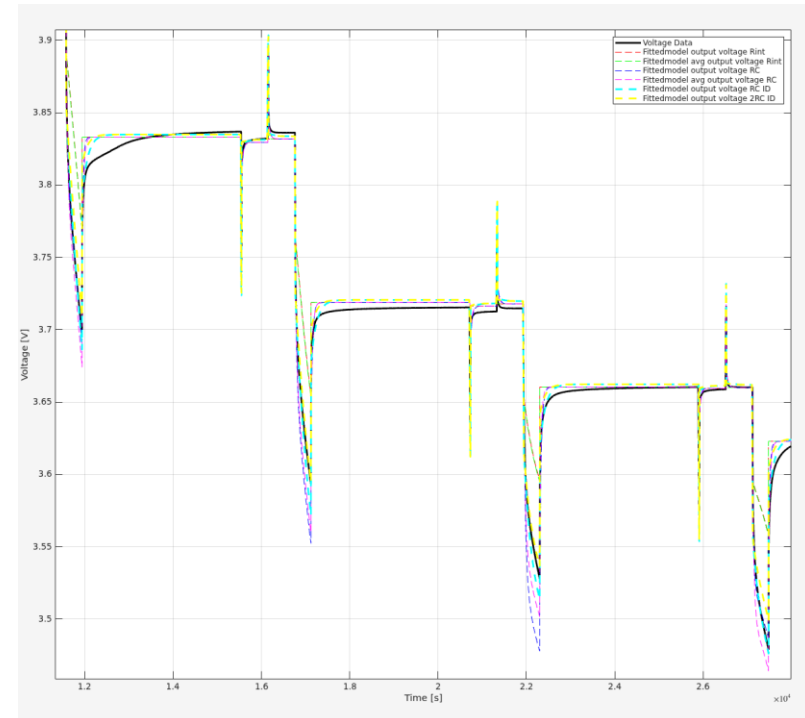
The resulting identification based on the HPPC data is shown below (example for cell #1) for the case of 2RC parallel.



Lab 7: results (contained in Lab7_in_class.m)

Finally, we compare the following models:

- R_{int}
- average R_{int}
- 1RC, manual identification
- average 1RC, manual identification
- average 1RC, System Identification Toolbox
- average 2RC, System Identification Toolbox



The average 2RC, Sys. Id. Tool. is the one that performs the best since it is calibrated exactly on the entire set of data (not only the jumps) and uses 2RC parallel. Clearly, the performance is still not perfect since we do not adopt a SOC-dependent set of parameters.

Note: what do we do for the time-varying case of R_{int} , R_1 and C_1 ? In this simulation we used the identified parameters at the closed jump in time $\rightarrow$ piecewise constant parameters.

Why state estimation is needed

Motivation

The adopted **ECM** is **only an approximation** of the real cell behavior.

In particular:

- the model has a simplified first-order structure,
- the internal resistance is represented by an average value,
- transient dynamics are not fully captured,
- current calibration errors may be present.

Even after current correction, a residual mismatch remains, especially during transients.

Therefore, a state observer is needed to combine:

- model prediction,
- voltage measurements.

Key idea

A filtering approach allows one to account for both model uncertainty and measurement uncertainty, and to properly **balance model-based prediction and experimental data**.

Why the Extended Kalman Filter

The standard Kalman filter applies to linear systems.

In battery ECMs, the main nonlinearity is typically due to the OCV–SOC relationship.

The Extended Kalman Filter (EKF) addresses this by locally linearizing the nonlinear model around the current estimate.

Main advantage

The EKF recursively combines:

- the model prediction,
- the measured terminal voltage,
- uncertainty information.

General EKF framework

Consider the nonlinear discrete-time system

$$x_{k+1} = f(x_k, u_k) + w_k$$

$$y_k = h(x_k, u_k) + v_k$$

where:

- x_k is the state vector,
- u_k is the input,
- y_k is the measured output,
- w_k is the process noise,
- v_k is the measurement noise.

Interpretation

w_k : model mismatch, parameter uncertainty, input uncertainty

v_k : sensor noise and output measurement uncertainty

Prediction and correction steps

Prediction

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1|k-1}, u_{k-1})$$
$$P_{k|k-1} = A_{k-1}P_{k-1|k-1}A_{k-1}^\top + Q_k$$

Correction

$$\hat{y}_k = h(\hat{x}_{k|k-1}, u_k)$$
$$r_k = y_k - \hat{y}_k$$
$$K_k = P_{k|k-1}C_k^\top (C_kP_{k|k-1}C_k^\top + R_k)^{-1}$$
$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k r_k$$
$$P_{k|k} = (I - K_k C_k)P_{k|k-1}$$

with

$$A_k = \left. \frac{\partial f}{\partial x} \right|_{\hat{x}_{k|k}, u_k}, \quad C_k = \left. \frac{\partial h}{\partial x} \right|_{\hat{x}_{k|k-1}, u_k}.$$

Interpretation of Q and R

Covariance matrices

Q_k : process noise covariance

R_k : measurement noise covariance

In this application

Q_k accounts for:

- simplified ECM structure,
- average parameter values,
- unmodeled transients,
- current uncertainty affecting the state evolution.

R_k accounts for:

- voltage sensor noise,
- possible uncertainty in the output prediction.

Key point

The EKF weights model prediction and voltage measurements according to their reliability.

Input uncertainty in cell model

In the ECM, the current I_k appears in both:

- the state equation,
- the output equation.

Current uncertainty has a double effect:

- it perturbs the state evolution,
- it also perturbs the predicted terminal voltage directly.

Practical treatment

In a basic EKF implementation, this effect is typically handled through suitable tuning of Q_k and R_k .

In a more refined formulation, current uncertainty can be modeled explicitly.

Battery ECM with one RC branch

State, input, and output

$$x_k = \begin{bmatrix} z_k \\ v_{RC,k} \end{bmatrix}, \quad u_k = I_k, \quad y_k = V_{t,k}$$

where:

- z_k : SOC,
- $v_{RC,k}$: RC branch voltage,
- $I_k < 0$ during discharge.

Continuous-time model

$$\dot{z} = \frac{I}{Q}$$

$$\dot{v}_{RC} = -\frac{1}{R_1 C_1} v_{RC} + \frac{1}{C_1} I$$

$$V_t = OCV(z) + v_{RC} + R_0 I$$

Discrete-time ECM model

Assuming constant current over one sampling interval T_s

$$z_{k+1} = z_k + \frac{T_s}{Q} I_k$$

$$v_{RC,k+1} = \alpha v_{RC,k} + R_1(1 - \alpha)I_k$$

with:

$$\alpha = e^{-T_s/(R_1 C_1)}$$

and

$$V_{t,k} = OCV(z_k) + v_{RC,k} + R_0 I_k$$

The model can be written as

$$\begin{aligned} x_{k+1} &= f(x_k, u_k) + w_k \\ y_k &= h(x_k, u_k) + v_k \end{aligned} \quad \Rightarrow \quad \begin{aligned} f(x_k, u_k) &= \begin{bmatrix} z_k + \frac{T_s}{Q} I_k \\ \alpha v_{RC,k} + R_1(1 - \alpha)I_k \end{bmatrix} \\ h(x_k, u_k) &= OCV(z_k) + v_{RC,k} + R_0 I_k \end{aligned}$$

- the state dynamics are affine
- the nonlinearity is introduced by the term $OCV(z_k)$.

Jacobian matrices for the EKF

State Jacobian

$$A_k = \frac{\partial f}{\partial x} = \begin{bmatrix} 1 & 0 \\ 0 & \alpha \end{bmatrix}$$

Output Jacobian

$$C_k = \frac{\partial h}{\partial x} = \left[\frac{d OCV}{dz}(\hat{z}_{k|k-1}) \quad 1 \right]$$

Why the OCV–SOC fit matters

The OCV–SOC map is required in the output equation.

Its derivative is required in the EKF linearization:

$$\frac{d OCV}{dz}$$

An analytical OCV–SOC expression is therefore particularly convenient.

Observability insight

The information on SOC carried by the voltage measurement depends on the local slope of the OCV–SOC curve:

$$\frac{d \text{OCV}}{dz}$$

Implication

- if the curve is steep, SOC is easier to estimate from voltage;
- if the curve is flat, SOC estimation becomes more difficult.

The local slope of the OCV–SOC characteristic strongly affects SOC observability.

Why EKF is useful in this case

Main reasons

- The ECM alone does not perfectly reproduce the cell behavior.
- The model is affected by parameter uncertainty and unmodeled dynamics.
- Current uncertainty may affect both state propagation and output prediction.
- The EKF provides a principled way to combine:
 - model prediction,
 - voltage measurements,
 - uncertainty information.

The EKF is well suited to SOC estimation for ECM-based battery models with nonlinear OCV–SOC characteristics.

Lab 8 Observer setup – case 1

- The model was discretized with a sampling time $T_s=0.1s$
- To assess convergence, the observer is initialized with an intentionally wrong initial condition:

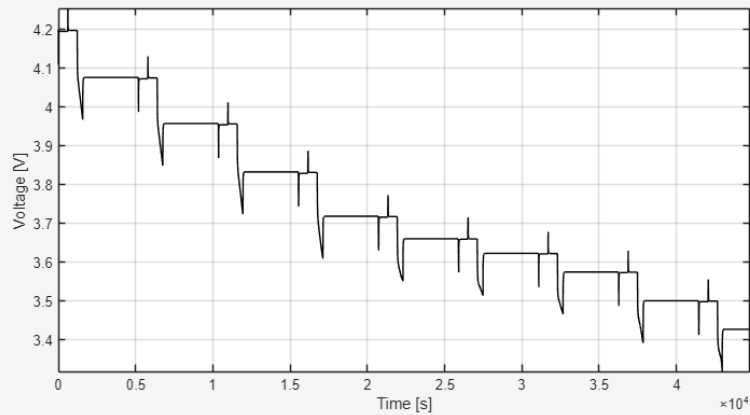
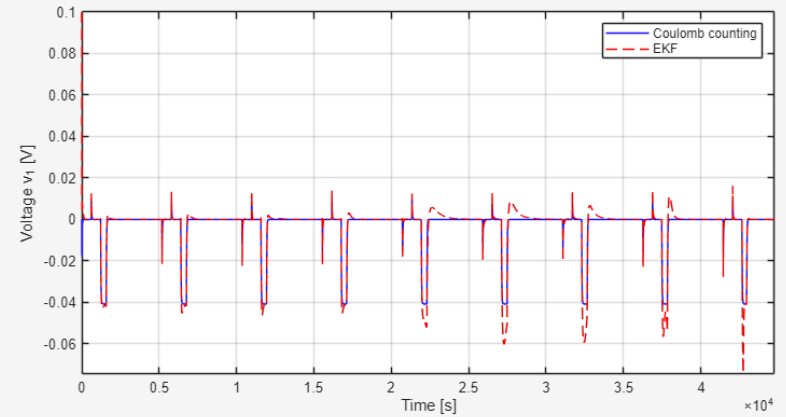
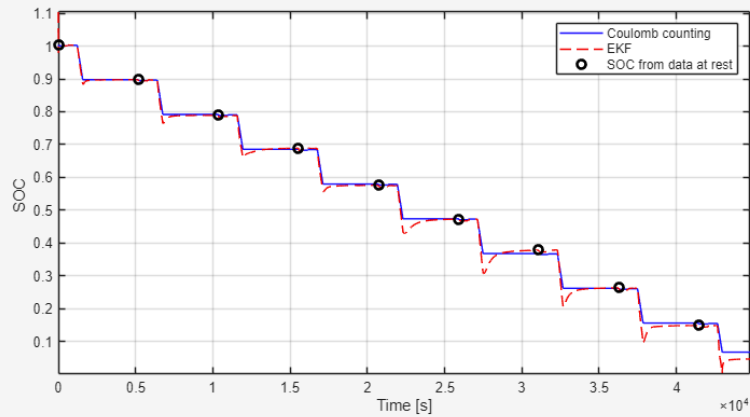
$$x_{\text{obs}}(0) = \begin{bmatrix} 1.1 \text{ SOC}_{\text{init}} \\ 0.1 \end{bmatrix}$$

- If the battery is at rest, SOC_{init} can be estimated fairly accurately from the measured voltage by inverting the **OCV–SOC curve** (e.g. using f_{zero})
- In ideal rest conditions, one would also expect $v_{RC}(0) = 0$
- Here, both **SOC** and v_{RC} are intentionally perturbed: the SOC estimate may be inaccurate if the battery is not exactly at rest, and $v_{RC}(0)$ may be nonzero due to incomplete relaxation

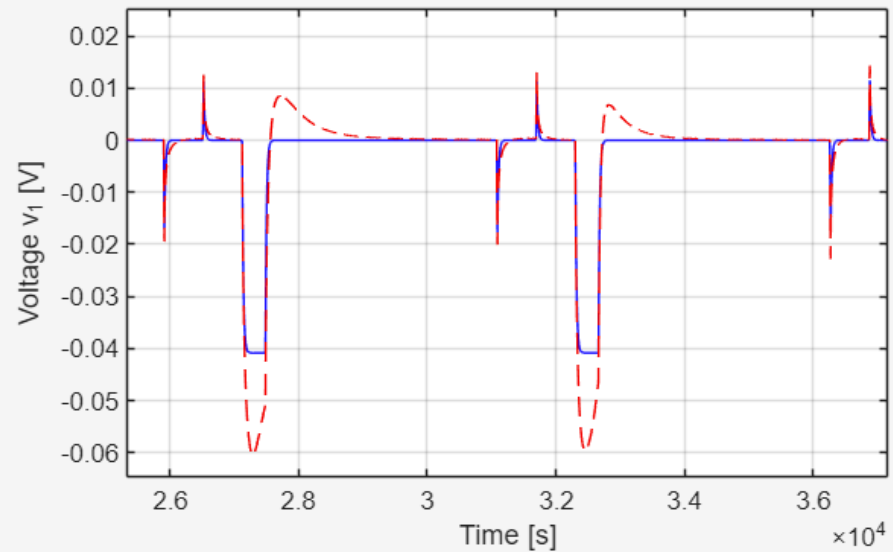
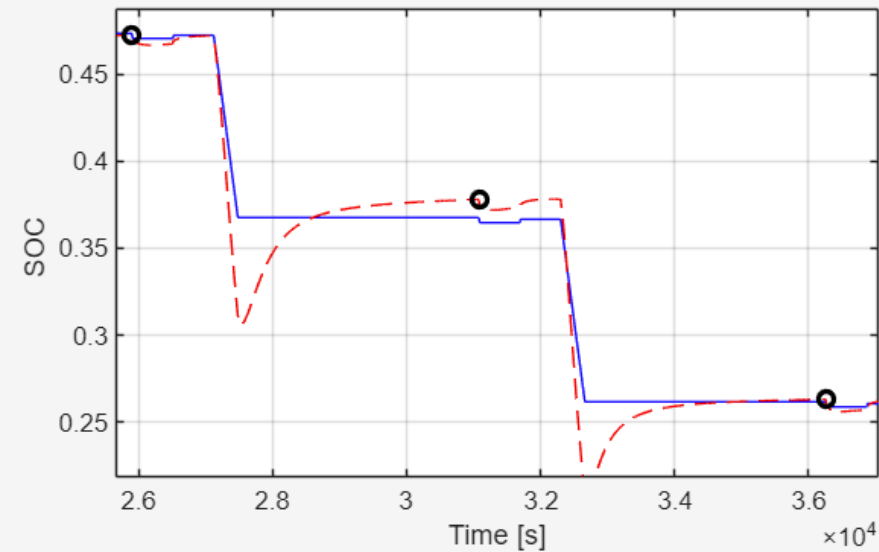
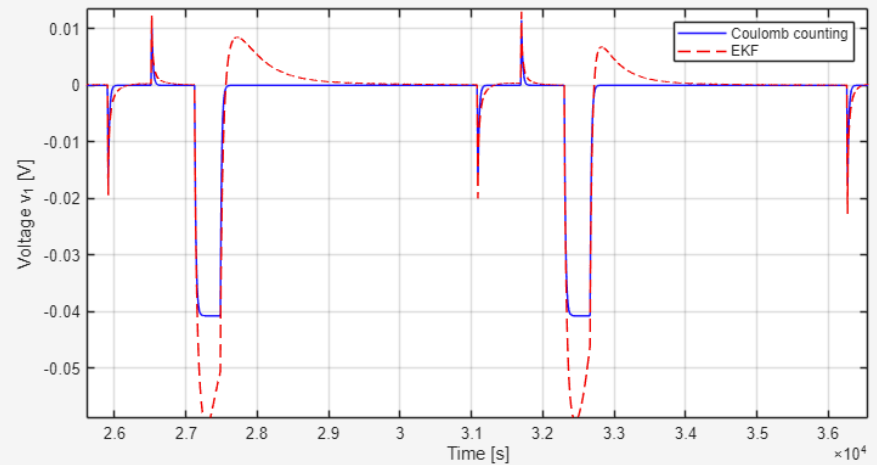
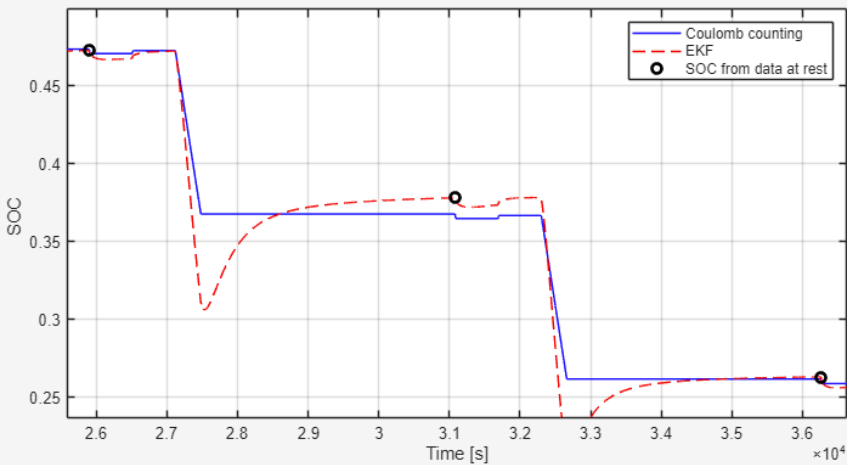
Covariance tuning

- Initial covariance: $P_0 = \text{diag}([0.1^2, 0.1^2])$
- Process noise covariance: $Q_{\text{obs}} = \text{diag}([0.01^2, 0.1^2])$
- A smaller variance is assigned to the SOC state, assuming its dynamics are more reliable than those of the RC voltage
- Measurement noise covariance: $R_{\text{obs}} = 10^{-6}$
- The terminal-voltage measurement is assumed to be highly accurate

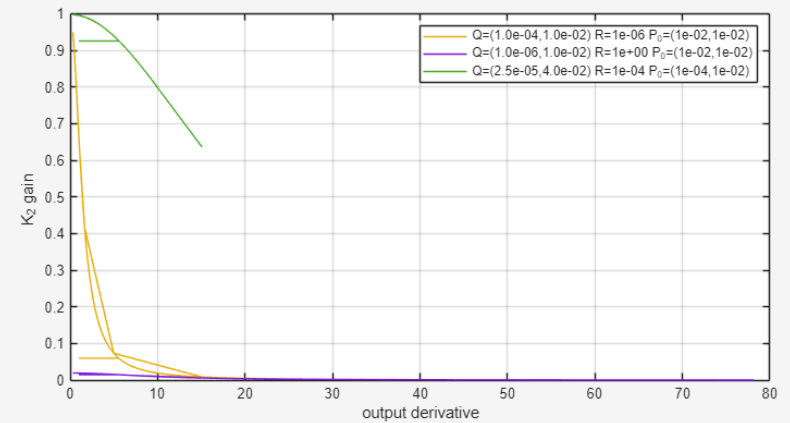
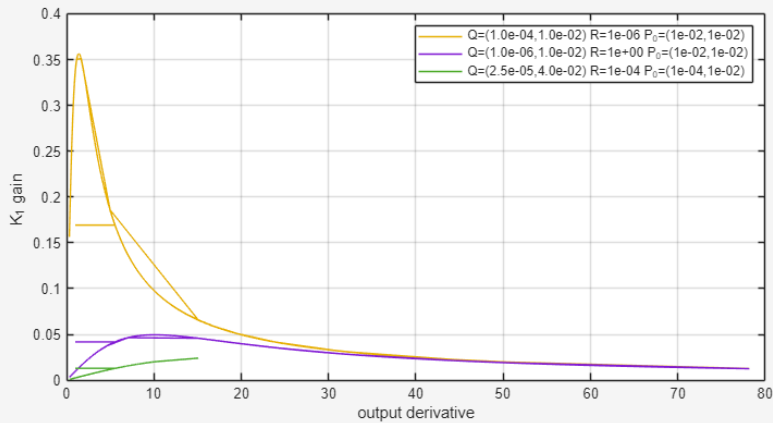
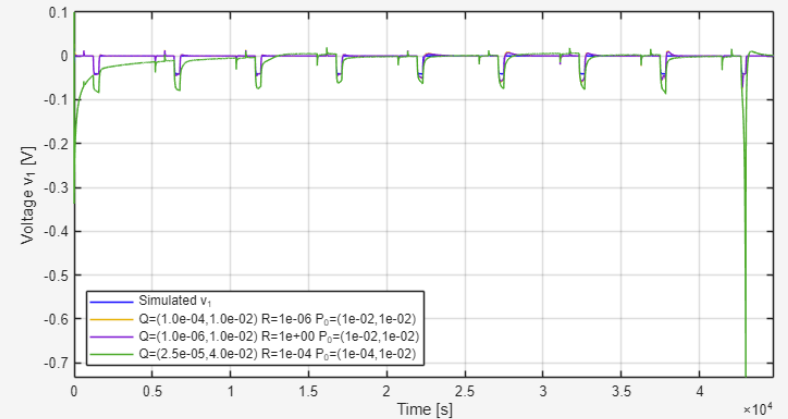
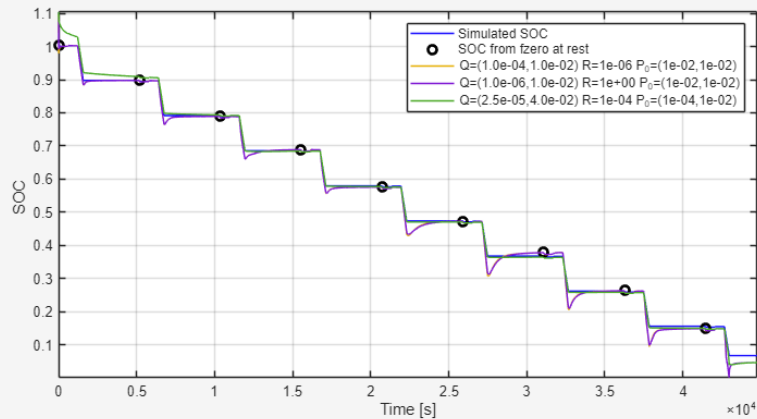
Lab 8 Observer results – case 1



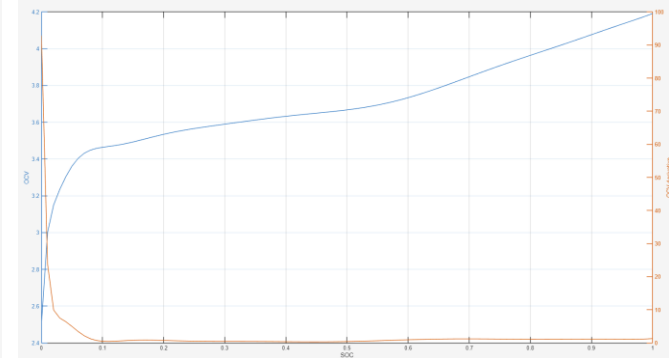
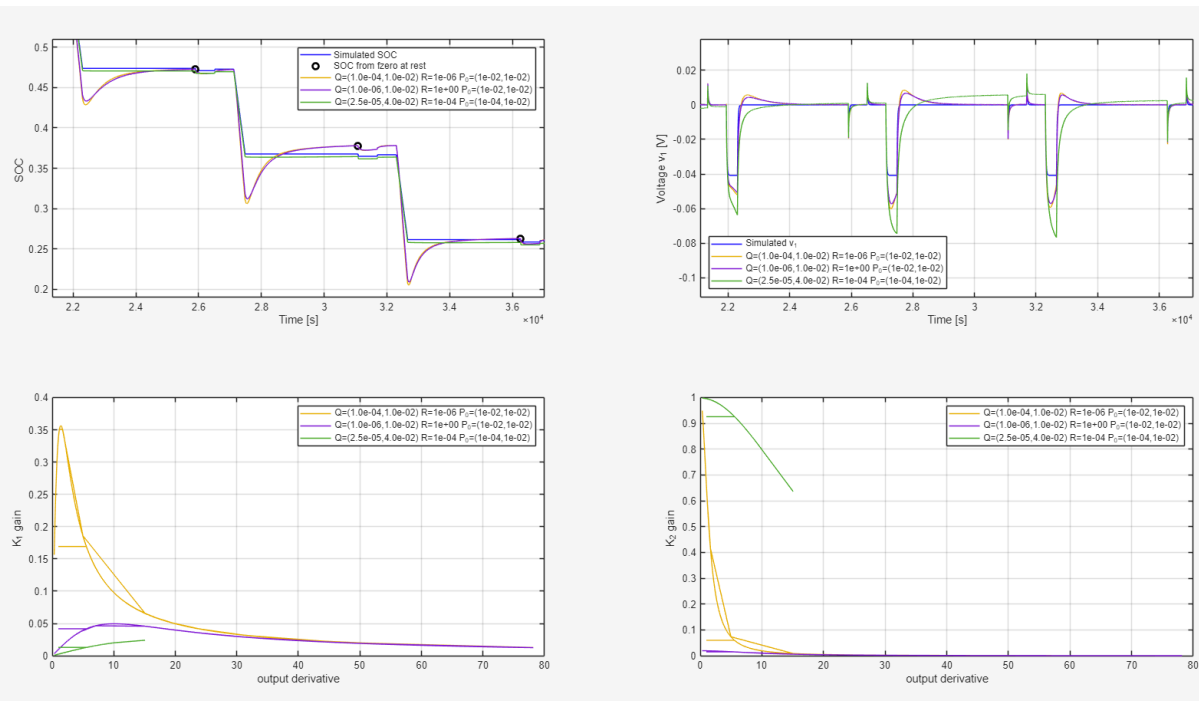
Lab 8 Observer results – case 1



Lab 8 Observer results – different tuning



Lab 8 Observer results – different tuning



Why is SOC accuracy so important?

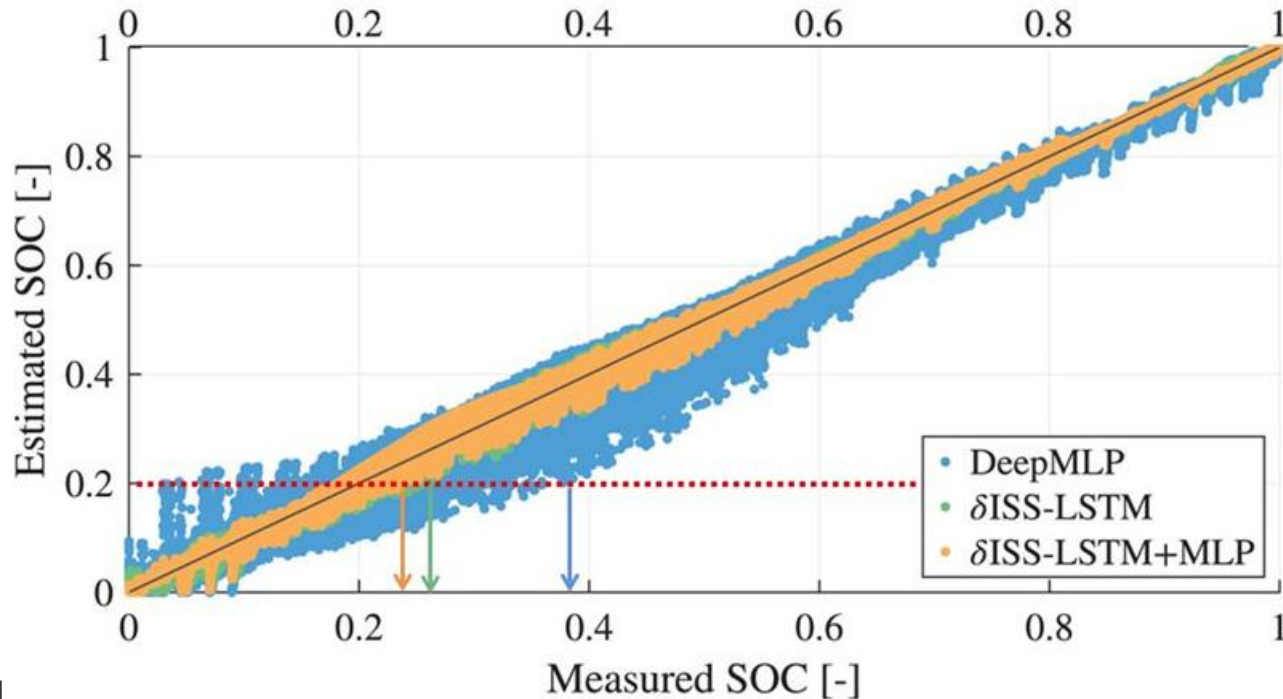


Figure 23 Comparison of steady-state SOC estimation performance among DeepMLP (blue), δ ISS-LSTM (green), and δ ISS-LSTM+MLP (orange) models for the testing dataset, and worst-case analysis in case of SOC underestimation.

Why is SOC accuracy so important?

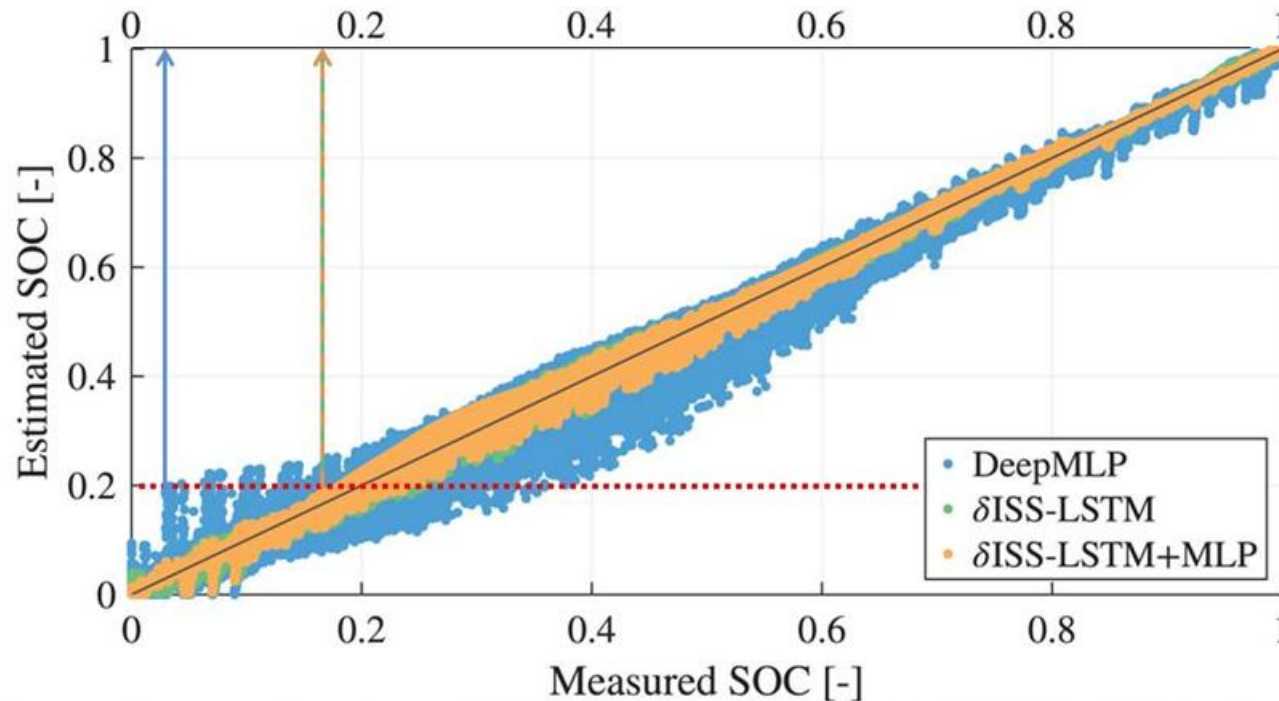


Figure 24 Comparison of steady-state SOC estimation performance among DeepMLP (blue), δ ISS-LSTM (green), and δ ISS-LSTM+MLP (orange) models for the testing dataset, and worst-case analysis in case of SOC overestimation.

EIS-based ECM parameter estimation

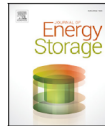
Journal of Energy Storage 82 (2024) 110389



Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est



Research papers

A guide to equivalent circuit fitting for impedance analysis and battery state estimation

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ARTICLE INFO

Dataset link: LiPo Battery LP-503562-IS-3 EIS, C capacity, ECM Data (Original data)

Keywords:

Lithium ion battery (Li-ion)
Lithium polymer battery (LiPo)
Electrochemical impedance spectroscopy (EIS)
Impedance analysis
Equivalent circuit model
Circuit parameter fitting
Analysis of uncertainty
Sensitivity analysis
State-of-charge (SOC) estimation
State-of-health (SOH) estimation
Machine learning
Gaussian process regression

ABSTRACT

In this study we define a comprehensive method for analyzing electrochemical impedance spectra of lithium batteries using equivalent circuit models, and for information extraction on state-of-charge and state-of-health from impedance data by means of machine learning methods. Estimation of circuit parameters typically implies a non-linear optimization problem. A detailed method for estimating initial values of the optimization algorithm is described, emphasizing short computation times and efficient convergence to global minimum. Parameters identifiability is investigated through an analysis of the injectivity of the model, Cramer-Rao lower bound, profile likelihood, and sensitivity analysis. An exploratory data analysis is presented to estimate the degree of correlation between impedance spectra (or circuit parameters) and battery state-of-charge or state-of-health, prior to the implementation of any machine learning algorithm. A publicly available dataset of impedance spectra of five lithium-polymer batteries is used to test the whole procedure. Estimation of state-of-charge and state-of-health is performed by implementing Gaussian process regression.

Journal of Energy Storage 14 (2017) 39–48



Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est



Parameterization of linear equivalent circuit models over wide temperature and SOC spans for automotive lithium-ion cells using electrochemical impedance spectroscopy

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ARTICLE INFO

Article history:

Received 6 October 2016
Received in revised form 29 August 2017
Accepted 29 August 2017
Available online 22 September 2017

Keywords:

Battery impedance
Electric impedance spectroscopy
Automotive lithium ion battery
Equivalent circuit model
Charge transfer

ABSTRACT

Three different types of automotive lithium-ion pouch cells are analyzed with electrochemical impedance spectroscopy over a wide range of operating points in SOC (10–90%) and temperature (–10 to +40°C) with the goal of establishing parameters for an accurate linear equivalent circuit model. The impedance vs frequency is analyzed and attributed to physical behaviors from current collectors, electrode and electrolyte, charge transfer and diffusion. The change in resistance is statistically analyzed as a function of SOC and temperature in an effort to quantify if there are any monotonic patterns. Results show that the charge transfer resistance versus temperature represent the largest relative change of resistance, followed by diffusion resistance versus temperature and the electrolyte resistance over temperature. Only weak correlations with resistance versus SOC are found regardless of frequency range. Frequency-domain fitted equivalent circuit model parameters are evaluated by comparison of physical measurements in long dynamic load cycles using root-mean-squared of the voltage error between model and physical measurements as performance index. A simple R+2RC model capturing diffusion dynamics performs best at RMSE 15.46 mV, better than most similar studies and better than the more complex R+6RC also evaluated.

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Learning-based SOH estimation

Journal of Power Sources 653 (2025) 237568



Contents lists available at ScienceDirect

Journal of Power Sources

journal homepage: www.elsevier.com/locate/jpowsour



TinyML models for SoH estimation of lithium-ion batteries based on Electrochemical Impedance Spectroscopy^{☆,☆☆}

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HIGHLIGHTS

- Developed TinyML models for battery SoH estimation from EIS data.
- Compared ECM-based and raw EIS data-driven approaches for model development.
- Optimized models converted to TinyML for real-time edge device deployment.
- CNN-GRU model achieved highest performance by using raw EIS data.

ARTICLE INFO

Keywords:

Electrochemical impedance spectroscopy
State of health
Deep learning models
Batteries
TinyML
Equivalent Circuit Model (ECM)

ABSTRACT

As battery systems become more widespread, the need for accurate and fast estimation of State of Health (SoH) in lithium-ion cells is increasingly critical. This study analyzes two data-driven model methods that leverage Electrochemical Impedance Spectroscopy (EIS) measurements to capture the internal electrochemical dynamics underlying battery degradation and estimate the cell's current SoH. The first approach, Method A, utilizes an Equivalent Circuit Model (ECM) from EIS data and uses it to train various state-of-the-art Deep Learning (DL) models, including LSTM, GRU, CNN-LSTM, and CNN-GRU. In contrast, Method B directly employs raw EIS data to train the same set of DL models, bypassing the need for ECM development. Both methods demonstrated strong performance, with the CNN-GRU model from Method B yielding the best results, achieving a Mean Absolute Error (MAE) of only 0.87% and a Root Mean Square Error (RMSE) of 1.20%. Additionally, both methods included an analysis of various input features, such as State of Charge (SoC), to evaluate their impact on model performance. Finally, the models of Method B were optimized for size and computational efficiency, making them suitable for deployment on low-power edge devices and applications requiring TinyML capabilities. The average latency and size reduction of the models were 99.61% and 88.61%, respectively.

Only high level
discussion

Optimal charging

IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, VOL. 66, NO. 9, SEPTEMBER 2017

7761

Optimal Charging of Li-Ion Batteries With Coupled Electro-Thermal-Aging Dynamics

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Satadru Dey, *Member, IEEE*, and Scott J. Moura, *Member, IEEE*

Abstract—Fast and safe charging protocols are crucial for enhancing the practicality of batteries, especially for mobile applications, such as smartphones and electric vehicles. This paper proposes an innovative approach to devising optimally health-conscious fast-safe charge protocols. A multiobjective optimal control problem is mathematically formulated via a coupled electro-thermal-aging battery model, where electrical and aging submodels depend upon the core temperature captured by a two-state thermal submodel. The Legendre–Gauss–Radau pseudospectral method with adaptive multi-mesh-interval collocation is employed to solve the resulting highly nonlinear six-state optimal control problem. Charge time and health degradation are, therefore, optimally traded off, subject to both electrical and thermal constraints. Minimum-time, minimum-aging, and balanced charge scenarios are examined in detail. Sensitivities to the upper voltage bound, ambient temperature, and cooling convection resistance are investigated as well. Experimental results are provided to compare the tradeoffs between a balanced and traditional charge protocol.

Index Terms—Battery management, battery aging, charge control optimization, electric vehicles, experimental validation.

tradeoffs between charging time and health degradation, which is the focus of this paper.

The traditional charging protocol for Li-ion batteries is constant-current/constant-voltage (CCCV) [2]. In the CC stage, the charging current is constant until a pre-specified voltage threshold is reached; in the CV stage the voltage threshold is maintained until the current relaxes below a pre-specified threshold value. This technique is simple and easily implemented. The current rate and voltage threshold are, however, almost universally selected in an ad-hoc manner.

Various methods were proposed to reduce charge times. Examples include multi-stage CC (high CC followed by low CC) plus CV (MCC-CV) [3], fuzzy logic [4], [5], neural networks [6], grey system theory [7], and an ant colony system algorithm [8]. Alternative protocols were reported to prolong the battery lifetime as well, such as MCC-CV (low CC followed by high CC plus CV) [9] and CCCV with negative pulse (CCCV-NP) [10]. These protocols are almost always heuristic. That is, they

Balancing

Review

A Comprehensive Review of the Art of Cell Balancing Techniques and Trade-Offs in Battery Management Systems

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Abstract

The battery pack is a critical component of electric vehicles, with lithium-ion cells being a frequently preferred choice. Lithium-ion cells are known for long life, high power and energy density, and are reliable for a broad range of temperatures. However, these batteries have a drawback of over-voltage, under-voltage, thermal runaway, and especially, state of charge or voltage imbalance. Among these, the cell imbalance is particularly important because it causes an uneven power dissipation in each cell, resulting in non-uniform temperature distribution. This uneven temperature distribution negatively affects the lifetime and efficiency of a battery pack. Cell imbalance is mitigated by cell balancing techniques, of which several methods have been presented over the last few years. These methods consider different power electronics circuits and control approaches to optimise cell balancing characteristics. This paper reviews basic to advanced cell balancing techniques and compares their circuit designs, costs, switching stresses, complexity, sizes, and control techniques to highlight the recent trends and future directions. This paper also compares the recent trend of machine learning integration with basic cell balancing topologies and provides a critical analysis of the outcomes.

Keywords: electric vehicles; cell balancing; active/passive cell balancing; battery management systems; state of charge

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Academic Editor: Giovanni Lattimerger
Received: 19 May 2025
Revised: 18 June 2025

Battery Balancing Methods: A Comprehensive Review

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Abstract—The trend toward more electric vehicles has demanded the need for high voltage, high efficiency and long life battery systems [1], [2]. A complete battery system consists of the following parts: protection, management and balancing. Of the three parts, balancing is the most important concerning the life of the battery system because without the balancing system, the individual cell voltages will drift apart over time. The capacity of the total pack will also decrease more quickly during operation which will result in the fail of the total battery system. This condition is especially severe when the battery has a long string of cells (high voltage battery systems) and frequent regenerative braking (charging) is done via the battery pack [3]–[5]. This paper presents the theory behind the proposed balancing methods for battery systems within the past twenty years. Comparison between the methods is carried out and different balancing methods are grouped by their nature of balancing.

The basic idea of active balancing is to use external circuits to actively transport energy among cells so as to balance the cells. The active balancing method can be used for most modern battery systems because they do not rely on the characteristic of cells for balancing. This method is the only applicable balancing method for Lithium based batteries since the temperature of the Lithium based batteries must be rigorously controlled in the safety operation range. Generally, active balancing method should be used for Lithium Iron battery pack which has three cells and up in series connected.

This paper will focus on the active balancing methods. Many active balancing methods are available. Sorted by energy flow, active balancing methods can be grouped into four big categories. They are dissipative method, single cell to pack method, pack to single cell method and single cell to single cell method.

Balancing-Aware Charging Strategy For Series-Connected Lithium-Ion Cells: A Nonlinear Model Predictive Control Approach

Andrea Pozzi, Massimo Zambelli *Student, IEEE*, Antonella Ferrara *Senior Member, IEEE*, and Davide M. Raimondo

Abstract—Charge unbalance is one of the key issues for series-connected Lithium-ion cells. Within this context, model-based optimization strategies have proven to be the most effective. In the present paper, an ad-hoc electrochemical model, tailored to control purposes, is firstly presented. Relying on this latter, a general nonlinear MPC for balancing-aware optimal charging is then proposed. In view of the possibility of a practical implementation, the concepts are subsequently specialized for an easily implementable power supply scheme. Finally, the nonlinear MPC approach is validated on commercial cells using a detailed battery simulator, with sound evidence of its effectiveness.

high temperature spikes, which also affect the self-discharge rates. Moreover, during operation, temperature gradients arise between the different cells and even inside of each of them. All of these facts, in turn, contribute to an increase in the differences among the cells characteristics, which eventually lead to greater unbalance.

In order to overcome such issues, a huge research effort has been made towards the development of algorithms implementing the idea of State of Charge (SOC) equalization. Two

Only high level
discussion

Frequency-domain parameter ID

Comparison of equivalent circuit model parameterization using GITT and EIS

Marek Sedlářík¹ · Petr Vyrůbal¹ · Jakub Lysák² · Martin Mačák³ · Martin Šedina¹ · Tomáš Kazda¹

Received: 19 October 2025 / Accepted: 13 January 2026
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Abstract

Accurate parameterization of equivalent circuit models (ECMs) is essential for reliable state estimation and dynamic simulation of lithium-ion (Li-ion) batteries. These models provide a representation of the electrochemical processes governing cell behavior and are widely applied across various fields, including electric mobility, stationary energy storage, avionics, and industrial applications. This work investigates two commonly employed approaches for ECM parameter identification: galvanostatic intermittent titration technique (GITT) and electrochemical impedance spectroscopy (EIS). Both methods were applied to Li-ion cells under controlled laboratory conditions and the parameters of a 3RC model were extracted and analyzed. This study compares time-domain and frequency-domain parameterization, addressing their methodological differences, measurement requirements, and resulting model characteristics. The identified models were validated using real racing data from the TU Brno Racing Formula Student electric vehicle. Both approaches achieved very good agreement between simulated and measured voltage responses, with root mean square errors of 9.86 mV for the EIS-based model and 9.61 mV for the GITT-based model. While the difference in accuracy is minor, the GITT-based model showed slightly better performance under highly dynamic operating conditions. The results indicate that the GITT-based approach can achieve reliable 3RC ECM parameterization without the need for specialized laboratory equipment, while providing voltage response accuracy comparable to that obtained using EIS, with a marginal improvement observed in this study.

Fault detection



Contents lists available at [ScienceDirect](#)

Journal of Power Sources

journal homepage: www.elsevier.com/locate/jpowsour



Internal short circuit detection for battery pack using equivalent parameter and consistency method



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HIGHLIGHTS

- Theoretical foundation and practical method of ISCr detection are introduced.
- The influence of ISCr on cells' OCV and internal resistance has been discussed.
- The effectiveness of the ISCr detection method is validated through experiments.
- The specific time consumed to detect ISCr is compared.

ARTICLE INFO

Article history:

Received 17 March 2015

Received in revised form

12 June 2015

Accepted 15 June 2015

Available online 21 June 2015

Keywords:

Internal short circuit

Detection

Characteristic parameter

Battery safety

Battery management system

Mean difference model

ABSTRACT

Internal short circuit (ISCr) detection of a battery is critical for preventing thermal runaway and enhancing electrical vehicle safety. In this paper, the electrical characteristics of the ISCr of a large format lithium ion battery are analyzed using the equivalent circuit model (ECM). An ISCr detection method is developed based on battery consistency within the battery pack. The ISCr detection method employs the recursive least square (RLS) algorithm based on the mean-difference model (MDM), which is derived from the ECMs consisting of the mean and difference value of cells' voltage and resistance. The algorithm first estimates the basic parameters of the MDM. Then the algorithm calculates the characteristic parameters, such as the differential of the voltage and the fluctuation function of the internal resistance, derived from the basic parameters of the MDM. These characteristic parameters obviously vary once ISCr occurs, thereby helping the battery management system to realize ISCr detection. The effectiveness of the ISCr detection method is validated through a series of experiments.

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Model Used for RLS Estimation

In the reference paper, the cell difference model is written as

$$\Delta U_i(k) = \Delta E_i(k) - I(k)\Delta R_i(k)$$

where

$$\Delta U_i(k) = U_i(k) - U_{\text{mean}}(k)$$

is the voltage deviation of cell i from the pack average.

This equation can be written in linear regression form as

$$z(k) = \theta_i(k)^T \phi(k)$$

with

$$z(k) = \Delta U_i(k),$$
$$\theta_i(k) = \begin{bmatrix} \Delta E_i(k) \\ \Delta R_i(k) \end{bmatrix},$$

and

$$\phi(k) = \begin{bmatrix} 1 \\ -I(k) \end{bmatrix}.$$

The objective of the RLS algorithm is to estimate the parameter vector

$$\theta_i(k) = \begin{bmatrix} \Delta E_i(k) \\ \Delta R_i(k) \end{bmatrix}$$

from the measured sequence of $\Delta U_i(k)$ and $I(k)$. Here, $\Delta E_i(k)$ represents the equivalent OCV deviation of cell i from the pack average, while $\Delta R_i(k)$ represents the equivalent internal-resistance deviation.

Under normal operating conditions, i.e., in the absence of faults, ΔE_i and ΔR_i are expected to vary slowly, mainly because of SOC drift, temperature changes, and aging. Therefore, RLS treats them as **slowly time-varying parameters**.

However, if an internal short circuit occurs, these quantities may change more rapidly because the faulty cell starts deviating from the pack average. The forgetting factor determines how quickly the estimator can track such changes.

At each time k , the estimate is obtained by solving, in recursive form, an exponentially weighted least-squares problem:

$$\hat{\theta}_i(k) = \arg \min_{\theta} \sum_{j=1}^k \lambda^{k-j} [z(j) - \theta^T \phi(j)]^2.$$

This formulation should be interpreted as a **local estimation problem**: although the true quantities $\Delta E_i(k)$ and $\Delta R_i(k)$ may change with time, the least-squares criterion estimates one parameter vector θ that best explains the recent data, with older samples progressively discounted.

The forgetting factor satisfies

$$0 < \lambda \leq 1.$$

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In the reference paper,

$$\lambda = 0.995.$$

Since $\lambda < 1$, older samples are progressively discounted:

$$\lambda^{k-j} \rightarrow 0 \quad \text{as} \quad k - j \rightarrow \infty.$$

Therefore, the estimate at time k is influenced mainly by recent data, while older data are gradually forgotten. This allows the algorithm to track slow variations of

$$\Delta E_i(k) \quad \text{and} \quad \Delta R_i(k),$$

rather than estimating a single constant value over the entire experiment.

Since

$$\lambda = 0.995$$

is close to one, the estimates are relatively smooth and robust to measurement noise. However, this also means that older data retain some influence for a relatively long time, which may delay the detection of abrupt changes associated with the onset or worsening of an internal short circuit.

Thus, the forgetting factor introduces a trade-off:

$\lambda \approx 1 \Rightarrow$ smoother estimates, but slower fault detection

λ smaller $\Rightarrow$ faster adaptation, but higher sensitivity to noise.

Since the model is **linear in the parameters**, the exponentially weighted least-squares cost is quadratic in θ :

$$J_k(\theta) = \sum_{j=1}^k \lambda^{k-j} [z(j) - \phi(j)^T \theta]^2.$$

Expanding the cost function gives

$$J_k(\theta) = \sum_{j=1}^k \lambda^{k-j} z(j)^2 - 2\theta^T \sum_{j=1}^k \lambda^{k-j} \phi(j) z(j) + \theta^T \left[\sum_{j=1}^k \lambda^{k-j} \phi(j) \phi(j)^T \right] \theta.$$

Therefore, the matrix multiplying the quadratic term in θ is

$$S(k) = \sum_{j=1}^k \lambda^{k-j} \phi(j) \phi(j)^T.$$

This is the **information matrix**.

Because the model is linear in θ , $S(k)$ depends only on the regressors $\phi(j)$. It describes the excitation structure of the data, i.e., whether the current profile contains enough information to distinguish ΔE_i from ΔR_i . The measured outputs $z(j)$ affect the numerical parameter estimate, but not the information matrix itself.

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Batch Weighted Least-Squares Estimate

The measured output $z(k)$ does not appear in the information matrix $S(k)$, but it appears in the linear term of the least-squares problem. To see this, set the gradient of the cost function with respect to θ equal to zero:

$$\frac{\partial J_k(\theta)}{\partial \theta} = 0.$$

This gives the normal equations

$$S(k)\hat{\theta}(k) = q(k).$$

At this stage, $\hat{\theta}(k)$ is still the **batch** weighted least-squares estimate, i.e., the parameter vector obtained by processing all data up to the current sample k together.

Here,

$$S(k) = \sum_{j=1}^k \lambda^{k-j} \phi(j) \phi(j)^T$$

and

$$q(k) = \sum_{j=1}^k \lambda^{k-j} \phi(j) z(j).$$

Therefore, if $S(k)$ is invertible, the batch weighted least-squares estimate is

$$\hat{\theta}(k) = S(k)^{-1} q(k).$$

Information Matrix for the Cell Difference Model

We now specialize the information matrix to the linear regression model used in the reference paper:

$$\Delta U_i(k) = \Delta E_i - I(k) \Delta R_i.$$

Equivalently,

$$z(k) = \phi(k)^T \theta,$$

with

$$z(k) = \Delta U_i(k), \quad \theta = \begin{bmatrix} \Delta E_i \\ \Delta R_i \end{bmatrix}, \quad \phi(k) = \begin{bmatrix} 1 \\ -I(k) \end{bmatrix}.$$

Since the model is **linear in the parameters**, the information relevant to parameter identifiability is entirely contained in the regressors $\phi(k)$. In this case, it depends fundamentally on the current profile $I(k)$.

The information matrix is therefore

$$S(k) = \sum_{j=1}^k \lambda^{k-j} \phi(j) \phi(j)^T,$$

and, substituting the expression of $\phi(k)$,

$$\phi(k) \phi(k)^T = \begin{bmatrix} 1 & -I(k) \\ -I(k) & I(k)^2 \end{bmatrix}.$$

Hence,

$$S(k) = \sum_{j=1}^k \lambda^{k-j} \begin{bmatrix} 1 & -I(j) \\ -I(j) & I(j)^2 \end{bmatrix}.$$

This expression makes clear that the richness of the current excitation determines whether ΔE_i and ΔR_i can be separated.

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Constant-Current Case: Loss of Identifiability

The previous expression also explains why a constant current profile is problematic.

If

$$I(j) = I_0 \quad \forall j,$$

then the information matrix becomes

$$S(k) = \left(\sum_{j=1}^k \lambda^{k-j} \right) \begin{bmatrix} 1 & -I_0 \\ -I_0 & I_0^2 \end{bmatrix}.$$

The matrix inside the product can be written as the outer product

$$\begin{bmatrix} 1 & -I_0 \\ -I_0 & I_0^2 \end{bmatrix} = \begin{bmatrix} 1 \\ -I_0 \end{bmatrix} \begin{bmatrix} 1 & -I_0 \end{bmatrix}.$$

Therefore, it has rank one. Equivalently, its determinant is zero:

$$\det \begin{bmatrix} 1 & -I_0 \\ -I_0 & I_0^2 \end{bmatrix} = 1 \cdot I_0^2 - (-I_0)^2 = 0.$$

Hence, $S(k)$ is singular and cannot be inverted.

This is a crucial point, because RLS relies on the inverse information matrix

$$P(k) = S(k)^{-1}.$$

Therefore, when the current is constant, the identification problem is not well posed: the two parameters cannot be estimated independently.

In the weighted least-squares problem, the information matrix is

$$S(k) = \sum_{j=1}^k \lambda^{k-j} \phi(j) \phi(j)^T.$$

and

$$q(k) = \sum_{j=1}^k \lambda^{k-j} \phi(j) z(j).$$

Therefore, if $S(k)$ is invertible, the batch weighted least-squares estimate is

$$\hat{\theta}(k) = S(k)^{-1} q(k).$$

This expression is **not yet the recursive RLS update**. It represents the estimate obtained by solving the weighted least-squares problem over the full dataset

$$\{z(j), \phi(j)\}_{j=1}^k.$$

The purpose of RLS is to recover this type of estimate recursively, without recomputing the full inverse $S(k)^{-1}$ from scratch at every new sample.

This matrix can be updated recursively as

$$S(k) = \lambda S(k-1) + \phi(k) \phi(k)^T.$$

Indeed,

$$S(k-1) = \sum_{j=1}^{k-1} \lambda^{k-1-j} \phi(j) \phi(j)^T,$$

hence, multiplying by λ ,

$$\lambda S(k-1) = \sum_{j=1}^{k-1} \lambda^{k-j} \phi(j) \phi(j)^T.$$

Adding the new contribution at time k gives

$$S(k) = \lambda S(k-1) + \phi(k) \phi(k)^T.$$

This recursion shows how the information available for parameter estimation evolves as new data become available.

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Similarly to $S(k)$, also $q(k)$ can be obtained recursively.

In the batch weighted least-squares formulation,

$$q(k) = \sum_{j=1}^k \lambda^{k-j} \phi(j) z(j).$$

At the previous time instant,

$$q(k-1) = \sum_{j=1}^{k-1} \lambda^{k-1-j} \phi(j) z(j).$$

Multiplying by λ , we obtain

$$\lambda q(k-1) = \sum_{j=1}^{k-1} \lambda^{k-j} \phi(j) z(j).$$

Adding the new contribution at time k ,

$$\phi(k) z(k),$$

gives

$$q(k) = \lambda q(k-1) + \phi(k) z(k)$$

Therefore, the two batch quantities can be updated as

$$S(k) = \lambda S(k-1) + \phi(k) \phi(k)^T$$

and

$$q(k) = \lambda q(k-1) + \phi(k) z(k).$$

Here, $S(k)$ accumulates information about the regressors, while $q(k)$ accumulates the information linking the regressors to the measured output. Both are needed to recover the batch weighted least-squares estimate

$$\hat{\theta}(k) = S(k)^{-1} q(k).$$

Since the batch solution requires the inverse of the information matrix, RLS introduces

$$P(k) = S(k)^{-1}$$

and updates $P(k)$ recursively instead of recomputing $S(k)^{-1}$ from scratch at every sampling instant.

The recursive update of $P(k)$ is obtained from the recursion of $S(k)$ using the Sherman–Morrison formula. This is the key computational advantage of RLS: once $P(k-1)$ is known, $P(k)$ can be updated directly, avoiding an explicit matrix inversion at each time step.

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$$S(k) = \lambda S(k-1) + \phi(k)\phi(k)^T.$$

RLS does not directly use $S(k)$, but its inverse:

$$P(k) = S(k)^{-1}.$$

Therefore,

$$P(k) = [\lambda S(k-1) + \phi(k)\phi(k)^T]^{-1}.$$

Since

$$P(k-1) = S(k-1)^{-1},$$

we can write

$$S(k-1) = P(k-1)^{-1}.$$

Thus,

$$P(k) = [\lambda P(k-1)^{-1} + \phi(k)\phi(k)^T]^{-1}.$$

Sherman–Morrison Formula

The Sherman–Morrison formula states that

$$(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1 + v^T A^{-1}u}.$$

In our case, identify

$$A = \lambda P(k-1)^{-1},$$

$$u = \phi(k),$$

$$v = \phi(k).$$

Therefore,

$$A + uv^T = \lambda P(k-1)^{-1} + \phi(k)\phi(k)^T.$$

This is exactly the matrix we need to invert.

Computing A^{-1}

Since

$$A = \lambda P(k-1)^{-1},$$

its inverse is

$$A^{-1} = \frac{1}{\lambda} P(k-1).$$

Substituting into Sherman–Morrison gives

$$P(k) = \frac{1}{\lambda} P(k-1) - \frac{\left(\frac{1}{\lambda} P(k-1)\right) \phi(k)\phi(k)^T \left(\frac{1}{\lambda} P(k-1)\right)}{1 + \phi(k)^T \left(\frac{1}{\lambda} P(k-1)\right) \phi(k)}.$$

Simplifying the numerator

The numerator of the fraction is

$$\left(\frac{1}{\lambda} P(k-1)\right) \phi(k)\phi(k)^T \left(\frac{1}{\lambda} P(k-1)\right)$$

so

$$= \frac{1}{\lambda^2} P(k-1) \phi(k)\phi(k)^T P(k-1).$$

Simplifying the denominator

The denominator is

$$1 + \phi(k)^T \left(\frac{1}{\lambda} P(k-1)\right) \phi(k)$$

that is,

$$1 + \frac{1}{\lambda} \phi(k)^T P(k-1) \phi(k).$$

Putting everything over a common denominator:

$$1 + \frac{1}{\lambda} \phi(k)^T P(k-1) \phi(k) = \frac{\lambda + \phi(k)^T P(k-1) \phi(k)}{\lambda}.$$

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Final expression

Therefore, the second term becomes

$$\frac{\frac{1}{\lambda^2} P(k-1) \phi(k) \phi(k)^T P(k-1)}{\frac{\lambda + \phi(k)^T P(k-1) \phi(k)}{\lambda}}$$

which simplifies to

$$\frac{1}{\lambda} \frac{P(k-1) \phi(k) \phi(k)^T P(k-1)}{\lambda + \phi(k)^T P(k-1) \phi(k)}.$$

Hence,

$$P(k) = \frac{1}{\lambda} P(k-1) - \frac{1}{\lambda} \frac{P(k-1) \phi(k) \phi(k)^T P(k-1)}{\lambda + \phi(k)^T P(k-1) \phi(k)}.$$

Factoring out $\frac{1}{\lambda}$, we obtain

$$P(k) = \frac{1}{\lambda} \left[P(k-1) - \frac{P(k-1) \phi(k) \phi(k)^T P(k-1)}{\lambda + \phi(k)^T P(k-1) \phi(k)} \right]$$

This is the recursive update of the inverse information matrix used in RLS.

From the batch weighted least-squares solution, we have

$$\hat{\theta}(k) = S(k)^{-1} q(k).$$

Since

$$P(k) = S(k)^{-1},$$

this can be written as

$$\hat{\theta}(k) = P(k) q(k)$$

This is still the weighted least-squares optimum at time k , written using the inverse information matrix.

From the recursive update of $q(k)$,

$$q(k) = \lambda q(k-1) + \phi(k) z(k),$$

we obtain

$$\hat{\theta}(k) = P(k) [\lambda q(k-1) + \phi(k) z(k)].$$

Now, using the normal equation at time $k-1$,

$$q(k-1) = S(k-1) \hat{\theta}(k-1),$$

and since

$$S(k-1) = P(k-1)^{-1},$$

we get

$$q(k-1) = P(k-1)^{-1} \hat{\theta}(k-1).$$

Therefore,

$$\hat{\theta}(k) = P(k) \left[\lambda P(k-1)^{-1} \hat{\theta}(k-1) + \phi(k) z(k) \right].$$

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Using the identity obtained from the information matrix update,

$$S(k) = \lambda S(k-1) + \phi(k)\phi(k)^T,$$

and multiplying by $P(k) = S(k)^{-1}$, we have

$$P(k)\lambda S(k-1) = I - P(k)\phi(k)\phi(k)^T.$$

Since

$$S(k-1) = P(k-1)^{-1},$$

this becomes

$$P(k)\lambda P(k-1)^{-1} = I - P(k)\phi(k)\phi(k)^T.$$

Substituting into the expression for $\hat{\theta}(k)$, we obtain

$$\hat{\theta}(k) = [I - P(k)\phi(k)\phi(k)^T] \hat{\theta}(k-1) + P(k)\phi(k)z(k).$$

Rearranging,

$$\hat{\theta}(k) = \hat{\theta}(k-1) + P(k)\phi(k) \left[z(k) - \phi(k)^T \hat{\theta}(k-1) \right].$$

The term in brackets is the prediction error:

$$\varepsilon(k) = z(k) - \phi(k)^T \hat{\theta}(k-1).$$

$$\varepsilon(k) = z(k) - \hat{z}(k),$$

$$\hat{z}(k) = \phi(k)^T \hat{\theta}(k-1)$$

is the model prediction of the new output based on the previous parameter estimate.

Hence,

$$\hat{\theta}(k) = \hat{\theta}(k-1) + P(k)\phi(k)\varepsilon(k)$$

Defining the correction gain as

$$K(k) = P(k)\phi(k),$$

the update becomes

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\varepsilon(k).$$

This shows that the RLS correction form is not heuristic: it is an algebraic rewriting of the batch weighted least-squares optimum at time k , using the recursive updates of $P(k)$ and $q(k)$.

Thus, the gain $K(k)$ is the specific correction factor that makes the recursive update reproduce the weighted least-squares solution that would be obtained by processing all data up to time k in batch form.